



Visualization of Data, Understanding and Interpretation

Unit 2 : Advanced Visualization & Storytelling

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

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Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview- static and interactive visualizations



UNIT 2: Advanced Visualization & Story Telling.

In this lecture, we will cover:

- Narrative vs. Exploratory Paradigms: Establish how **static and interactive visualizations** partition control of the cognitive thread between the presenter and the viewer.
- The Physical and Cognitive Interaction Loops: Model selection mechanics **via Fitts' Law**, evaluate the performance penalty of lag, and construct a drill-down epistemic cost hierarchy.
- **Visual Thinking Design Patterns**: Analyze and implement brushing, coordinated multiple views, dynamic queries, and near-neighbor highlighting to navigate high-dimensional spaces.

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Static vs Interactive Visualizations (TB 3, Sec 9.4)

- Static representations restrict the **visual display to pre-computed, unalterable visual structures.**
- Interactive environments construct a dynamic interface that **allows the user to perform physical operations to manipulate the displayed structure in real-time.**
- **Static** visualizations rely heavily on accompanying **spoken or written language to establish the primary narrative context.**
- Interactive visualizations rely on automated, computed responses to user inputs, transforming the interface into an **external cognitive support tool.**
- The transition to interaction becomes necessary when the complexity of the dataset exceeds the spatial resolution limits of a single static monitor.



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Static vs Interactive Visualizations



What are Static Visualizations?

- Static visualizations are commonly seen as **infographics** posted on the **web or printed as handouts**. Usually focused on a specific data story, **users can't go beyond a single view** to explore additional stories beyond what's in front of them. The story is specifically captured in an engaging single page layout.

What are Interactive Visualizations?

- Interactive visualizations are commonly seen on the web only as applications. **Users can select specific data points to build a visualized story of their choosing**. Better known as Data Visualization Apps, these visualizations allow the user to be part of the data visualization process by building a story of their choosing.

Which is better?

The target customer, data story, and **ROI** are 3 important indicators that can help figure out which is best.

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Static vs Interactive Visualizations

Target Customer – This is your audience. How do they prefer to receive and interact with their information? Some audiences prefer static reports where others want to be engaged with their visualized outcome.

Data Story – How complex is the story? Isolating on a specific data story is required for static visualizations. With an abundance of data, having just one static representation may not be enough to really visualize the whole story. An interactive visualization provides the solution for larger data sets with complex stories and can provide relevant information for extended periods of time.

ROI is difficult to measure because the value of a visualization depends on its impact. It can be assessed using three factors:

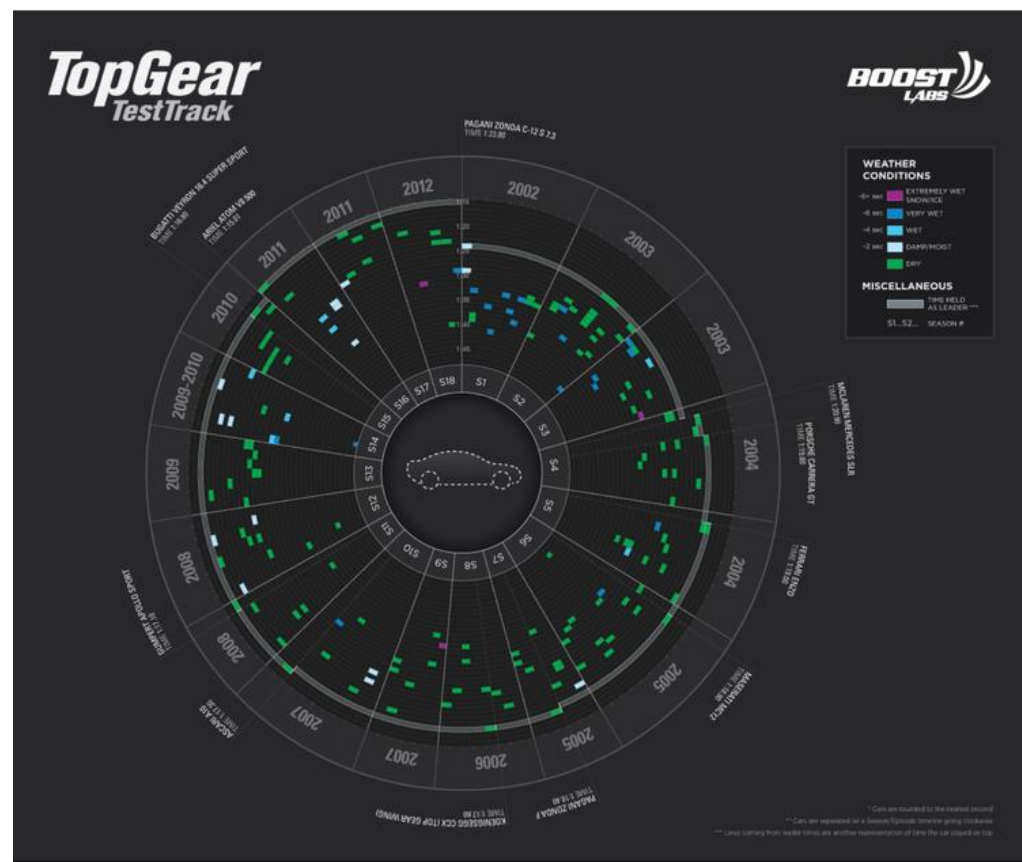
- **Customer Engagement:** Interactive visualizations increase engagement, capture more insights, and improve conversions.
- **Content Quality:** Shareable and engaging visualizations can reach a wider audience and generate more interaction.
- **Content Schedule:** Interactive dashboards are easier to update over time, while static visuals often require rebuilding, increasing costs.

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Static vs Interactive Visualizations

Static visualizations statistically have a lower engagement time frame than Interactive Visualizations for obvious reasons.

An experiment to see which had the longest average session time. The test used two portfolio items:



This Top Gear Selector Wheel (static), contains 185 car lap times.



Same infographic but allow the enthusiast to get more specific information faster and be able to take control over how the data is being digested (interactive).

Static Visualization = 15 seconds

Interactive Visualization = 1 minute and 4 seconds

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Static vs Interactive Visualizations

The test supported our assumptions that **interactive will promote longer engagement times**. In addition, the **volume of unique visits for the interactive visualization were twice the amount compared to the static visualization unique visit count**.

- Interactive Visualization Applications can be built to **display on all devices (web and mobile) and be able to be print friendly**
- Static Visualizations are **at least 5x cheaper** to build than Interactive visualizations
- **Interactive Visualizations can be built so that all users**, regardless of disability status, can access the information

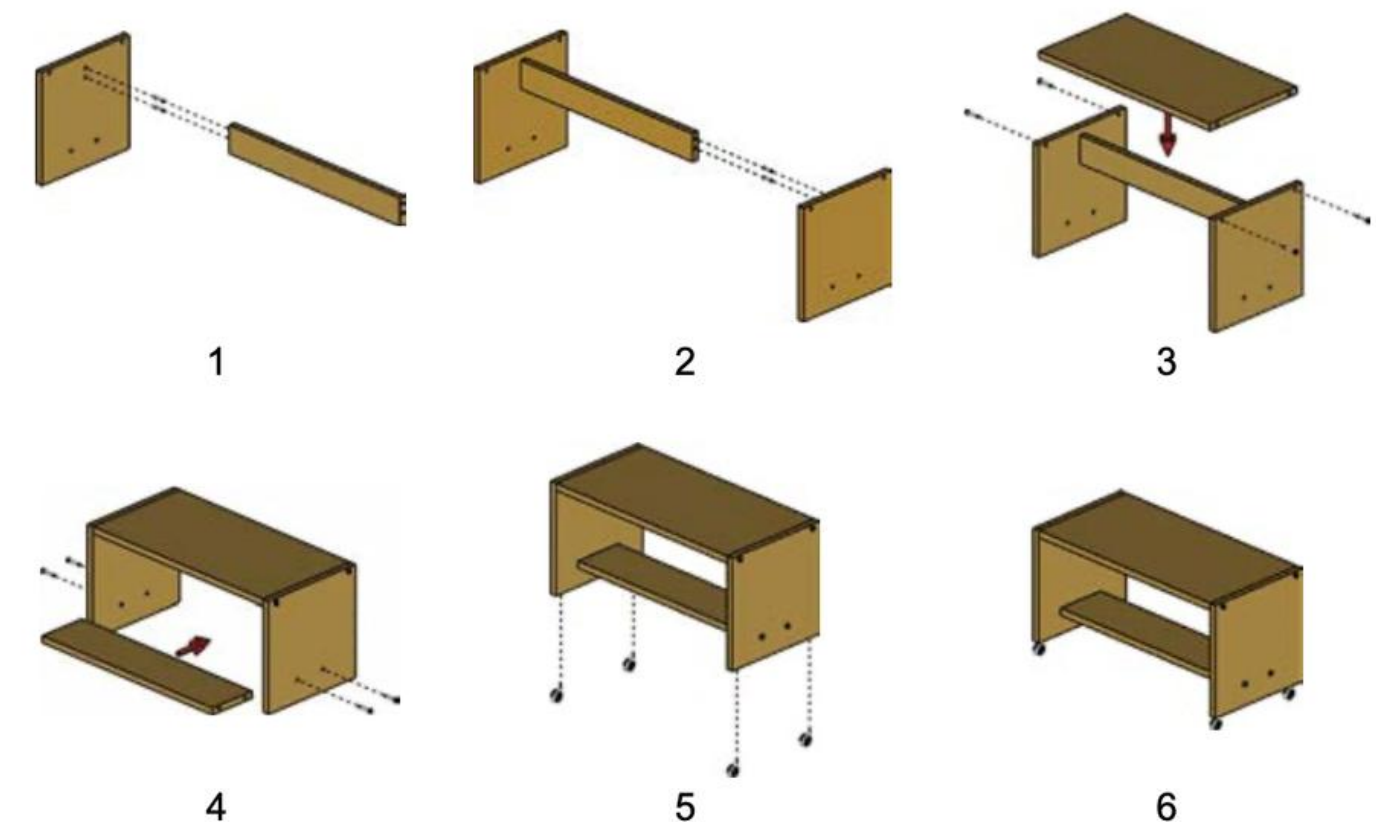
There isn't a one size that fits all solution.

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Narrative Control and the Cognitive Thread-static

(TB 3, Sec 9.4)

- In static data presentation, the author maintains **absolute control over the sequence of visual events and cognitive framing**.
- In **exploratory interaction**, the cognitive sequence is driven dynamically by the analyst's active search goals.
- Slide decks and formal reports function in a hybrid mode, where the narrative structure is authored but navigation remains reader-driven.
- **Overloading an active presentation with unnecessary interactive elements disrupts attention and breaks the presenter's narrative thread.**
- The level of user control should match the context: presentation-oriented visuals require author-driven constraint, whereas diagnostic applications require high user autonomy.



Sequence of static frames that maintain step-by-step narrative continuity.

Narrative control = controlling what the viewer notices, in what order, and what conclusion they are likely to draw.

Cognitive thread = the mental path the viewer follows while understanding the visualization.

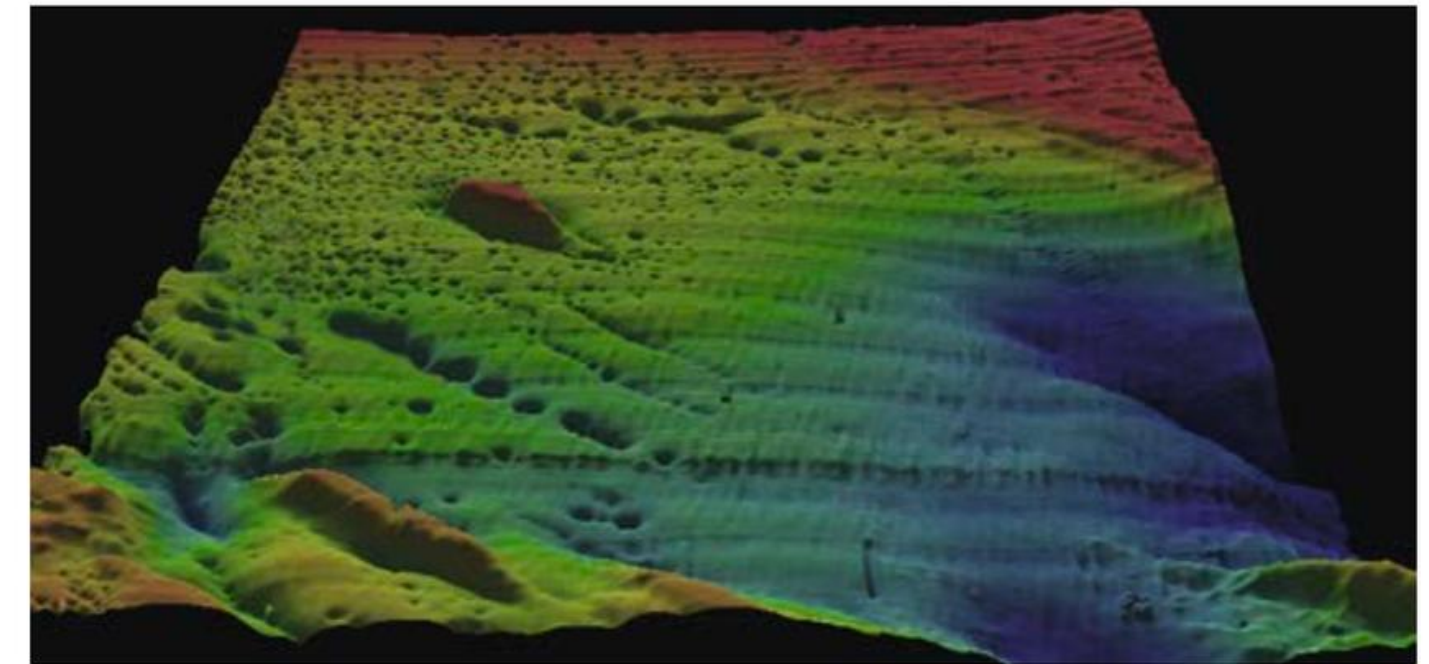
Think of it as a chain of questions and answers in the viewer's mind.

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Limitations of Static Visualizations

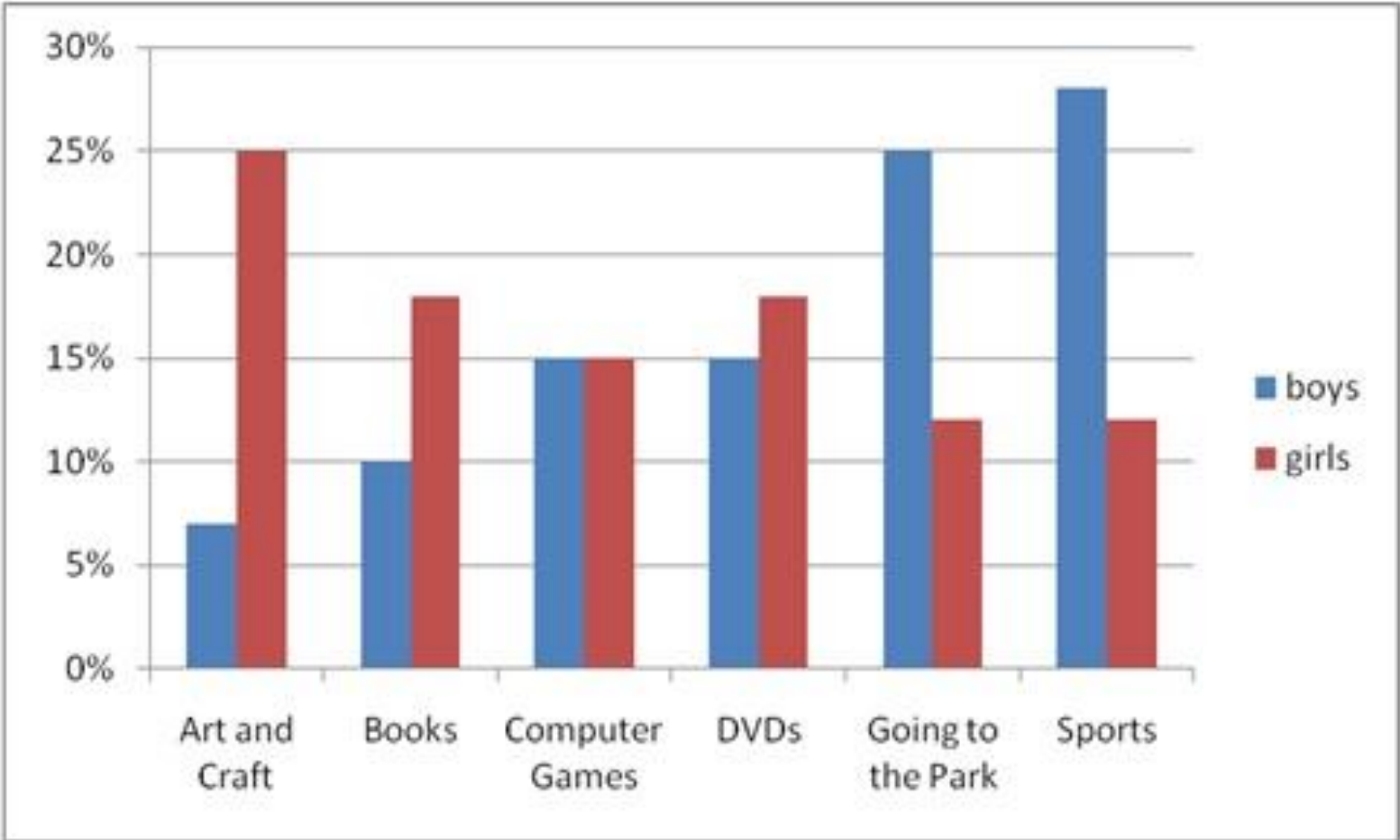
(TB 3, Sec 10.1)

- Static visual channels hit a physical layout wall when trying to display **high-dimensional datasets with dozens of attributes**.
- Representing massive datasets statically causes extreme **visual clutter**, making individual data points and categorical structures impossible to resolve.
- Static graphics **fail to naturally represent complex temporal changes or multi-step causal relationships** without resorting to dense multi-frame arrays.
- Attempting to **display all data simultaneously** forces the eye to **perform exhaustive, high-cost manual scans** across a **cluttered canvas**.
- Interactive scaling resolves these limits by hiding non-essential context and serving detailed parameters strictly on demand.



Passamaquoddy Bay heights field. The visual emphasizes how a static visual projection reveals patterns but cannot support direct attribute querying without interactive tools.

3D height/elevation field of Passamaquoddy Bay. The colors and surface shape represent variations in the data.



Two ways to present data — and when each one wins

S Static

A fixed image that doesn't change once created.

- Examples: printed charts, PDF reports, PNG/JPG images
- Quick to make, easy to share
- Looks the same for every viewer
- Great for print or offline use
- Shows only one fixed view of the data

I Interactive

Lets the viewer explore — hover, click, filter, zoom.

- Examples: Tableau, Power BI, D3.js, Plotly dashboards
- Viewer can filter, zoom, and drill into details
- Reveals patterns a static chart would miss
- Needs more build effort and a suitable platform
- Best for large or complex datasets

At a Glance: Choosing the Right Approach

Aspect	Static	Interactive
Viewer control	None	High — filter, zoom, hover, click
Best for	Reports, presentations, print	Dashboards, exploratory analysis
Common tools	Excel, Matplotlib, Illustrator	Tableau, Power BI, D3.js, Plotly
Build complexity	Lower	Higher
Data depth shown	One fixed view	Multiple views, on demand

Takeaway

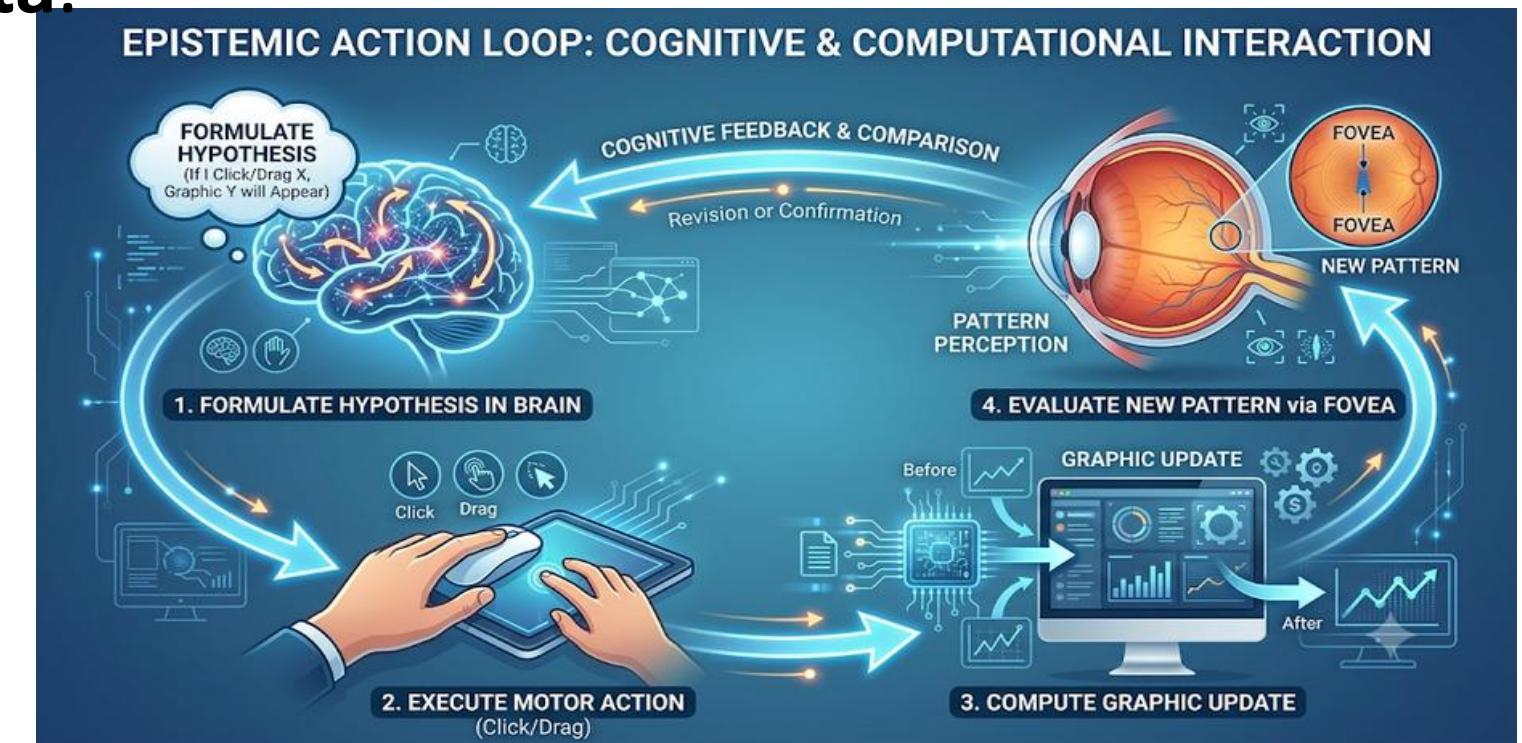
Use static visuals for simple, one-time storytelling — like a report or slide. Use interactive visuals when you want your audience to explore the data themselves, or when the dataset is too rich to capture in a single image.

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Epistemic Actions and Visual Exploration -interactive

(TB 3, Sec 10.1)

- An epistemic action is any physical manipulation of the visualization interface designed to uncover previously hidden information. **An epistemic action is any physical action someone takes on a visualization's interface (clicking, dragging, zooming, etc.) specifically to *reveal information that was hidden*, rather than to change the underlying data.**
- pragmatic actions that physically alter the underlying database, **epistemic actions** alter the visual state to help human cognitive processing.
- Basic epistemic actions include **eye movements, cursor hovers, window selections, and dynamic zoom operations.**
- The primary goal of interaction design is to **reduce the temporal and cognitive costs** of executing these epistemic actions.
- Interactive systems allow analysts to **dynamically execute visual queries, transforming abstract datasets into solvable visual patterns.**



why interactivity in visualizations matters — because small, low-effort user actions (not changing data, just changing its display) let people discover insights faster.

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Epistemic Actions and Visual Exploration

(TB 3, Sec 10.1)



Google Maps / Instagram scrolling

- Pinch-to-zoom on a map is a **pragmatic vs epistemic action** in action: you're not changing the city's actual layout (pragmatic), you're just changing what's *visible* to help your brain understand it (epistemic).

Scrolling through a messy Excel sheet or a dashboard (e.g., checking cricket stats, stock prices, or Spotify Wrapped-style data)

Hovering over a bar in a chart to see its tooltip value = a textbook **epistemic action** — you didn't change the data, you revealed hidden information.

YouTube Analytics

A content creator has a dashboard showing views, watch time, and subscribers.

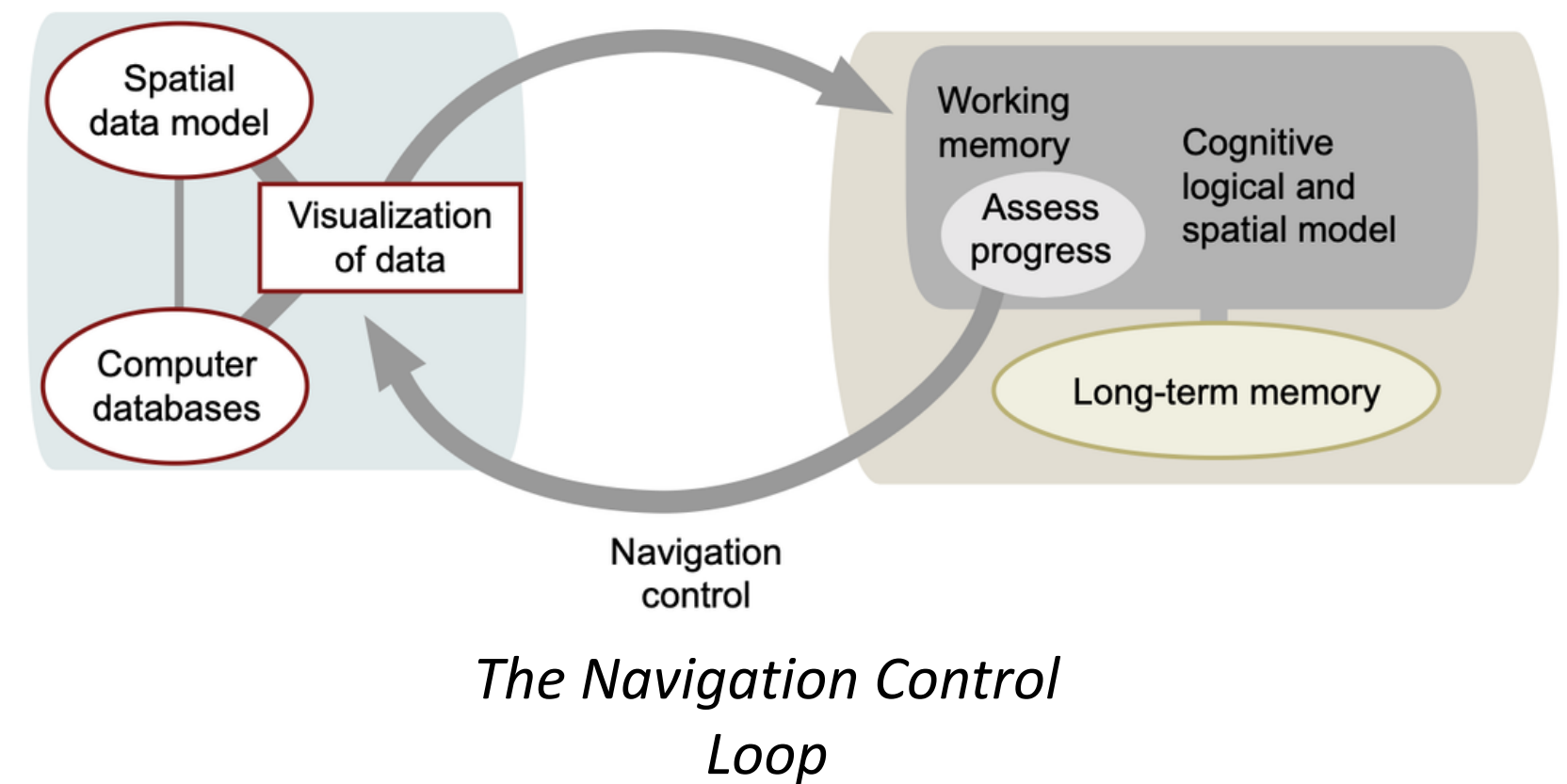
Select **Last 28 days** → **Epistemic**

Edit/delete an incorrect analytics record → **Pragmatic**

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The Human-Computer Navigation Control Loop (TB 3, Sec 10.3)

- Human visual navigation is modeled as a **set of interlocking feedback loops** that operate across distinct cognitive timescales.
- The **lowest loop** coordinates physical **eye-hand movements**, demanding instant, sub-second visual updates to maintain tracking stability.
- The **intermediate loop** coordinates **spatial navigation**, allowing users to build a cognitive map of the data landscape.
- The **highest loop** drives **analytical problem-solving**, where hypotheses are formed, tested, and revised through active visual manipulation.
- Delays or disruptions in any of these feedback loops instantly degrade the user's analytical performance and break their train of thought.



model that explains how humans interact with an interactive visualization.

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The Human-Computer Navigation Control Loop (TB 3, Sec 10.3)



Google Maps to navigate to a new café in an unfamiliar part of the city

Loop 1 — Eye-Hand Coordination (fastest, sub-second)-As you ride, your eyes are locked on the little blue arrow (your position) and the road ahead. Your hand makes tiny adjustments to your phone mount angle or taps to keep the screen active.

Loop 2 — Spatial Navigation (intermediate, seconds)

As you keep riding, you start building a **mental map** of the area — "okay, I already passed the big yellow building, so the café should be after this next signal." You're not staring at the phone constantly anymore; you glance at it occasionally because your brain is now holding a spatial model of the route.

Loop 3 — Analytical Problem-Solving (slowest, deliberate)

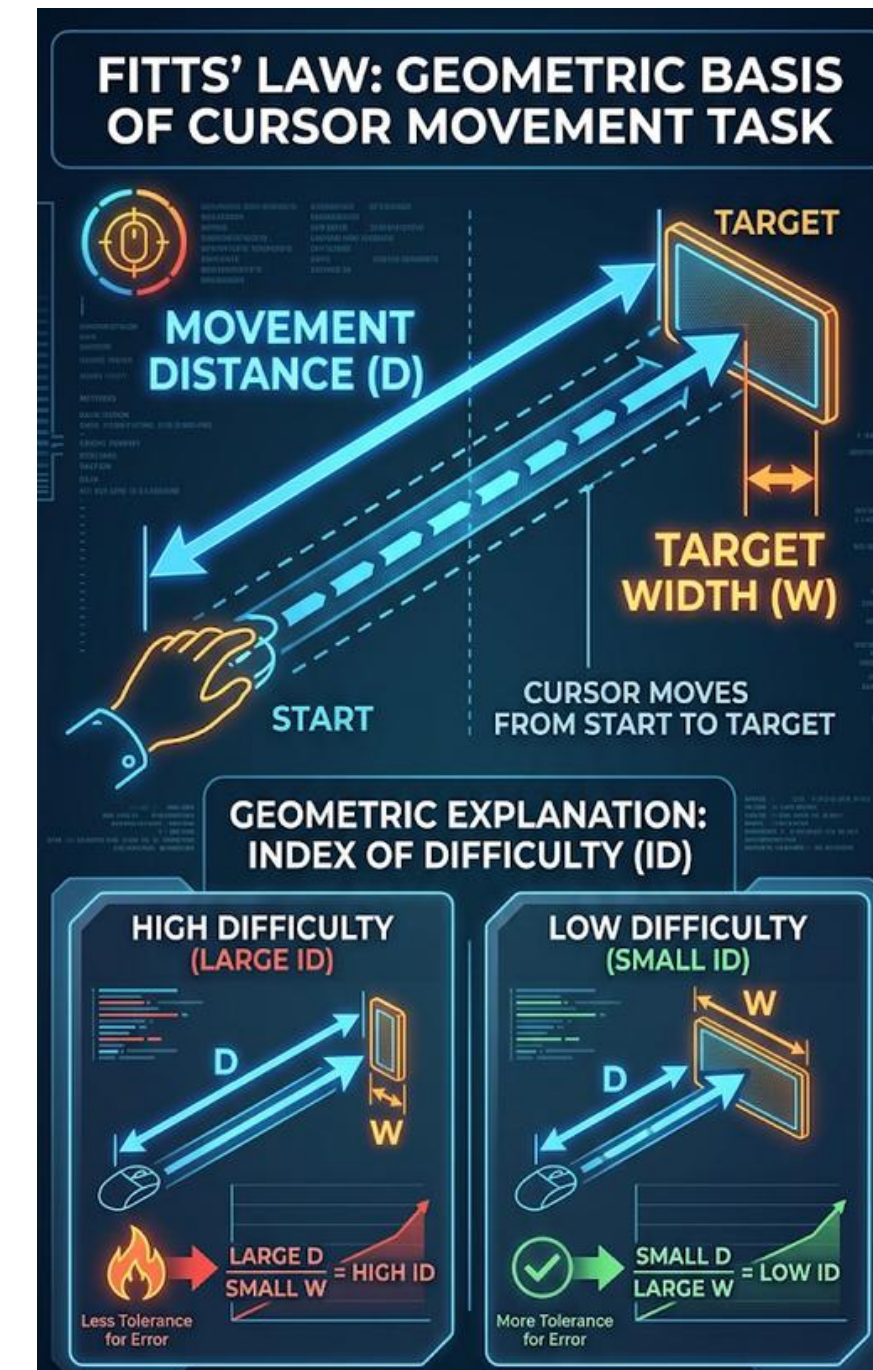
Suddenly Maps says "arriving in 2 minutes" but you don't see any café — just a residential building. Now you're actively reasoning: *"Maybe the entrance is around the back? Maybe the pin is slightly off, like it usually is for small shops? Let me zoom in and check street view."* You form a hypothesis, test it (zoom, rotate map, look at satellite view), and revise based on what you see.

Fitts' Law provides a rigorous mathematical model of the **time required to move a cursor to a target on a screen**:

$$\text{Movement Time} = a + b \log_2 \left(\frac{D}{W} + 1.0 \right)$$

- The index of difficulty is determined by the ratio of target distance (D) to target width (W).
- The index of performance (1/b) represents the human motor system's processing throughput, typically averaging 4 bits per second for mouse
- To minimize selection time, critical interactive targets must be designed with larger target areas or placed closer to the user's current cursor position.

This law provides a standardized framework for evaluating and optimizing the physical usability of visualization interfaces.



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Fitts' Law

(TB 3, Sec 10.2)



Fitts' Law provides a rigorous mathematical model of the time required to move a cursor to a target on a screen:

$$\text{Movement Time} = a + b \log_2 \left(\frac{D}{W} + 1.0 \right)$$

a and **b** are **empirically determined constants** — they aren't fixed universal numbers, but are found experimentally for a *specific input device* (mouse, touchscreen, trackpad, stylus, etc.)

a (the intercept)

Represents the **fixed start/stop overhead** of a movement — the small, constant delay that exists regardless of how far or how small the target is (e.g., reaction time, the time to initiate movement, or the mechanical/device-specific delay before motion even starts).

b (the slope)

Represents how much **extra time is added per unit of difficulty** — essentially, how "sensitive" the device/user is to increasing distance or decreasing target size.

Its inverse, **1/b**, is called the **index of performance** — measured in **bits per second**, representing the throughput of the human motor system for that device.

A smaller *b* means the person/device can handle increasing difficulty more efficiently (less time penalty per added bit of difficulty).

An interface designer is testing a new mouse on a computer screen.

The user needs to click a button located 16 cm away from the current cursor position. The button width is 2 cm. Through previous testing, the designer established the system constants as $a = 0.2$ seconds and $b = 0.1$ seconds/bit.

Using Fitts's Law, what is the predicted Movement Time (MT) required for the user to click the button?

Solution

Formula for Index of Difficulty:

- $ID = \log_2(D/W + 1)$

ID Calculation:

- $\log_2(16/2 + 1) = \log_2(9) \approx 3.17$ bits

Formula for Movement Time:

- $MT = a + b(ID)$

MT Calculation: $0.2 + 0.1 \times 3.17 = 0.517$ seconds

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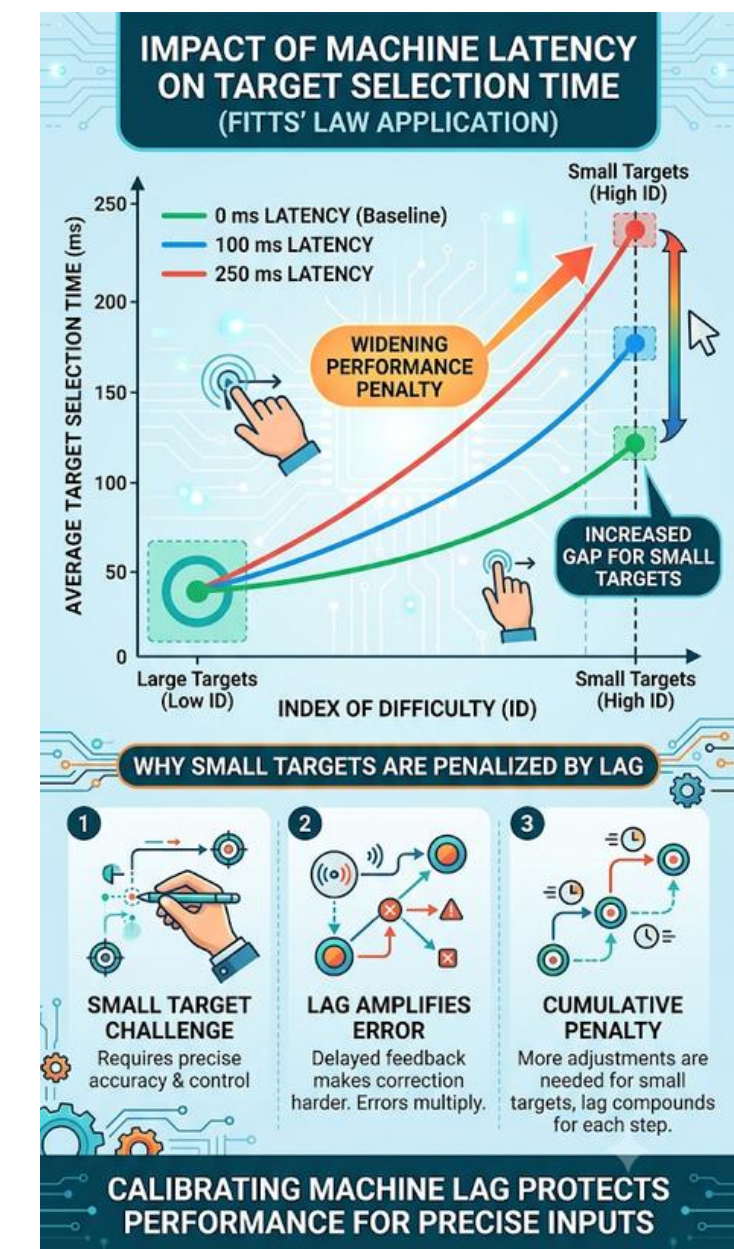
The Perceptual Penalty of Machine Lag (TB 3, Sec 10.2)

Machine lag is the physical delay between a user's motor action and the corresponding visual update on the screen.

Incorporating lag modifies the Fitts' Law movement time equation to account for system latency:

$$\text{Movement Time} = a + b(\text{HumanTime} + \text{MachineLag}) \log_2 \left(\frac{D}{W} + 1.0 \right)$$

- The performance penalty of lag scales non-linearly, causing disproportionately larger delays as interaction targets get smaller. (Non-linear penalty, worse for smaller targets)
- Even a fraction of a second of lag can cause users to overcorrect their cursor movements, breaking natural eye-hand coordination.
- Real-time interfaces must prioritize **keeping latency below 100 milliseconds** to preserve smooth interaction during highly analytical tasks.

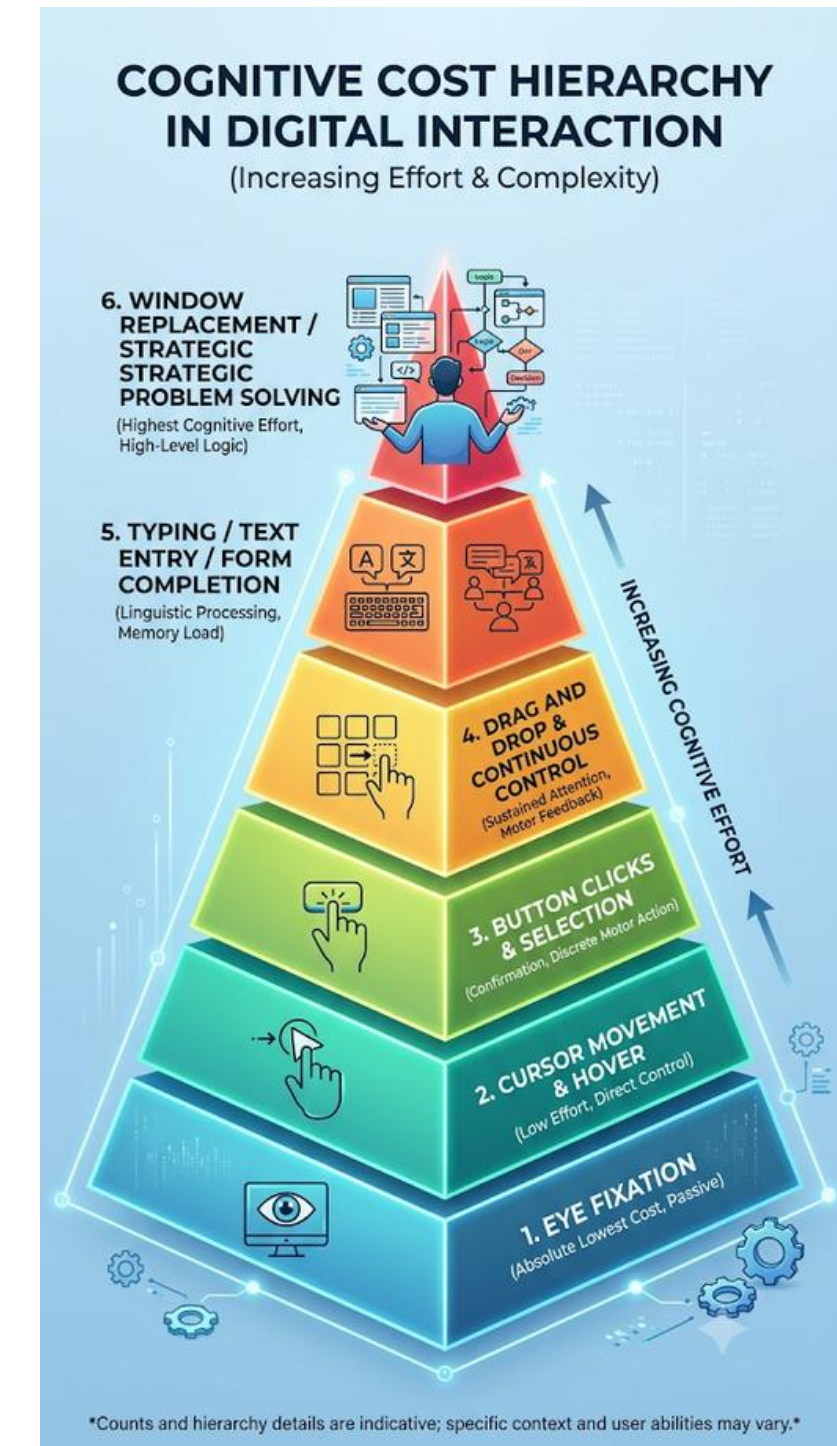


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The Drill-Down Epistemic Cost Hierarchy

(TB 2, Sec 10.4)

- Eye Fixation (lowest cost): Moving attention to an adjacent screen element takes only 100 to 200 milliseconds, allowing up to 3 queries per second.
- Hover Query (low cost): Simply moving the cursor over a data element instantly reveals pop-up details, averaging about 1 second of user time.
- Click-to-Open Panel (moderate cost): Clicking on an element opens a persistent side or overlay panel, costing approximately 2 seconds.
- Window Replacement (highest cost): Opening a new window that entirely replaces the current view disrupts context and forces a heavy cognitive restart.
- Design Rule: Always prioritize lower-cost interaction patterns to preserve the analyst's short-term visual working memory.



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The Focus-Context Problem

(TB 3, Sec 10.3)



- The focus-context problem arises from the **mismatch between the need to inspect fine-grained details and the need to maintain an overall structural view of the dataset.**
- In large information spaces, detail views must be visually tied to their broader environmental context to remain interpretable.
- This cognitive challenge spans multiple disciplines, including geographic maps, temporal datasets, and software system architectures.
- A perfect design allows users to smoothly transition between local details and global overviews with minimal cognitive disruption.
- We can solve this problem using three primary interaction models: **multiple linked windows, geometric distortion, or rapid zooming.**

You can't see fine detail and the big picture at the same time on one screen — you need a way to switch between "zoomed in" and "zoomed out" without getting lost.

Google Maps — when you zoom into your street (detail), you lose sight of the city (context). A mini-map in the corner, or smooth zoom animation, solves this by keeping you oriented as you switch between the two.

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Spatial Navigation Metaphors for 3D Interfaces (TB 3, Sec 10.2)- solution 1

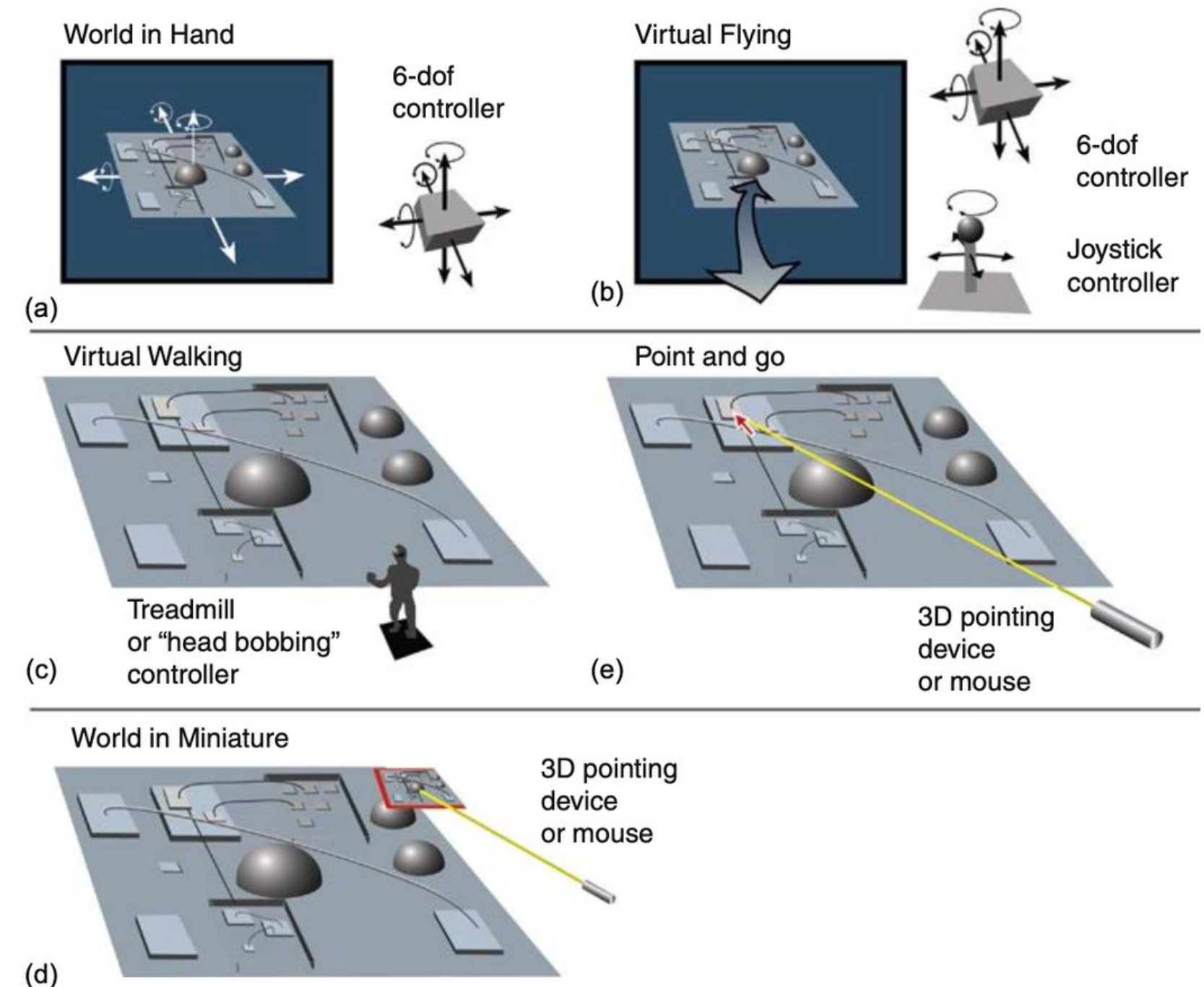
- The Power of Metaphors: Interactive metaphors provide users with familiar mental models to help them navigate complex 3D virtual spaces efficiently.

- World-in-Hand: The user metaphorically grabs and manipulates a compact 3D data object as if holding it, which is highly efficient for localized inspection.

- Flying: The user flies through a large terrain in an unconstrained way, which works best when speed automatically scales to the surrounding detail.

- Point and Go: The user points directly to a target destination, and the camera teleport-walks them there smoothly along a safe path.

- Degrees of Freedom Constraints: Minimizing the degrees of freedom to only what is necessary (e.g., locking roll and yaw) drastically reduces navigation errors.



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Spatial Navigation Metaphors for 3D Interfaces (TB 3, Sec 10.2)- solution 1



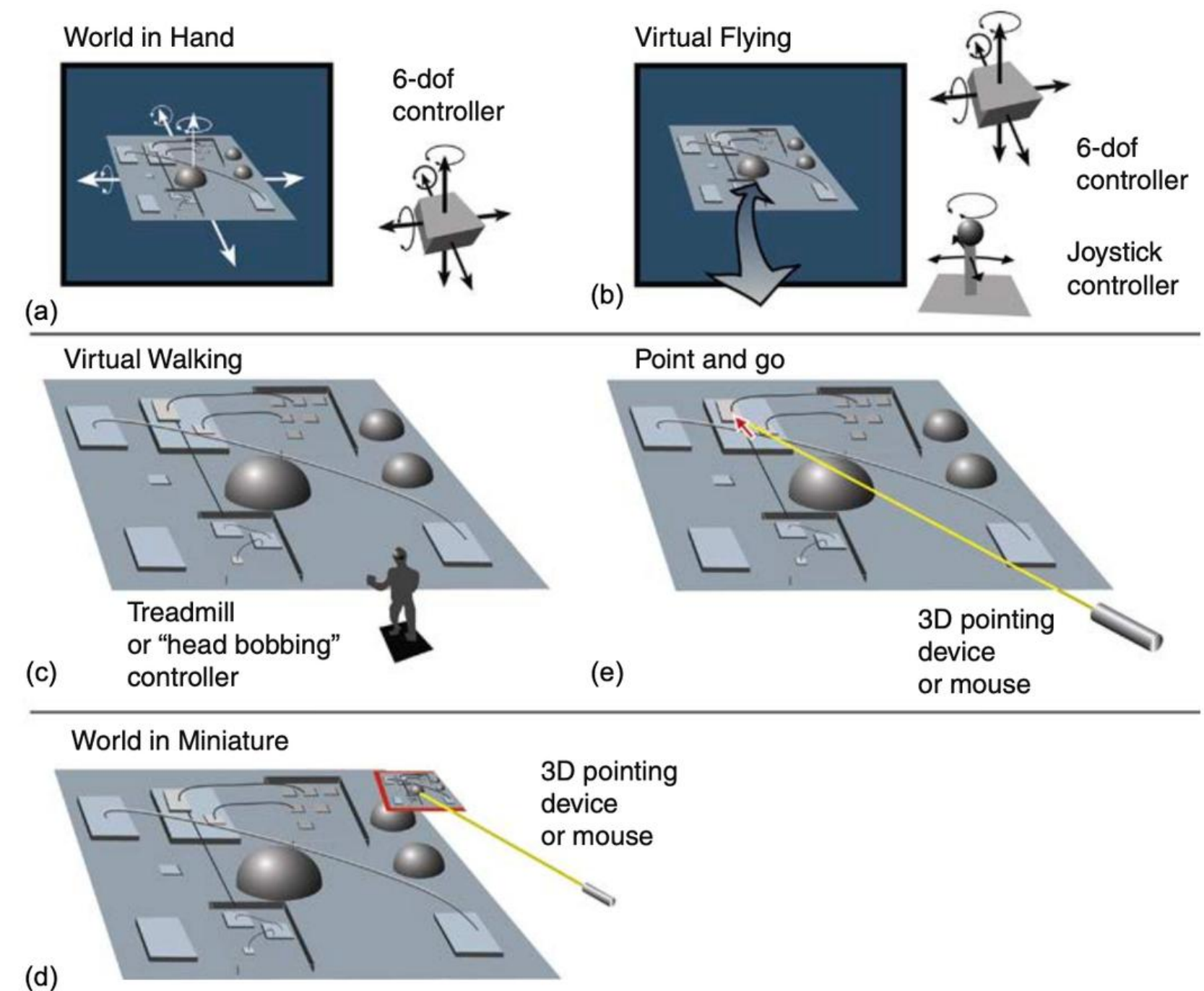
World-in-Hand — you hold the whole object in your hand and rotate it. Great focus (you see detail up close), but you lose the surrounding environment (no context).

Flying — you fly freely through the space. Good context (you see the whole terrain), but hard to focus on one specific detail while moving fast.

Walking — you move at ground level like a real person. Natural, but slow and limited view.

Point and Go — you point where you want to go, and it smoothly takes you there. Balances both — you get moved to your focus point without losing track of the path.

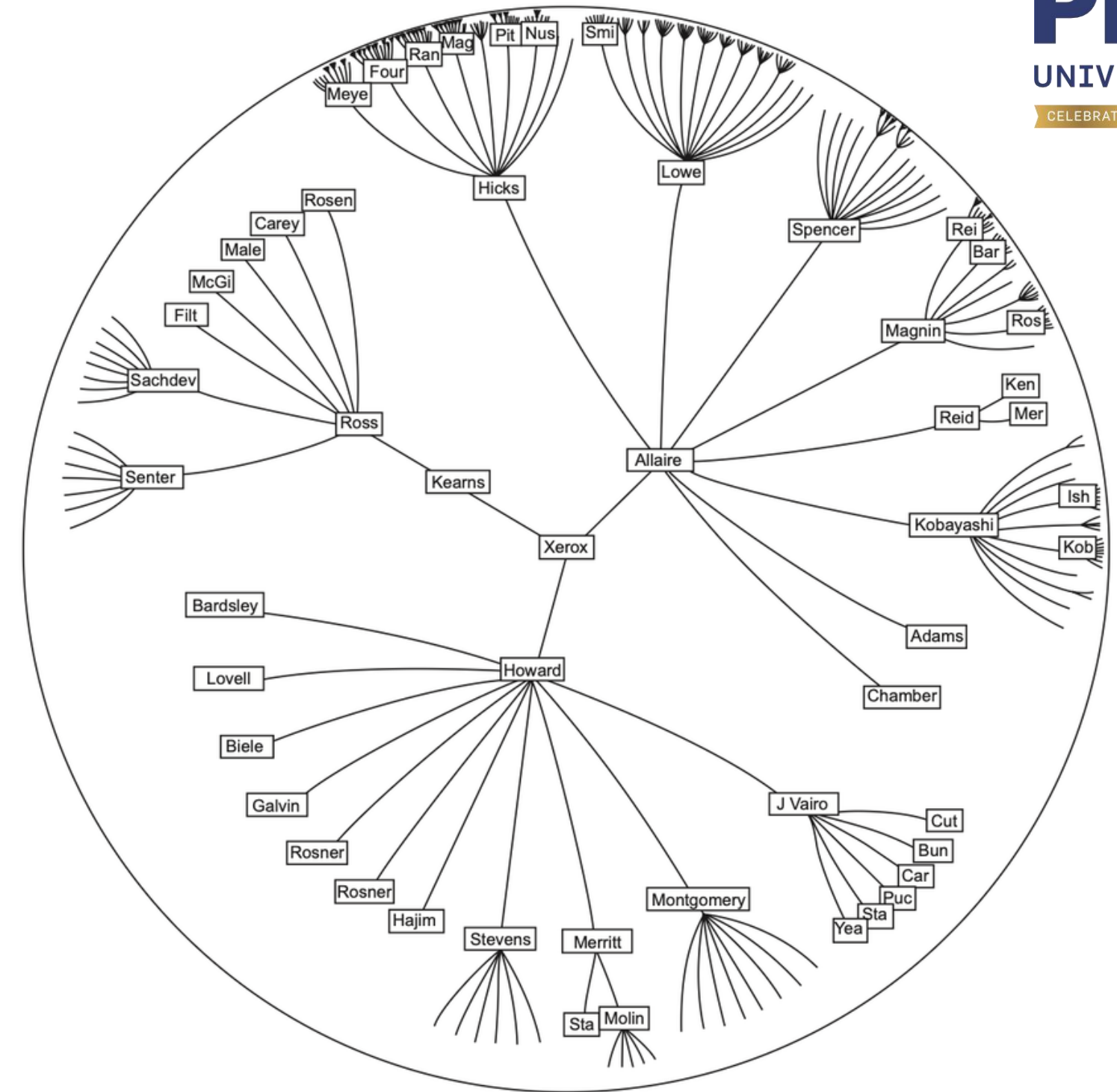
World in Miniature — you get a tiny model of the whole space in your hand to navigate the big one. Strong context (see everything), weaker focus (details are tiny).



Good 3D interface design means **picking (or combining) the metaphor that matches the user's current goal**

Spatial Distortion: Geometric Fisheye Views (TB 3, Sec 10.2) – solution 2

- Geometric Distortion: Fisheye distortion expands the visual space allocated to a focal target while compressing the surrounding data.
- Hyperbolic Browsing: Complex hierarchies can be projected onto a hyperbolic plane **where dragging a node to the center automatically expands its branches.**
- Perceptual Limitations: Distorting spatial layouts can disorient users by shifting landmarks to completely new locations on the screen.
- Scale Limits: **Geometric fisheye distortion should be constrained to a maximum scaling factor of five to ensure patterns remain recognizable.**
- Visual Preservation: The system must preserve topological features so that spatial relationships remain interpretable during distortion.



Hyperbolic tree browser from Lamping et al. (1995). The focus can be changed by dragging a node from the periphery to the center.

macOS Dock magnification

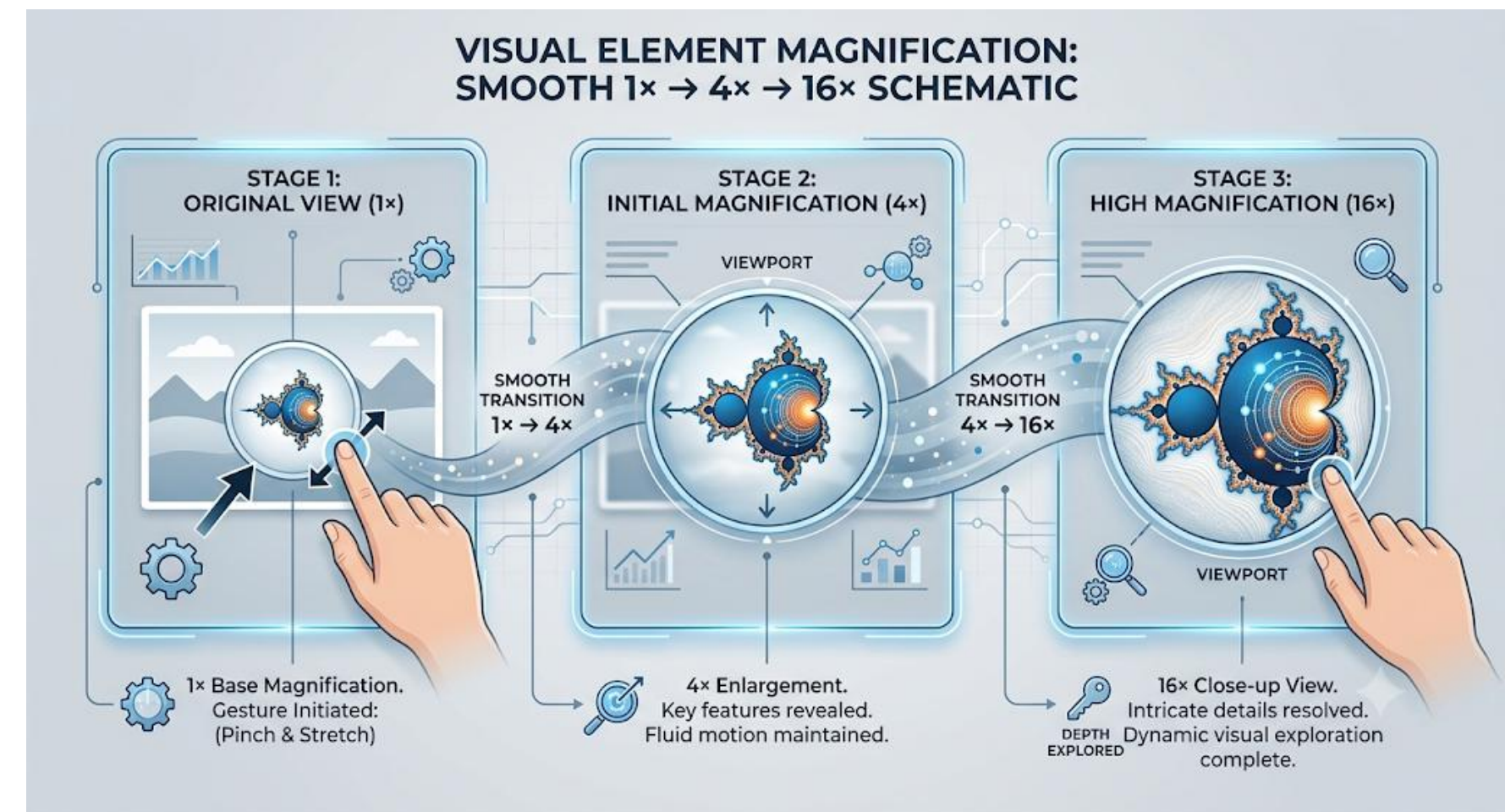
When you move your mouse over the row of app icons at the bottom of a Mac, the icon directly under your cursor grows large (easy to identify and click), while the icons on either side shrink slightly and compress toward the edges — without any icon actually disappearing or the dock changing its overall length.

- **Focal point (cursor position)** = magnified, like the center of a fisheye lens
- **Surrounding icons** = compressed but still visible, like the edges of a fisheye photo
- You always keep the full context (every app is still there, still in order) while getting detail (a big, clear icon) exactly where you're looking

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Rapid Zooming and Google Earth Mechanics (TB 3, Sec 10.3)- solution 3

- Rapid zooming bypasses the spatial distortions of geometric fisheye views by enabling smooth, continuous scale transitions on a single viewport canvas.
- Smooth temporal animations are required during zoom transitions to help the observer maintain object identity and global context across scale boundaries.
- A transition speed of under one second prevents visual search fatigue and supports the seamless integration of detailed views and broad contexts.
- Standard zoom scaling rates should default to a factor of three to four times magnification per second to optimize human perceptual tracking.
- Providing visual coordinate grids or a persistent background minimal prevents the user from becoming disoriented in large virtual spaces.



Instead of distorting the layout (like fisheye did) or showing two separate synced windows, this approach keeps one single continuous view and just smoothly zooms in/out through scale — pinch-to-zoom style (as shown in the image: 1x → 4x → 16x).

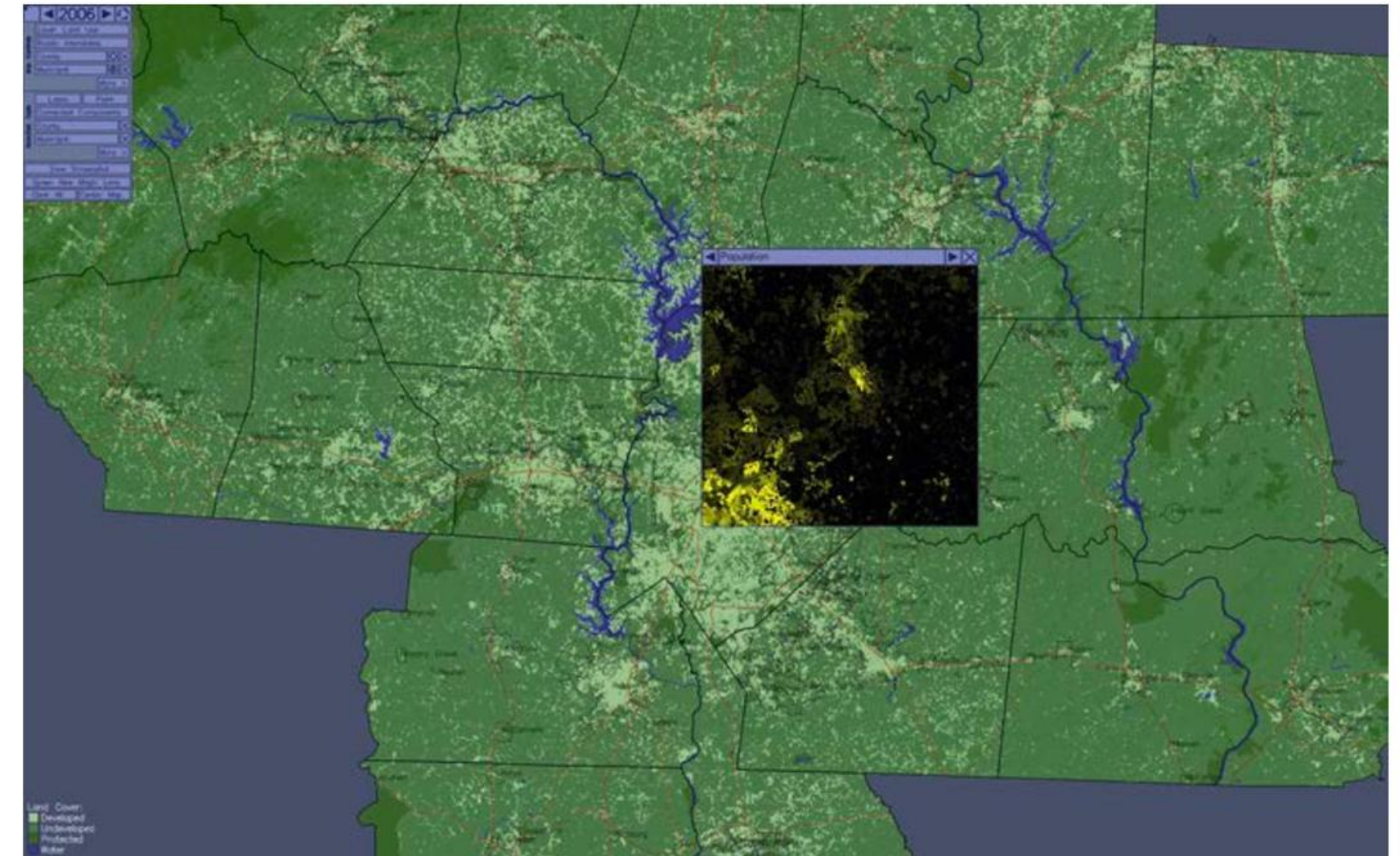
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Interactive Magic Lenses

(TB 3, Sec 10.3)

- A magic lens functions as a dynamic, coordinate-linked visual filter that overlays and reveals hidden data attributes over the current focal area.
- Coordinated two-handed interfaces increase productivity by dedicating lens positioning to the non-dominant hand and detailed selection to the dominant hand.
- Transparent toolglass palettes minimize cognitive friction by showing interactive filtering options directly over the visual elements being modified.
- Offset lenses linked by connector lines prevent the active focus region from directly obscuring adjacent or surrounding environmental context.
- Overlaying population densities directly over physical geographic maps illustrates how lenses seamlessly integrate abstract variables with underlying spatial layouts.

A "magic lens" is a movable window you drag over a visualization that reveals a *different layer* of data underneath the same spot — without replacing or leaving the main view.

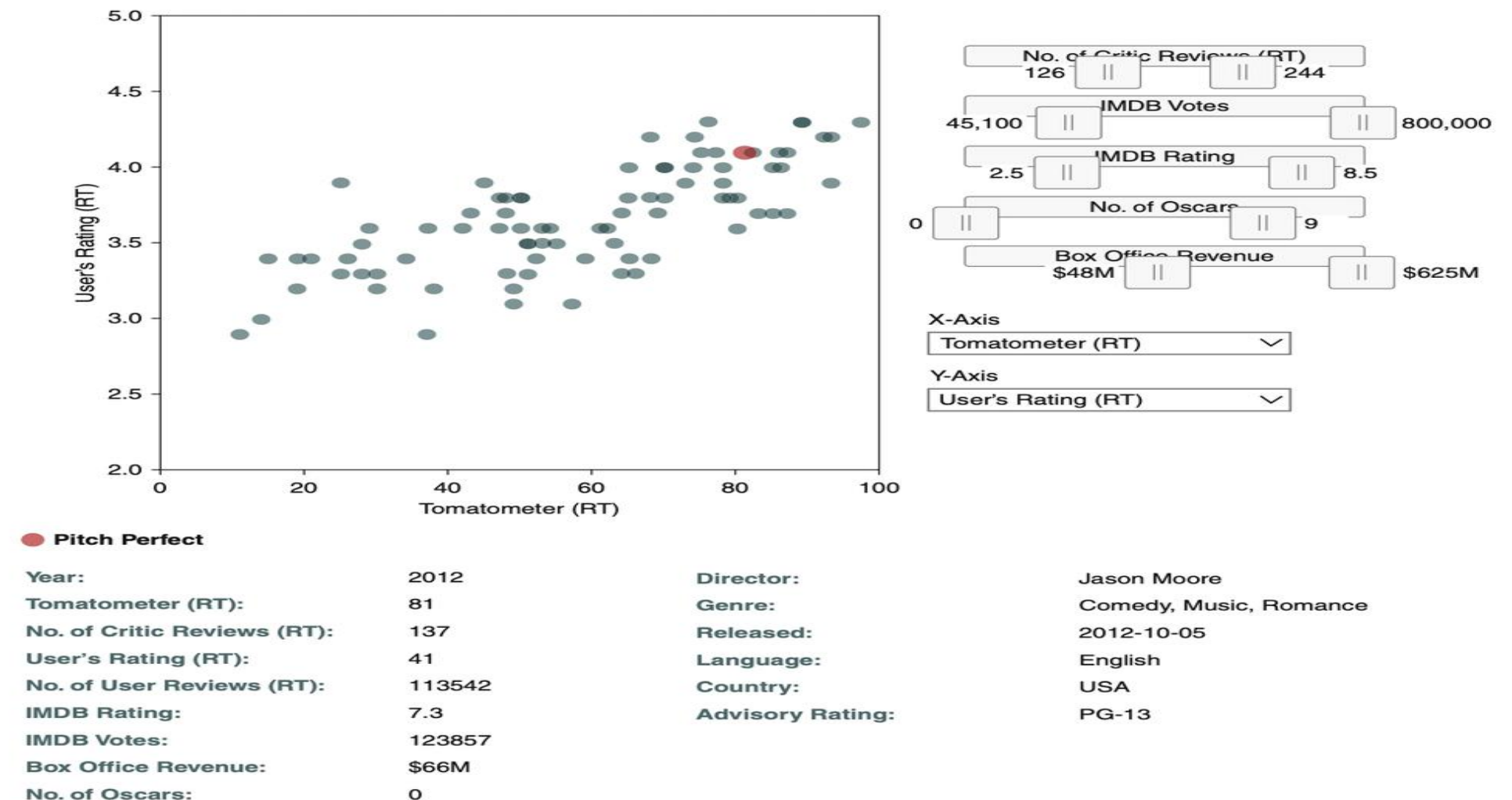


A magic lens showing population density with a background map showing land-use patterns.

Visualization of Data, Understanding and Interpretation

Interacting with Non-Spatial Data: Dynamic Queries (TB 3, Sec 10.4)

- Dynamic querying allows real-time interactive partitioning of highly multidimensional datasets using double-ended range sliders.
- Continuous slider manipulation filters out unwanted data points instantly, allowing target visual structures to emerge from the noise.
- Display update response times must remain below 100 milliseconds to preserve the user's cognitive flow during continuous visual scanning.
- Up to 100,000 multi-attribute data points can be navigated effectively using five or fewer synchronized query sliders.
- Visual information scent must be embedded into slider tracks to indicate where dense clusters of data exist before filtering occurs.



Pitch Perfect (highlighted in red) has a Tomatometer of 81 — meaning 81% of critic reviews were positive — paired with a User's Rating of 4.1, showing fairly strong agreement between critics and audiences.

Aperture JS movie database dashboard with double-ended range query sliders filtering ratings and box office revenue in real-time.

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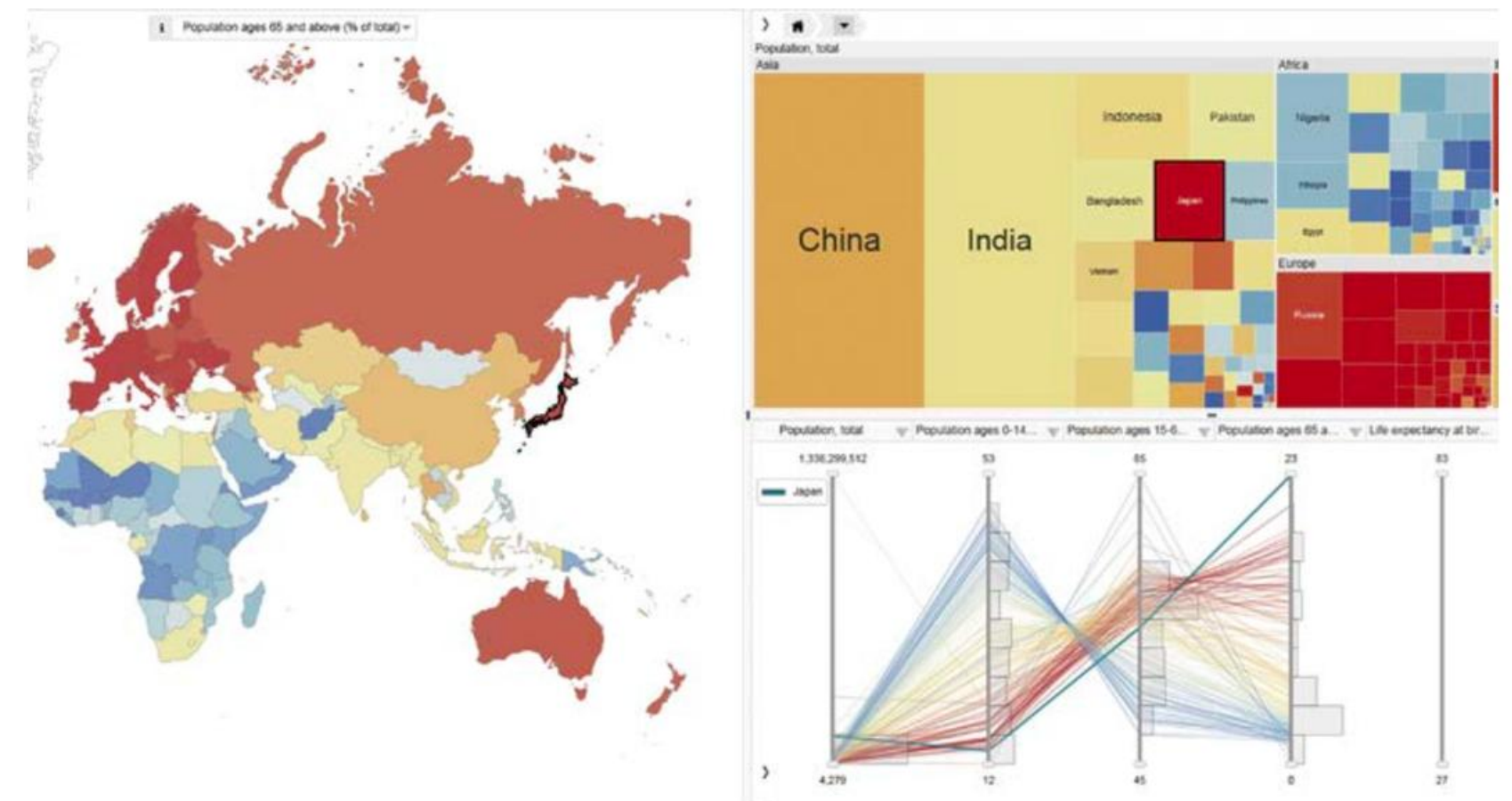
Coordinated Multiple Views and Cross-View Brushing (TB 3, Sec 12.5)

- **Coordinated Multiple Views –same dataset in several chart formats**

Composite visualizations display the same multi-dimensional dataset across several different adjacent chart formats simultaneously.

- Cross-view brushing highlights a user-selected subset of data points across all open windows to reveal complex multi-attribute correlations.(interaction that *links* those views together)
- Side-by-side eye movements replace heavy cognitive working memory loads when synchronized views show related parameters simultaneously.
- **Muting(dimming) the contrast of unselected categories** is a highly effective highlighting strategy that makes the selected subset pop out preattentively.
- **Maintaining absolute consistency in category colors and shapes** across all charts ensures immediate mapping of concepts during transitions.

Pattern that works for both spatial and non-spatial data simultaneously,



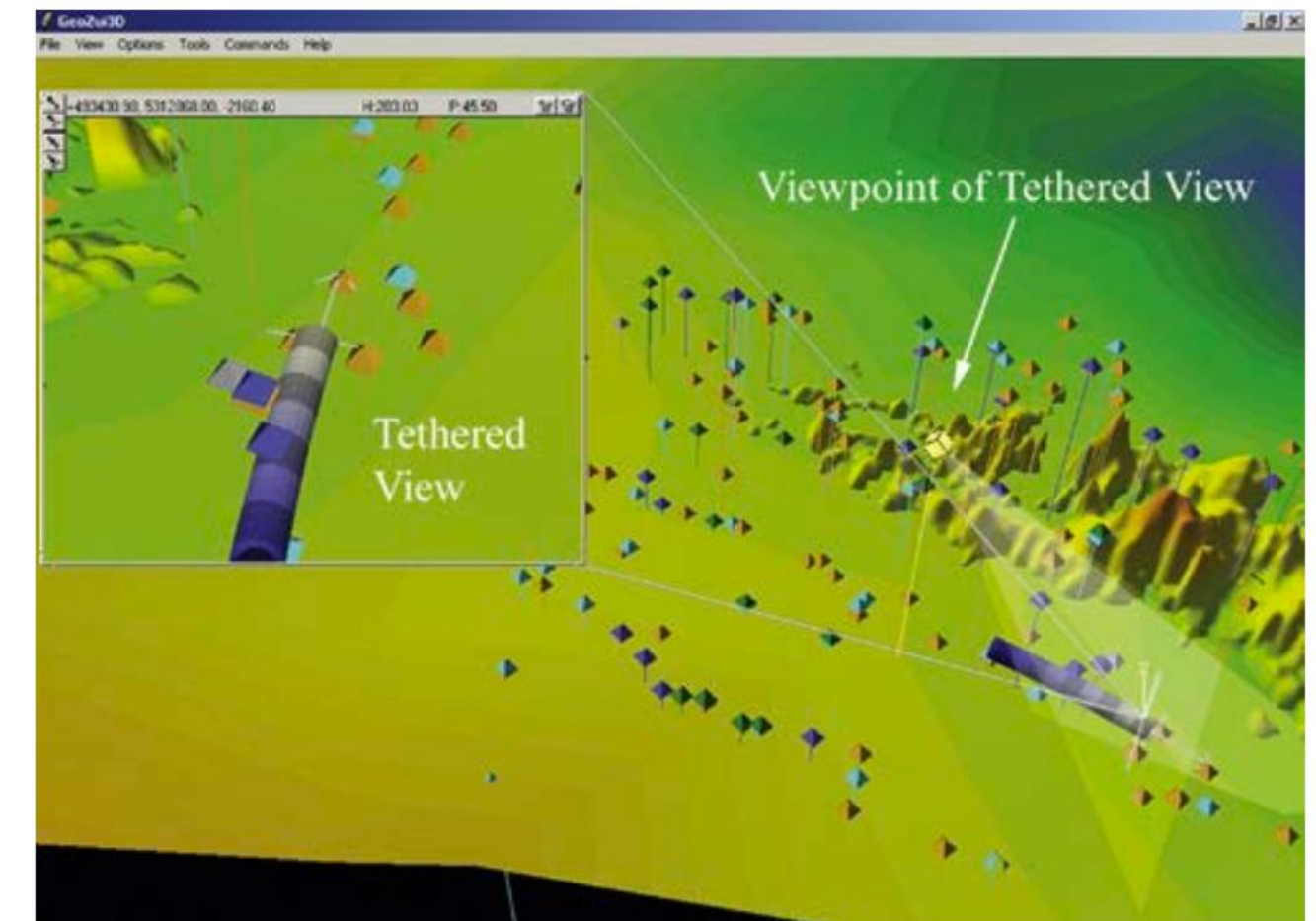
OECD regional dataset where selecting Japan highlights its metrics in the map, treemap, and parallel coordinate views.

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Visual Momentum and Slide Transitions

(TB 3, Sec 9.4)

- Visual momentum measures how easily an observer can link information across consecutive camera cuts or sudden display transitions.
- Grouping **identical visualization formats** sequentially minimizes the cognitive re-orientation cost of learning new coordinates and axes.
- Using persistent visual anchors like constant scales, grids, and labeled axes helps the brain track visual entities across slides.
- **Inset overview (minimaps)** provide continuous spatial context and orientation cues when users zoom deeply into detailed focal areas.
- Smooth camera transition paths prevent disorientation in 3D data spaces by maintaining directionality and scale changes during movement.



Tethered magnifier window utilizing coordinate frames to preserve spatial momentum.

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Network Zooming and Degree-of-Relevance Highlighting

(TB 3, Sec 12.5)



- **The solution,**

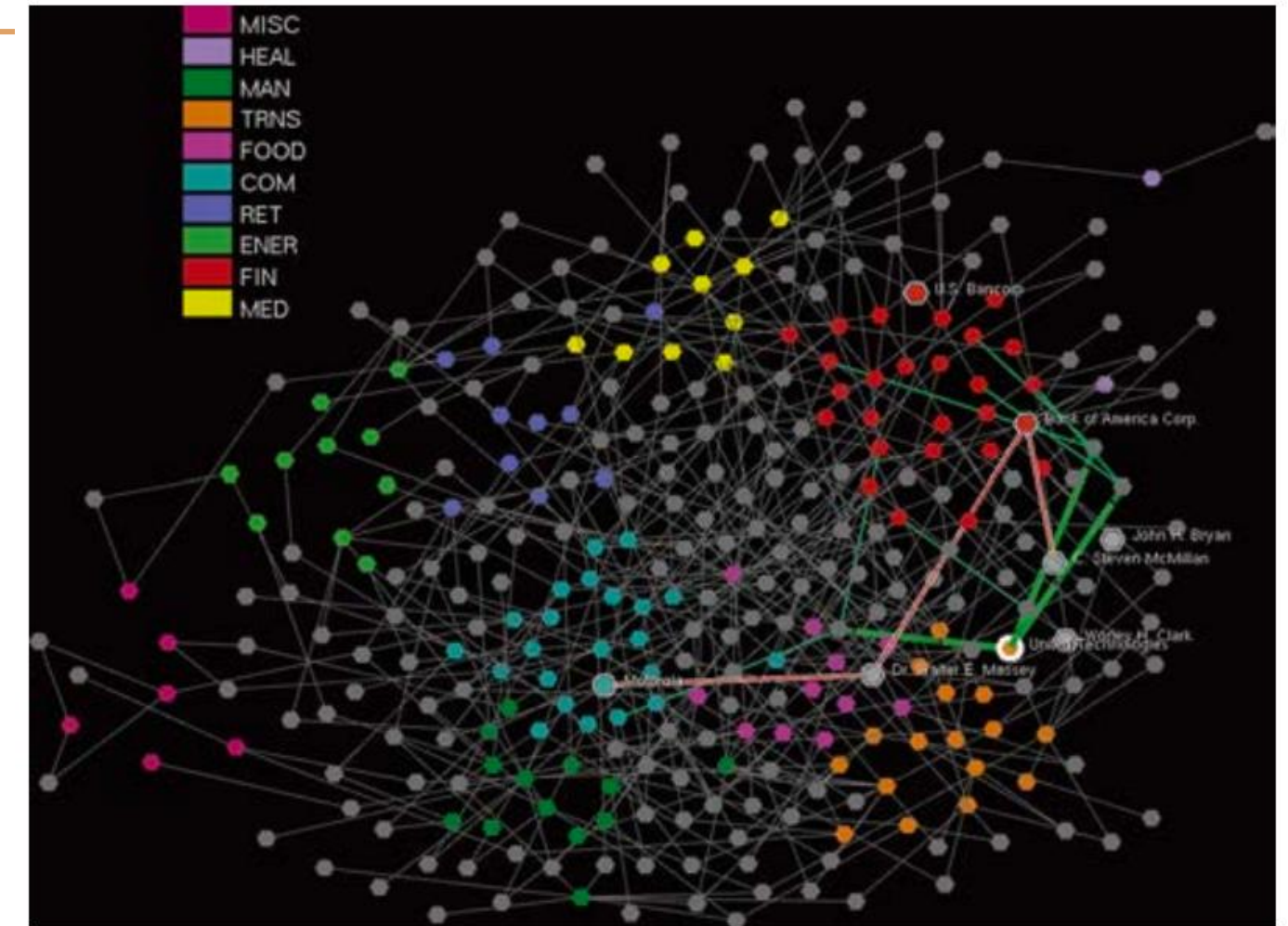
Degree-of-relevance highlighting enables the analysis of dense network diagrams by dynamically **fading out currently irrelevant nodes and links.**

- **Selecting or hovering** over a node triggers a breadth-first search to highlight connected entities within a small path radius.

- **Limiting highlighted path lengths to two or three links prevents** the screen from becoming overloaded with secondary relationships.

- **High-degree nodes with thousands of edges** should be isolated using database search inputs rather than direct manual expansion.

- **Dynamic highlighting bypasses static layout limits** by allowing edge crossings to remain readable through localized user interactions.



MEgraph network display utilizing motion and edge-highlighting to trace board-of-director connections.

When a network graph has thousands of nodes and edges, it turns into an unreadable tangled mess ("hairball"). You can't tell what's connected to what.?

DoR- Degree-of-relevance is basically a score for "how closely related is this node to the thing I'm currently focused on?"

Visualization of Data, Understanding and Interpretation

Table Lens: Making Tabular Data Interactive (TB 3, Sec 10.4)

- The Table Lens maps raw spreadsheet rows into visual columns of mini bar charts, showing both local details and global trends.
- **Row-wise expansion** allows users to view specific text details on demand while keeping the surrounding numeric trends in view.
- **Sorting column values** dynamically reorganizes the table to reveal multi-attribute correlations and categorical clustering instantly.
- This spreadsheet-inspired interface beats traditional parallel coordinate plots in user testing for complex decision-making tasks.
- Mapping quantitative values to bar widths leverages preattentive human shape processing to scan thousands of records in seconds.

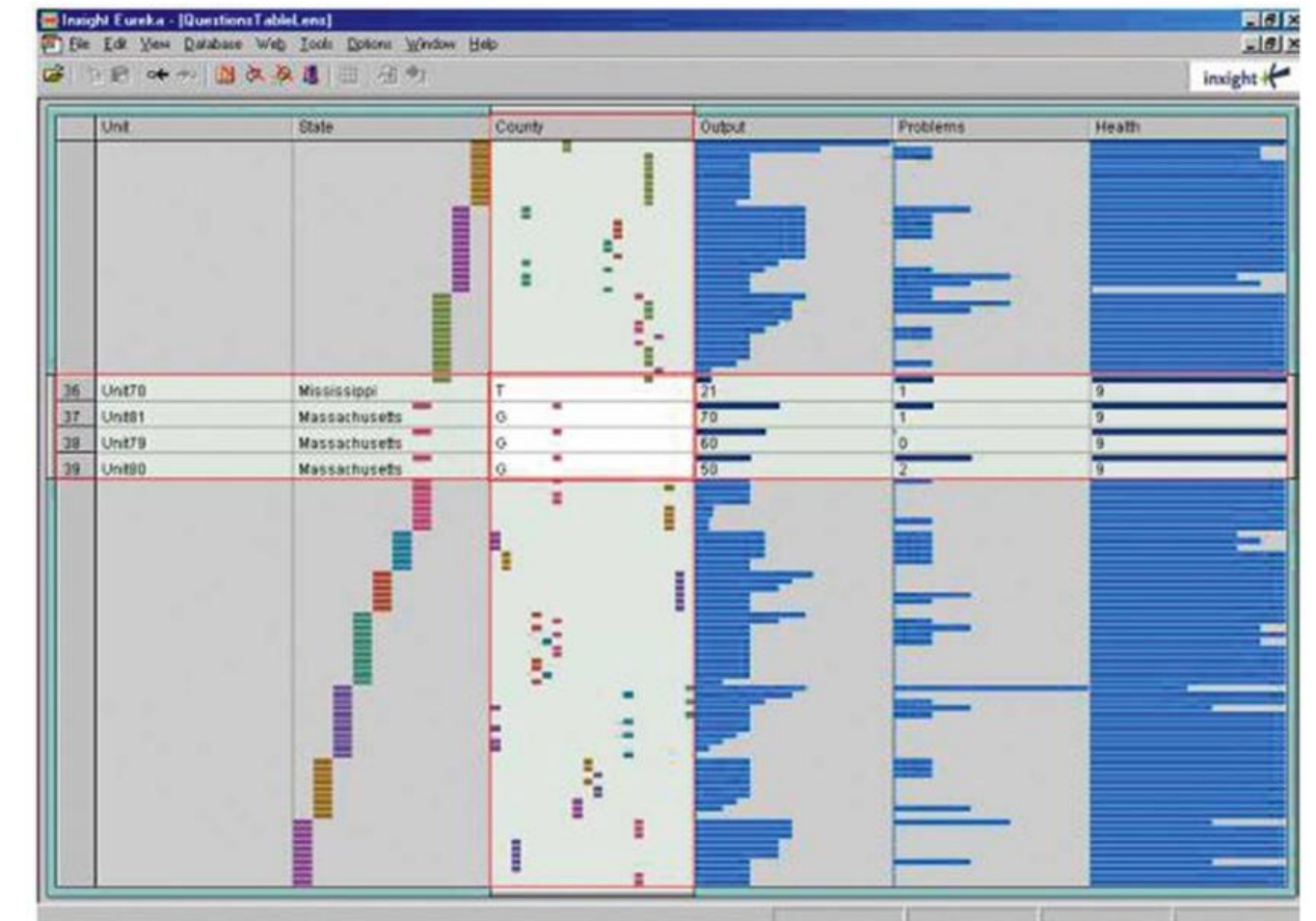
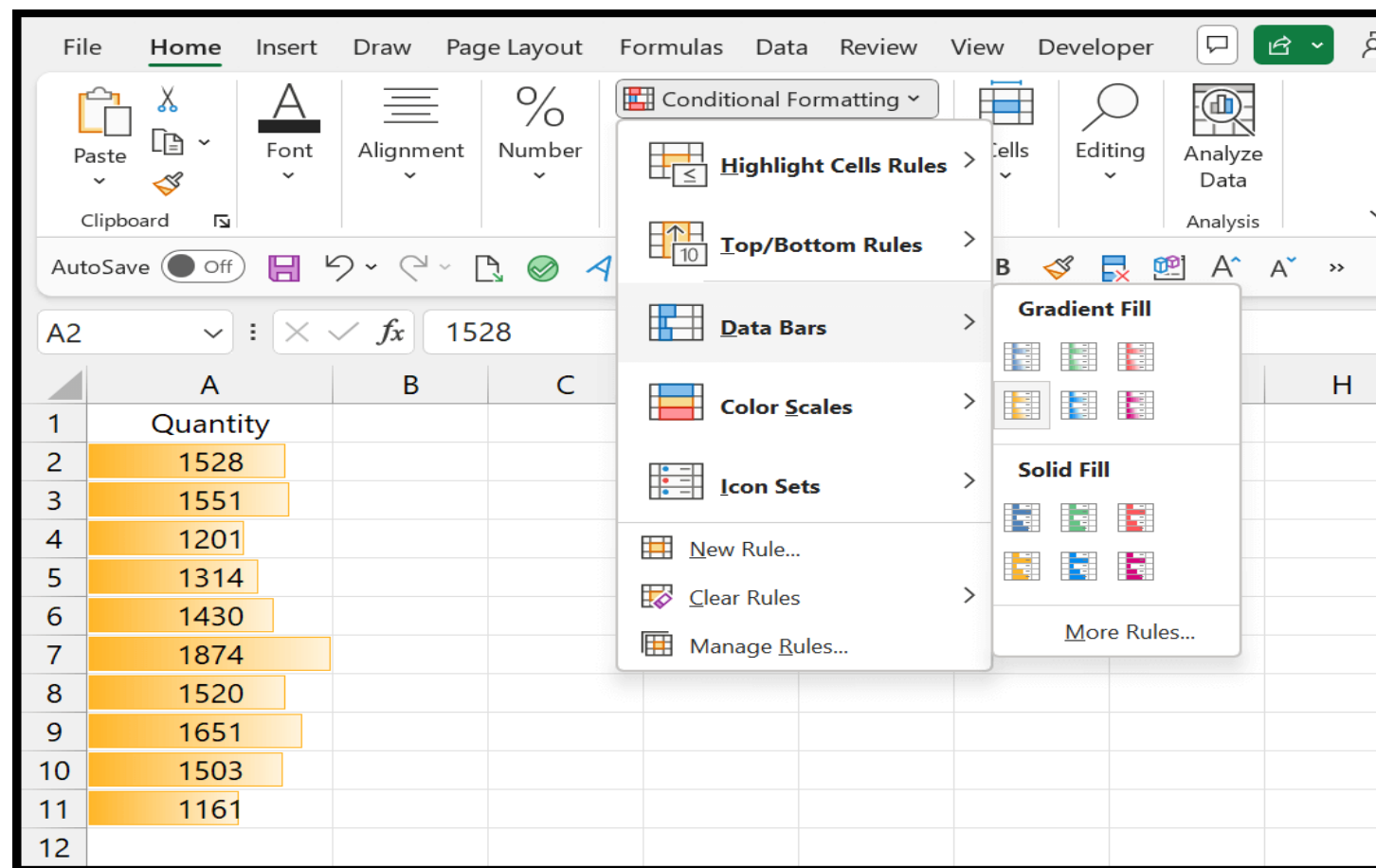


Table Lens interface with columns sorted and selected rows expanded to display text.



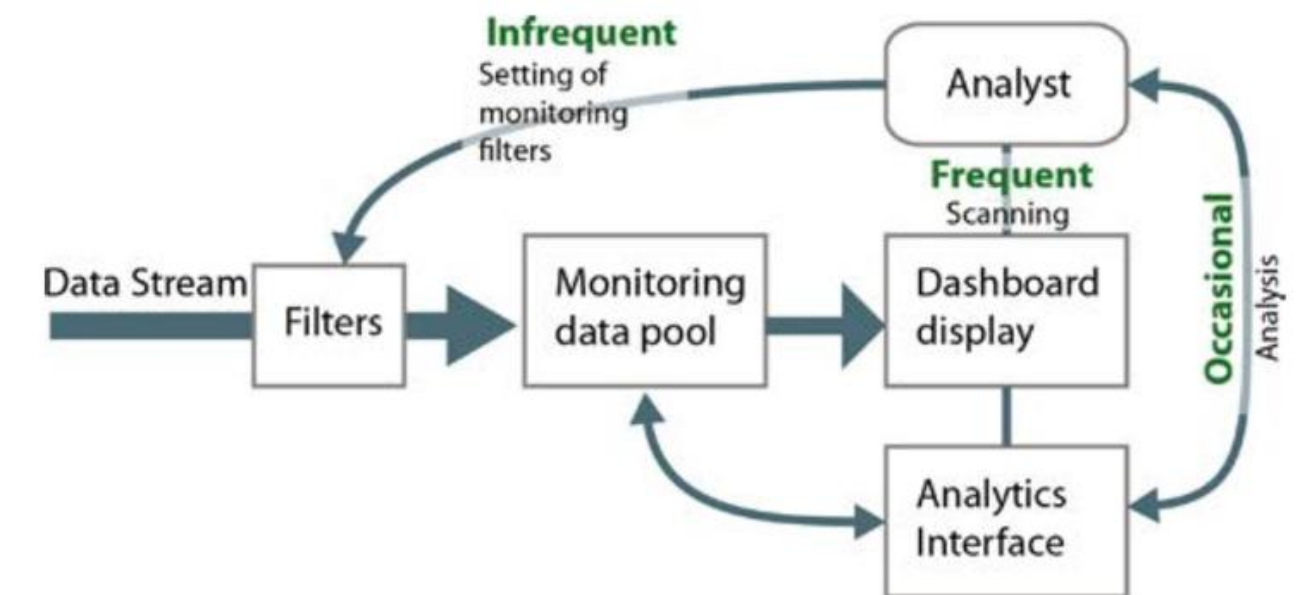
Excel or Google Sheets and applied **conditional formatting** → **Data Bars** to a column of numbers, that's a Table Lens in action:

- A column of raw sales figures (e.g., ₹12,000 to ₹85,000) gets a colored horizontal bar drawn *inside each cell*, proportional to its value
- You instantly spot the highest and lowest performers just by scanning bar lengths — no need to read every number
- You can still click into any cell to see the exact value (row-wise detail on demand)
- Sorting the column re-orders the bars, instantly showing you the ranking visually

Visualization of Data, Understanding and Interpretation

Visual Monitoring and Analytics Dashboard Patterns (TB 3, Sec 12.5)

- Operational monitoring dashboards must **display all critical indicators on a single screen without requiring user interaction**.
- Dashboard processes operate across three timescales spanning **long-term filter editing, mid-term analysis, and continuous peripheral scanning**.
- Subtle motion or blinking highlights should be used for critical alerts because the human peripheral visual system is highly sensitive to motion.
- Using a second screen for deep analytical drills prevents operators from hiding or losing track of the primary monitoring overview.
- **Introducing periodic false alerts** into repetitive monitoring workflows helps prevent change blindness and maintains operator focus.



Interlocking timescales of monitoring workflows

- **Infrequent** — setting up monitoring filters (done rarely, e.g., once a shift)
- **Frequent** — continuous scanning of the dashboard display (the operator's eyes constantly glancing over it)
- **Occasional** — deeper analysis when something looks off (switching to the Analytics Interface)

Visualization of Data, Understanding and Interpretation

Visual Monitoring and Analytics Dashboard Patterns (TB 3, Sec 12.5)



hospital ICU patient monitor

The screen next to a patient's bed continuously displays vital signs — heart rate, blood pressure, oxygen levels — all visible **at a glance, with zero interaction needed**. The nurse doesn't click anything to see this; it's just always there (matches: *"display all critical indicators without requiring user interaction"*).

If a vital sign goes dangerously low (like oxygen dropping), the screen **flashes/blinks red** and beeps

The main monitor stays focused on **live vitals** (frequent scanning); if the nurse needs to review the patient's history or run deeper analysis, they check a **separate computer/second screen**

Hospitals also periodically run **test alarms** (a similar concept to the "deliberate false alerts") to make sure staff stay alert and responsive, rather than tuning out the constant beeping over a long shift

continuous passive monitoring (vitals) + occasional deep analysis (records) + motion-based alerts (blinking/beeping) for anything critical.

Visualization of Data, Understanding and Interpretation

Model-Based Interactive Planning

(TB 3, Sec 12.5)



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- Model-based planning combines human visual intuition with computational simulations in a real-time predictive loop.

A technique for using visualizations to interact with predictive/simulation models — not just displaying past/current data, but letting users explore "what will happen if..."

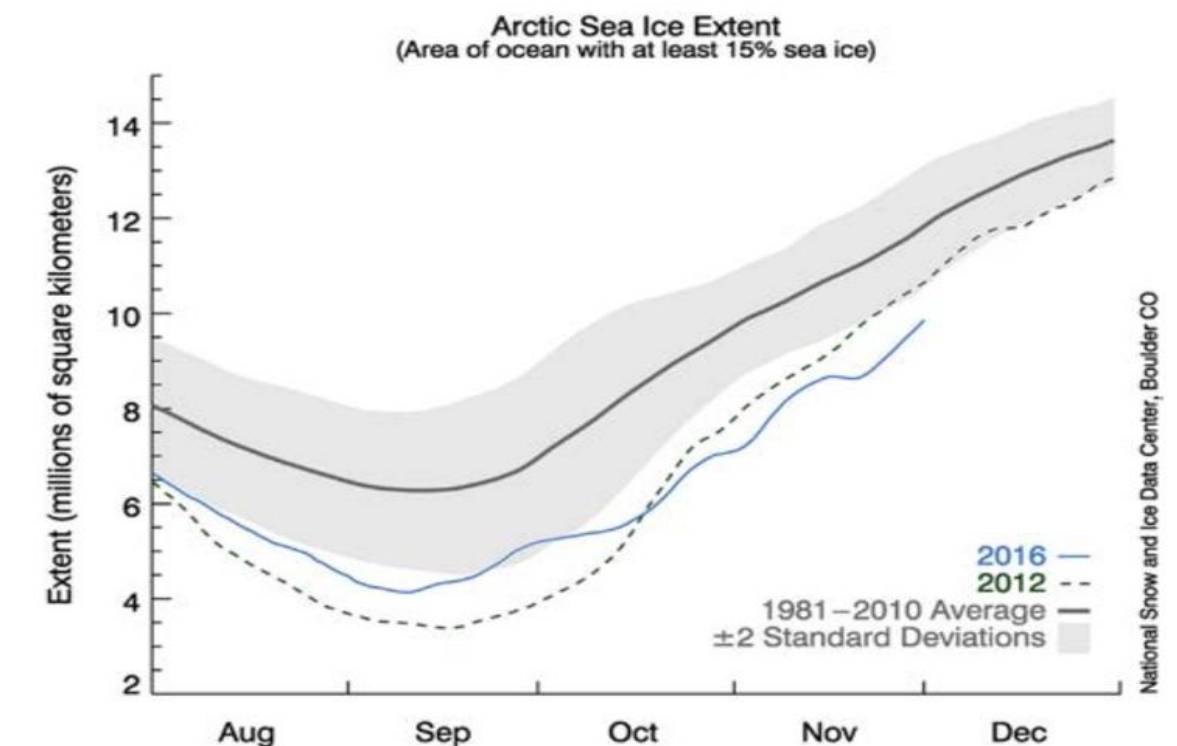
- Adjusting parameter sliders allows users to immediately observe how simulated systems evolve over years or decades.

- Visualizing error bands or multiple shaded uncertainty ranges prevents users from assuming predictive simulations are perfectly precise.

- Making model rules transparent allows engineers and decision-makers to verify, trust, and debug the projected scenarios.

- Comparing baseline and alternative scenarios side-by-side helps identify complex cause-and-effect relationships within abstract models.

- A gray shaded band = the historical average range (± 2 standard deviations) — representing normal/expected variability
- A solid gray line = the 1981–2010 average
- A dashed line (2012) and blue line (2016) = actual observed years, both falling well *below* the historical average band — visually making it obvious these years were abnormally low



Forecasting line chart overlaid with shaded bands representing predictive uncertainty ranges.

COVID-19 case projection dashboards

These dashboards let public health officials and the public interact with a live epidemic model: **Parameter sliders** — you could adjust inputs like "mask compliance rate," "vaccination rate," or "social distancing level," and the projected case/death curve for the coming weeks or months would instantly redraw

Shaded uncertainty bands — nearly every COVID forecast chart showed a shaded region around the projected line (best-case to worst-case range),

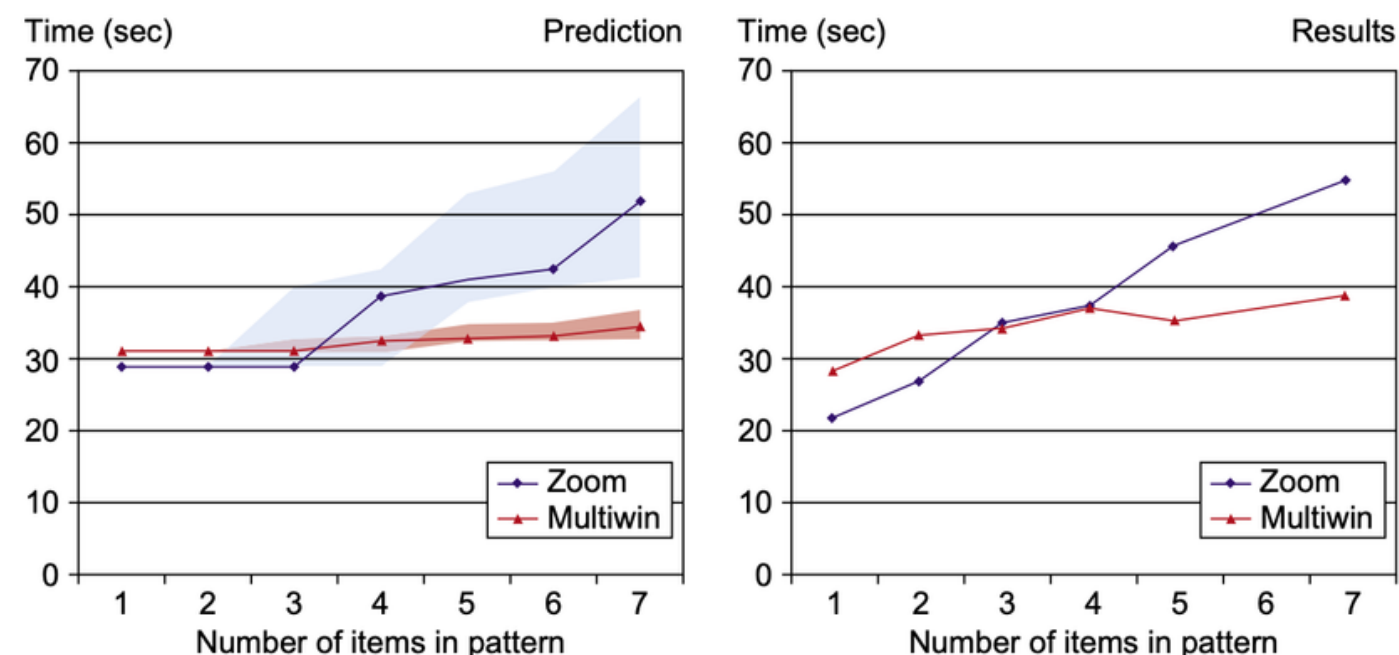
Baseline vs. alternative scenarios — dashboards commonly showed "if no intervention" vs. "if lockdown continues" as two overlaid lines, so policymakers could visually compare the cause-and-effect impact of different decisions before committing to one

Model transparency — reputable dashboards published their underlying assumptions (transmission rate, recovery time, etc.), so epidemiologists could verify and debug the model's logic rather than blindly trusting a black-box number

Visualization of Data, Understanding and Interpretation

Cognitive Bottlenecks: Working Memory Limits (TB 3, Sec 11.2)

- Visual working memory is extremely limited, storing only 3 to 5 simple visual elements at any given moment.
- Visual search patterns are planned and executed using **top-down attentional maps**, moving sequentially from one fixation point to another.
- Change blindness occurs because our internal memory of a scene is highly simplified; **if a visual transition is not animated or highlighted, our brains often miss massive changes.**
- **Static diagrams support visual comparisons** by letting the eye scan back and forth between elements on a single high-resolution screen.
- **Interactive transitions must happen quickly and smoothly** to prevent the user's working memory from dropping key context during the transition.



Model predictions are shown on the left. Measured task performance is shown on the right. Multiple windows speed performance relative to the use of a zooming interface when the number of objects to be compared is five or more.

predicts that Zoom gets progressively slower than Multiwin as the number of items increases (LHS)

Once you need to compare **5 or more items**, using multiple separate windows is actually *faster* than zooming back and forth

Data type	Focus-context solution
Spatial/geographic	Fisheye distortion, Rapid Zooming, 3D navigation metaphors, Magic Lenses
Hierarchical	Hyperbolic/fisheye browsing (tree structures)
Network/graph	Degree-of-relevance highlighting
Tabular/spreadsheet	Table Lens (compact bar-chart overview + row expansion for detail)
Non-spatial multi-attribute	Dynamic Queries (filtering technique, not officially named a focus-context solution)
Mixed spatial + non-spatial	Coordinated Multiple Views / Cross-View Brushing

1. A dashboard allows an analyst to freely select data points, filter attributes, and change the displayed view while investigating an unfamiliar dataset. Which characteristic best describes this interaction?

- A. Author-controlled narrative
- B. Exploratory interaction
- C. Static visual framing
- D. Precomputed presentation

Answer: B

Exploratory = you're driving the car, deciding the route as you go

Author-controlled narrative = you're a passenger on a guided tour with fixed stops

Static/Precomputed = you're just looking at a photo of someone else's trip

2. According to Fitts' Law, what happens to movement time when you reduce both target distance (D) and target width (W) by 50%?

- A. Movement time doubles
- B. Movement time is cut in half
- C. Movement time remains unchanged
- D. Movement time increases exponentially

Answer: C

4. A user selects a subset of countries in a map, and the same countries are automatically highlighted in a treemap and parallel-coordinate plot. What technique is being demonstrated?

- A. Dynamic querying
- B. Cross-view brushing
- C. Geometric distortion
- D. Rapid zooming

Answer: B

Visualization of Data, Understanding and Interpretation

MCQs

5. A visualization contains thousands of records, and the analyst wants to identify patterns by interactively filtering numerical ranges. Which technique is most appropriate?

- A. Static framing
- B. Window replacement
- C. Dynamic queries
- D. Narrative sequencing

Answer: C



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 2 : Advanced Visualization & Storytelling

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

In this lecture, we will cover:

- Understand networks as relationships between nodes and links, along with topology- and attribute-based network tasks.
- Explore node-link diagrams, their visual grammar, design principles, strengths, and limitations.
- Learn how adjacency matrices represent networks and how ordering reveals clusters, hubs, cliques, and other structural patterns.
- Compare node-link, adjacency matrix, and hybrid NodeTrix representations and understand when each is most effective.

Visualization of Data, Understanding and Interpretation

Introduction to Networks

Networks:

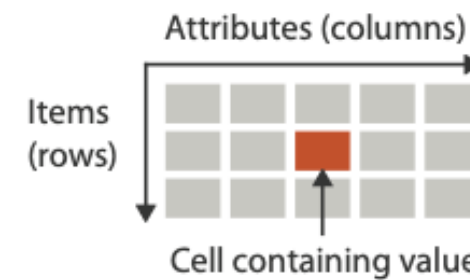
- Model relationship between things (aka graphs).
- Two kinds of items, nodes and links. Both can have attributes.

Tree:

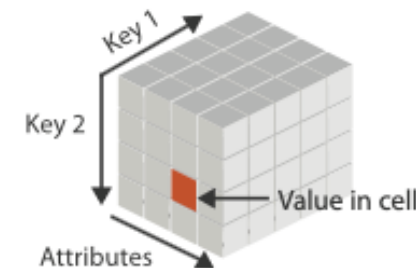
- Special case of networks.
- No cycles, one parent per node.

→ Dataset Types

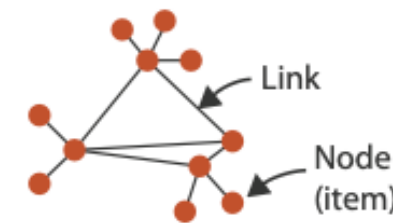
→ Tables



→ Multidimensional Table



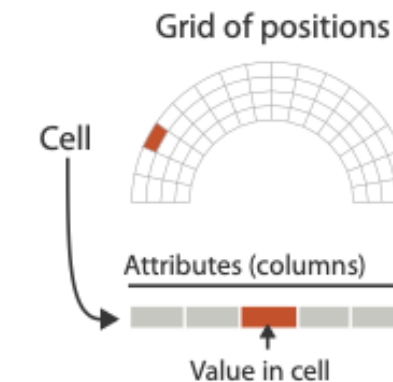
→ Networks



→ Trees



→ Fields (Continuous)



→ Geometry (Spatial)



Visualization of Data, Understanding and Interpretation

Network Tasks

Topology based tasks:

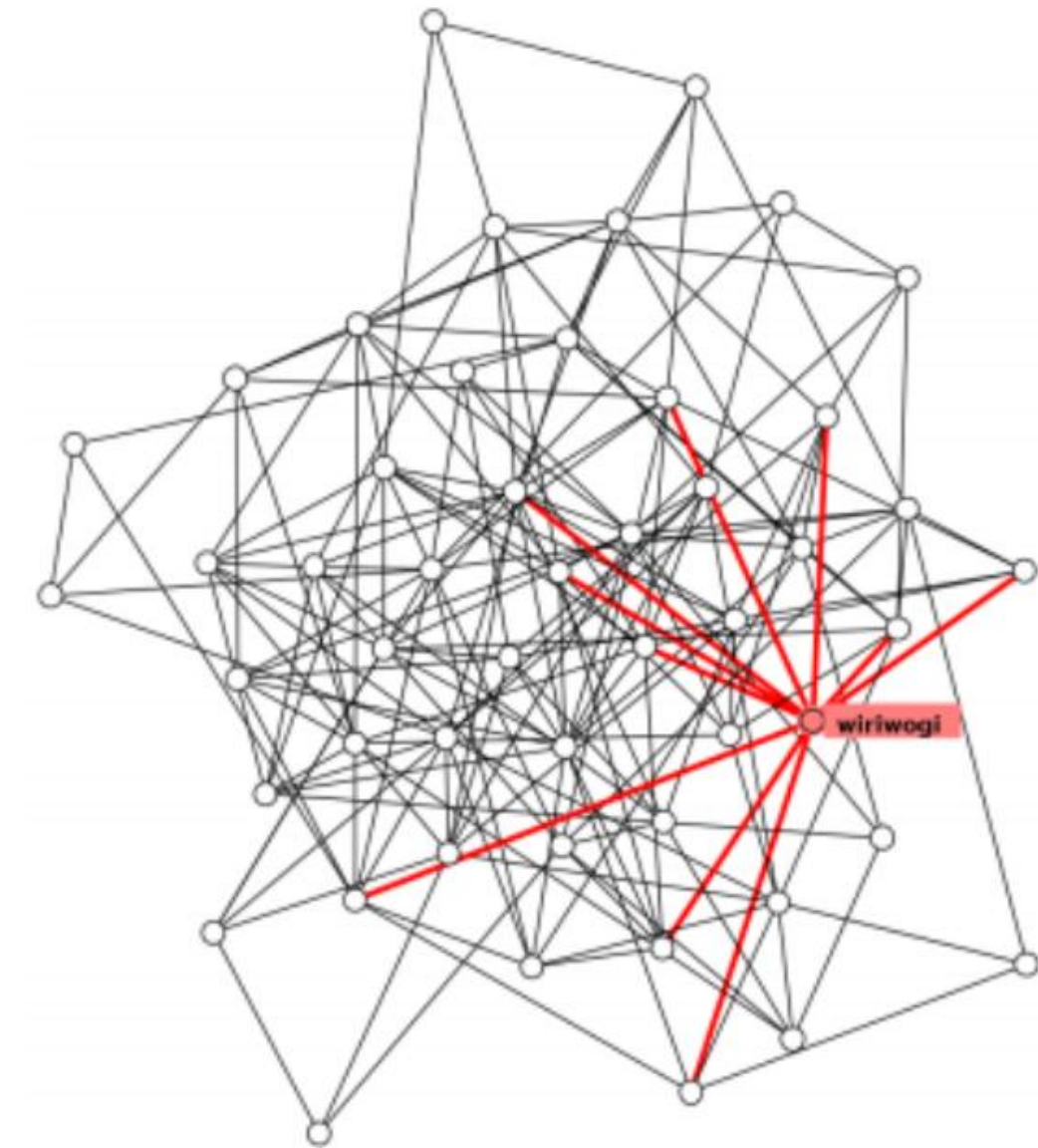
- Find paths
- Find (topological) neighbours
- Compare centrality/importance measures
- Identify clusters/communities

Attribute based tasks:

- Similar to table data (find distributions,.. etc)

Combination tasks, incorporating both:

- Example: Find friends of friends who like cats
- Topology: Find all adjacent nodes of a given node
- Attribute: Check if has-pet (node attribute) == cat



Visualization of Data, Understanding and Interpretation

Node-Link Diagrams

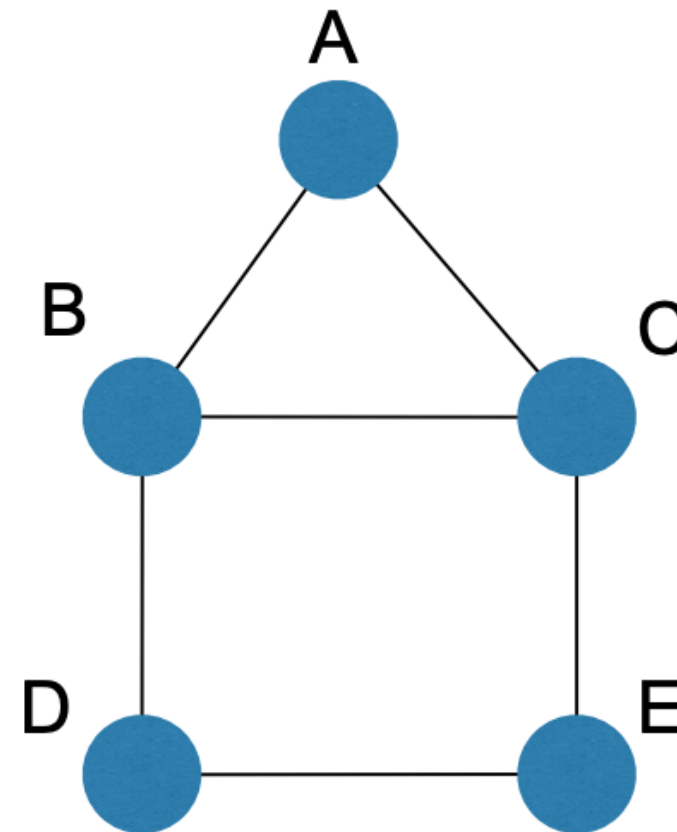
Nodes: Point marks

Links: Line marks

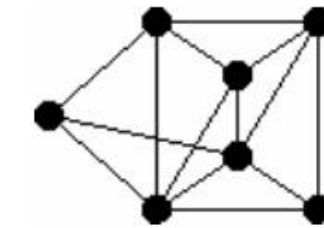
- Straight lines or arcs
- Connections between nodes

Intuitive and familiar

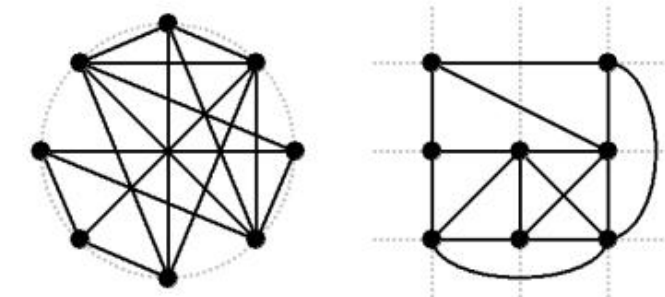
- Most common
- Many variants



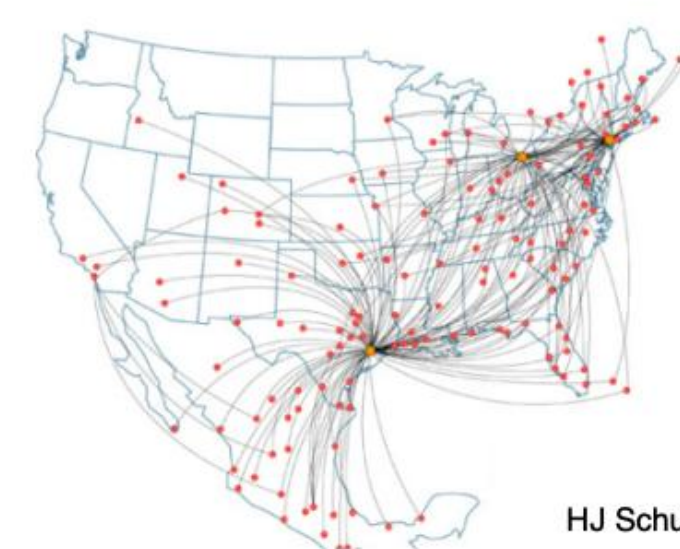
Free



Styled



Fixed



HJ Schulz 2006

Visualization of Data, Understanding and Interpretation

Node-Link Diagrams – Visual Grammar



What are Node-Link Diagrams?

- Represent entities as nodes and relationships as links (edges).
- One of the most widely used visualization techniques for displaying networks.
- Commonly used in social networks, software engineering, databases, transportation, and communication networks.

Visual Grammar

- Closed shapes (circles, rectangles): Represent entities or objects.
- Connecting lines: Represent relationships or interactions.
- Node attributes: Color, size, and shape encode additional information (type, category, importance).
- Edge attributes: Thickness, color, style, and arrows represent relationship type, strength, or direction.

Key Principle

Effective node-link diagrams leverage Gestalt principles (proximity, connectedness, and enclosure), allowing users to naturally perceive objects and their relationships.

Visualization of Data, Understanding and Interpretation

Node-Link Diagrams – Visual Grammar

Graphical code	Visual instantiation	Semantics
1. Closed contour		Object, entity
2. Compact shapes		Entity types
3. Color of region		Entity types
4. Size of region		Entity value: larger = more

The basic visual grammar of entity representations for node-link diagrams.

Visualization of Data, Understanding and Interpretation

Node-Link Diagrams – Design Principles



Why Node-Link Diagrams Work

- Closely match how humans naturally think about objects and their connections.
- Excellent for exploring paths, connectivity, hierarchies, and network topology.
- Provide intuitive visual representations for complex relationships.

Design Guidelines

- Minimize edge crossings to improve readability.
- Reduce unnecessary bends in edges.
- Position related nodes close together to reveal communities.
- Use arrows or tapered edges for directed relationships.
- Encode attributes using color, size, and line thickness instead of excessive text labels.

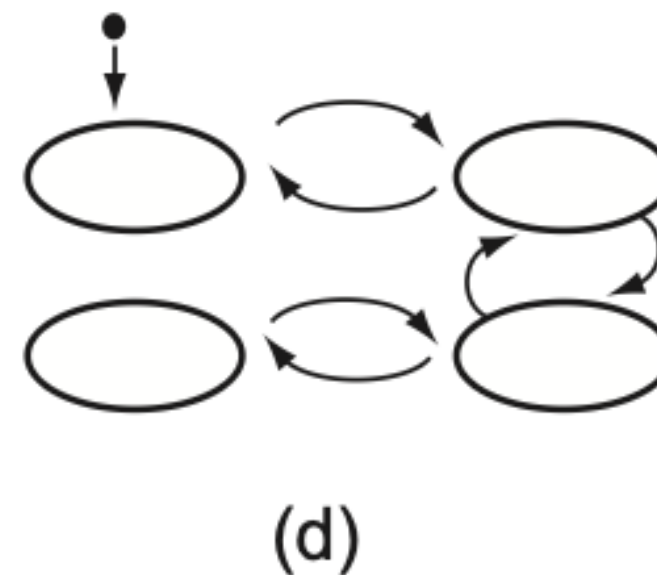
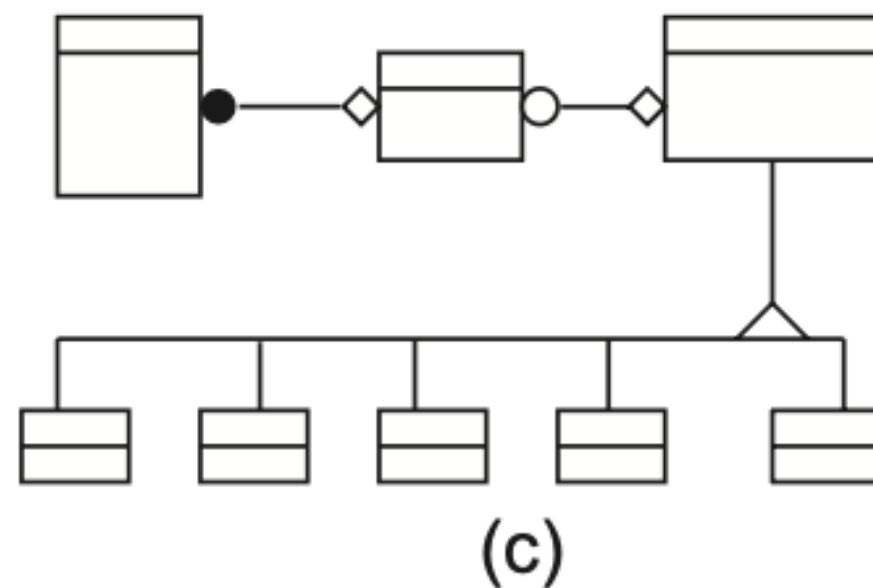
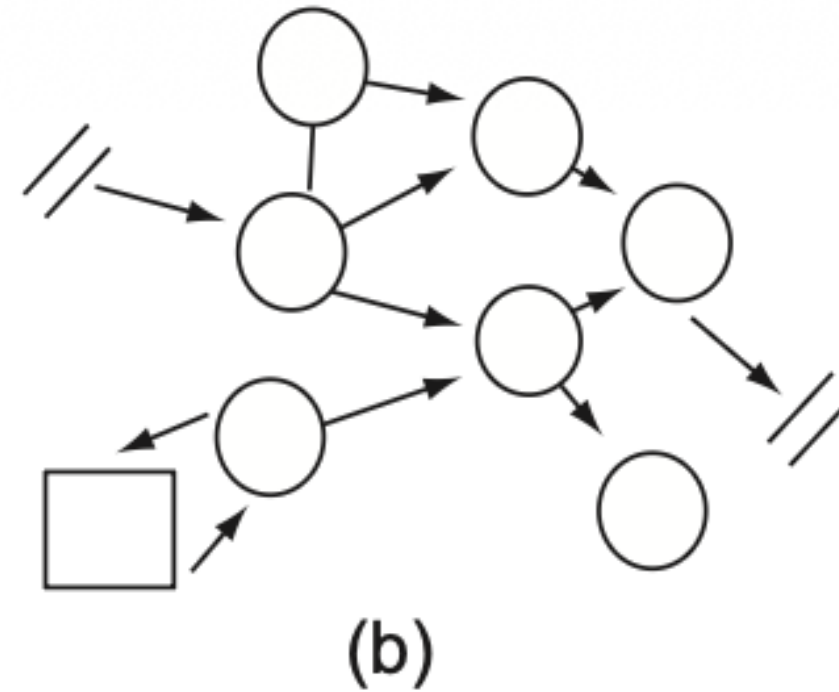
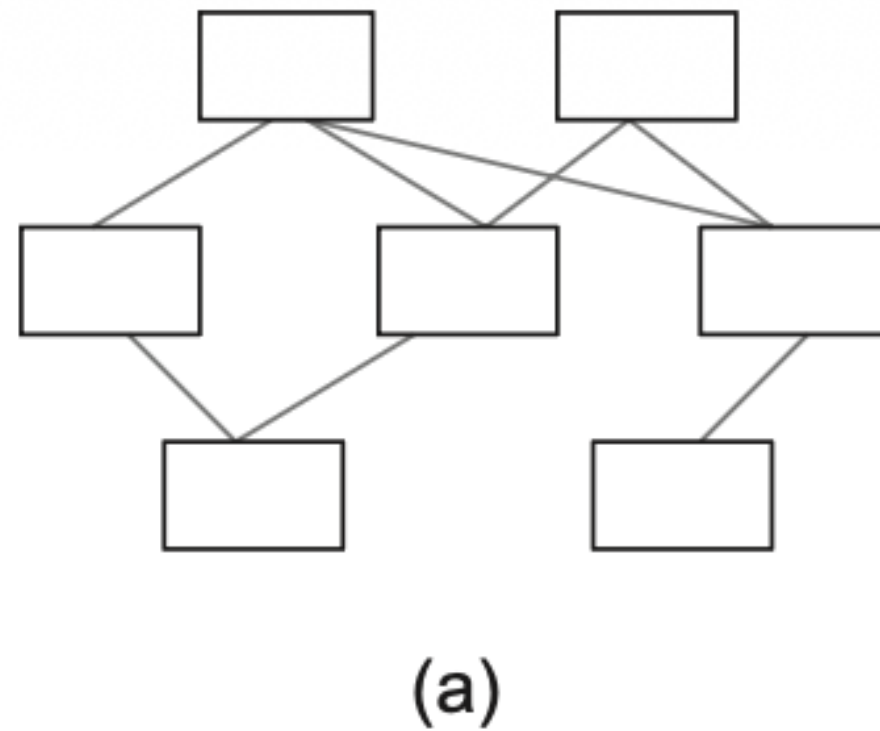
Limitations

- Become cluttered as the number of nodes and edges increases (hairball effect).
- Dense networks may require alternatives such as adjacency matrices or NodeTrix visualizations.

Takeaway: A well-designed node-link diagram reduces cognitive effort, making relationships and network structures easier to understand.

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Node-Link Diagrams – Design Principles



Four different kinds of node-link diagrams used in software engineering:

- (a) A code module diagram.
- (b) A data-flow diagram.
- (c) An object modeling diagram.
- (d) A state-transition diagram.

Each of these diagrams would normally contain text labels on the nodes and the arcs.

Visualization of Data, Understanding and Interpretation

Criteria for Good Node-Link Layouts

Minimise:

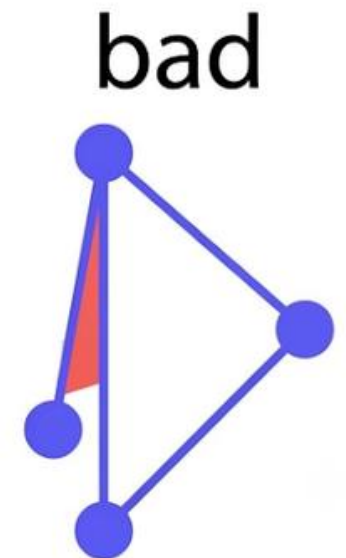
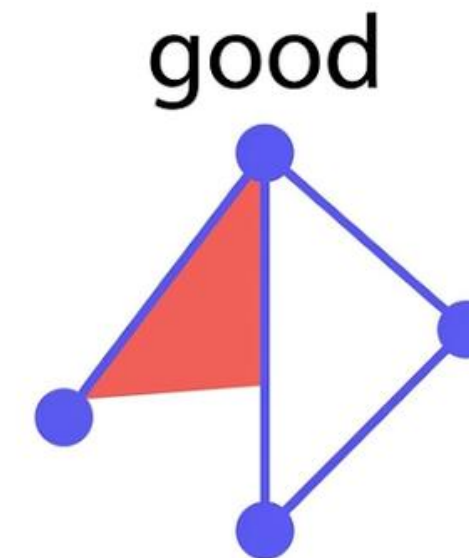
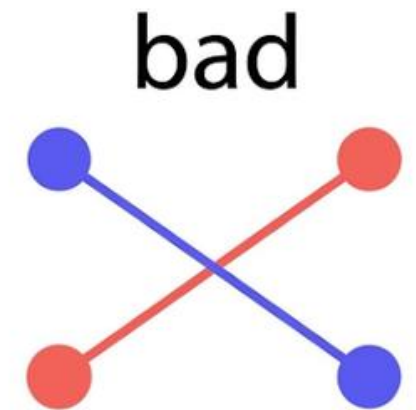
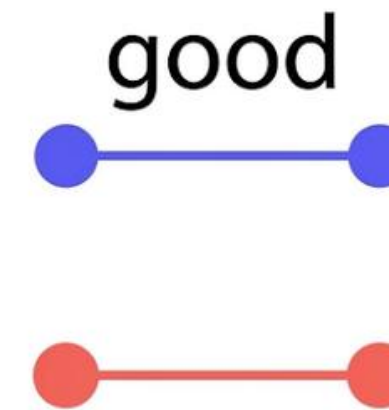
- Edge crossings
- Distances between topological neighbour nodes
- Total drawing area
- Edge bends

Maximise:

- Angular distance between different edges
- Aspect ratio disparities

Emphasize symmetry:

- Similar graph structures should look similar in layout



Visualization of Data, Understanding and Interpretation

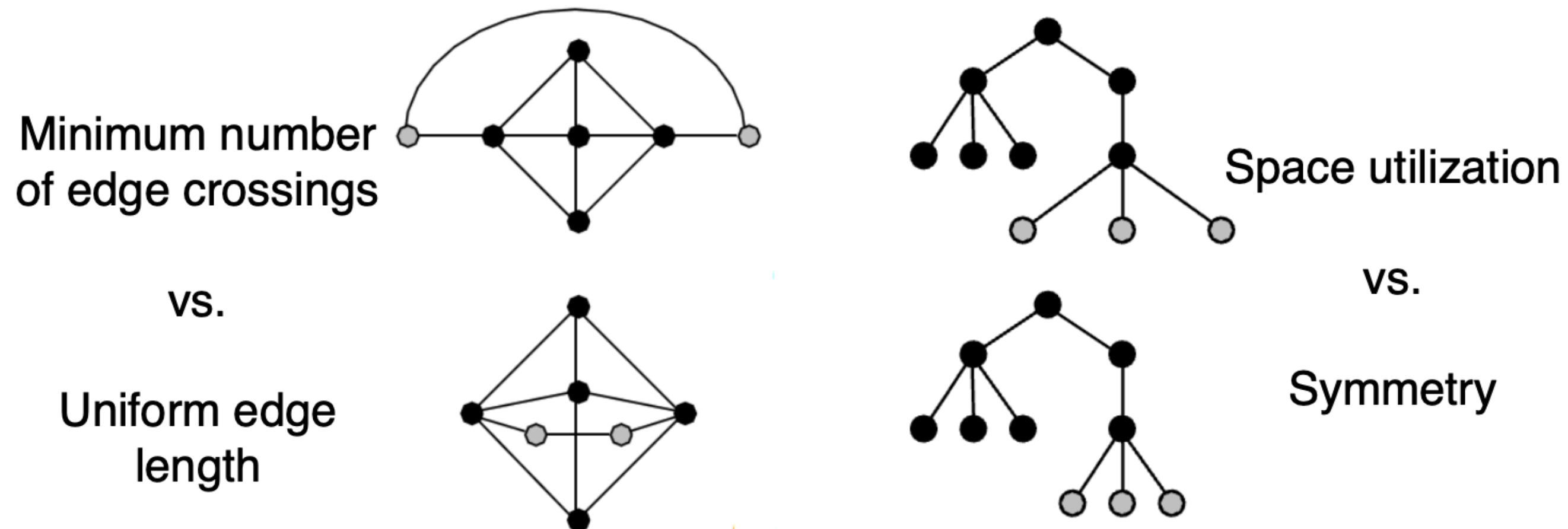
Criteria Conflict

Most graph drawing criteria are NP-hard to optimize individually.

- Finding the mathematically optimal layout for criteria such as minimum edge crossings or perfect symmetry is computationally difficult, especially for large graphs.

Many aesthetic criteria directly conflict with one another.

- Improving one visual property often makes another property worse, so graph drawing is usually a trade-off rather than an optimization of a single objective.



Visualization of Data, Understanding and Interpretation

Optimization Based Layouts

Formulate layout problem as an optimization problem.

Convert criteria into weighted cost function:

- $F(\text{layout}) = a * [\text{crossing counts}] + b * [\text{drawing space used}] + \dots$

Use known optimization techniques to find layout at minimal cost:

- Energy-based physics models
- Force-directed placement
- Spring embedders

Visualization of Data, Understanding and Interpretation

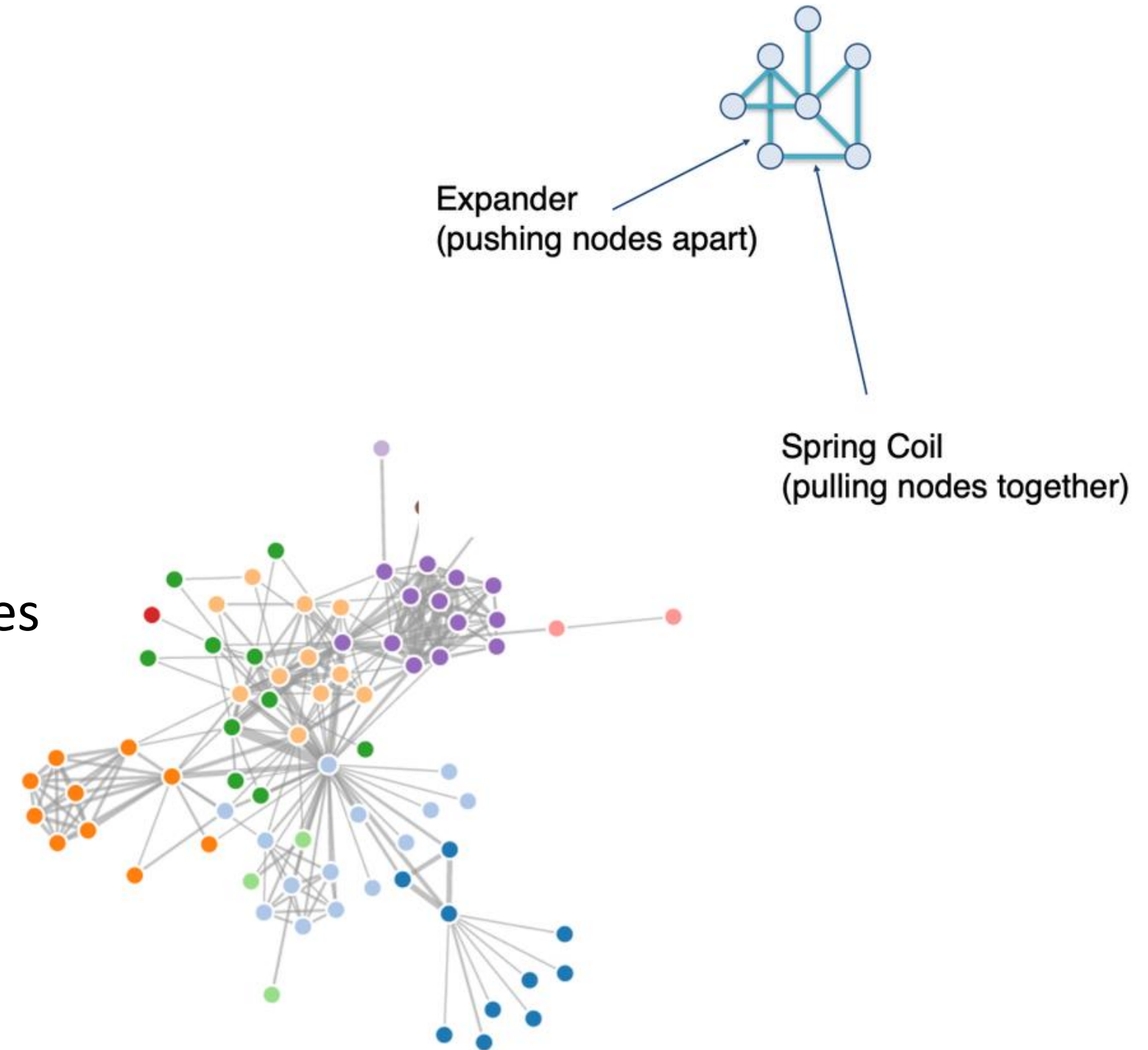
Force-directed Placement

Physics model:

- Links = springs pull together
- Nodes = magnets repulse apart

Algorithm:

- Place vertices in random locations
- While not equilibrium:
 - Calculate force on vertex
 - Sum of
 - Pairwise repulsion of all nodes
 - Attraction between connected nodes
 - Move vertex by $c * \text{vertex_force}$



Visualization of Data, Understanding and Interpretation

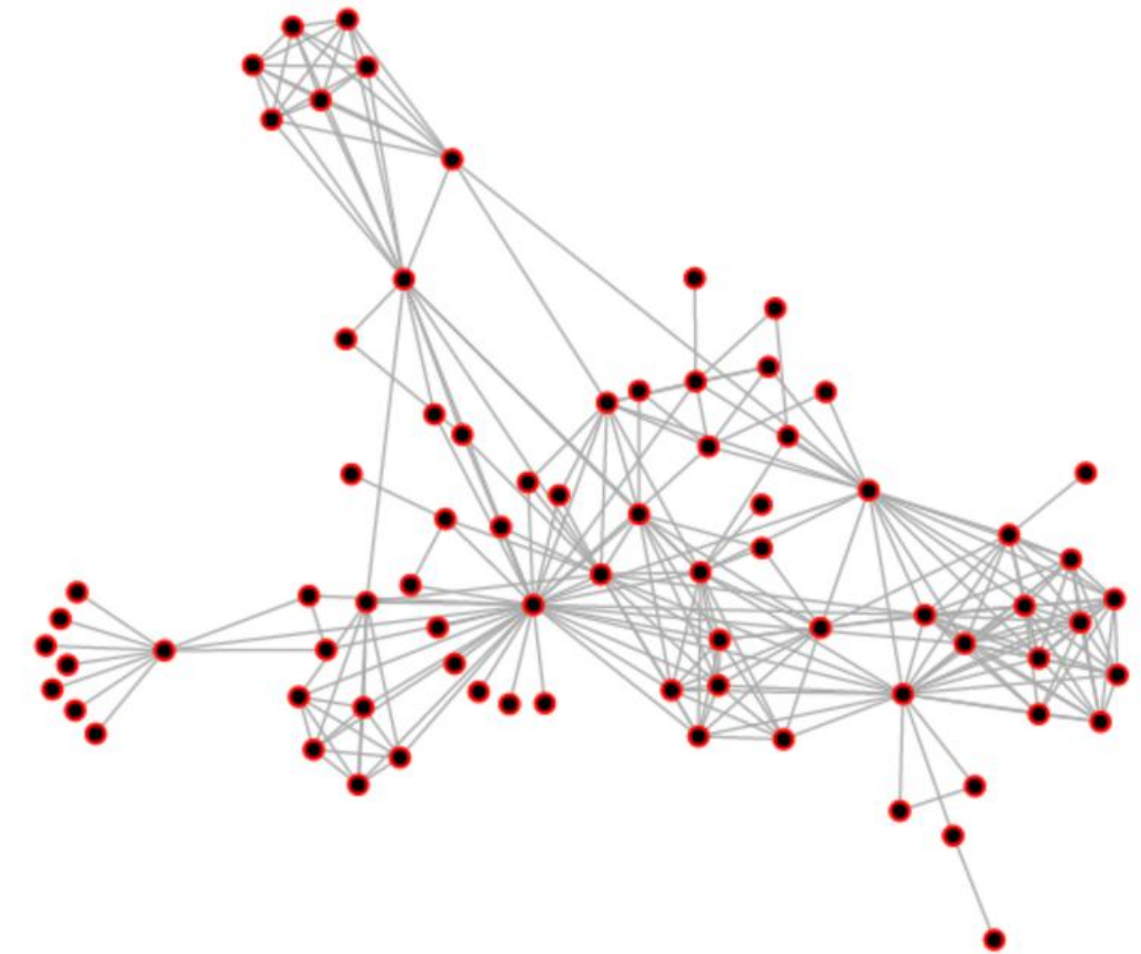
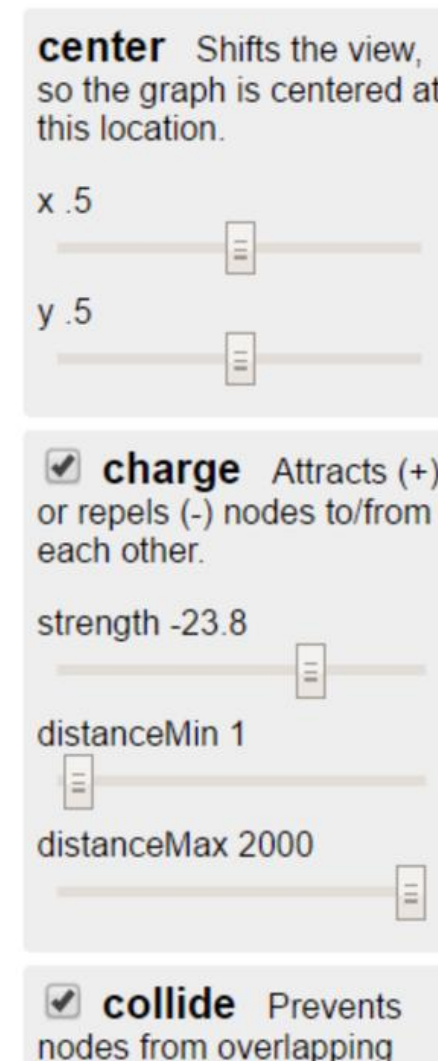
Force-directed Placement Properties

Strengths:

- Reasonable layout for small, sparse graphs
- Clusters typically visible
- Edge length uniformity

Weaknesses:

- Nondeterministic
- Computationally expensive: $O(n^3)$ for n nodes
- Naive FD doesn't scale well beyond 1000 nodes
- Iterative progress: engaging but distracting

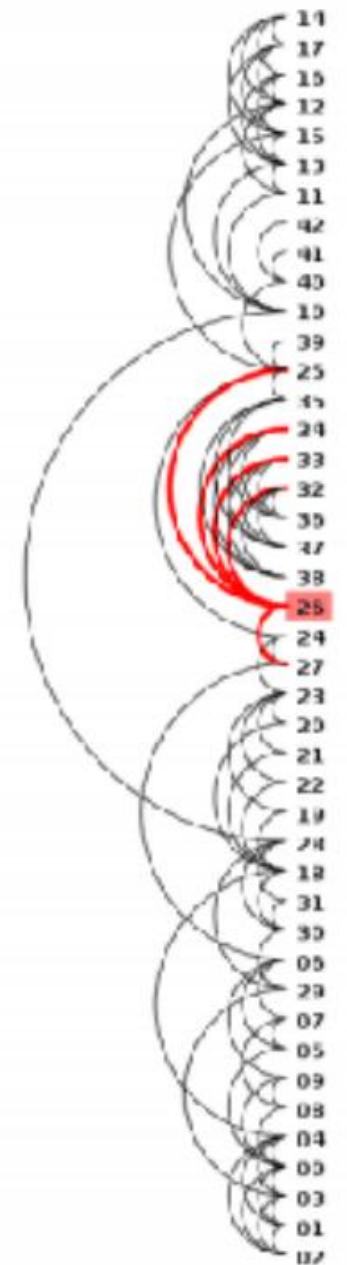
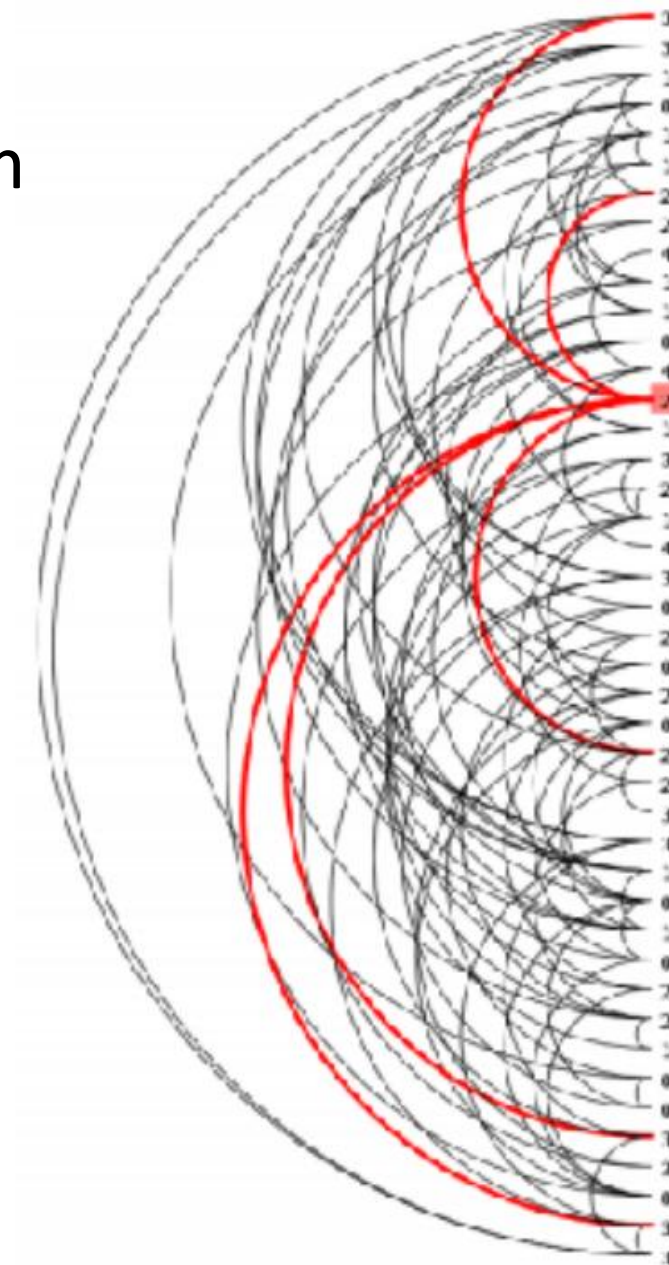
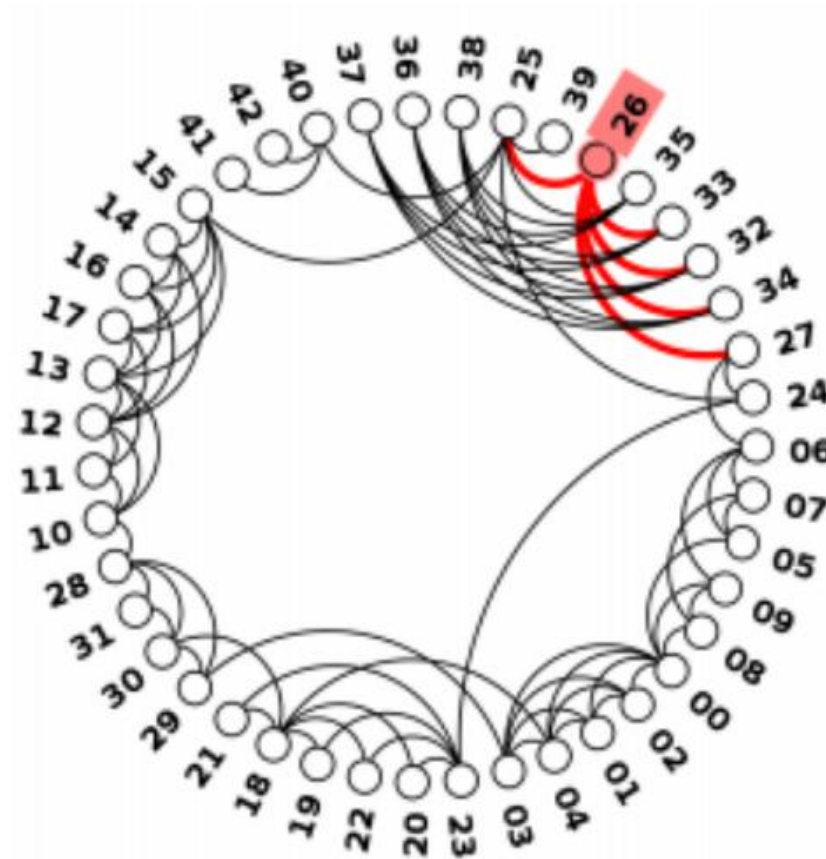
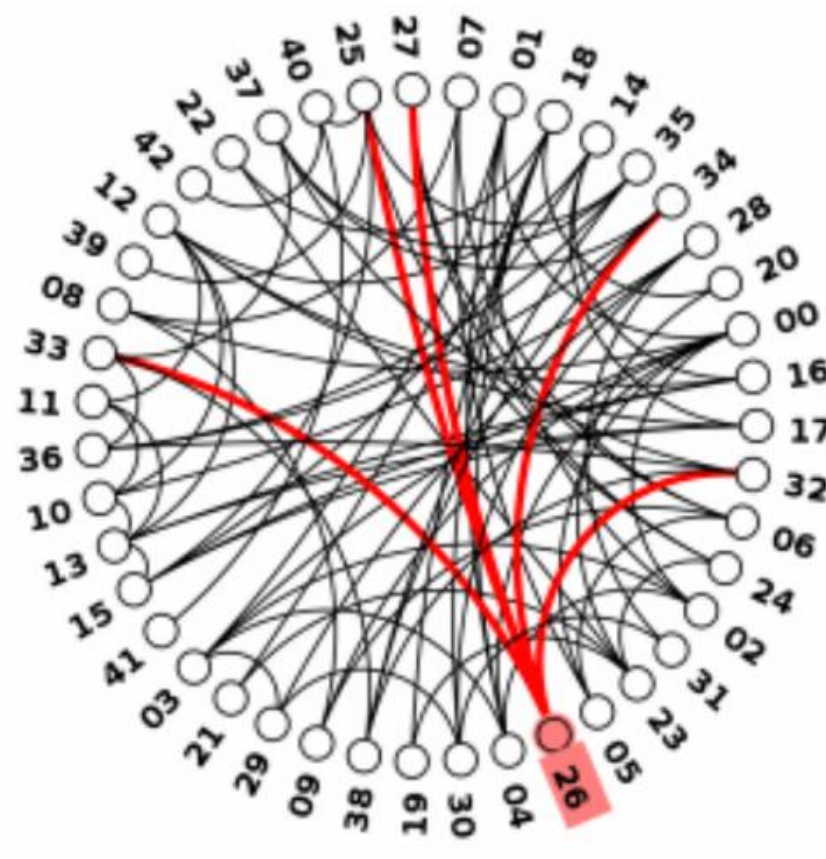


<https://bl.ocks.org/steveharoz/8c3e2524079a8c440df60c1ab72b5d03>

Visualization of Data, Understanding and Interpretation

Circular Layouts / Arc Diagrams

- Restricted node-link layouts: Lay out nodes around circle or along line
- Data:
 - Original: Network
 - Derived: Node ordering attribute (Global Computation)
- Considerations: Node ordering crucial to avoid excessive clutter from edge crossings
 - Examples: Before and after barycentric ordering

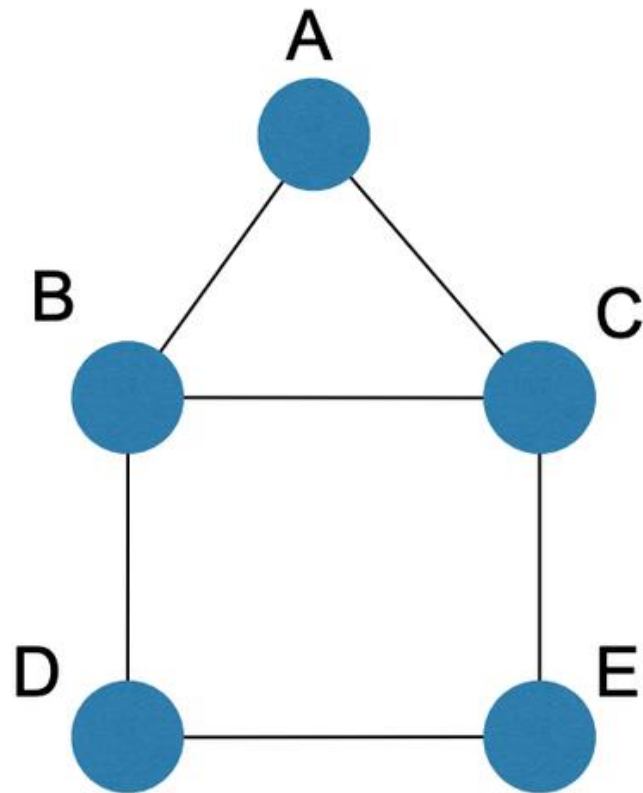


<https://profs.etsmtl.ca/mmcguffin/research/2012-mcguffin-simpleNetVis/mcguffin-2012-simpleNetVis.pdf>

Visualization of Data, Understanding and Interpretation

Adjacency Matrix Representations

- Derive adjacency matrix from network

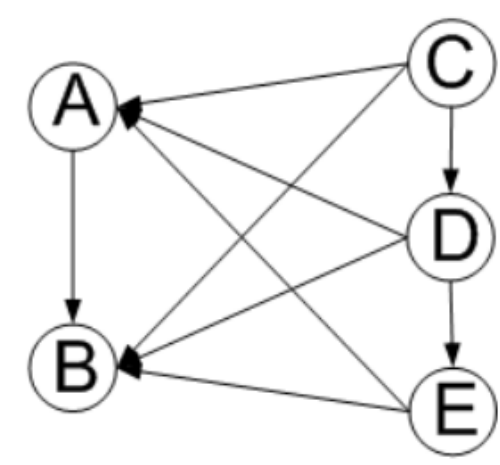
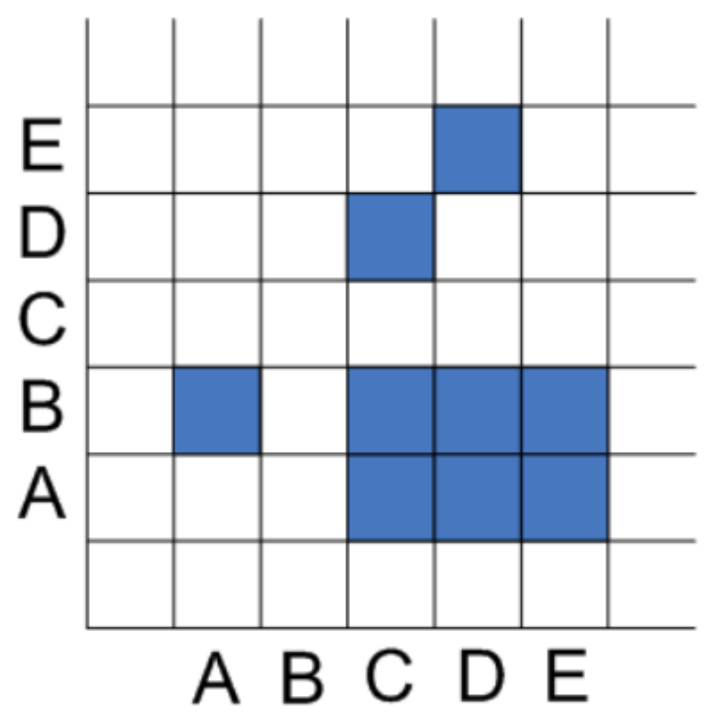
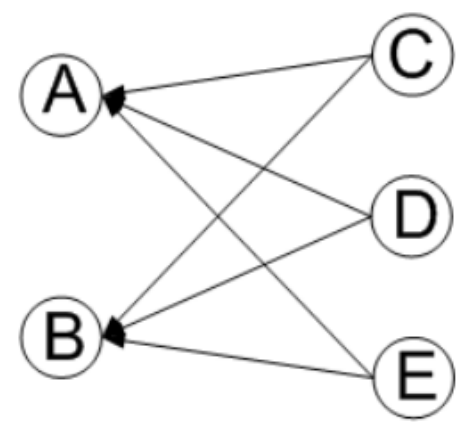
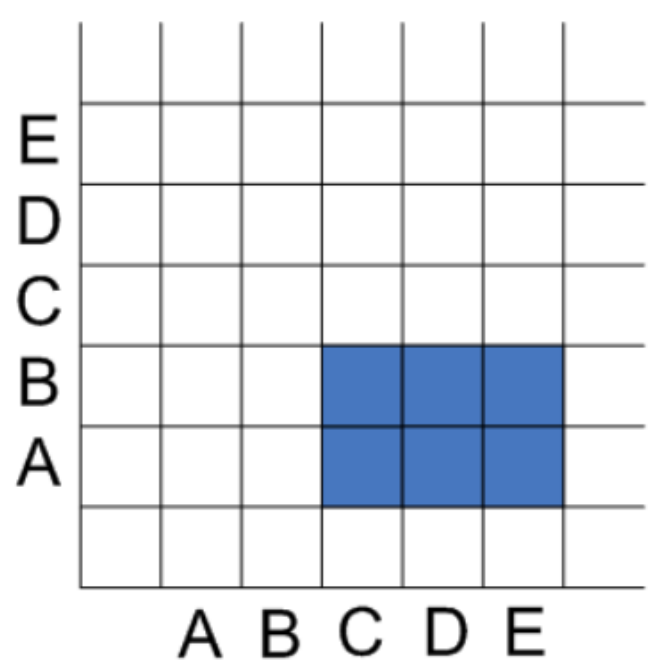
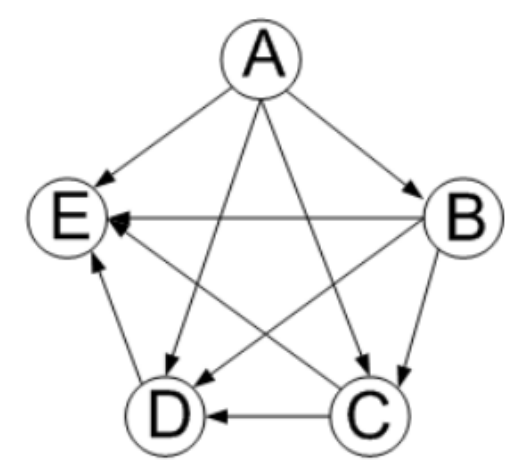
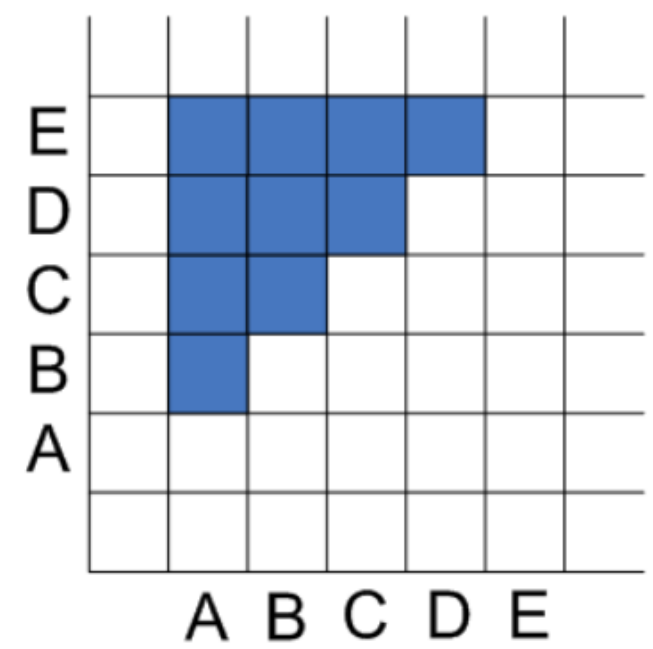
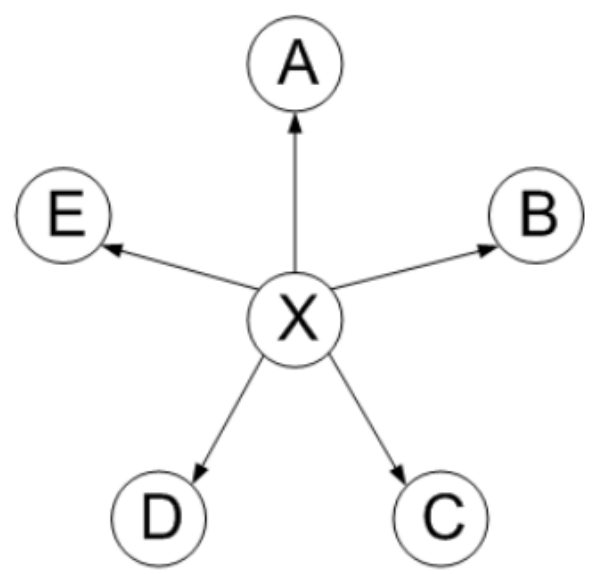
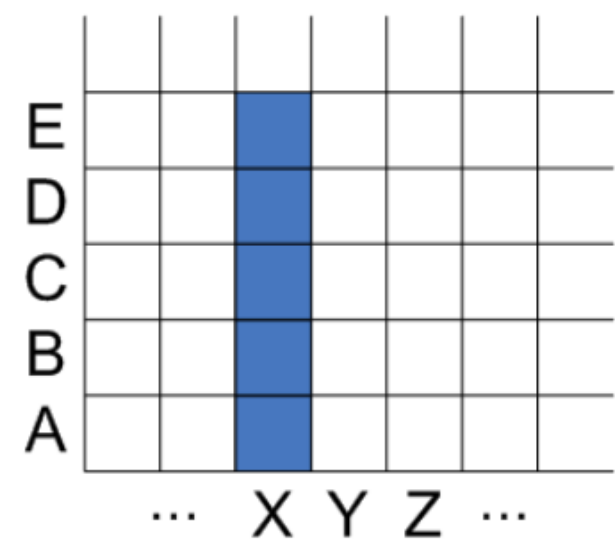


	A	B	C	D	E
A		1	1	0	0
B	1		1	1	0
C	1	1		0	1
D	0	1	0		1
E	0	0	1	1	

An adjacency matrix is a tabular representation of a graph that shows which nodes are directly connected. Each row and column corresponds to a node (A–E). If two nodes share an edge, the corresponding cell is marked 1 (blue); otherwise, it is 0 (white). Since this is an undirected graph, the matrix is symmetric about the main diagonal, and the diagonal entries are 0 because a node is not connected to itself. This matrix makes it easy to analyze network structure and perform graph algorithms.

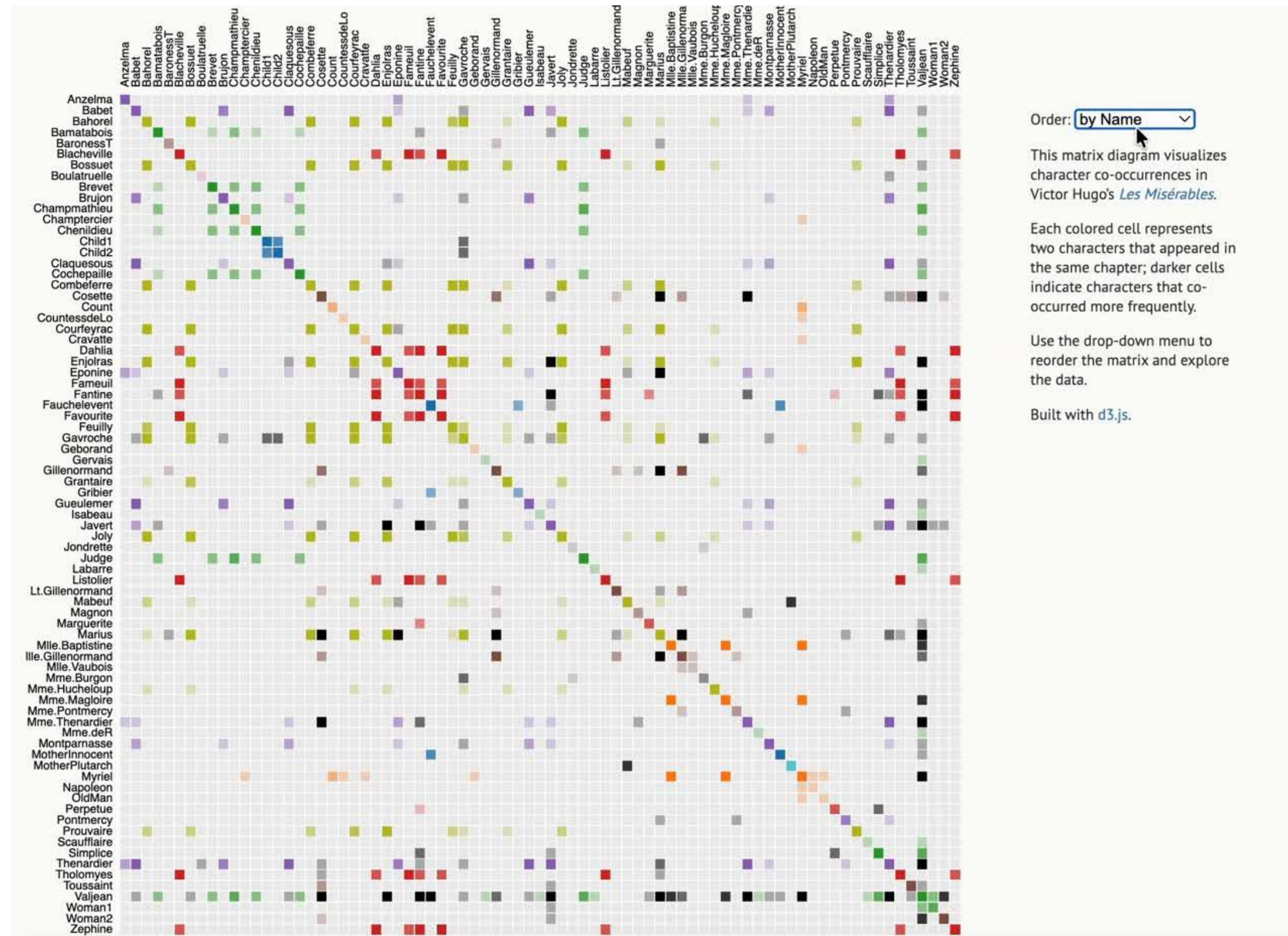
Visualization of Data, Understanding and Interpretation

Adjacency Matrix Examples



Visualization of Data, Understanding and Interpretation

Adjacency Matrix : Node Order is Crucial



The animation demonstrates that visualization design choices can dramatically influence insight without changing the underlying data. Proper organization reveals hidden patterns and communities. It reinforces the importance of thoughtful data arrangement as a key step in creating effective and interpretable visualizations.

Visualization of Data, Understanding and Interpretation

Adjacency Matrix View

Data: Network

- Transform into same data/encoding as heatmap

Derived data: Table from network

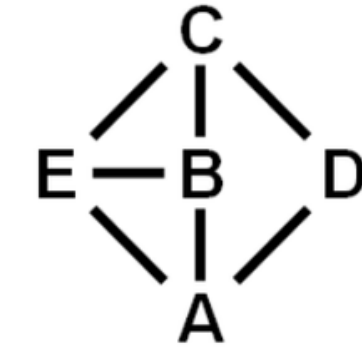
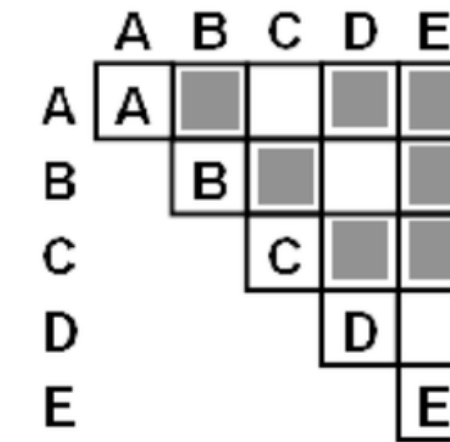
- 1 quantitative attribute
 - Weighted average between nodes
- 2 categorical attributes
 - Node list * 2

Visual encoding

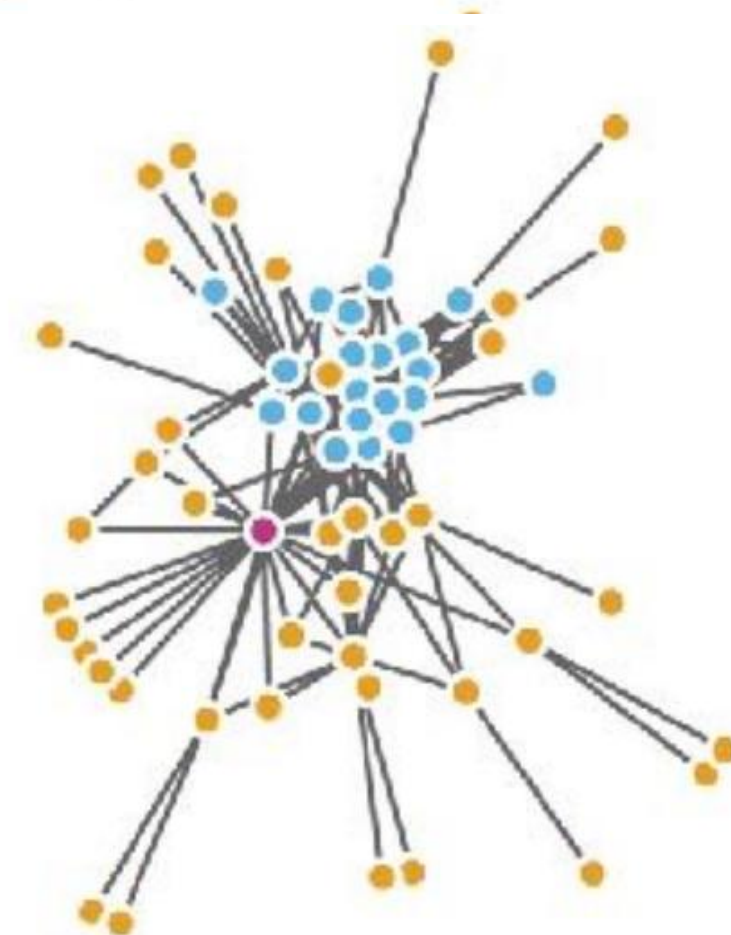
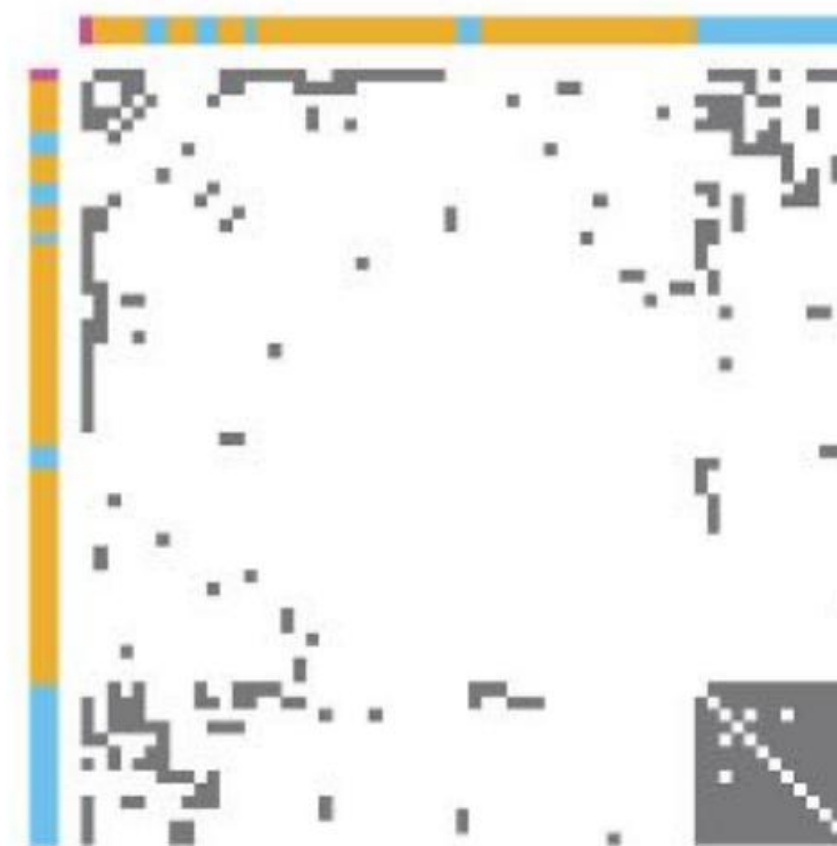
- Cell shows presence/absence of edge

Scalability

- 1k nodes, 1M edges



[NodeTrix: a Hybrid Visualization of Social Networks.
Henry, Fekete, and McGuffin. IEEE TVCG (Proc. InfoVis)
13(6):1302-1309, 2007.]



[Points of view: Networks. Gehlenborg and Wong. Nature Methods 9:115.]

Visualization of Data, Understanding and Interpretation

Why use Adjacency Matrices ?

Adjacency matrices excel when you need to:

- Detect clusters: Reordering rows and columns reveals tightly connected groups as dense blocks along the diagonal.
- Analyze density: Instantly assess how densely connected different parts of your network are.
- Compare subgroups: See connection patterns between different categories or communities at a glance.
- Handle dense networks: Avoid the "hairball" problem—matrices remain readable even when node-link diagrams become cluttered.
- Find structural patterns: Identify recurring motifs, bipartite structures, or hierarchical organization.

Visualization of Data, Understanding and Interpretation

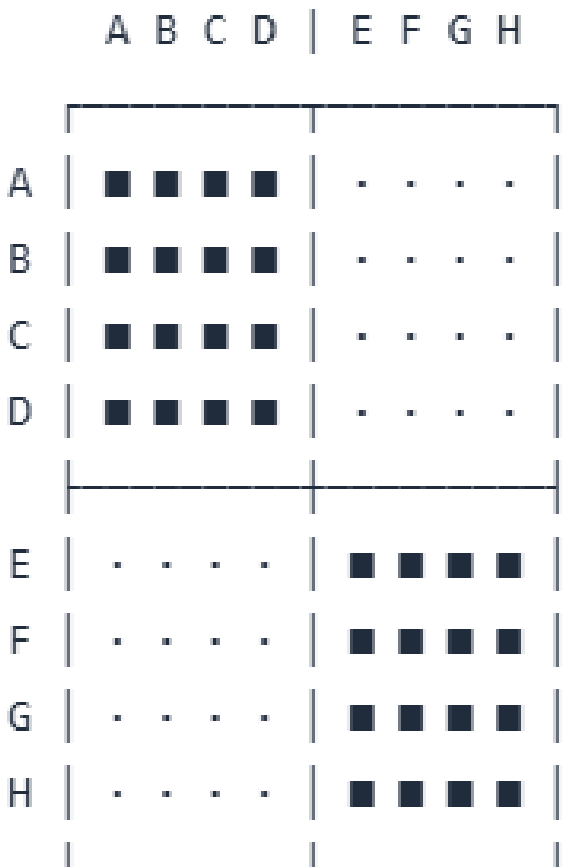
Why use Adjacency Matrices ?

Understanding common visual patterns in adjacency matrices helps you quickly interpret network structure.



Diagonal Blocks (Clusters):

Dense blocks along the diagonal indicate clusters—groups of nodes that are highly connected to each other. When nodes are properly ordered, communities appear as dark squares on the diagonal.

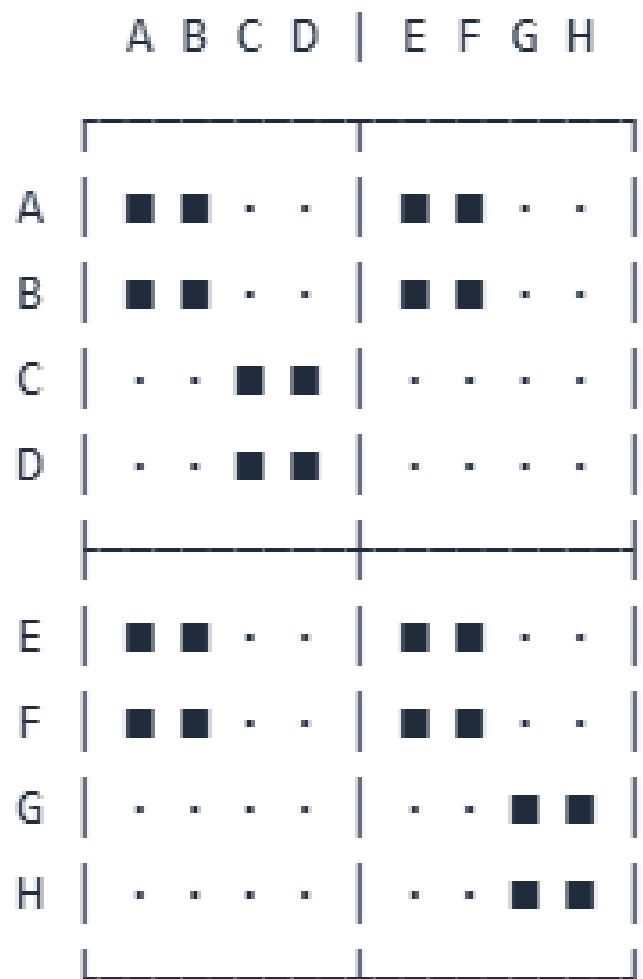


What to look for: Dark squares along the main diagonal. Larger blocks = larger communities. Well-separated blocks = distinct groups.

Why use Adjacency Matrices ?

Off-Diagonal Blocks (Inter-cluster Connections):

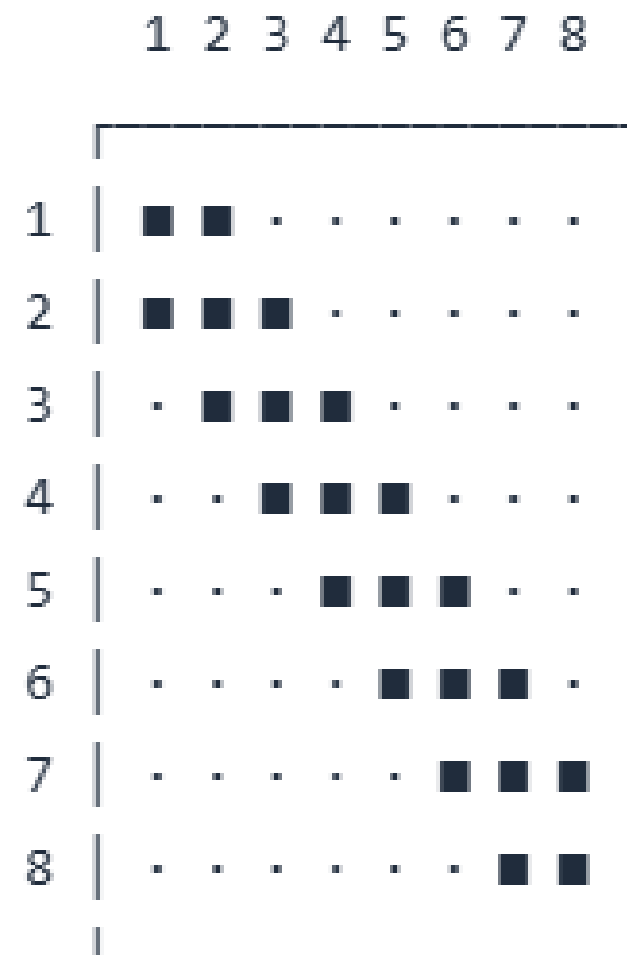
Dense regions away from the diagonal show connections between different clusters. These reveal how communities interact with each other.



What to look for: Dense rectangles in off-diagonal positions. Symmetric blocks (mirrored across diagonal) in undirected networks.

Bandwidth Pattern (Sequential Structure):

When connections primarily occur between nearby nodes (in some ordering), the matrix shows a band pattern around the diagonal. This indicates sequential, hierarchical, or temporal structure.

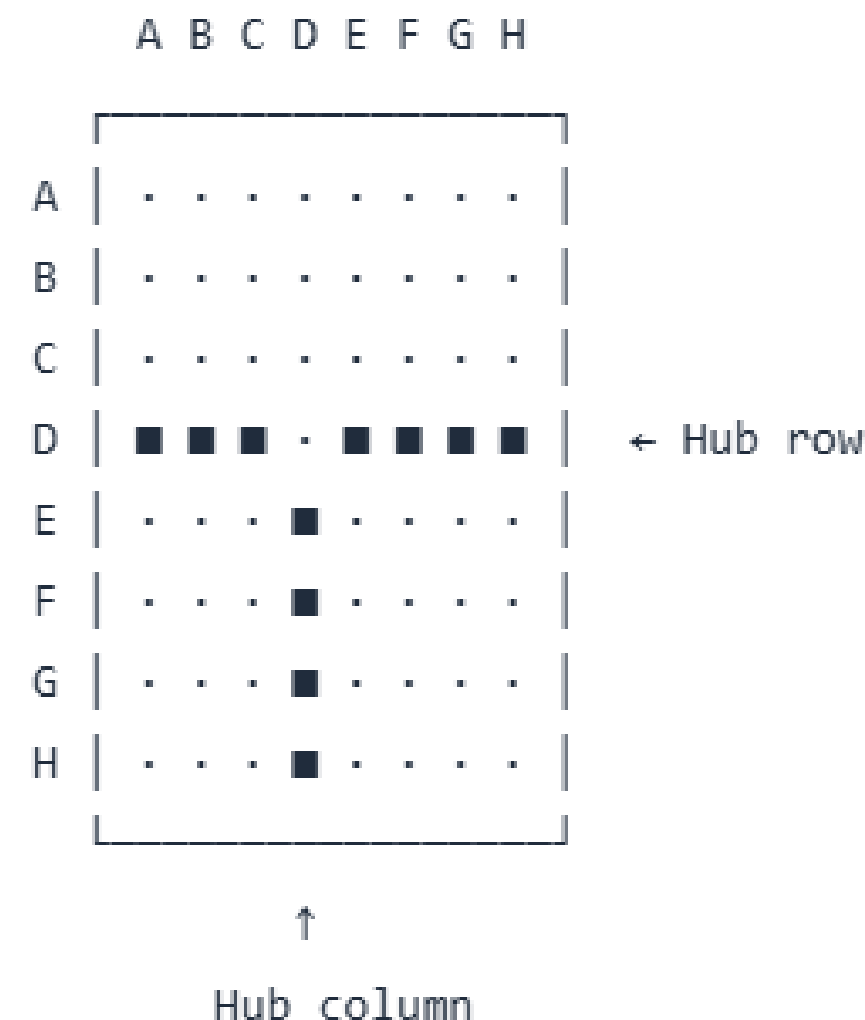


What to look for: Connections concentrated near the diagonal forming a band. Narrow band = local connections only. Wide band = longer-range connections.

Why use Adjacency Matrices ?

Star Pattern (Hub Row/Column):

A single row and column with many filled cells indicates a hub node—an entity connected to many others. Hubs appear as crosses in the matrix.



What to look for: A row and its corresponding column filled with many connections. Multiple hubs = multiple crossing lines.

Why use Adjacency Matrices ?

Bipartite Pattern (Two-mode Network):

When nodes divide into two groups where connections only occur between groups (not within), the matrix shows an off-diagonal block pattern with empty diagonal blocks.

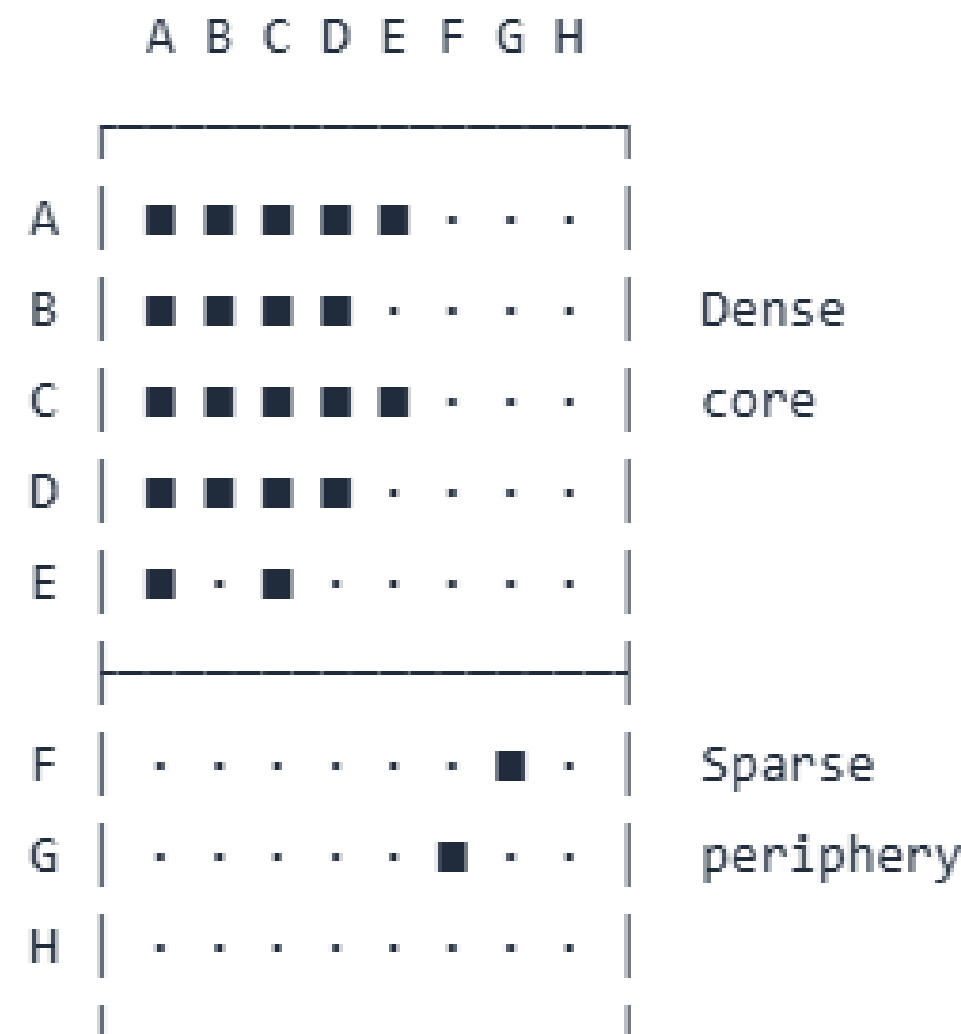
	A	B	C	D	E	F	G	H
A	.	.	.	.	■	■	■	.
B	.	.	.	.	■	.	■	■
C	.	.	.	.	.	■	■	■
D	.	.	.	.	■	■	.	■
E	■	■	.	■	.	.	.	.
F	■	.	■	■	.	.	.	.
G	■	■	■	.	.	.	.	.
H	.	■	■	■	.	.	.	.

What to look for: Empty diagonal blocks (no within-group connections) with filled off-diagonal blocks (all between-group connections).

Why use Adjacency Matrices ?

Spars vs Dense Regions:

Varying density across the matrix reveals which parts of your network are more or less connected. Sparse rows/columns indicate peripheral nodes; dense regions indicate active cores.



What to look for: Contrast between heavily filled and sparsely filled regions. Completely empty rows/columns = isolates.

Why use Adjacency Matrices ?



Clique (Fully Connected Block):

A completely filled square block indicates a clique—every node in that group connects to every other. Cliques appear as solid dark squares.

	A	B	C	D	E	F	G	H
A	.	.	.	.	.	.	.	.
B	.	■	■	■	■	.	.	.
C	.	■	■	■	■	.	.	.
D	.	■	■	■	■	.	.	.
E	.	■	■	■	■	.	.	.
F	.	.	.	.	.	.	.	.
G	.	.	.	.	.	.	.	.
H	.	.	.	.	.	.	.	.

← Clique

What to look for: Completely filled square blocks (all cells filled). Perfect cliques have no gaps.

Visualization of Data, Understanding and Interpretation

Node-Link vs Matrix Representations

Node-Link Diagram Strengths:

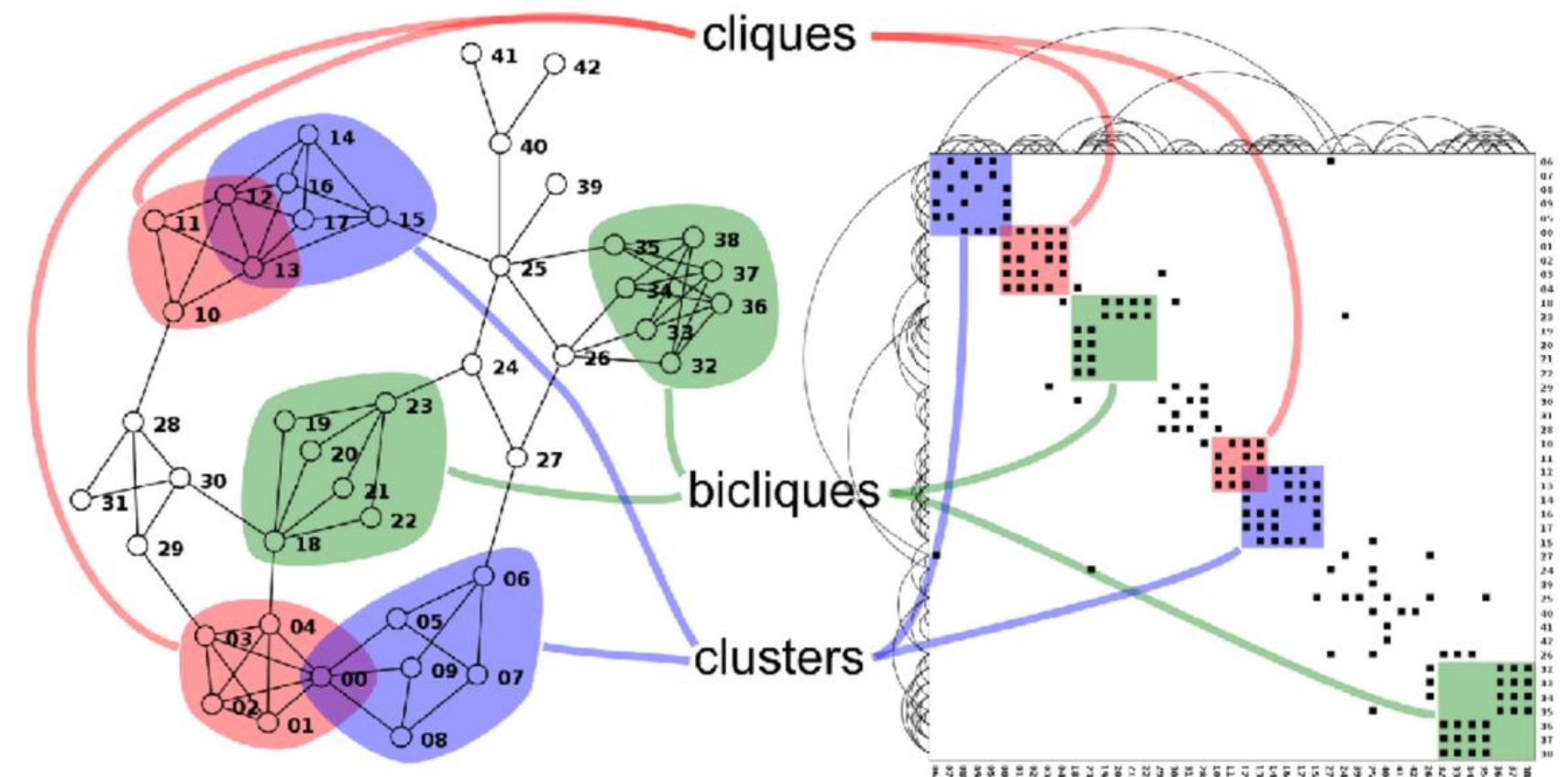
- Topology understanding, path tracing
- Intuitive, flexible, no training needed

Adjacency Matrix Strengths:

- Focus on edges rather than nodes
- Layout straightforward (reordering needed)
- Predictability, scalability
- Some topology tasks trainable

Empirical Study:

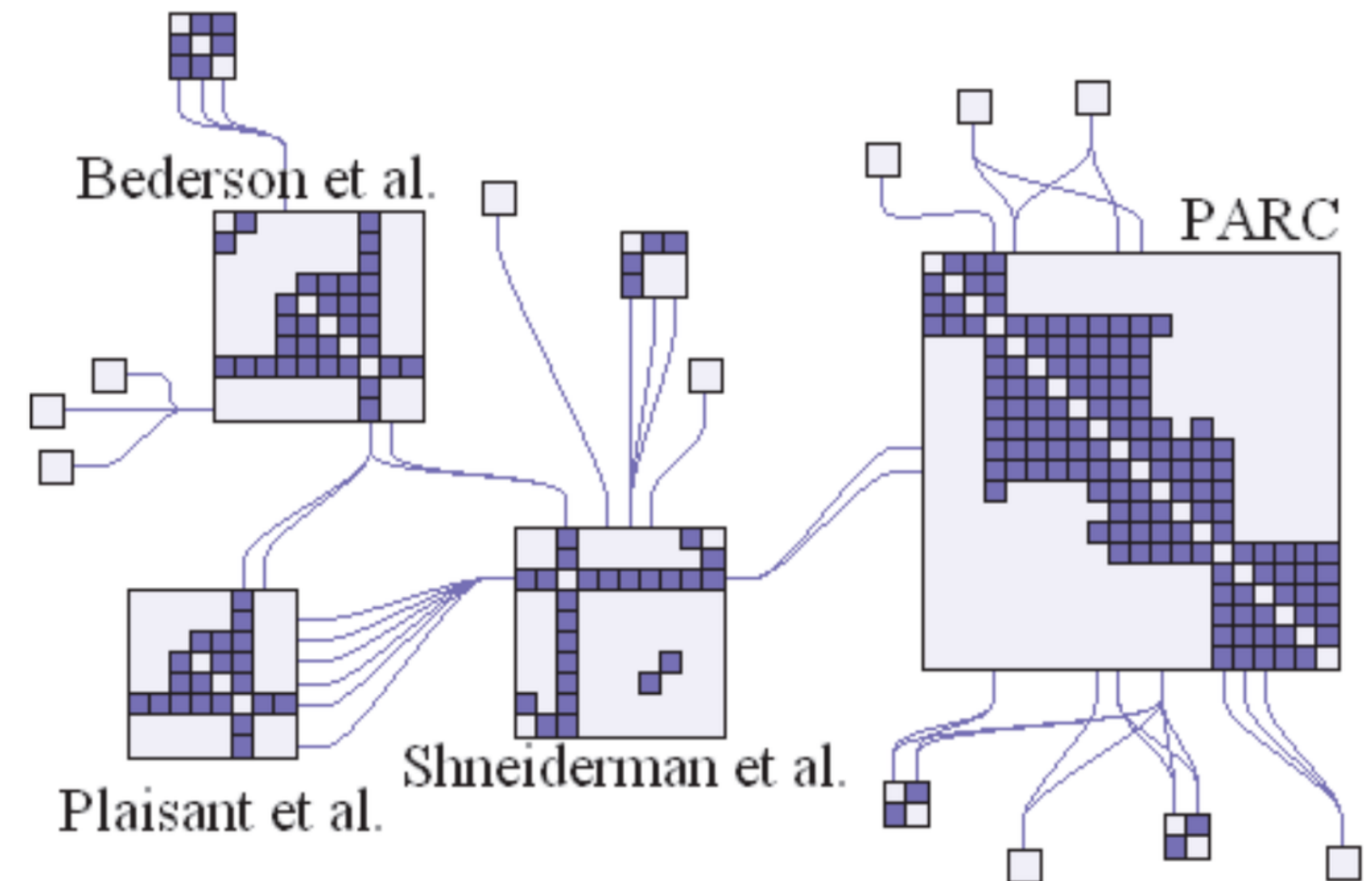
- Node-link best for small networks
- Matrix best for large networks (if tasks don't involve path tracing)



<http://www.michaelmcguffin.com/courses/vis/patternsInAdjacencyMatrix.png>

NodeTrix:

- Combines node-link diagrams and adjacency matrices into a single visualization.
- Communities are shown as matrices, making dense internal connections easy to analyze.
- Connections between communities are displayed as traditional node-link edges, preserving the overall network structure.
- Reduces visual clutter caused by overlapping edges in large graphs.
- Ideal for networks with distinct clusters, such as social, biological, and collaboration networks.
- Provides both global context and local detail, making it easier to study relationships within and between groups.



NodeTrix: a Hybrid Visualization of Social Networks. Henry, Fekete, and McGuffin. IEEE TVCG (Proc. InfoVis) 13(6):1302-1309, 2007

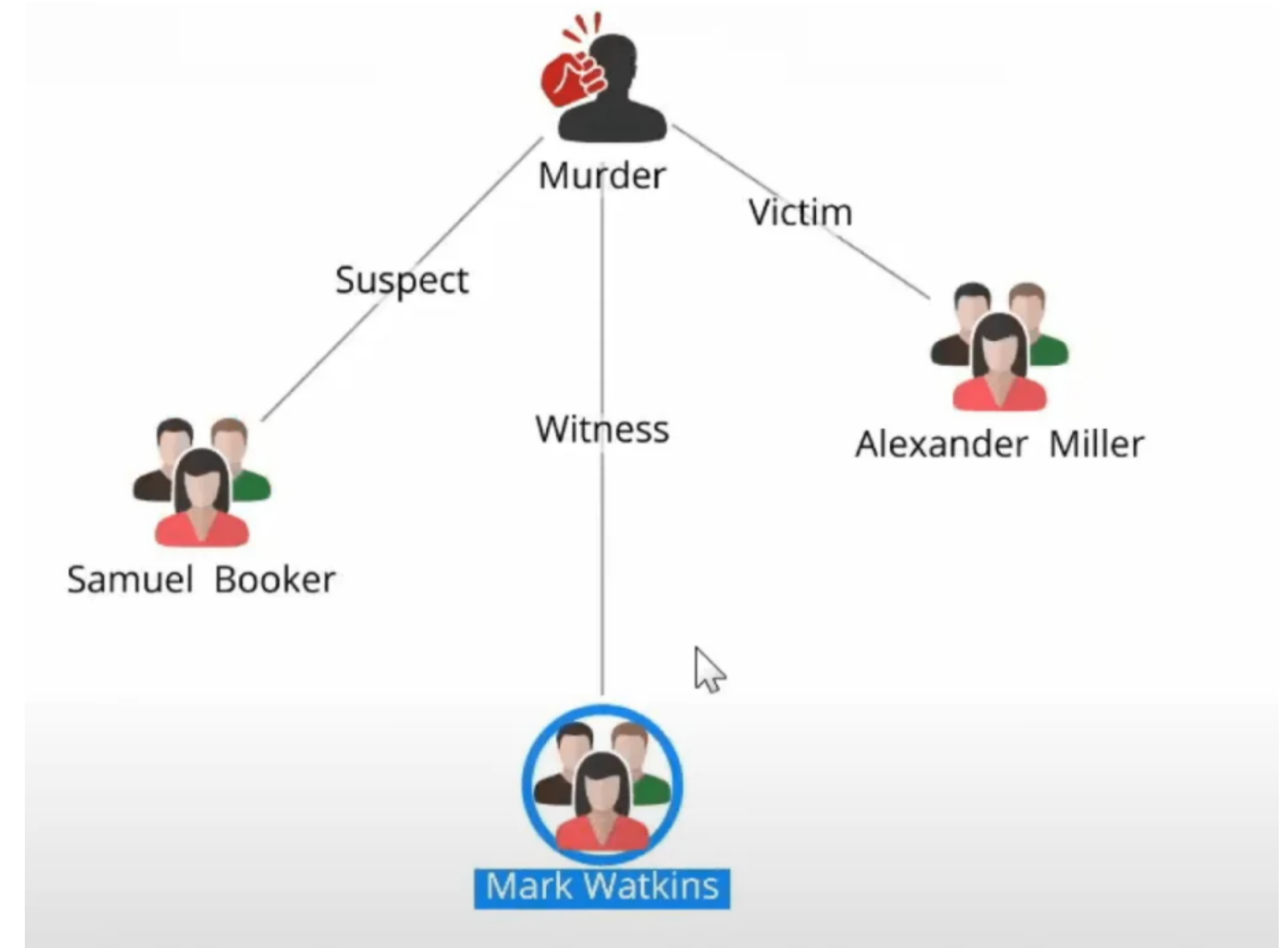
Visualization of Data, Understanding and Interpretation

Node-Link Diagram - Example

Below is an example of a node link diagram for a fictitious law enforcement use case. In this case, the node link diagram visualizes how several people relate to a crime, where Alexander Miller was a murder victim, Mark Watkins was a witness to the murder, and Samuel Booker has been identified as a suspect.

A node link diagram is an excellent construct for the many use cases where the key questions are around identifying and understanding how and whether entities are connected.

Node link diagrams are extremely useful in use cases such as intelligence analysis (e.g., one person is an associate of a suspect or known criminal), fraud detection (e.g., the same social security number was used by different people), and many others.



Visualization of Data, Understanding and Interpretation

MCQs

1. In a network visualization task such as “find friends of friends who like cats,” which combination is being used?

- A. Geometry and topology
- B. Topology and attributes
- C. Attributes and geometry
- D. Centrality and clustering

Answer: B

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MCQs

2. Why can a node-link diagram become less effective for a very dense network?

- A. Nodes lose their attributes
- B. Paths cannot be represented
- C. Excessive edges create visual clutter
- D. Directed edges become impossible

Answer: C

3. Suppose an undirected network is reordered so that several highly connected groups appear as dark blocks along the diagonal of its adjacency matrix. What does this most likely indicate?

- A. Sequential structure
- B. Isolated vertices
- C. Multiple directed paths
- D. Distinct network communities

Answer: D

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MCQs

4. In force-directed placement, what is the intended effect of the two main forces?

- A. Links attract and nodes repel
- B. Links repel and nodes attract
- C. Both nodes and links repel
- D. Both nodes and links attract

Answer: A

5. What is the main advantage of NodeTrix compared with using only a node-link representation?

- A. It removes all node ordering
- B. It avoids using network edges
- C. It combines matrices with node-link structure
- D. It guarantees an optimal graph layout

Answer: C



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 2 : Advanced Visualization & Storytelling

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

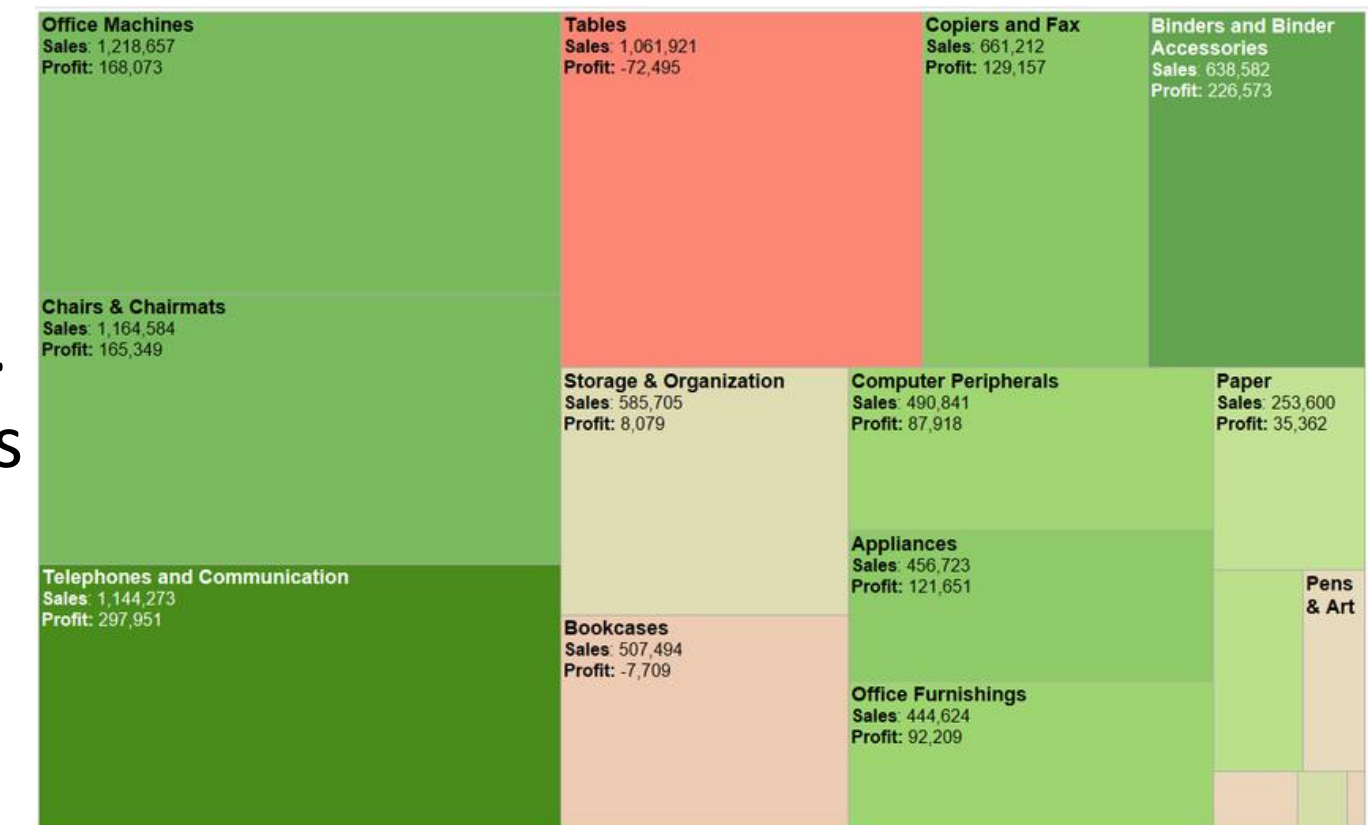
In this lecture, we will cover:

- Understand treemaps and their use in representing hierarchical data
- Explore treemap layouts, area perception, labeling, and visual encoding
- Examine visual saliency, preattentive processing, and efficient visual search
- Study attention, highlighting, redundant coding, and visual dimensions

Visualization of Data, Understanding and Interpretation

Treemaps: Space-Filling Hierarchical Representation

- A treemap represents a **tree structure using nested rectangular regions**, providing an alternative to conventional node-link diagrams.
- Construction of the original treemap:
 - The initial rectangle is divided using vertical partitions according to the branches of the tree.
 - Each resulting subrectangle is divided using horizontal partitions.
 - This alternating process continues recursively until the leaf nodes are reached.
- The area of each leaf rectangle corresponds to the amount of information stored at that node.
- **Major advantage:** The space-filling design allows large trees containing thousands of branches to be visualized while showing the amount of information associated with each branch.
- **Major limitation:** Non-leaf nodes are not directly shown, making the hierarchical structure less clear than in conventional tree diagrams.

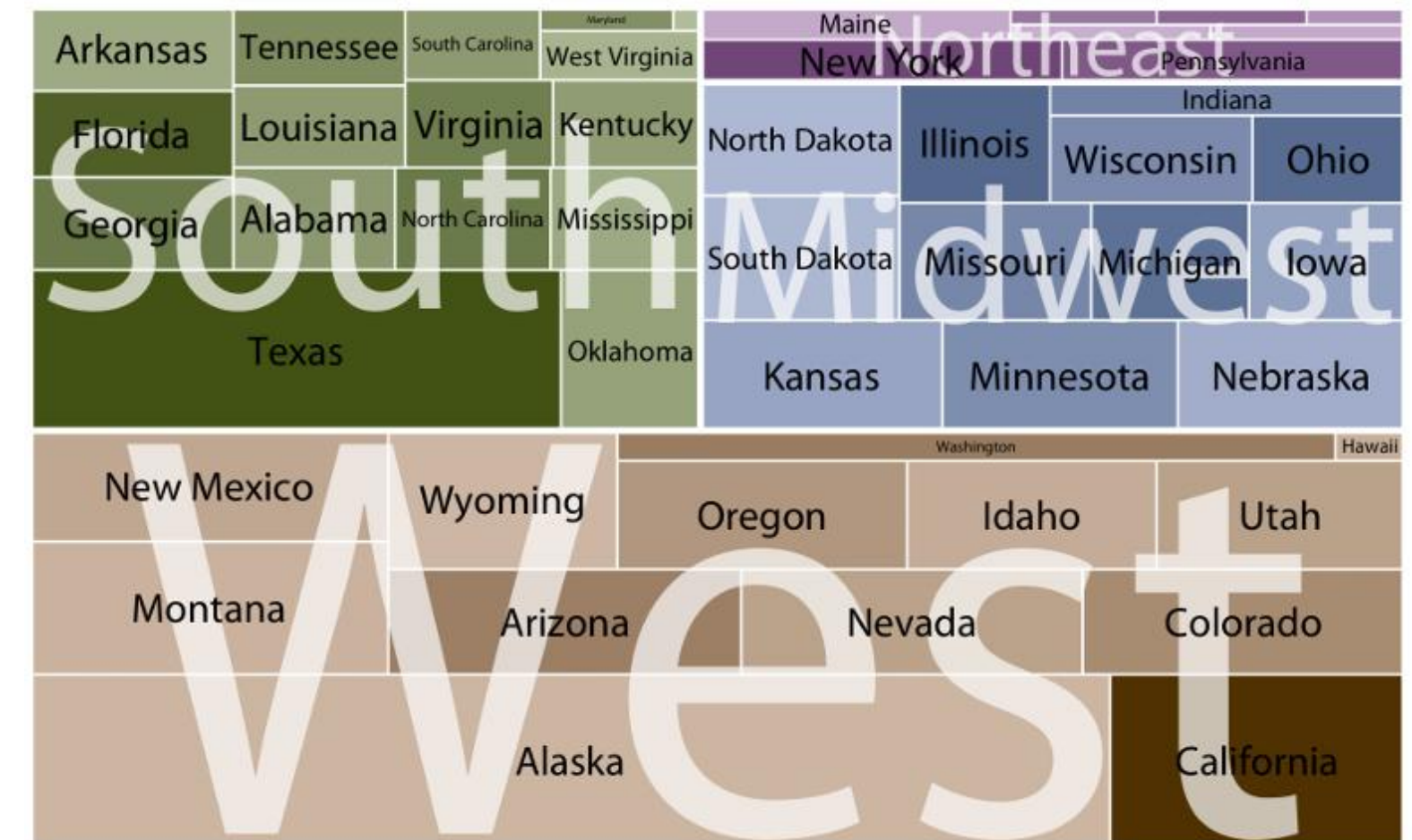


- The construction of the original treemap begins with a **single rectangle representing the root** of the tree.
- The root rectangle is divided into smaller rectangles according to the branches originating from the tree.
- At the first level, the division is made using **vertical partitions**, creating a separate region for each branch.
- Each resulting sub rectangle represents a branch of the hierarchy and is then **subdivided further** according to its own child branches. At the next level, the subdivision uses **horizontal partitions**.
- The process continues by **alternating vertical and horizontal partitions** at successive levels of the hierarchy.
- This recursive subdivision continues until the leaf nodes of the tree are reached.
- The result is a **space-filling layout** in which the entire available rectangular region is progressively allocated to different parts of the hierarchy.
- This construction allows the complete tree to be represented within a **compact 2D space**, making it possible to display very large hierarchical structures.

Visualization of Data, Understanding and Interpretation

Layout Algorithms: The Aspect Ratio Problem

- The original slice-and-dice approach can produce rectangles that are very long and thin, resulting in high aspect ratios.
- These are harder to compare by area, difficult to select and provide less space for labels.
- Higher aspect ratios indicate increasingly elongated rectangles.
- **Squarified treemaps** were developed to address this problem.
 - Their layout algorithm attempts to produce rectangles that are as close to square as possible.



States in the U.S. visualized as a treemap. Each rectangle represents one state, and the area of each rectangle is proportional to the state's land surface area. The states are grouped into four regions, West, Northeast, Midwest, and South. The coloring is proportional to the number of inhabitants for each state, with darker colors representing larger numbers of inhabitants.

Data source: 2010 U.S. Census

Key Takeaway

Treemap layout is a trade-off between preserving area while producing rectangles that are easy to compare, select, and label.

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Labeling in Treemaps: The Space Constraint



- Treemaps make efficient use of space, allowing labels to be placed **directly inside** the corresponding rectangles. Inside labels create a **stronger visual connection** between the text and the data region than labels placed outside the visualization.
- The **space-efficient rectangular layout** also makes internal labeling more practical than in alternatives such as nested pie charts, where thin regions may be difficult to label.
- However, as the hierarchy becomes deeper, individual rectangles can become very small, leaving insufficient space for labels. Consequently, labels may need to be shortened, omitted, or revealed interactively when the available space is insufficient.
- This creates an important design trade-off between showing hierarchical detail and maintaining readability.

Key Takeaway

Treemaps use their space efficiently for internal labels, but increasingly small regions can make labels difficult to display and read.

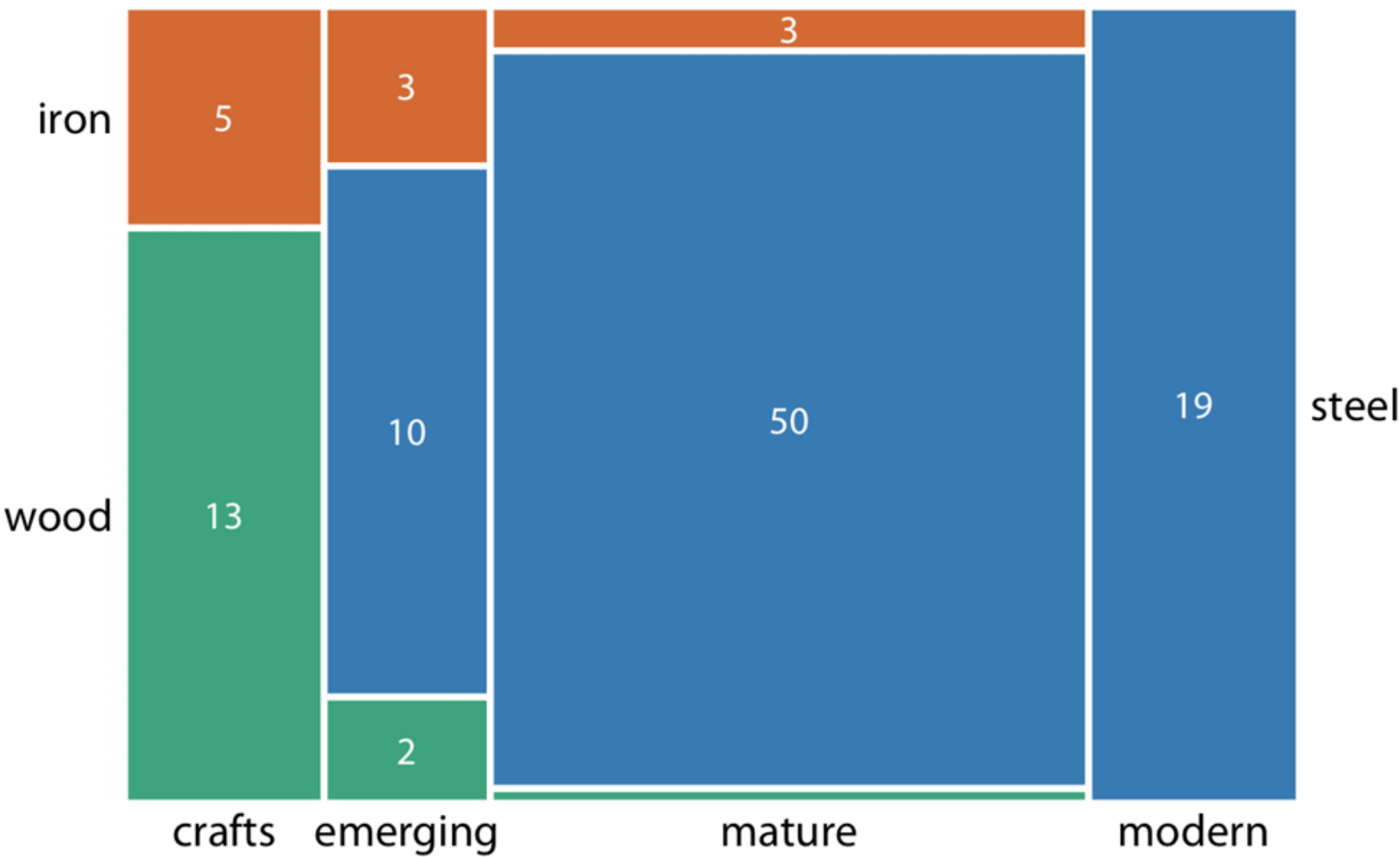
- Mosaic plots and treemaps both represent **proportions using the areas of rectangles** within an enclosing rectangle.
- Their key difference is how the rectangles are organized:
 - **Mosaic plot:** subdivides the x- and y-axes according to categorical variables.
 - **Treemap:** recursively nests rectangles inside one another to represent hierarchical relationships.
- In a treemap, the hierarchy can be represented recursively:
 - First divide the total area into major categories.
 - Then subdivide each category into its corresponding subcategories.
- Mosaic plots require categorical variables that collectively cover all observations, with combinations of the variables defining the possible groups.
- The two visualizations therefore have different emphases:
 - Mosaic plots: useful for **examining relationships** between categorical variables and their combinations.
 - Treemaps: useful for **emphasizing hierarchical proportions and the relative size** of major categories.
- Both have a common limitation: comparing areas directly can be difficult because rectangles may have different shapes and baselines.

Key Takeaway

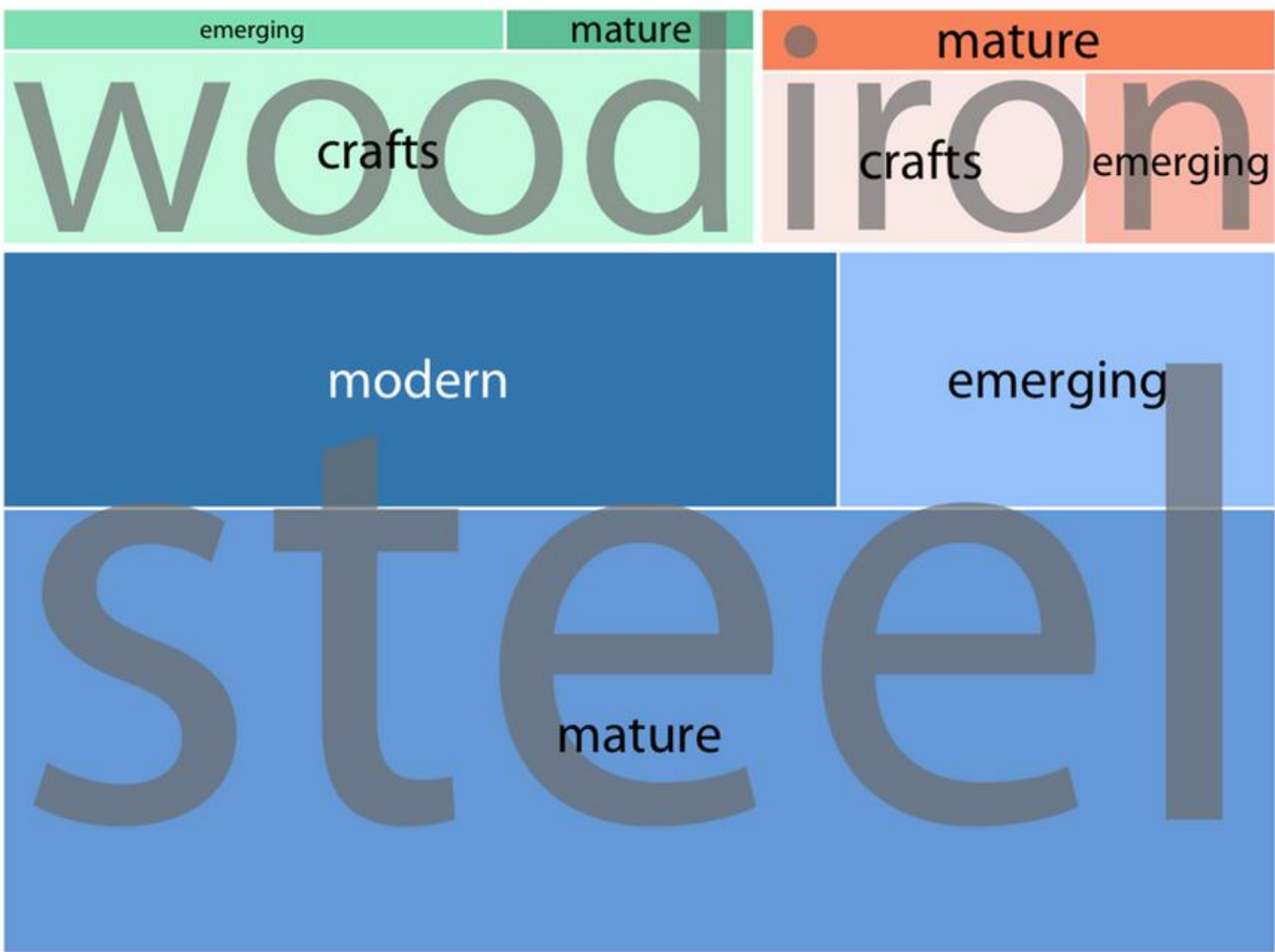
Mosaic plots emphasize relationships between categorical variables, while treemaps emphasize hierarchical proportions through recursive nesting.

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Treemaps vs. Mosaic Plots



Mosaic Plot



Tree Map

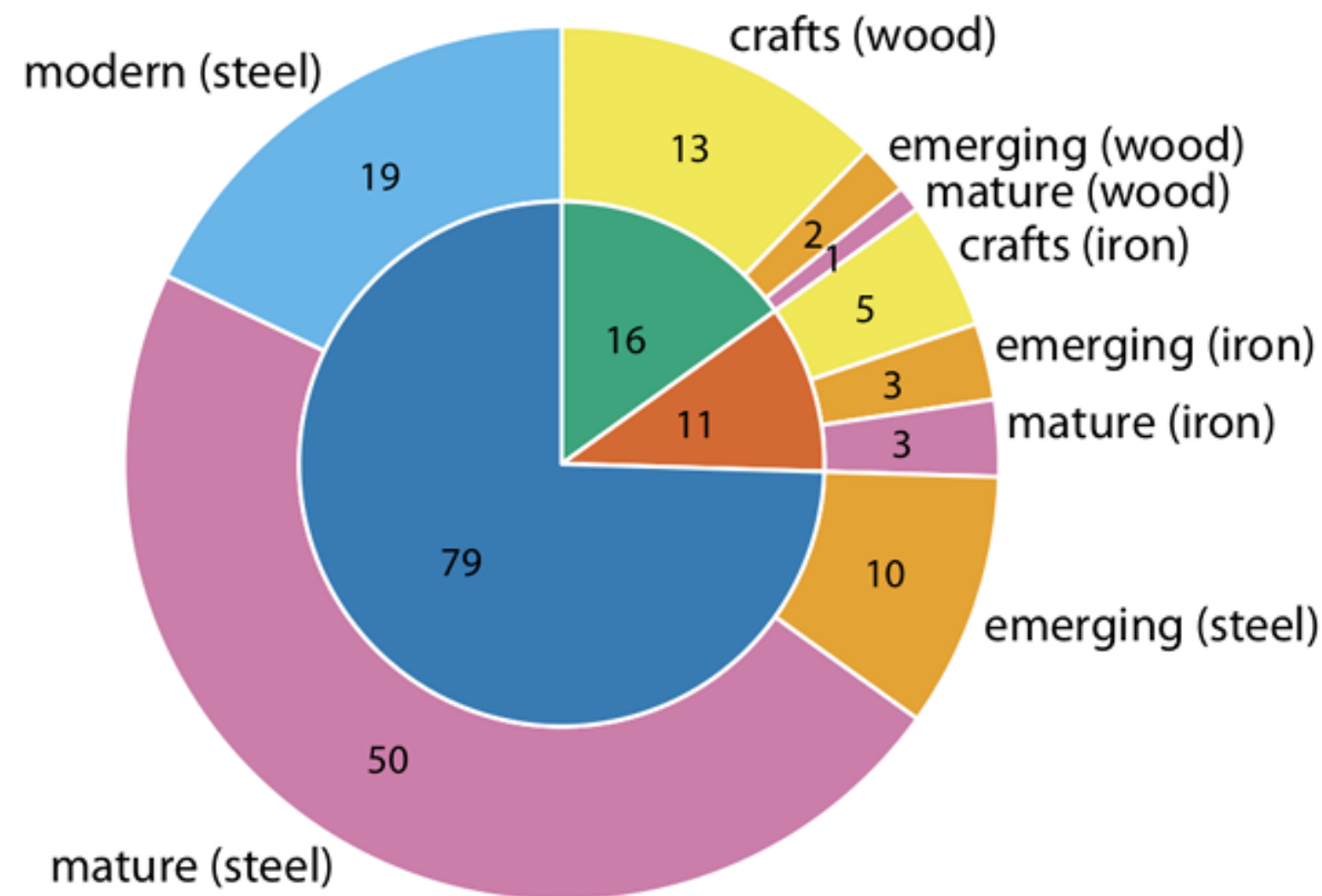
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Nested Pie Charts vs. Treemaps

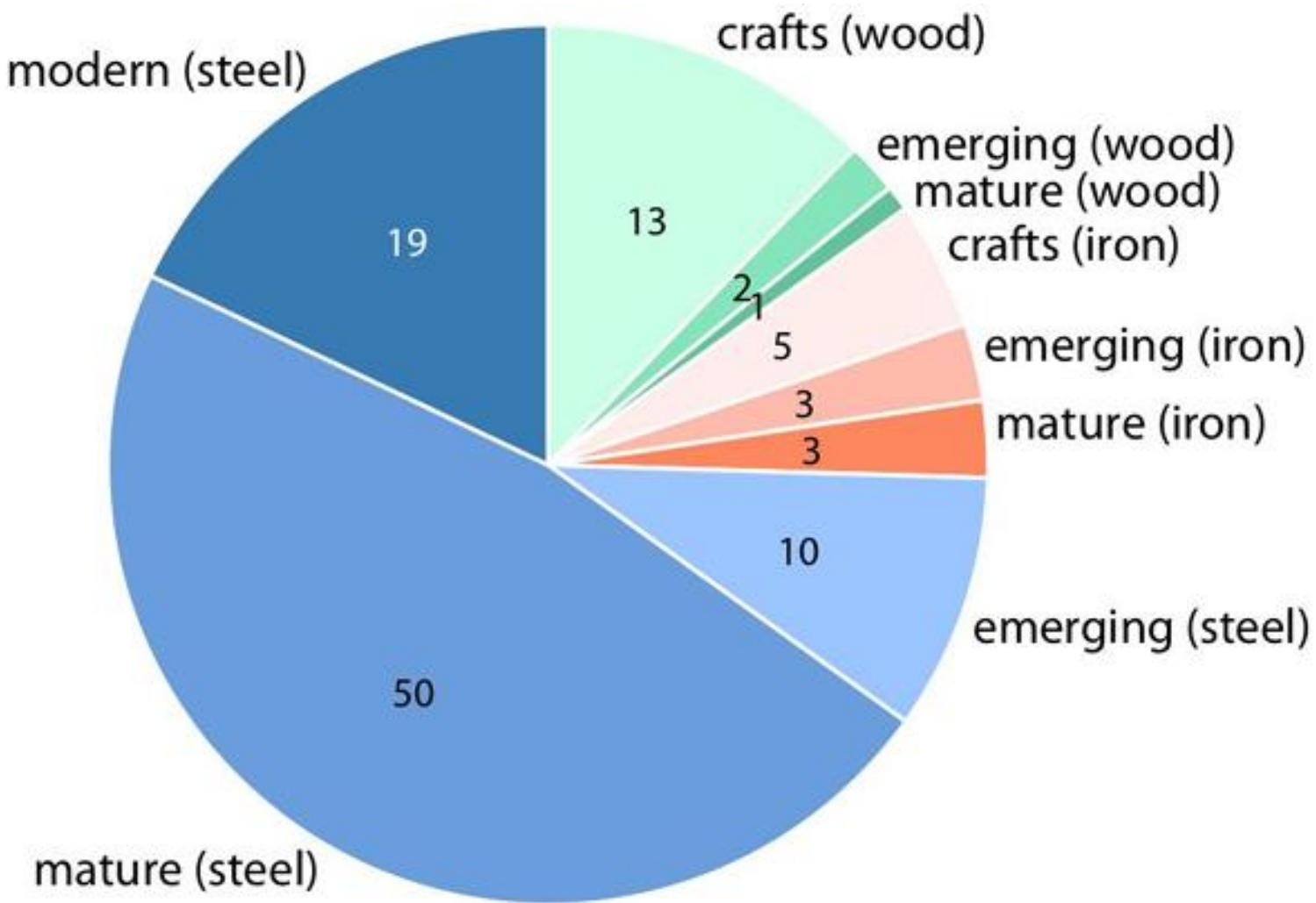
- Nested pie charts provide an alternative for visualizing hierarchical proportions, using area to represent data values. Like mosaic plots and treemaps, they differ from ordinary pie charts mainly in their use of a **polar coordinate system** rather than Cartesian coordinates.
- One approach uses concentric circles:
 - The inner circle **represents the breakdown by a primary variable**.
 - The outer circle **further subdivides each category** according to a second variable.
- A second approach recursively subdivides the pie into smaller slices and uses nested colors to indicate the hierarchy.
 - The primary variable determines the **major slices**.
 - The secondary variable determines **subdivisions within each slice**.
 - Color intensity can distinguish different levels of the secondary variable.
- Although it represent the data reasonably well, treemaps are preferable in several respects:
 - Rectangles make more efficient use of available space.
 - Treemaps have little superfluous white space, leaving more room for labels.
 - Very thin pie slices can be difficult to see, whereas treemap rectangles can provide more usable regions.

Visualization of Data, Understanding and Interpretation

Nested Pie Charts vs. Treemaps



Breakdown of bridges in Pittsburgh by construction material (steel, wood, iron, inner circle) and by era of construction (crafts, emerging, mature, modern, outer circle). Numbers represent the counts of bridges within each category.

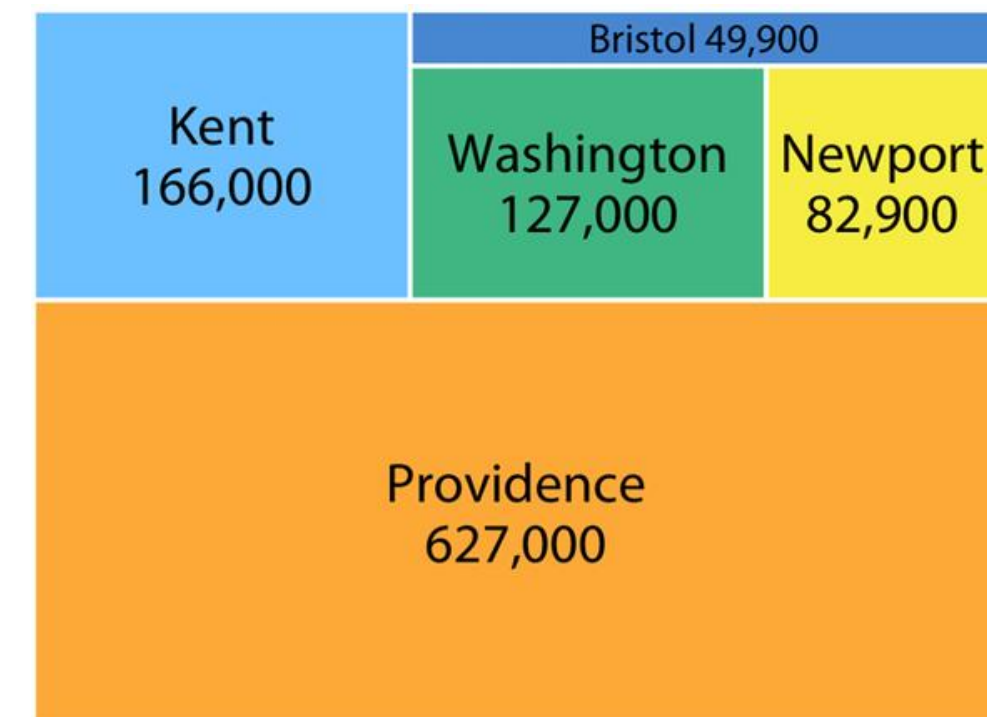
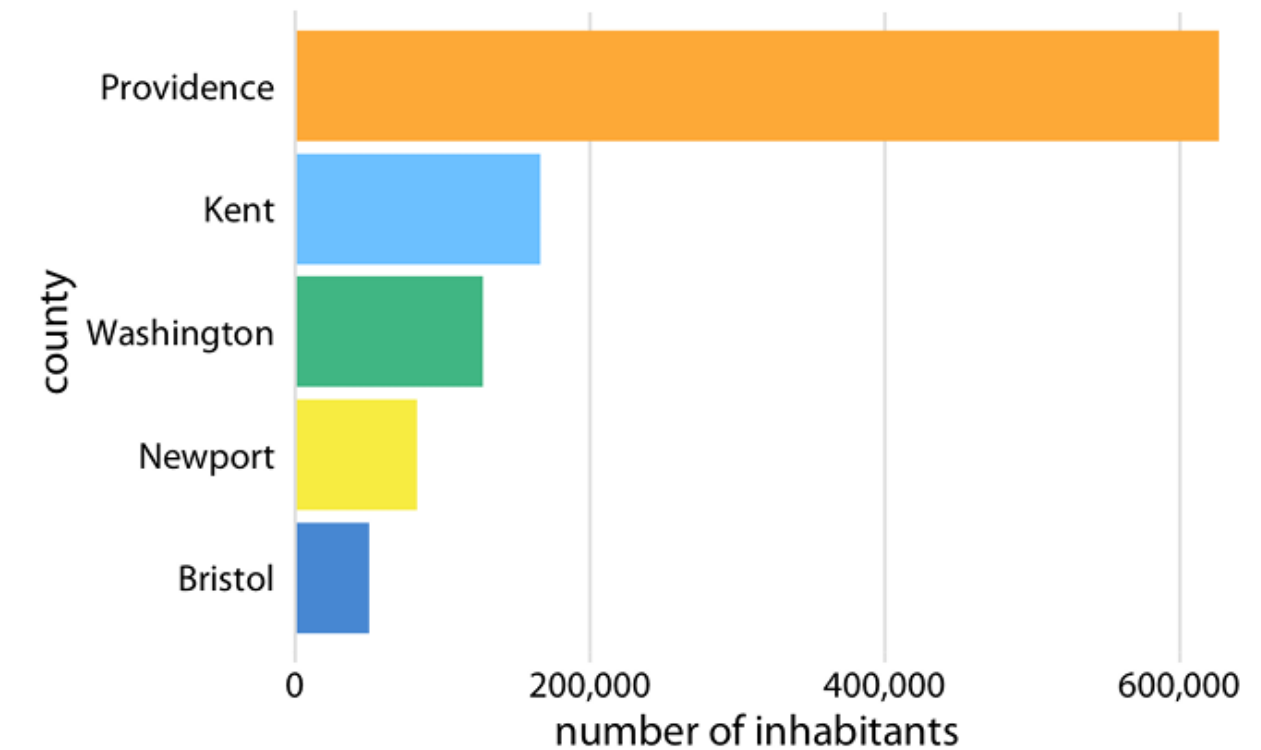


Breakdown of bridges in Pittsburgh by construction material (steel, wood, iron) and by era of construction (crafts, emerging, mature, modern). Numbers represent the counts of bridges within each category.

Visualization of Data, Understanding and Interpretation

Treemaps and the Perception of Area

- Treemaps encode quantitative values primarily through the **area of rectangular regions**. Although the areas are correctly proportional to the underlying data values, area is not perceived as precisely as distance.
- Human perception is generally better at comparing **linear distances than comparing areas**. Therefore, differences between values can appear less pronounced in a treemap than when the same values are represented using bar lengths.
- Treemaps are therefore particularly useful for **understanding relative proportions and hierarchical structure**, rather than making highly precise comparisons between individual values.



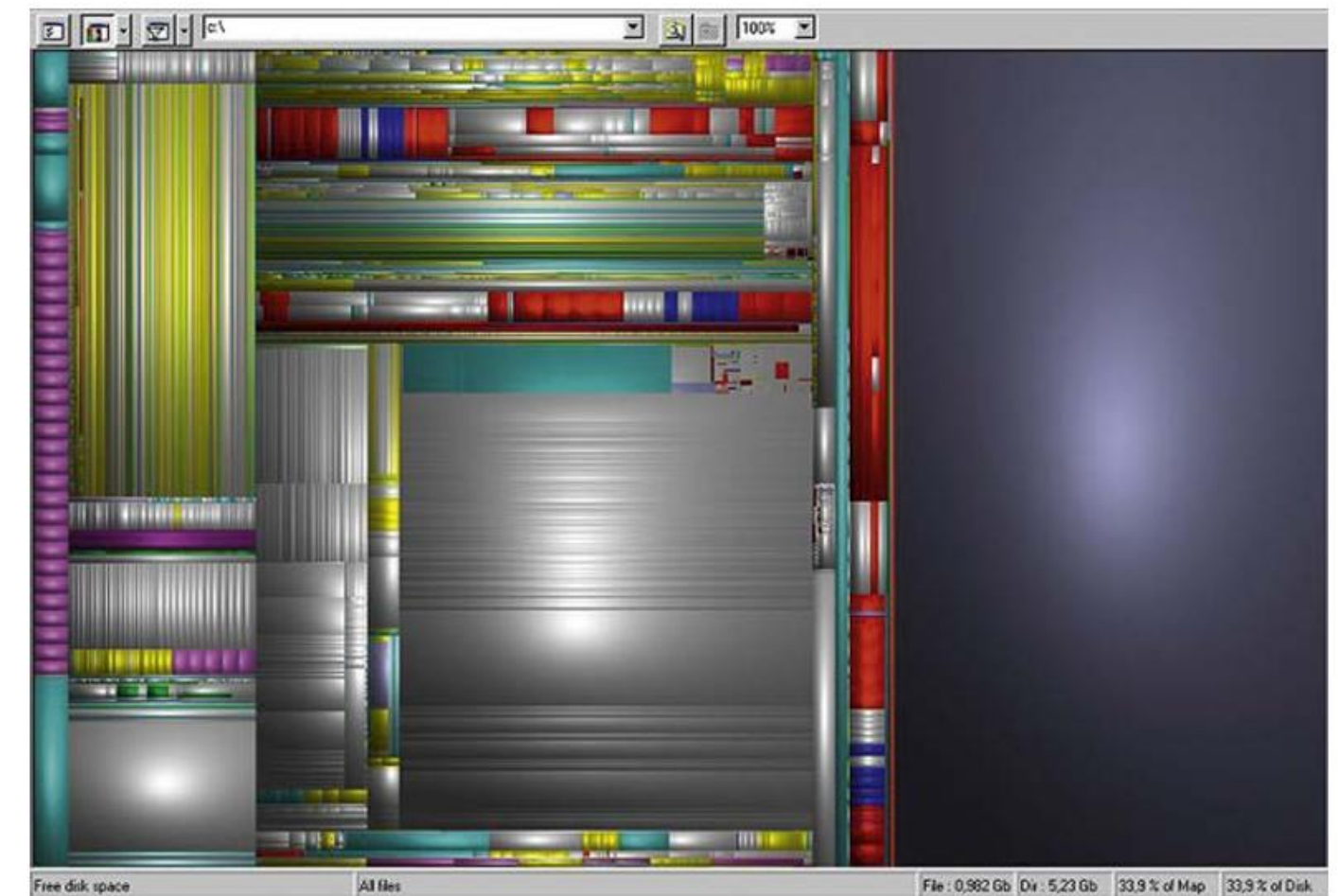
Key Takeaway

Treemaps accurately encode values through area, but human perception makes area-based comparisons less precise than distance-based comparisons.

Visualization of Data, Understanding and Interpretation

Cushion Maps: Enhancing Treemap Hierarchy with Shading

- A major limitation of conventional treemaps is that they are highly effective at showing the size and grouping of leaf nodes, but they do not convey the hierarchical structure particularly clearly.
- **Cushion maps** address this limitation by adding shading to the treemap.
- A hierarchical shading model is applied recursively to the treemap:
 - Large branches receive an overall shading pattern.
 - Smaller branches receive their own shading within the shading of their parent.
- The resulting changes in shading provide shape-from-shading cues, making different levels of the hierarchy more visually distinguishable. This allows the hierarchical structure to become more apparent than in an unshaded treemap.



The cushion map is a variation of a treemap that uses shape-from-shading information to reveal hierarchical structure

Key Takeaway

Cushion maps enhance conventional treemaps with **hierarchical shading**, making parent–child structure more visually apparent.

Visualization of Data, Understanding and Interpretation

Multi-Channel Encoding: Area, Color & Luminance

- Treemaps can use **multiple visual channels** to represent different data attributes simultaneously.
- **Area** → represents a **quantitative size** attribute.
- **Color** → distinguishes different categories or directions of change.
- **Luminance** → represents the magnitude of a value or deviation.
- A **diverging/double-ended color scale** can represent values above and below a neutral reference point.
- Combining these channels allows a treemap to communicate multiple dimensions of data within the same hierarchical layout.



A color sequence with black representing zero. Increasing positive values are shown by increasing amounts of red. Increasing negative values are shown by increasing amounts of green. The map itself is a form of treemap.

Key Takeaway

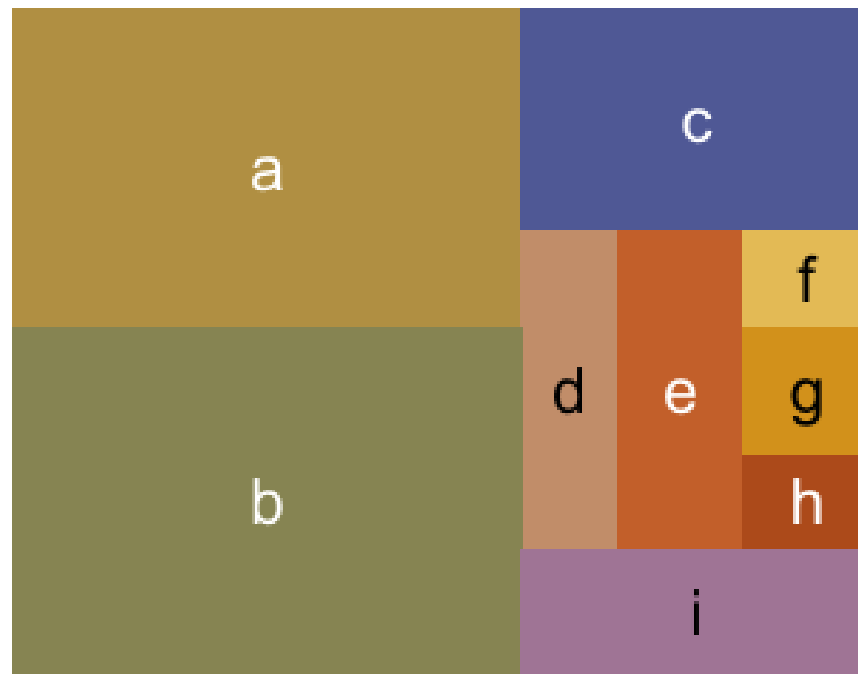
Multiple visual channels allow treemaps to encode several data attributes simultaneously.

Visualization of Data, Understanding and Interpretation

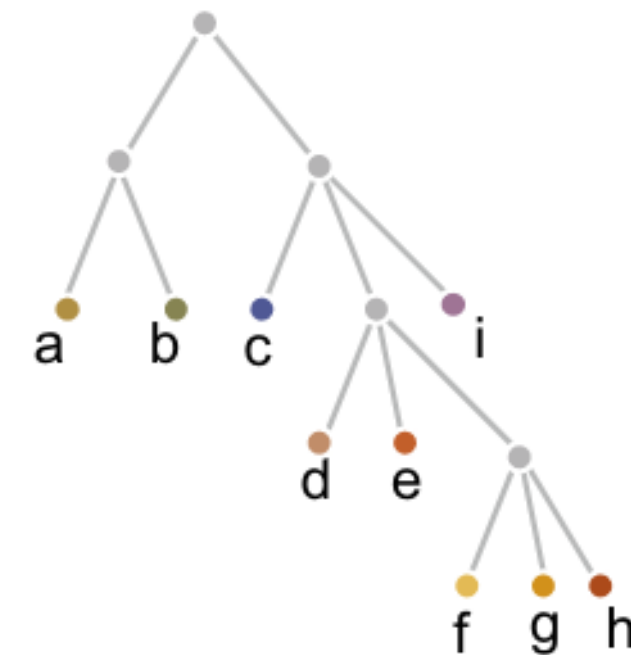
Treemaps vs. Node-Link Diagrams: Scalability

- **Node-link diagrams:** Make hierarchical relationships and connections explicit.
- **Treemaps:** Use a space-filling layout to display large numbers of hierarchical elements compactly.
- Treemaps are particularly effective for showing the size and grouping of leaf nodes.
- The trade-off is that treemaps sacrifice some clarity of the internal tree structure for greater space efficiency and data density.

Node-link diagrams **emphasize structure**; treemaps **emphasize space-efficient representation** of quantities.



A treemap representation of hierarchical data.



The same tree structure, represented using a conventional node-link diagram.

Visualization of Data, Understanding and Interpretation

Interactive Treemaps: Drill-Down & Elision



- Treemaps can display large and deeply nested hierarchies, but showing every level simultaneously can make individual regions difficult to examine.
- Drill-down allows the user to select a parent rectangle and explore its children in greater detail.
 - Example: selecting a sector can reveal the companies belonging to that sector.
 - Selecting a company or subgroup can reveal its next level of information.
- Elision addresses the opposite problem: instead of displaying the entire hierarchy, irrelevant portions can be hidden or collapsed to conserve screen space.
- These interactions allow a treemap to support both:
 - **Overview** → see the overall hierarchy and proportions.
 - **Detail** → focus on a selected part of the hierarchy.

Key Takeaway

Drill-down provides detail within a selected treemap region, while elision reduces visual complexity by hiding unnecessary hierarchy.

Visualization of Data, Understanding and Interpretation

Limitations of Treemaps

- **Hierarchical structure:** Treemaps are excellent at showing the size and grouping of leaf nodes, but the underlying tree structure is less explicit than in node-link diagrams.
- **Area comparison:** Comparing individual rectangles can be difficult because they do not share a common baseline.
- **Shape variation:** Different rectangles can have very different shapes, making differences in their areas difficult to judge accurately.
- **Small regions:** Deep hierarchies can produce small rectangles, creating problems with visibility and labeling.
- **Perceptual limitation:** Area-based comparison is inherently less precise than comparing linear dimensions, such as bar lengths.

Key Takeaway

Treemaps provide **excellent space efficiency and hierarchical overview**, but this comes at the cost of precise comparison, explicit hierarchy, and readability of small regions.

Visualization of Data, Understanding and Interpretation

Saliency Visualization

- Visual search begins with a visual query — a mental question about **what we need to find** in a visualization.
- The visual system then searches the display for patterns that can resolve that query.
- How quickly something is found depends on how **visually distinct** it is from surrounding information.
- Three factors that determine how easily a visual element can be found:

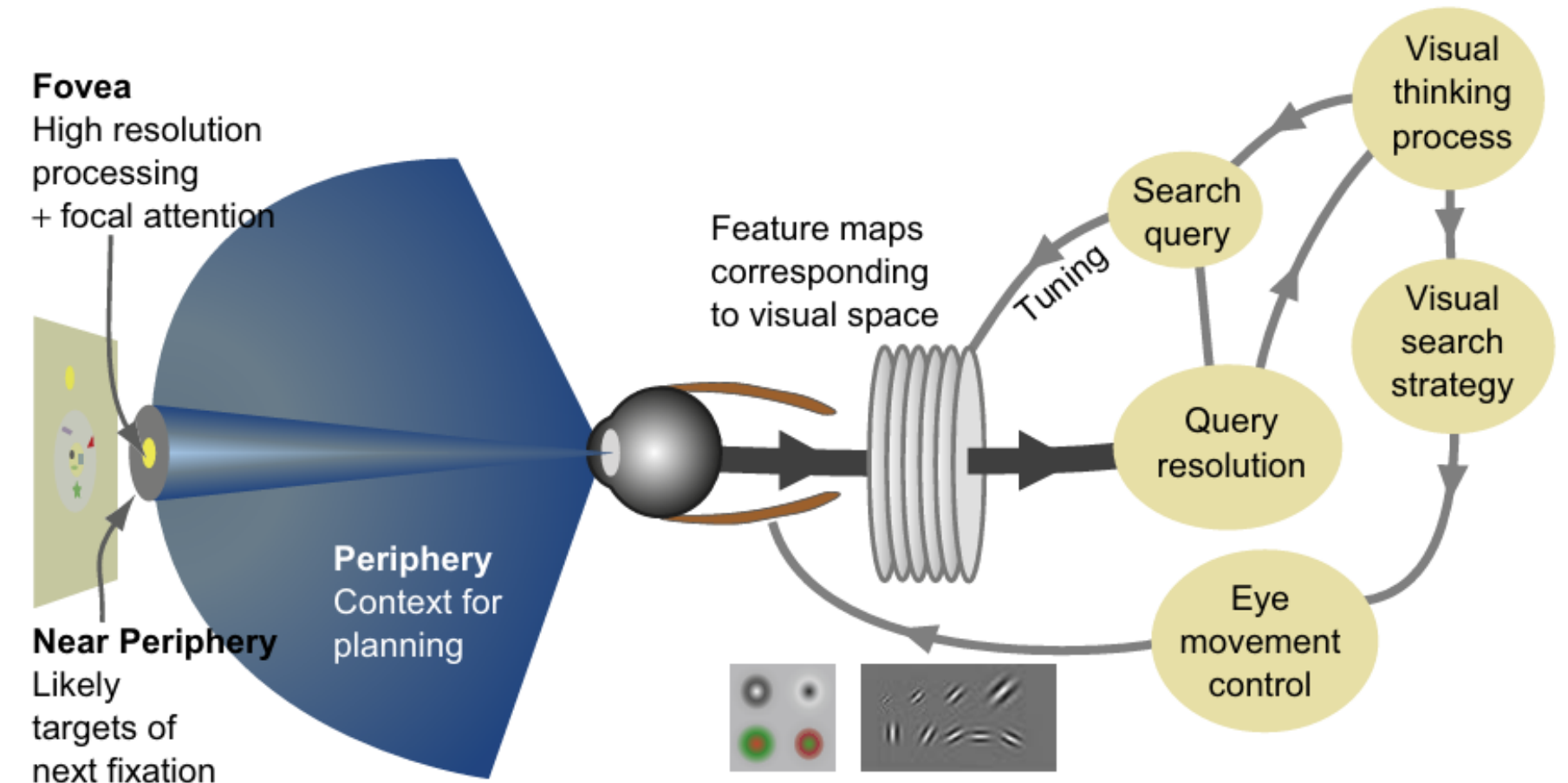
1. A priori salience — some visual patterns naturally generate stronger neural activity than others.

2. Top-down salience modification — attention can tune the visual system toward features relevant to the current task.

3. Scene gist — rapid recognition of the overall type of scene or visualization helps activate appropriate search strategies.

Key Takeaway

Visual saliency determines how easily important information stands out during visual search.



The visual search process

Visualization of Data, Understanding and Interpretation

Visual Queries: Turning Problems into Visual Searches

- A visual query transforms an aspect of a problem into a **visual pattern search**.
- The user first determines what needs to be found, then searches the visualization for the corresponding visual pattern.
- The visual pattern may be a **shape, color, position, size**, or more complex combination of features.
- Visual queries are a critical part of visual thinking because they determine **how efficiently information can be extracted** from a display.
- A good visualization **maps data to graphical properties** so that likely visual queries can be executed quickly and accurately.
- Visual thinking is constrained by visual working memory, making **efficient visual encoding** especially important.
- In interactive visualizations, actions such as **zooming and selecting** can support visual queries by providing additional information when required.

Key Takeaway

Good visualization design turns information-seeking problems into efficient visual searches.

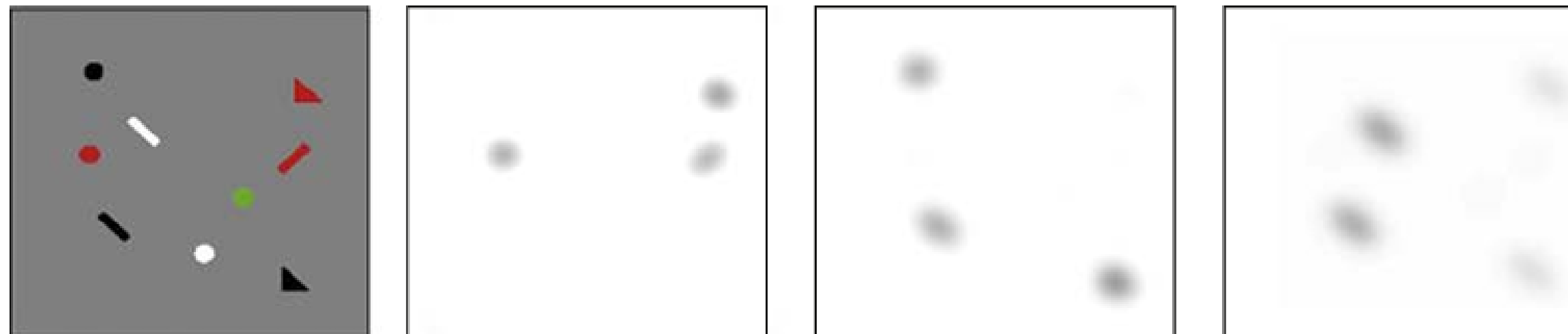
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Parallel Processing & Feature Maps

- Once a visual query defines what we are looking for, the visual system begins searching the display for the relevant features.
- The first stage of this process is rapid parallel processing: basic visual properties are extracted simultaneously across the entire visual field.
- Different properties are processed through separate feature maps, such as color — e.g., redness or greenness, orientation — e.g., vertical or horizontal edges, motion — e.g., vertical or horizontal movement and texture and spatial-frequency characteristics.
- Matching regions become candidate targets for attention and eye movements.

Key Takeaway

Visual queries specify what to look for; feature maps provide the low-level mechanisms that help locate it.



The symbols shown on the left are processed via a set of feature maps and the result directs eye movements

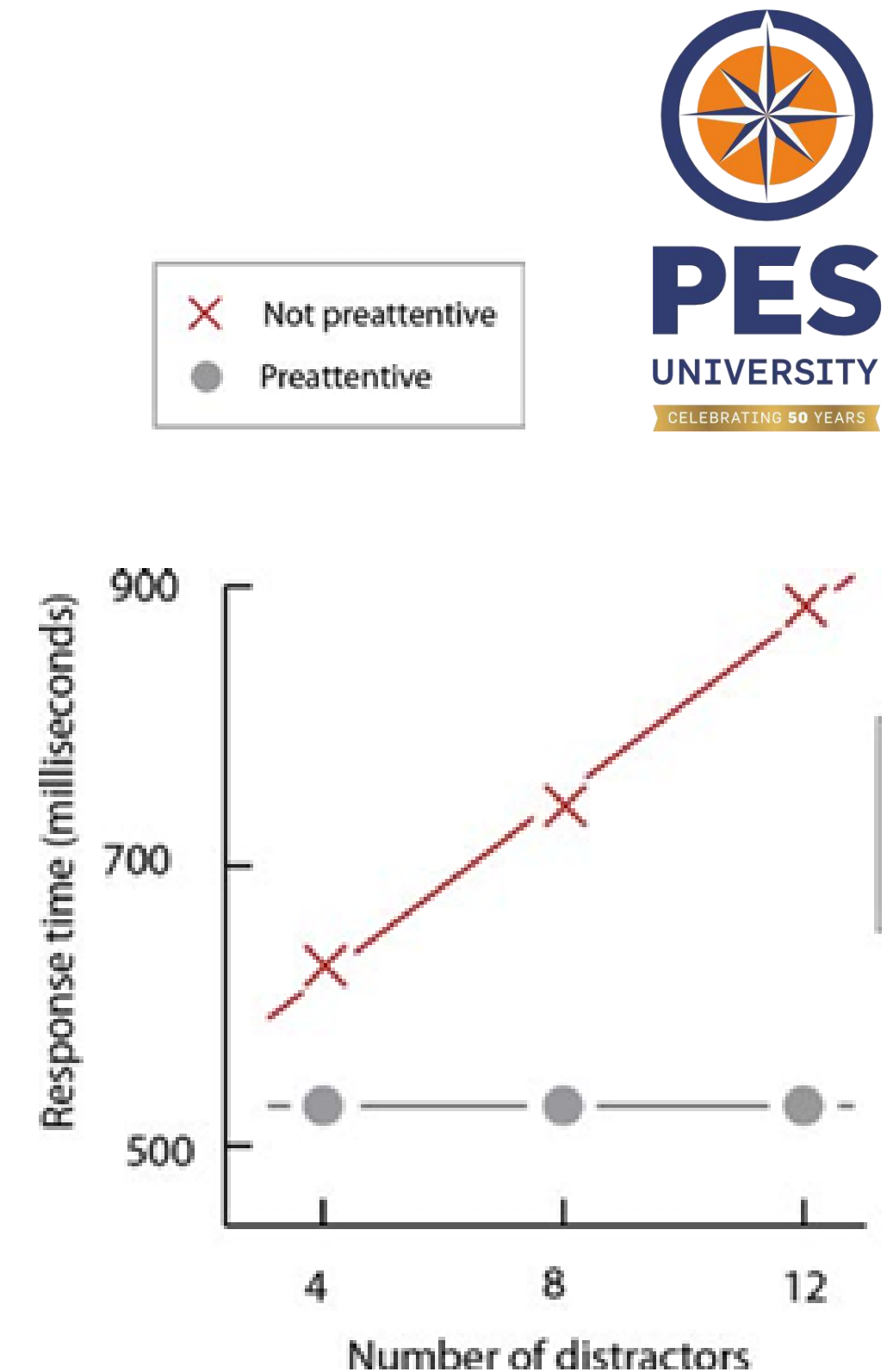
Visualization of Data, Understanding and Interpretation

Preattentive Processing & Visual “Pop-Out”

- Some visual features are processed so efficiently that they immediately stand out from their surroundings. This phenomenon is called **preattentive processing**.
- This creates a **pop-out effect**: the target can be found quickly even when the number of irrelevant distractors increases.
- Two search patterns:
 - **Preattentive / parallel search**: detection time remains largely independent of the number of distractors.
 - **Non-preattentive / sequential search**: search time increases as distractors increase.
- Strong preattentive cues include **color, orientation, size, contrast/luminance, and motion**. These features help guide attention and determine what the viewer looks at next.
- **Design implication**: When ease of search is important, use strong preattentive cues before weaker visual cues.

Key Takeaway

Preattentive features make important elements “pop out,” allowing them to be found quickly without a sequential search.

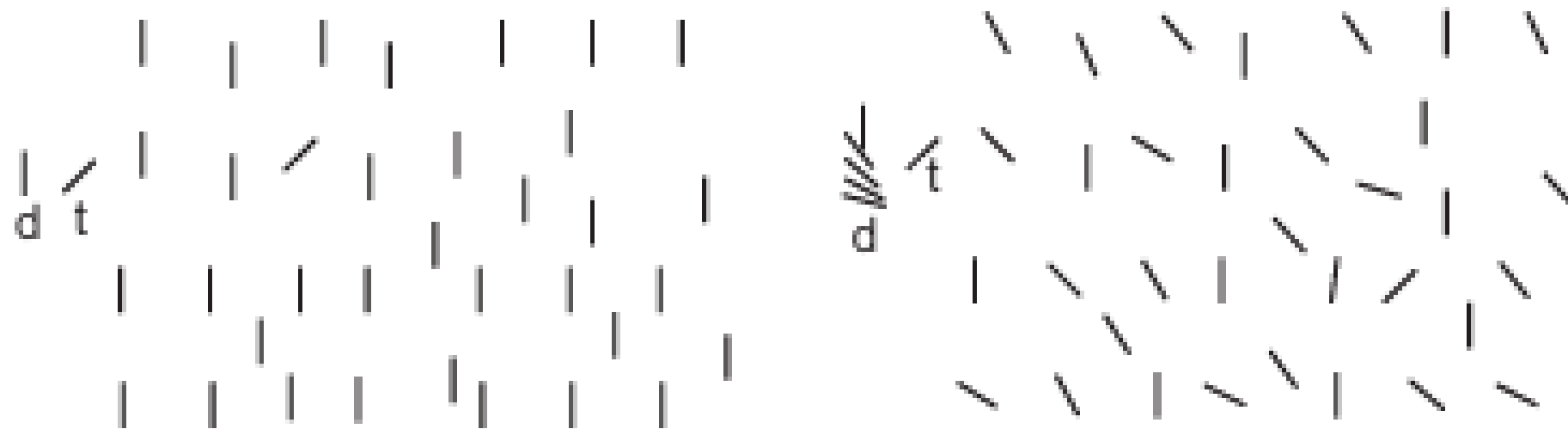


The circles show time to perceive an object that is preattentively distinct from its surroundings. In this case, time to process is independent of the number of irrelevant objects (distractors). The Xs show how time to process nonpreattentively distinct targets increases with the number of distractors.

Visualization of Data, Understanding and Interpretation

Contrast and Context

- Saliency depends on difference: A visual target is easier to find when **it differs clearly** from surrounding elements.
- **Orientation:** A differently oriented line can pop out when the surrounding lines have a consistent orientation.
- **Distractor variety matters:** As the variety among surrounding elements increases, the target becomes less distinctive.
- **Key principle:** The effectiveness of a visual cue depends on both the target–distractor difference and the similarity among distractors.

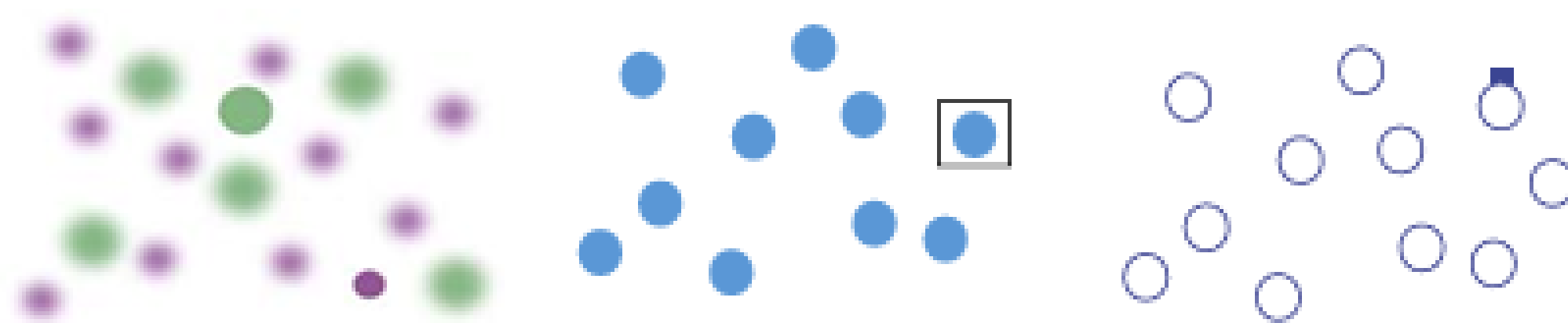


On the left, the right-slanted bar pops out; on the right, it does not. Yet, most of the distractors on the right have an orientation that is more different from the target orientation than the distractors on the left.

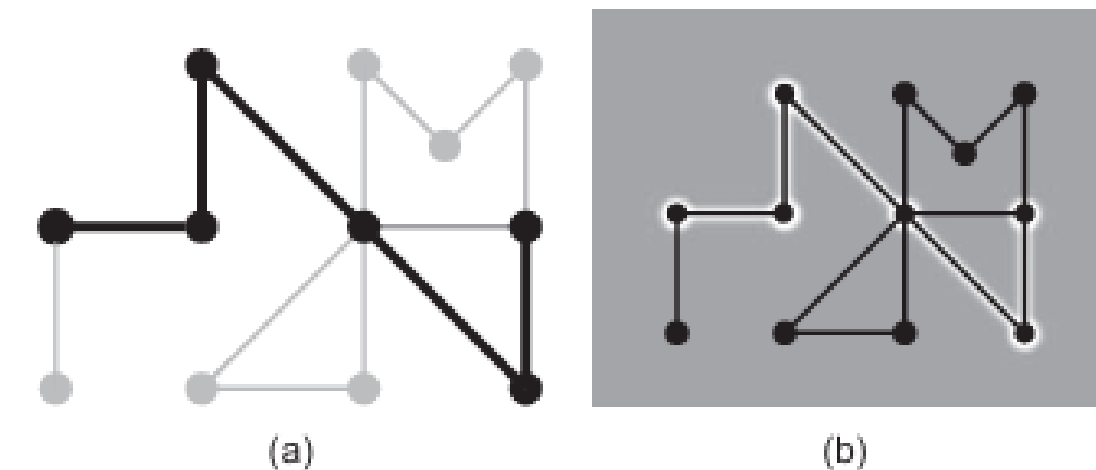
Visualization of Data, Understanding and Interpretation

Highlighting Strategies

- **Use distinctive cues:** Highlight important information using visual features that are not heavily used elsewhere in the display.
- **Positive asymmetry:** Adding a mark is generally more effective for highlighting than removing one.
- **Multiple channels:** Color, shape, size, texture, motion, and blur can be used to direct attention.
- **Subtle highlighting:** Motion or blinking can be effective when other visual channels are already occupied, but should remain subtle.
- **Semantic depth of field:** Blurring less important information can make relevant elements stand out, but excessive blur may reduce readability.



A number of highlighting methods that use positive asymmetric preattentive cues: sharpness, added surrounding feature, added shape.



Two methods for highlighting a node-link diagram. (a) The contrast is reduced for the less important parts of the network. (b) The background contrast is increased using haloing to emphasize important parts.

Visualization of Data, Understanding and Interpretation

Redundant Coding for Visual Search

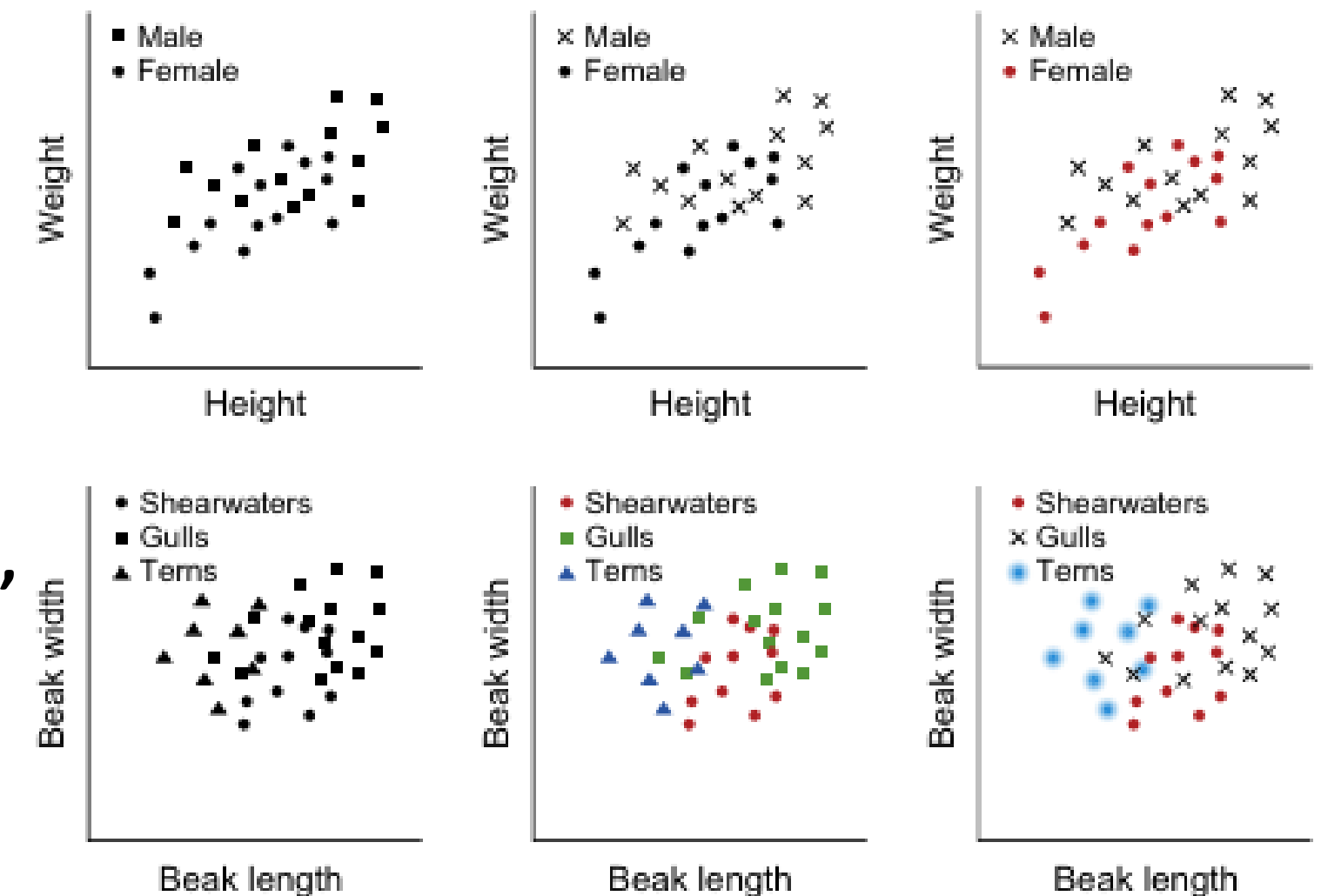
- Encode the same data distinction using **multiple visual channels**, such as color + shape + size.
- Different visual channels are processed independently, so combining them can make symbols more visually distinct.
- Particularly useful when one visual channel is **already heavily used elsewhere** in the visualization.
- Example: distinguish two categories using different shapes and colors rather than color alone.
- Redundant coding can **improve visual search, identification, and robustness** when a single cue is difficult to perceive.

Design Guideline

Make symbols distinct in terms of spatial frequency, orientation, and color for maximum visibility.

Example:

A scatterplot can distinguish groups using shape + color, making the categories easier to locate than using shape or color alone.



Feature channels can be used to make symbols more distinct from one another. The graphs on the right use redundant color coding in addition to more distinctive shapes

Visualization of Data, Understanding and Interpretation

Bottom-Up and Top-Down Attention



Bottom-Up Processing

- Begins with the visual properties of the display.
- Feature maps detect properties such as color, orientation, size, and motion across the visual field.
- Salient features can attract attention even before detailed conscious analysis.

Top-Down Processing

- Attention is influenced by the current task or visual query.
- The visual system becomes more sensitive to features relevant to what we are looking for.
- Example: Searching specifically for vertical objects increases sensitivity to vertical orientation.

Interaction Between the Two

- Bottom-up processing provides candidate visual features.
- Top-down attention determines which features and patterns are relevant to the current task.
- Thus, visual search is not purely stimulus-driven; it is continuously tuned by the observer's goals.

Key Takeaway

Bottom-up processing tells us what stands out; top-down attention determines what matters.

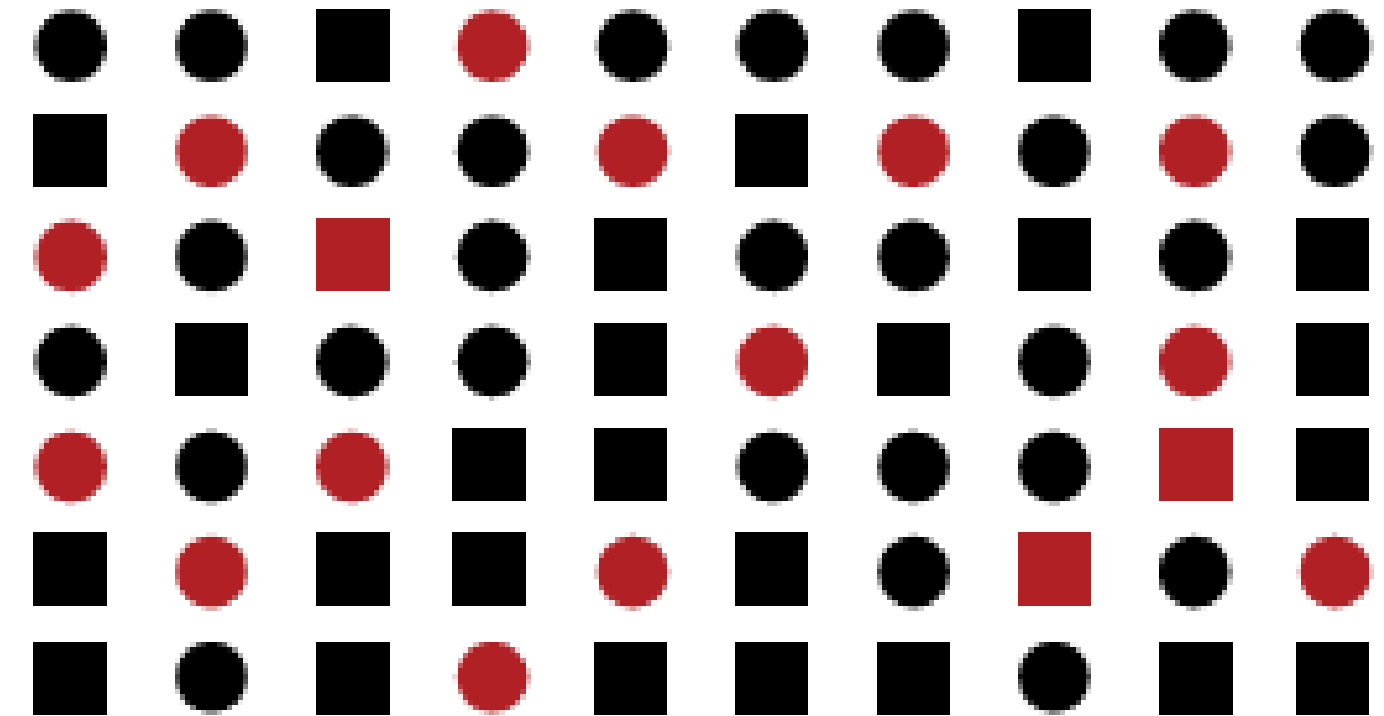
Visualization of Data, Understanding and Interpretation

Conjunction Search — The Processing Bottleneck

- A conjunction search requires finding an object defined by **two or more visual features** simultaneously.
- Example: finding a red square among red circles and black squares.
- Neither color alone nor shape alone identifies the target.
- Therefore, the visual system cannot simply use a single feature map to locate it.

Why is it slower?

- Unlike single-feature pop-out, conjunction searches are generally not preattentive.
- The observer must effectively examine candidate objects serially, making search time increase as the number of distractors increases.
- Even extensive experience does not make complex conjunctions instantly findable.



Searching for the red squares is slow because they are identified by a conjunction of shape and color.

Integral Dimensions

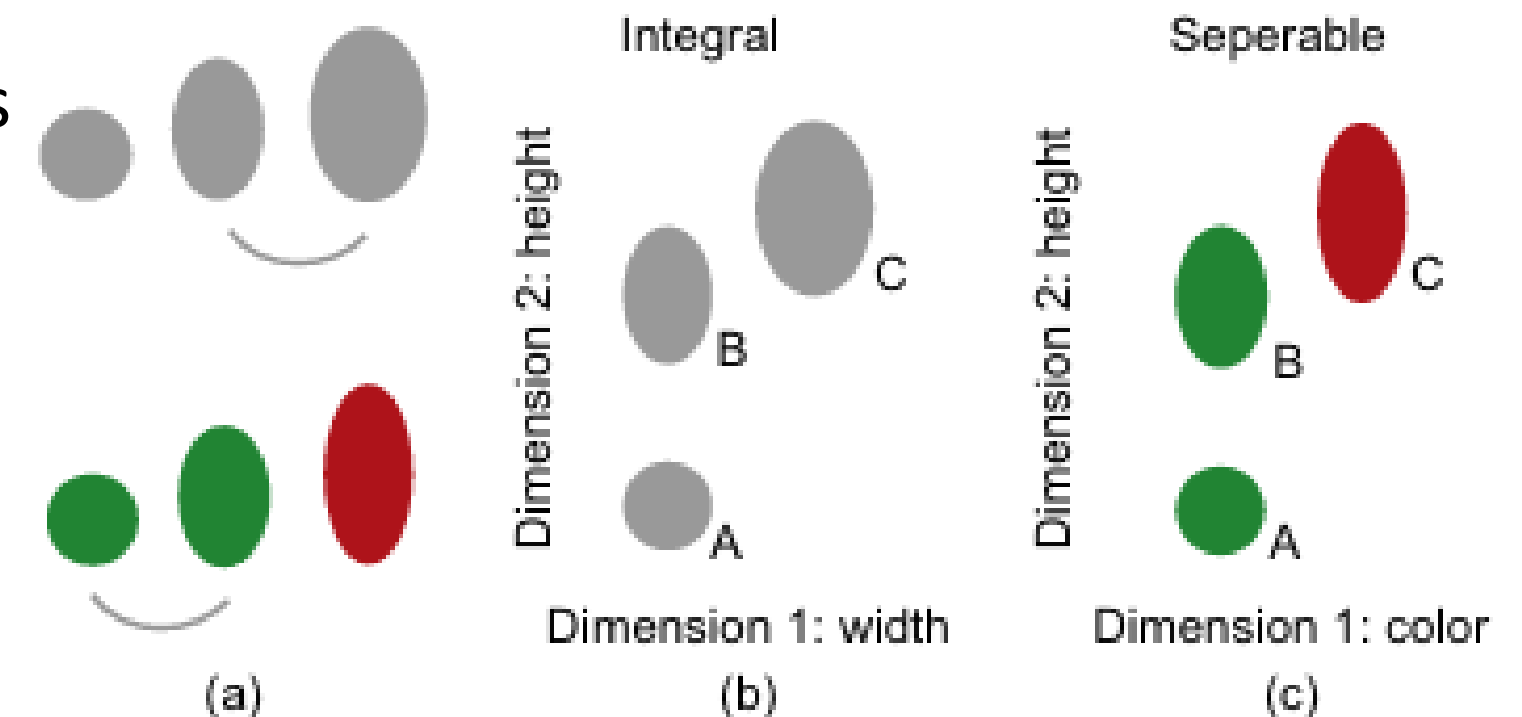
- Multiple visual attributes are perceived holistically as a single pattern.
- Example: width + height of an ellipse — changing one affects the perceived overall shape.
- Individual dimensions are therefore harder to judge independently.

Separable Dimensions

- Visual attributes are perceived independently.
- Example: color + size of a circle.
- Users can attend to one dimension without substantially interfering with the other.
- Sorting or comparing a single dimension is therefore easier.

Key Takeaway

Use separable visual dimensions when users need to independently compare multiple data attributes.



(a) The width and height of an ellipse are perceived integrally, so the ellipses are seen as more similar to each other (because they have the same shape) than the pair having the same width. The color and height of a shape are perceived separably, so the two green shapes are seen as most similar. (b, c) Space plots of the two examples.

Visualization of Data, Understanding and Interpretation

Summary — Saliency & Efficient Visual Search



1. Make Important Information Distinct

- Important data should be represented using visually distinctive features.
- Strong preattentive channels include color, orientation, size, and motion.
- Avoid relying on subtle differences when rapid search is required.

2. Avoid Visual Interference

- Similar visual properties can make objects harder to distinguish.
- Symbols become less salient when distractors share similar visual characteristics.
- Increasing the difference between target and surrounding elements improves search.

3. Use Appropriate Visual Channels

- Choose graphical properties that allow the required visual query to be performed efficiently.
- When multiple attributes must be encoded, separable dimensions such as color and size allow more independent judgments.
- Use redundant coding when additional distinction is useful.

4. Design for the User's Task

- Saliency is not purely a property of the display.
- Bottom-up saliency comes from the visual stimulus itself.
- Top-down attention tunes perception according to what the user is looking for.

Visualization of Data, Understanding and Interpretation

Summary — Saliency & Efficient Visual Search



Core Principle

Design visualizations so that important information is easy to find, distinguish, and interpret with minimal perceptual effort.

Principle
Preattentive proce
Saliency
Redundant codi

Visualization of Data, Understanding and Interpretation

MCQ

Q1. A treemap contains 150 rectangles. One rectangle is significantly different in color from all others. As the number of surrounding rectangles increases, the time required to detect it remains nearly constant.

Which explanation best fits this observation?

- A. The target is processed through serial scanning
- B. The target is identified through conjunction search
- C. The target exploits a preattentive feature channel
- D. The target requires additional working-memory capacity

Visualization of Data, Understanding and Interpretation

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Answer: C

Visualization of Data, Understanding and Interpretation

MCQ

Q2. A visualization uses red squares, red circles, blue squares, and blue circles. A user is asked to locate the red squares.

Which design change would most directly make the target easier to locate preattentively?

- A. Increase the number of red circles
- B. Add more blue squares
- C. Encode the target with a distinct feature
- D. Reduce the overall number of displayed objects

Visualization of Data, Understanding and Interpretation

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Answer: C

Visualization of Data, Understanding and Interpretation

MCQ

Q3. A treemap initially uses a slice-and-dice layout, but many resulting rectangles are extremely thin. The designer replaces it with a squarified layout.

What is the main trade-off?

- A. Better aspect ratios, less order preservation
- B. More hierarchy, less area encoding
- C. More labels, less scalability
- D. Better paths, weaker containment

Visualization of Data, Understanding and Interpretation

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Answer: A

Visualization of Data, Understanding and Interpretation

MCQ

Q4. Consider two visualizations containing identical graphical elements. In Visualization A, users search for vertical bars; in Visualization B, they search for red bars.

Why can the same display produce different search behavior?

- A. Feature maps are permanently fixed
- B. Attention can retune feature sensitivity
- C. Serial search replaces parallel processing
- D. Working memory changes the display structure

Visualization of Data, Understanding and Interpretation

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Answer: B

Visualization of Data, Understanding and Interpretation

MCQ

Q5. A treemap is being used to visualize a deeply nested hierarchy. The analyst needs to preserve the overall hierarchy and proportions, but some lower-level regions become too small to inspect. Which interaction would best address this problem?

- A. Elision
- B. Drill-down
- C. Reordering
- D. Squarification

Visualization of Data, Understanding and Interpretation

MCQ

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Answer: B



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 2 : Advanced Visualization & Storytelling

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

In this lecture, we will cover:

- Understanding high-dimensional data and the need for dimensionality reduction
- Mathematical foundations of PCA, including centering, standardization, covariance, eigenvalues, eigenvectors, projection, and explained variance
- Understanding t-SNE and UMAP through neighborhood preservation, probability distributions, embeddings, and parameter effects
- Comparing PCA, t-SNE, and UMAP for different visualization objectives
- Exploring real-world applications in image analysis, text, genomics, and machine-learning embeddings

Visualization of Data, Understanding and Interpretation

The High-Dimensional Visualization Problem

A dataset may contain dozens, hundreds, or thousands of variables for each observation.

For example, a single patient may be described by:

Age · Height · Weight · BMI · Blood Pressure · Glucose · Heart Rate · ... · 100+ measurements

Each patient is therefore a point in a high-dimensional feature space: $\mathbf{x}_i = (x_{i1}, x_{i2}, \dots, x_{ip})$
where p is the number of dimensions.

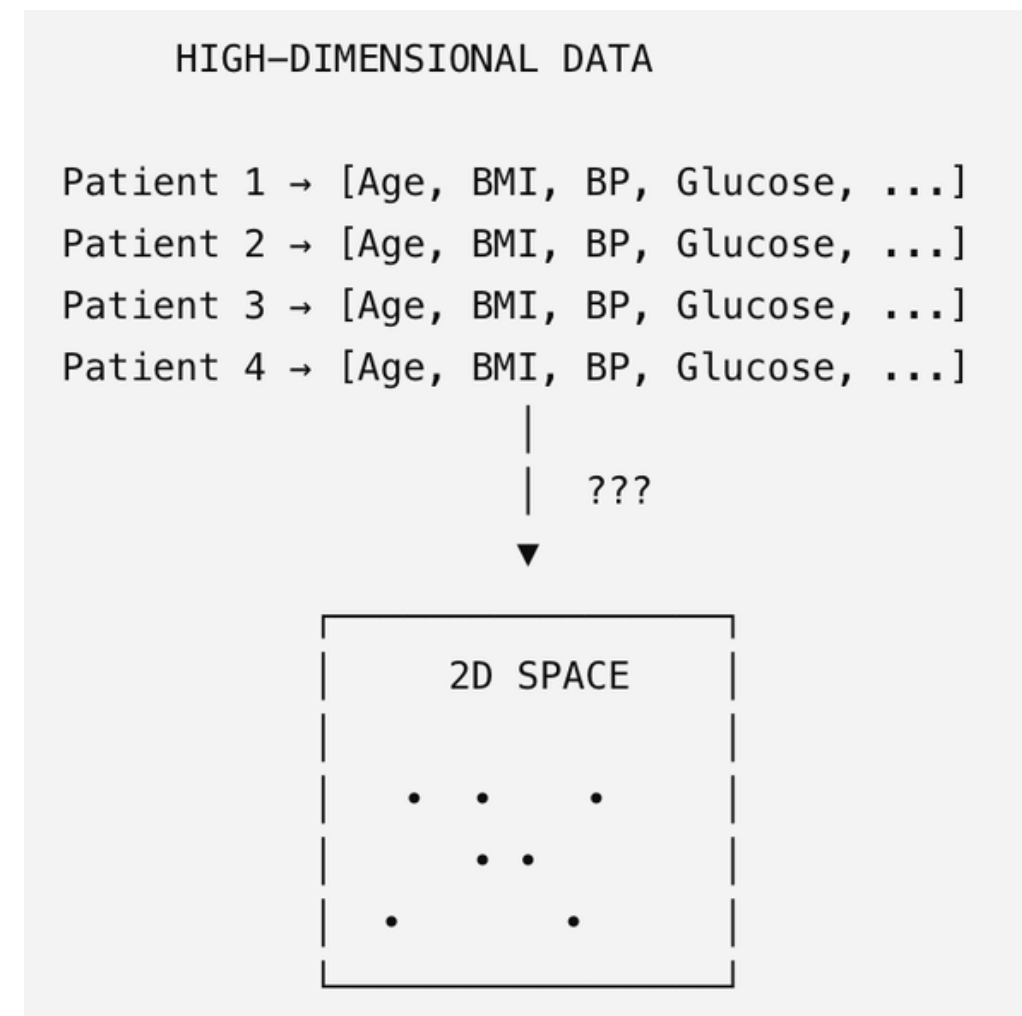
The visualization problem

A scatter plot can directly show: 2 dimensions $\rightarrow$ 2D

but what about: 50 dimensions $\rightarrow$?

We need a way to transform high-dimensional data into a lower-dimensional representation that can be visually explored.

Dimensionality reduction converts high-dimensional measurements into a low-dimensional representation that can be visualized and interpreted.



Visualization of Data, Understanding and Interpretation

The High-Dimensional Visualization Problem

What Does “Dimension” Actually Mean?

A dimension is a measurable attribute of an observation.

In a dataset, each dimension corresponds to one attribute, variable, or feature used to describe an observation.

Consider a dataset of patients:

Patient	Age	BMI	Glucose	Blood Pressure	Cholesterol
P1	42	27.4	108	128	196
P2	55	31.2	145	142	231
P3	36	23.8	92	118	174

Here:

- Rows → observations / data entities
- Columns → dimensions / attributes
- Values → measurements of each dimension

Visualization of Data, Understanding and Interpretation

The High-Dimensional Visualization Problem

Therefore, this dataset has 5 dimensions: $\mathbf{x}_i = (\text{Age, BMI, Glucose, BP, Cholesterol})$
Each patient can be viewed as one point in a 5-dimensional feature space.

Why Dimensions Matter?

- With 1 dimension, we can visualize a distribution directly.
- With 2 dimensions, we can use a scatter plot.
- With 3 dimensions, we can attempt a 3D representation.
- Beyond 3 dimensions, direct spatial visualization becomes difficult.

So a dataset with p features can be represented mathematically as: $X \in \mathbb{R}^{n \times p}$
where:

- n = number of observations
- p = number of dimensions/features
- Each observation is a point in p -dimensional space.

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Visualization of Data, Understanding and Interpretation

Dimension Reduction

(TB 1, Sec 12.3)

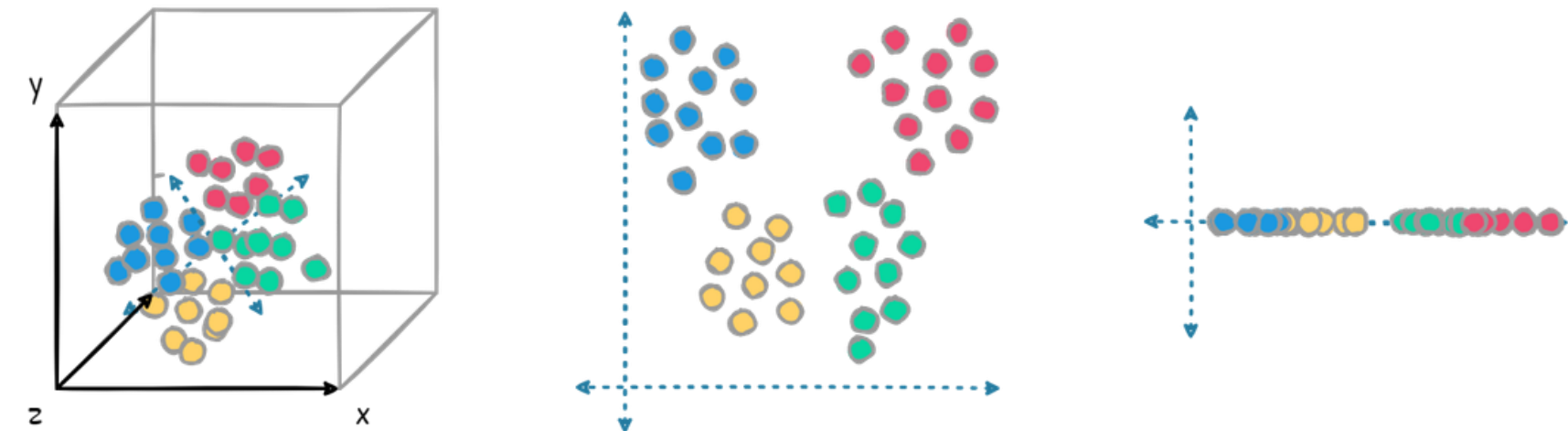
Why Do We Need Dimension Reduction?

- So far, we have explored relationships between two variables using scatter plots and multiple variables using correlation matrices (correlograms).
- However, real-world datasets often contain dozens or even hundreds of correlated variables, making them difficult to visualize and interpret.
- Dimension reduction transforms these high-dimensional datasets into a small number of new variables that preserve most of the important information, allowing patterns, clusters, and relationships to be visualized more easily.

Why is it Important?

- Removes redundant information caused by highly correlated variables.
- Simplifies complex datasets while retaining most of the underlying structure.
- Enables visualization of high-dimensional data in two or three dimensions for easier interpretation.

(Typical visualization methods: PCA, t-SNE, UMAP)



Visualization of Data, Understanding and Interpretation

The Curse of Dimensionality

As the number of dimensions increases, the data space grows extremely rapidly, making it harder to sample, compare, and visualize observations effectively.

For a grid with 10 intervals per dimension: Number of regions = 10^d

Example: Suppose each dimension is divided into 10 equal bins.

Dimensions (d)

As dimensions increase, the space grows exponentially.

Why is this a problem?

- Data points become increasingly sparse in the available space.
- Distances and similarities can become less informative as dimensions increase.
- Direct visualization becomes difficult because the display has limited spatial dimensions and perceptual bandwidth.

Visualization of Data, Understanding and Interpretation

The Curse of Dimensionality

Another Example:

For p variables, the number of unique pairwise relationships is:

$$\frac{p(p-1)}{2}$$

Therefore,

- $p = 5 \rightarrow 10$
- $p = 10 \rightarrow 45$
- $p = 30 \rightarrow 435$
- $p = 100 \rightarrow 4,950$

This creates the following problems:

- Too many variables
- Too many pairwise relationships
- Human visual space is limited

Visualization of Data, Understanding and Interpretation

Redundant Information

Many Variables Can Describe the Same Underlying Pattern

High-dimensional datasets often contain correlated variables, meaning that several measurements change together and therefore carry overlapping information. Consider measurements of a person's physical characteristics: **Height · Weight · Arm Length · Leg Length · Waist · Hip · Chest**

A larger person will generally tend to be:

- taller
- heavier
- longer in the arms and legs
- larger in waist, hip, and chest circumference

Therefore, these 7 dimensions do not necessarily represent 7 independent pieces of information.

Instead of thinking: *7 variables → 7 independent dimensions*

we can think of the measurements as being driven largely by a smaller number of underlying factors:

$$\text{Body measurements} \rightarrow \underbrace{\text{Overall size}}_{\text{major source of variation}} + \underbrace{\text{Sex / body shape}}_{\text{additional variation}} + \dots$$

Visualization of Data, Understanding and Interpretation

What Does Dimension Reduction Actually Do?

From High-Dimensional Measurements to a Low-Dimensional Representation

Dimensionality reduction does not simply delete columns from a dataset. Instead, it derives a new, lower-dimensional representation from the original high-dimensional data.

Suppose we have: $X \in \mathbb{R}^{n \times p}$

where n = observations and p = measured dimensions.

The goal is to construct: $Z \in \mathbb{R}^{n \times k}$ where: $k \ll p$

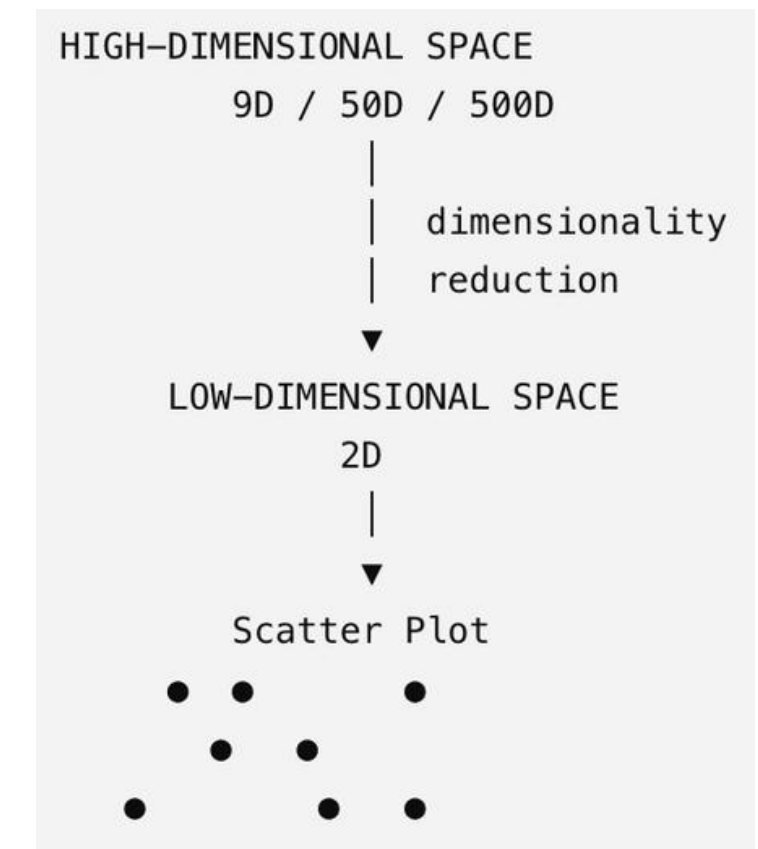
For example: *100 measured dimensions $\rightarrow$ 2 derived dimensions*

What happens to the information?

The original data contains many patterns, relationships, and variations. A reduction method tries to retain the most useful structure while representing the data with fewer dimensions.

A useful way to think about the process is:

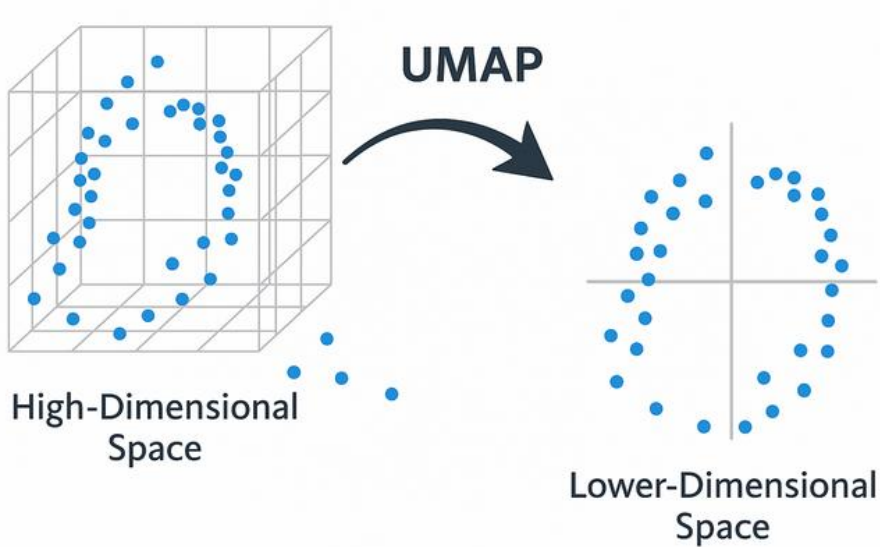
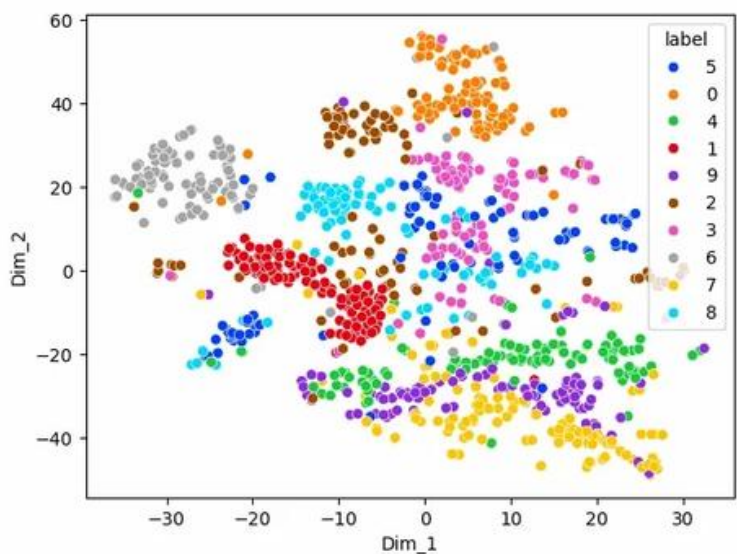
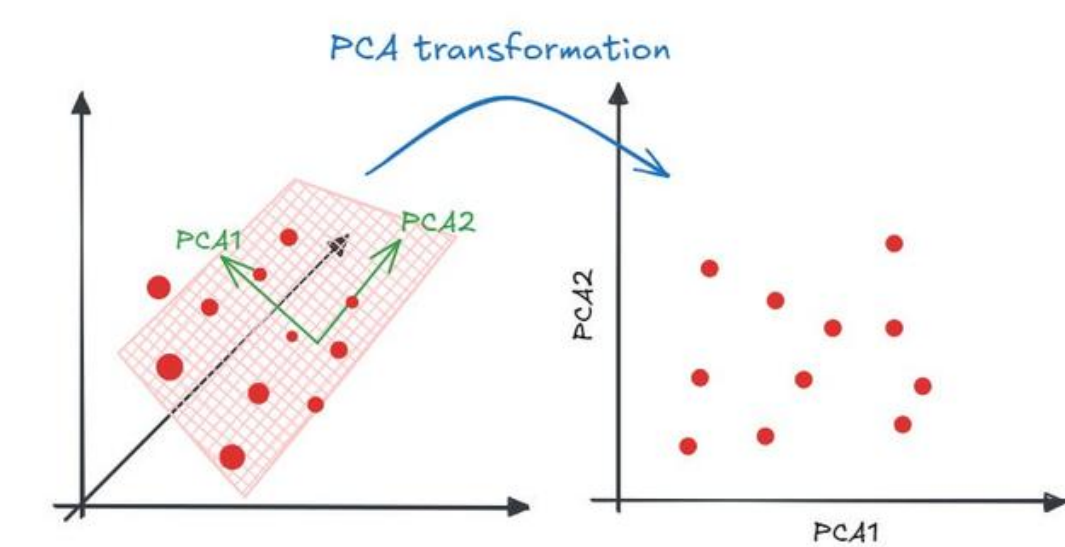
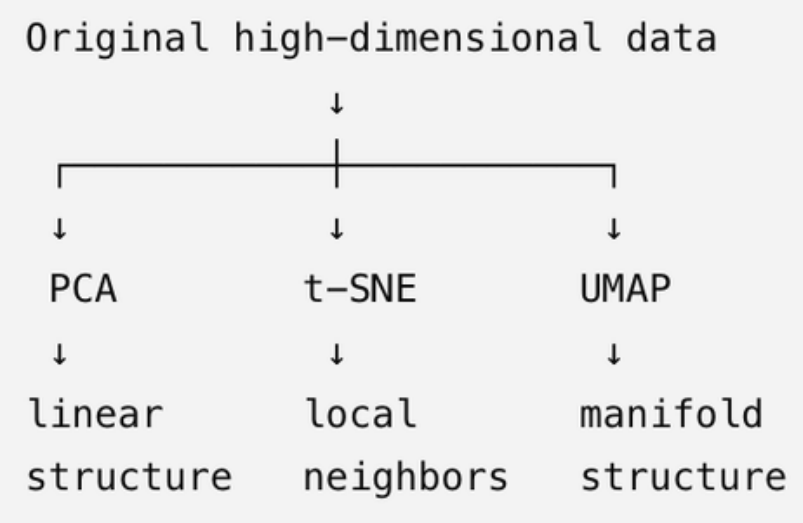
*High-dimensional data $\rightarrow$ Find important structure $\rightarrow$ Construct a lower-dimensional representation
 $\rightarrow$ Visualize in 2D / 3D*



Visualization of Data, Understanding and Interpretation

Common Dimension Reduction Techniques

Technique	Purpose
PCA (Principal Component Analysis)	Finds new axes that capture the maximum variation in the data.
t-SNE (t-distributed Stochastic Neighbor Embedding)	Preserves local neighborhoods to reveal clusters in complex datasets.
UMAP (Uniform Manifold Approximation and Projection)	Preserves both local and global structure while providing fast, scalable visualization.



Visualization of Data, Understanding and Interpretation

Principle Component Analysis

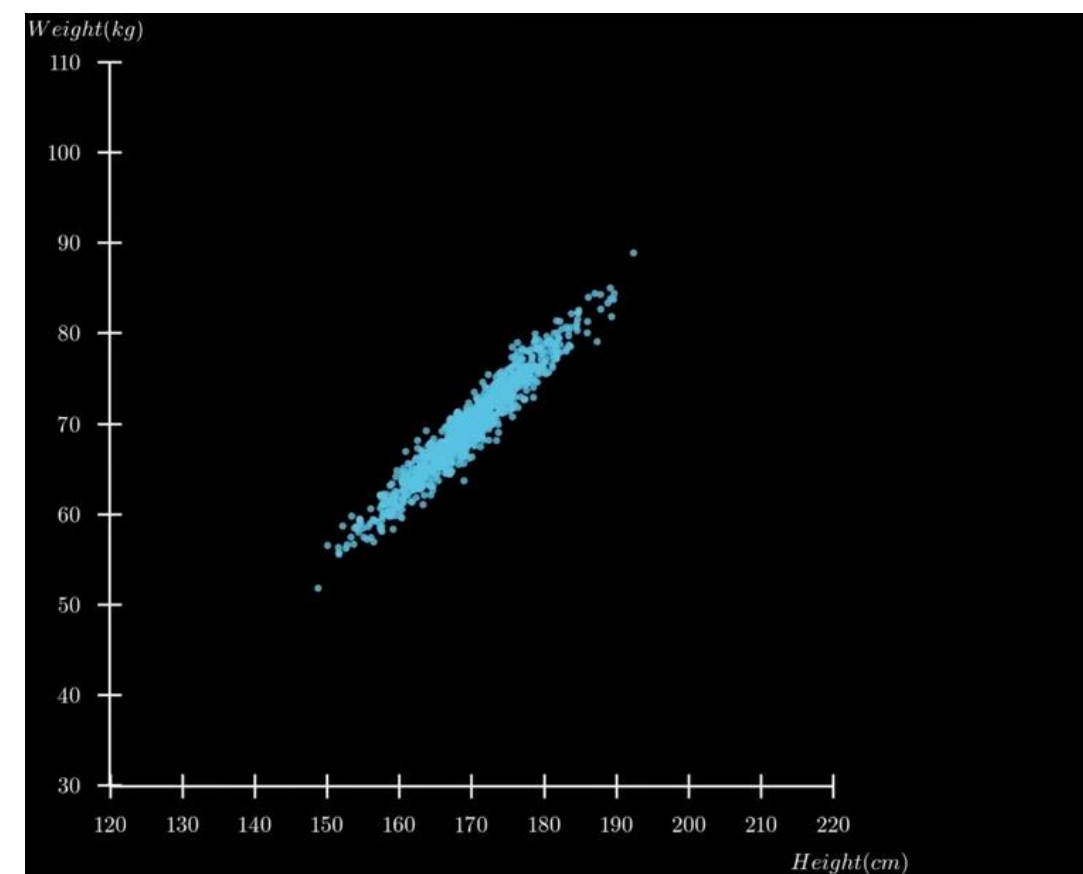
Principal Component Analysis (PCA) is a dimension reduction technique that transforms correlated variables into a smaller set of uncorrelated variables called principal components, while preserving most of the variation in the data.

Let us understand this visually with the help of an example:

- Imagine we have gathered information (*weight, height*) of a 1000 people
- We can plot this data in the form of a graph to observe how this data is distributed (*scatter plot*)

Source: https://youtu.be/o_cAOa5fMhE?si=cliXzpQ5YuL8NYnQ

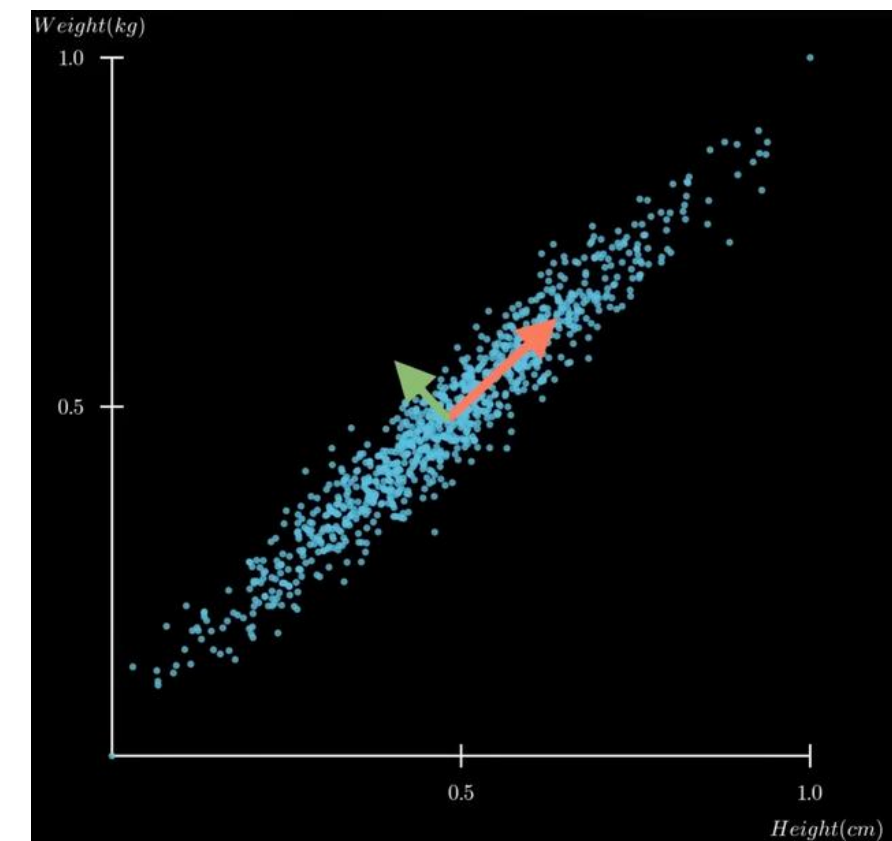
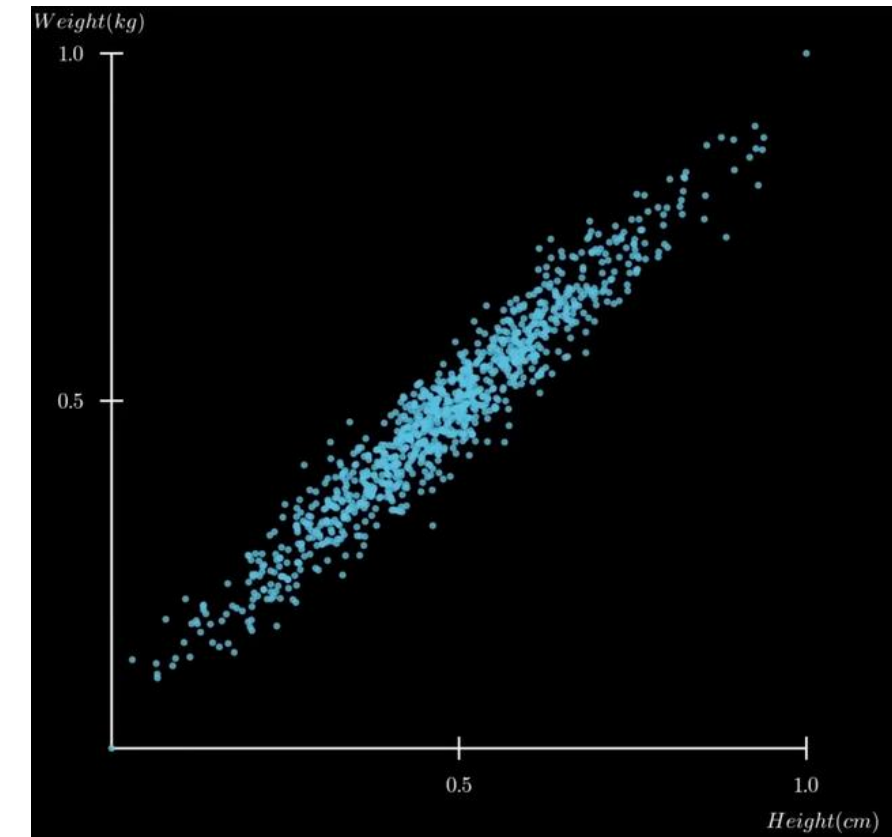
Name	Height (cm)	Weight (kg)
Alice	170 cm	68 kg
Bob	185 cm	82 kg
Charlie	178 cm	77 kg
...	...	...



Visualization of Data, Understanding and Interpretation

Principle Component Analysis

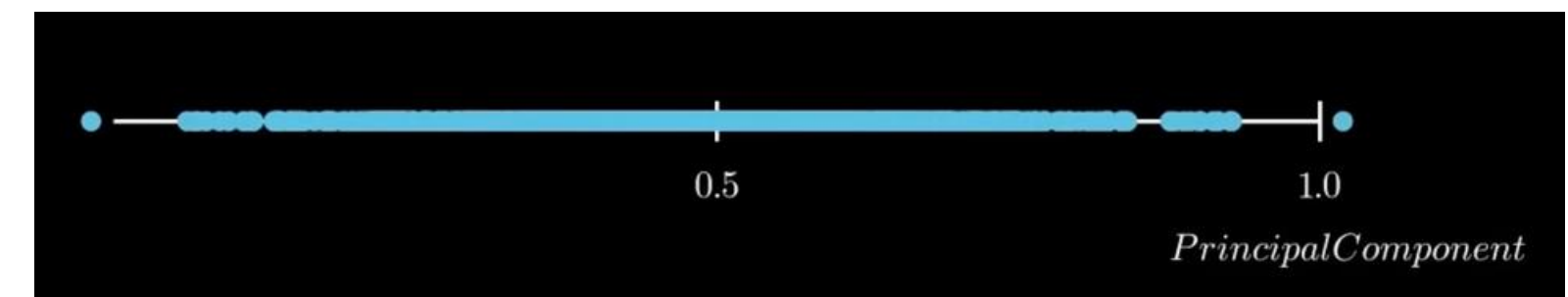
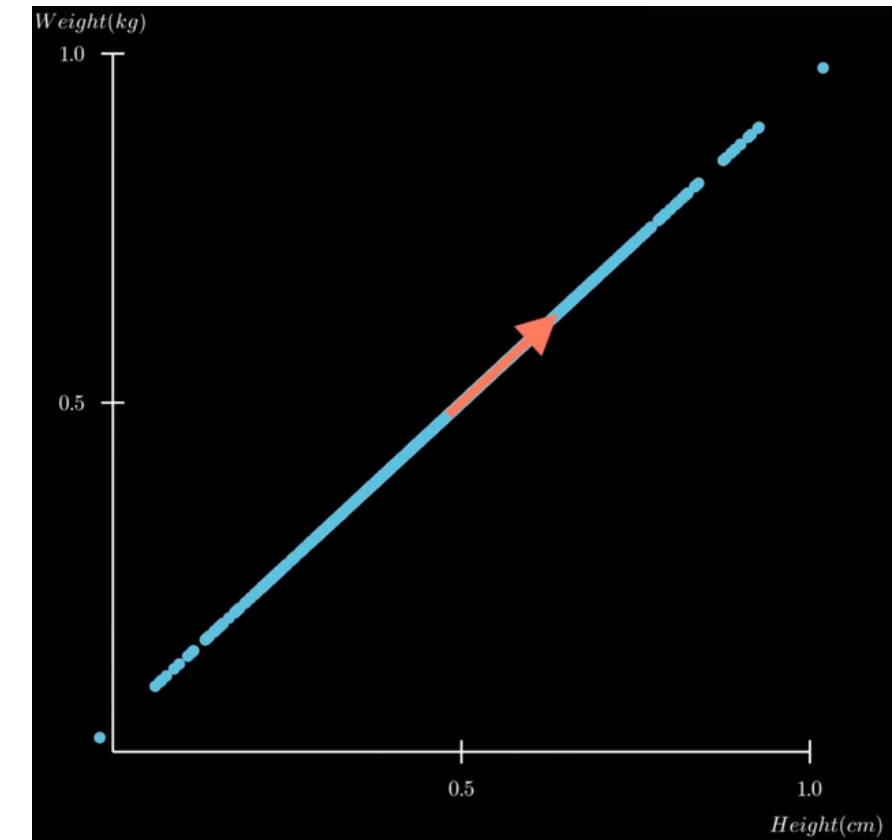
- We can then normalize this data. Here, it means bringing the data to a range between 0-1. The result of normalization gives the first figure
- We can notice that the data varies along 2 directions represented by the two arrows in the second figure
- We can also say that the data varies greatly along the red arrow in comparison to the green. Hence, the variation along the red arrow is more significant than the green arrow
- These arrows are called *principal components*



Visualization of Data, Understanding and Interpretation

Principle Component Analysis

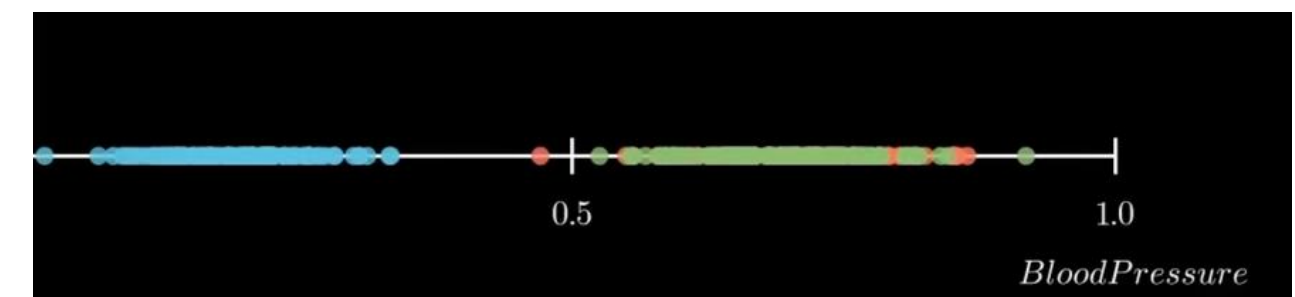
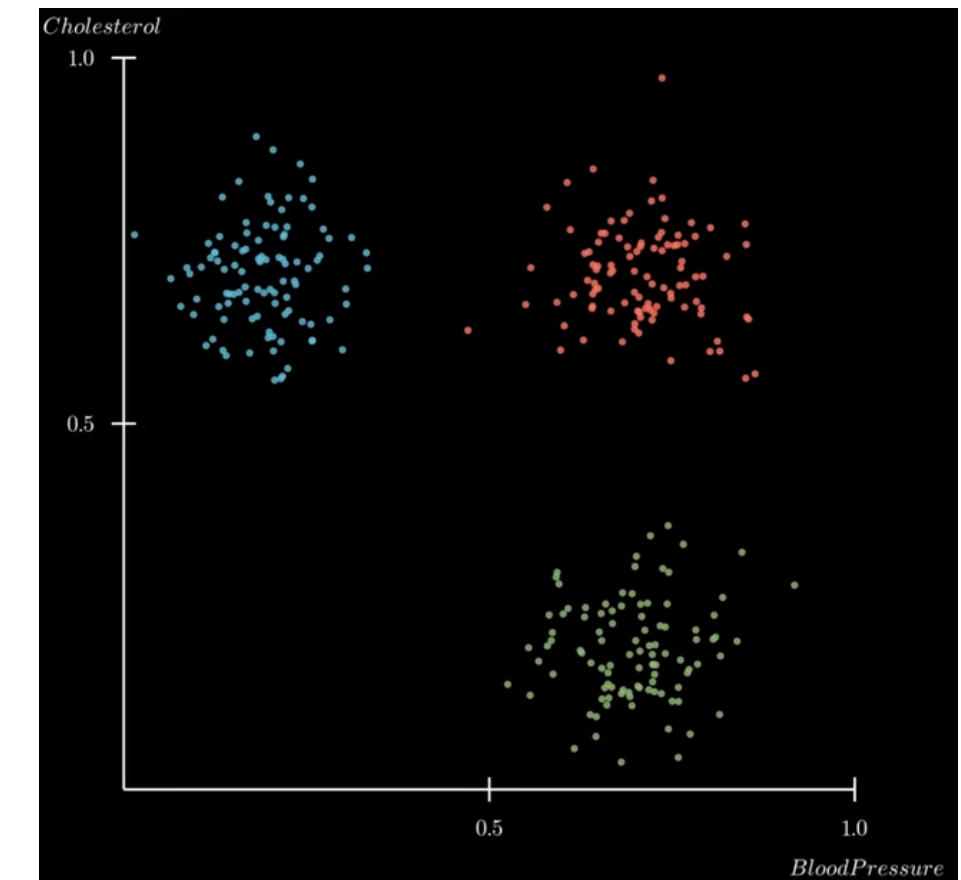
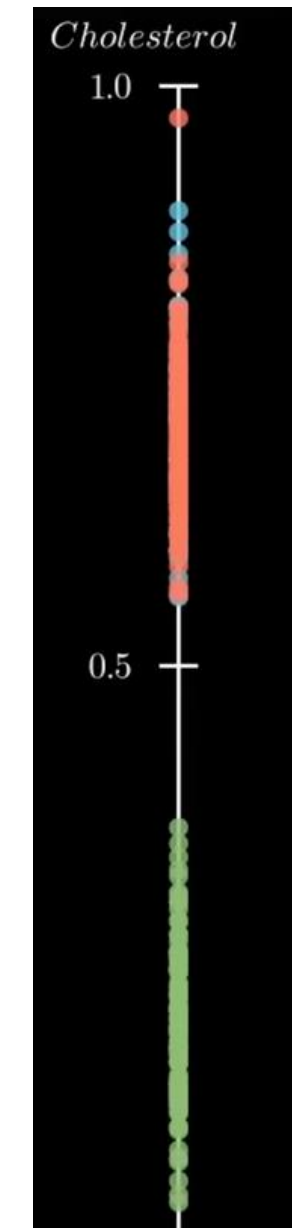
- Suppose we want to represent this 2D data on a *single axis*
- One way to do that is to project the data onto the largest principal component axis (*red axis*)
- The final 1D graph is given in the second figure
- However, you might wonder why we can't project the data directly as given onto the horizontal axis
- When you revert back to the 2D graph, it seems that just projecting to this axis yields a very similar result
- To understand this better, let's change the dataset



Visualization of Data, Understanding and Interpretation

Principle Component Analysis

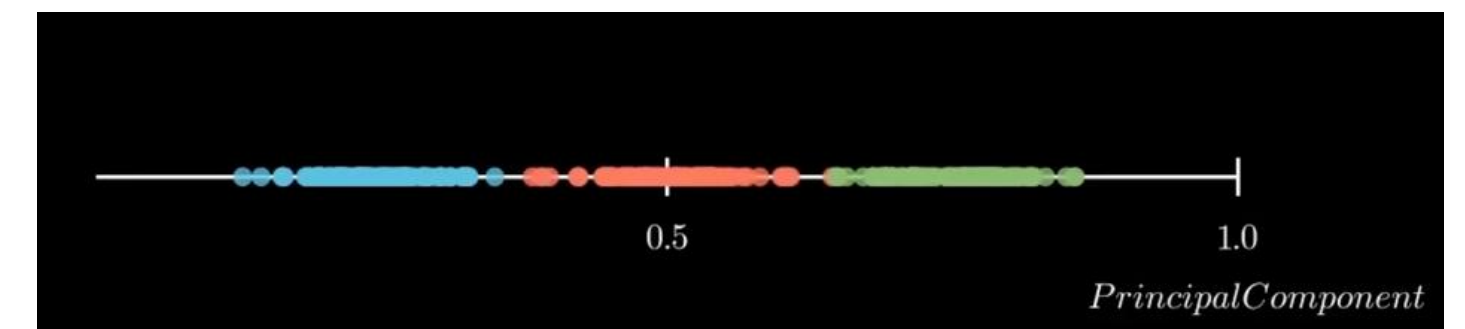
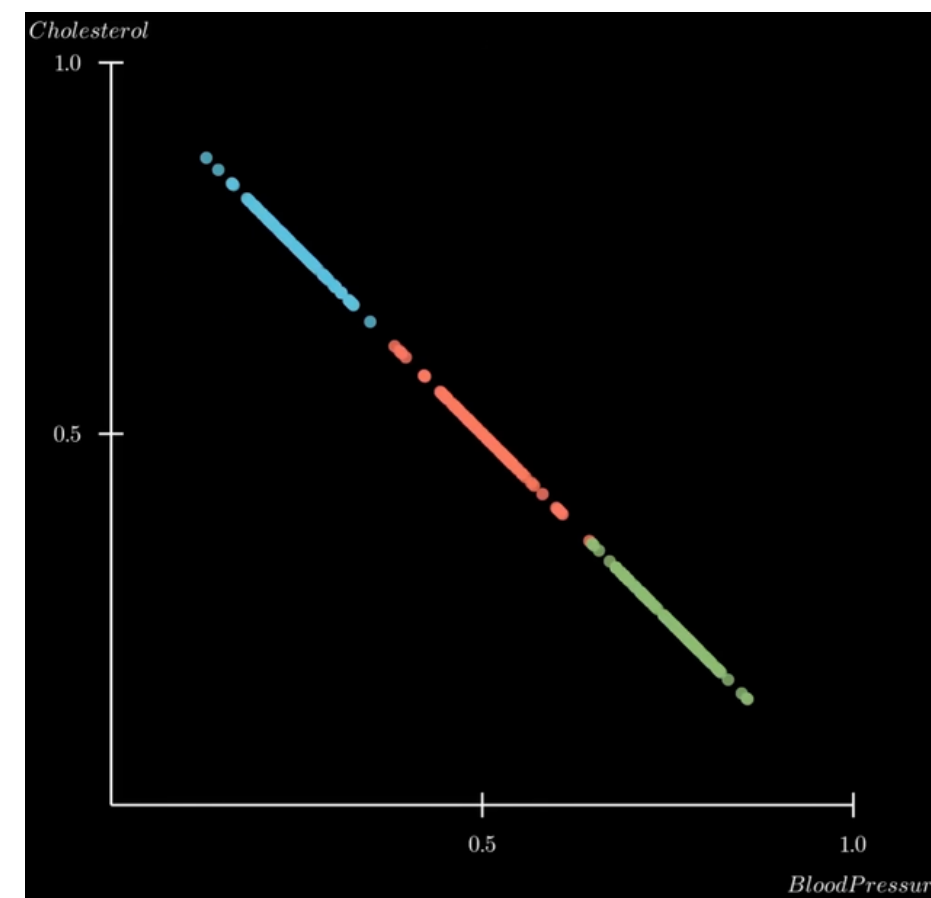
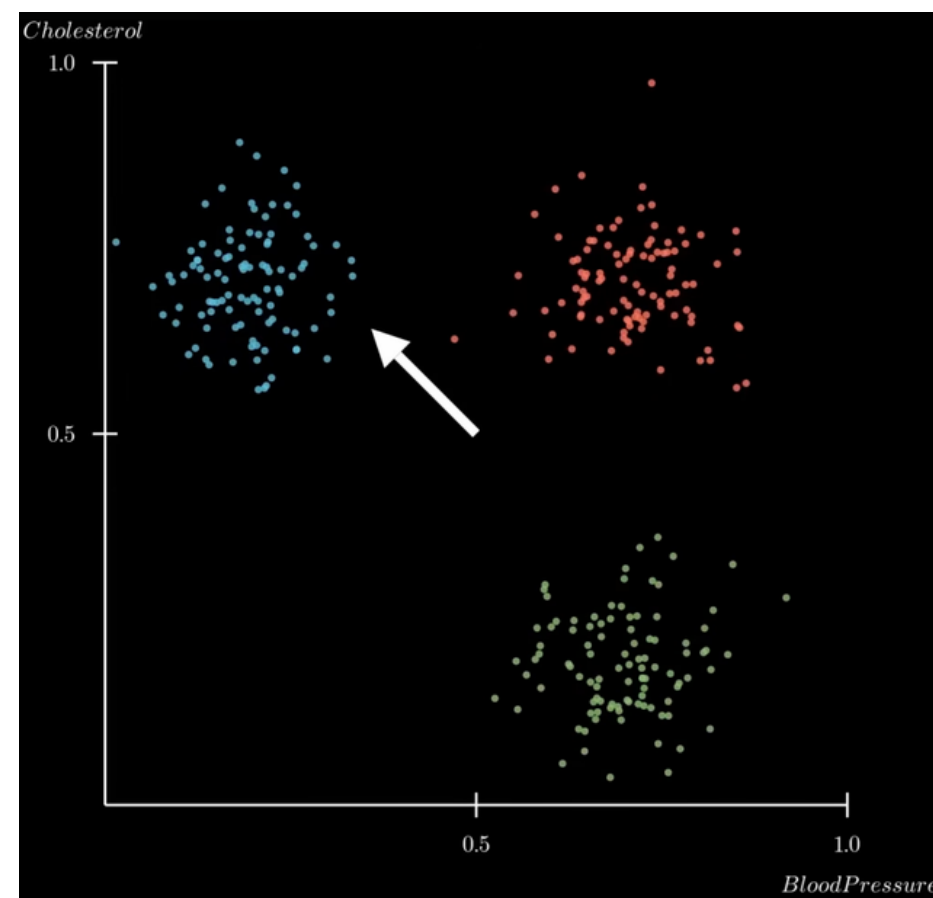
- The new dataset records blood pressure and cholesterol levels of several patients
- When we plot this distribution, we find that the data forms 3 distinct clusters
- If we project the data onto the x-axis or y-axis, we see only 2 distinct clusters
- This means that we've combined 2 different populations in our low dimensional representation, which is incorrect
- Now, let's see what happens when we project the data along the principal component



Visualization of Data, Understanding and Interpretation

Principle Component Analysis

- This time, we successfully retain all 3 clusters in our low dimensional representation
- This low dimensional representation preserves the main features of our original data, which is much better than the simple projections we tried previously
- Now, lets try and understand how we can compute these principle components mathematically



Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Suppose we have n observations measured across p variables:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

Each row represents an observation, while each column represents a measured variable.

PCA transforms these original variables into a new set of variables called principal components (PCs). Wilke describes these PCs as linear combinations of standardized original variables, ordered so that PC1 captures the greatest possible variation, PC2 the next greatest, and so on.

Steps: $X \rightarrow \text{Center} \rightarrow \text{Scale} \rightarrow \text{Covariance} \rightarrow \text{Eigenvectors} \rightarrow \text{Projection}$

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Step 1: Center the Data

For each variable, calculate its mean:

$$\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$$

Then subtract the mean from every observation:

$$x_{ij}^{(c)} = x_{ij} - \bar{x}_j$$

x_i = original observation

$\bar{x}$ = mean of the variable

$x_i^{(c)}$ = centered value

This moves the center of the data cloud to the origin: **Mean of every variable = 0**
Centering does not change the spread of the data. It only changes its location.

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components



Step 1: Worked Example

Suppose the values of a variable are: $X = [2, 4, 6, 8]$

1. Calculate the mean $\bar{x} = \frac{2 + 4 + 6 + 8}{4}$

$\bar{x} = 5$

1. Subtract the mean from every value

Original (xi)	Calculation	Cer
2	(2-5)	
4	(4-5)	
6	(6-5)	
8	(8-5)	

Therefore: $X_c = [-3, -1, +1, +3]$

Check

The mean of the centered values must be zero:

$$\frac{-3 - 1 + 1 + 3}{4} = \frac{0}{4} = 0$$

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Step 2: Standardize the Variables

When variables are measured on different scales, their variances are not directly comparable. Standardize each variable using:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}$$

where s_j is the standard deviation of variable j
*note: the numerator implies mean centering

After standardization:

$$\mu = 0, \quad \sigma^2 = 1$$

This prevents a variable with numerically large units from dominating PCA simply because of its scale.

$$\text{Sample: } s = \sqrt{\frac{\sum (x_i - \bar{x})^2}{n - 1}}$$

$$\text{Population: } \sigma = \sqrt{\frac{\sum (x_i - \mu)^2}{N}}$$

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Step 2: Worked Example

Using the same data from the previous example, we already calculated:

$$\bar{x} = 5 \quad X_c = [-3, -1, +1, +3]$$

1. Calculate the standard deviation

Sample Standard Deviation Formula:

$$s = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

s = sample standard deviation

x_i = individual observation

$\bar{x}$ = sample mean

n = number of observations

$n - 1$ = **degrees of freedom**

Expanded Form:

$$s = \sqrt{\frac{(x_1 - \bar{x})^2 + (x_2 - \bar{x})^2 + \cdots + (x_n - \bar{x})^2}{n - 1}}$$

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

For the example $X = [2, 4, 6, 8]$:

$$\bar{x} = 5$$

$$s = \sqrt{\frac{(2 - 5)^2 + (4 - 5)^2 + (6 - 5)^2 + (8 - 5)^2}{4 - 1}}$$
$$= \sqrt{\frac{9 + 1 + 1 + 9}{3}} = \sqrt{\frac{20}{3}} \approx \boxed{2.582}$$

Note!

- Sample standard deviation: Used when the data represents a sample from a larger population. It divides by $n-1$ to account for the fact that the population mean is being estimated from the sample.
- Population standard deviation: Used when the data contains the entire population of interest. It divides by N because the true population mean is known.

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

1. Divide every centered value by s

For $x = 2$:

$$z = \frac{2 - 5}{2.582}$$
$$= \frac{-3}{2.582} \approx \boxed{-1.162}$$

For $x = 4$:

$$z = \frac{-1}{2.582} \approx \boxed{-0.387}$$

For $x = 6$:

$$z = \frac{1}{2.582} \approx \boxed{+0.387}$$

For $x = 8$:

$$z = \frac{3}{2.582} \approx \boxed{+1.162}$$

Therefore:

$$Z = [-1.162, -0.387, +0.387, +1.162]$$

The standardized values now have:

$$\text{Mean} = 0, \quad \text{Standard deviation} = 1$$

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Why Is Standardization Important?

PCA is sensitive to the scale of the variables. Consider two variables describing people:

- Age: 20 – 60 years
- Annual income: ₹20,000 – ₹80,000

Their numerical scales are very different.

If PCA is performed directly on the raw measurements, a variable with a much larger numerical scale can have a much larger variance and therefore dominate the principal components.

Example: $\text{Var}(\text{Age}) = 100$, but $\text{Var}(\text{Income}) = 400,000,000$

The variance of income is enormously larger because of its units and scale, not necessarily because income contains more meaningful structure.

After standardization, both variables are transformed so they are expressed in the same unit:

How many standard deviations is this observation from the mean?

- $z = 2$ means the observation is 2 standard deviations above the mean.
- $z = -1.5$ means the observation is 1.5 standard deviations below the mean.

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How to Compute Principal Components



Why PCA benefits from this

PCA compares variables based on their patterns of variation and relationships, rather than allowing measurement units to determine which variable dominates.

Operation	Formula	Effect
Centering	$x - \bar{x}$	Mean becomes 0
Standardization	$\frac{x - \bar{x}}{s}$	Mean becomes 0 and SD becomes 1

- Centering answers: “Where is the data centered?”
- Standardization answers: “How far is each value from the mean relative to the variable's spread?”

This distinction will make the subsequent covariance-matrix calculation much easier for students to follow.

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Step 3: Mean Centering in Matrix Form

Mean centering shifts every variable so that its mean becomes zero. Suppose our dataset contains n observations and p variables:

$$X = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{bmatrix}$$

The mean of variable j is: $\bar{x}_j = \frac{1}{n} \sum_{i=1}^n x_{ij}$

The vector containing the means of all variables is: $\boldsymbol{\mu} = [\bar{x}_1 \quad \bar{x}_2 \quad \cdots \quad \bar{x}_p]$

We subtract these column means from every observation: $X_c = X - \mathbf{1}\boldsymbol{\mu}$

where $\mathbf{1}$ is an $n \times 1$ vector of ones.

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Step 3: Worked Example

Consider: $X = \begin{bmatrix} 2 & 10 \\ 4 & 14 \\ 6 & 18 \end{bmatrix}$ There are 3 observations and 2 variables.

1. Calculate the column means

For X_1 :

$$\bar{x}_1 = \frac{2 + 4 + 6}{3} = 4$$

For X_2 :

$$\bar{x}_2 = \frac{10 + 14 + 18}{3} = 14$$

Therefore:

$$\mu = [4 \quad 14]$$

1. Subtract the row means from every row

$$X_c = \begin{bmatrix} 2 - 4 & 10 - 14 \\ 4 - 4 & 14 - 14 \\ 6 - 4 & 18 - 14 \end{bmatrix}$$

$$X_c = \begin{bmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{bmatrix}$$

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1. Verify the result

For X_1 :

$$\frac{-2 + 0 + 2}{3} = 0$$

For X_2 :

$$\frac{-4 + 0 + 4}{3} = 0$$

Therefore:

Mean of every centered variable = 0

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Matrix Interpretation

The operation can also be explicitly written as:

$$\mathbf{1}\mu = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 4 & 14 \end{bmatrix} = \begin{bmatrix} 4 & 14 \\ 4 & 14 \\ 4 & 14 \end{bmatrix}$$

Hence:

$$\begin{aligned} X_c &= \begin{bmatrix} 2 & 10 \\ 4 & 14 \\ 6 & 18 \end{bmatrix} - \begin{bmatrix} 4 & 14 \\ 4 & 14 \\ 4 & 14 \end{bmatrix} \\ &= \begin{bmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{bmatrix} \end{aligned}$$

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Important!

Relationship With Step 2 — Standardization:

The standardization formula from **Step 2** was:

$$z_{ij} = \frac{x_{ij} - \bar{x}_j}{s_j}$$

Notice that the numerator already contains:

$$x_{ij} - \bar{x}_j$$

Therefore, **standardization already performs mean centering as part of the calculation.**

So if Step 2 has produced:

$$X_s = \begin{bmatrix} z_{11} & z_{12} \\ z_{21} & z_{22} \\ \vdots & \vdots \end{bmatrix}$$

we do not center X_s again.

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How to Compute Principal Components

Step 4: Construct the Covariance Matrix

After centering/standardizing the variables, PCA needs to determine how the variables vary together. Covariance measures whether two variables tend to increase or decrease together.

For two variables X and Y:
$$\text{Cov}(X, Y) = \frac{1}{n-1} \sum_{i=1}^n (x_i - \bar{x})(y_i - \bar{y})$$

- Positive covariance: both variables tend to increase/decrease together.
- Negative covariance: one tends to increase when the other decreases.
- Covariance near 0: little linear co-variation.

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How to Compute Principal Components

Constructing the Covariance Matrix

For p variables, all pairwise covariances are arranged into a $p \times p$ matrix:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) & \cdots \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}$$

The diagonal contains **variances**:

$$\Sigma_{jj} = \text{Var}(X_j)$$

The off-diagonal elements contain **covariances**:

$$\Sigma_{jk} = \text{Cov}(X_j, X_k)$$

Because:

$$\text{Cov}(X_j, X_k) = \text{Cov}(X_k, X_j)$$

the covariance matrix is **symmetric**.

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Step 4: Worked Example

Use the centered matrix from Step 3:

- $n = 3$ observations
- $p = 2$ variables

$$X_c = \begin{bmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{bmatrix}$$

1. Calculate variance of X_1

$$X_1 = [-2, 0, 2]$$

$$\begin{aligned} \text{Var}(X_1) &= \frac{(-2)^2 + 0^2 + (2)^2}{3 - 1} \\ &= \frac{4 + 0 + 4}{2} \end{aligned}$$

$$\boxed{\text{Var}(X_1) = 4}$$

1. Calculate variance of X_2

$$X_2 = [-4, 0, 4]$$

$$\begin{aligned} \text{Var}(X_2) &= \frac{(-4)^2 + 0^2 + (4)^2}{2} \\ &= \frac{16 + 0 + 16}{2} \end{aligned}$$

$$\boxed{\text{Var}(X_2) = 16}$$

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

1. Calculate covariance between X_1 and X_2

$$\begin{aligned}\text{Cov}(X_1, X_2) &= \frac{(-2)(-4) + (0)(0) + (2)(4)}{2} \\ &= \frac{8 + 0 + 8}{2}\end{aligned}$$

$$\boxed{\text{Cov}(X_1, X_2) = 8}$$

The positive value makes sense:
when X_1 increases, X_2 also increases.

1. Build the covariance matrix

We now place these values into the matrix:

$$\Sigma = \begin{bmatrix} \text{Var}(X_1) & \text{Cov}(X_1, X_2) \\ \text{Cov}(X_2, X_1) & \text{Var}(X_2) \end{bmatrix}$$

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How to Compute Principal Components

Therefore: $\Sigma = \begin{bmatrix} 4 & 8 \\ 8 & 16 \end{bmatrix}$

Notice the symmetry: $\Sigma_{12} = \Sigma_{21} = 8$

Matrix Shortcut

Instead of calculating every element individually, the covariance matrix can be obtained directly from the centered data matrix:

$$\Sigma = \frac{1}{n-1} X_c^T X_c$$

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How to Compute Principal Components

For our example:

$$X_c^T = \begin{bmatrix} -2 & 0 & 2 \\ -4 & 0 & 4 \end{bmatrix}$$

Therefore:

$$X_c^T X_c = \begin{bmatrix} -2 & 0 & 2 \\ -4 & 0 & 4 \end{bmatrix} \begin{bmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{bmatrix}$$

Multiplying:

$$= \begin{bmatrix} 8 & 16 \\ 16 & 32 \end{bmatrix}$$

Since $n - 1 = 2$:

$$\Sigma = \frac{1}{2} \begin{bmatrix} 8 & 16 \\ 16 & 32 \end{bmatrix}$$

$$\Sigma = \begin{bmatrix} 4 & 8 \\ 8 & 16 \end{bmatrix}$$

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How to Compute Principal Components

Why Is This Step Important?

The covariance matrix summarizes the variation and relationships among all variables in a single mathematical object.

PCA uses this matrix to find the directions in which the data has the greatest variance.

The next step is therefore: *Covariance Matrix* $\rightarrow$ *Eigenvalues + Eigenvectors*

The eigenvectors will give us the new principal directions, while the eigenvalues will tell us how much variance each direction captures.

Important: $X_c \rightarrow \Sigma \rightarrow \lambda, v \rightarrow PCs \rightarrow \text{Projection}$

PCA convention: if we use the standardized matrix from Step 2 rather than merely the centered matrix, then the same calculation

$$\Sigma = \frac{1}{n-1} X_s^T X_s$$

produces the correlation matrix, because every standardized variable has variance 1. This is the convention used in Wilke's PCA treatment, where variables are standardized to zero mean and unit variance before determining the principal directions.

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How to Compute Principal Components

Step 5: Eigenvalues and Eigenvectors

The covariance matrix tells us how the variables vary together, but PCA still needs to find the new directions in which the data should be viewed. PCA obtains these directions by calculating the eigenvectors of the covariance matrix. At the same time, the corresponding eigenvalues tell us how much variance exists along each direction.

1. Eigenvector $\rightarrow$ direction of a principal component
2. Eigenvalue $\rightarrow$ variance captured along that direction

The eigenvectors are ordered by decreasing eigenvalue: $\lambda_1 > \lambda_2 > \lambda_3 > \dots$

so that: $v_1 \rightarrow PC_1, \quad v_2 \rightarrow PC_2, \quad v_3 \rightarrow PC_3, \dots$

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How to Compute Principal Components

Step 5: Worked Example

From Step 4, we obtained:

$$\Sigma = \begin{bmatrix} 4 & 8 \\ 8 & 16 \end{bmatrix}$$

Part A: Calculate the Eigenvalues

Eigenvalues satisfy:

$$\Sigma v = \lambda v$$

Rearranging:

$$(\Sigma - \lambda I)v = 0$$

For a non-zero eigenvector to exist:

$$|\Sigma - \lambda I| = 0$$

where I is the identity matrix.

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How to Compute Principal Components

For our covariance matrix:

$$\Sigma - \lambda I = \begin{bmatrix} 4 - \lambda & 8 \\ 8 & 16 - \lambda \end{bmatrix}$$

Therefore:

$$\begin{vmatrix} 4 - \lambda & 8 \\ 8 & 16 - \lambda \end{vmatrix} = 0$$

For a 2×2 matrix:

$$\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - bc$$

Therefore:

$$(4 - \lambda)(16 - \lambda) - (8)(8) = 0$$

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How to Compute Principal Components

Expand:

$$64 - 4\lambda - 16\lambda + \lambda^2 - 64 = 0$$

$$\lambda^2 - 20\lambda = 0$$

Factor:

$$\lambda(\lambda - 20) = 0$$

Therefore:

$$\lambda_1 = 20$$

$$\lambda_2 = 0$$

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How to Compute Principal Components

Part B: Interpret the Eigenvalues

We have:

$$\lambda_1 = 20, \quad \lambda_2 = 0$$

The total variance is:

$$\lambda_1 + \lambda_2 = 20 + 0 = 20$$

Therefore, the proportion of variance explained by PC1 is:

$$\frac{\lambda_1}{\lambda_1 + \lambda_2} = \frac{20}{20} = \boxed{100\%}$$

and PC2 explains:

$$\frac{0}{20} = \boxed{0\%}$$

Interpretation

All variation in this particular example lies along one direction. That makes sense because our original variables were perfectly linearly related: $X_2 = 2X_1 + 6$. So although we started with 2 variables, the data effectively contains only 1 direction of variation.

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How to Compute Principal Components

Part C: Calculate the Eigenvector for $\lambda_1 = 20$

We use:

$$(\Sigma - \lambda I)v = 0$$

Substitute $\lambda = 20$:

$$\begin{bmatrix} 4 - 20 & 8 \\ 8 & 16 - 20 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Therefore:

$$\begin{bmatrix} -16 & 8 \\ 8 & -4 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Using the first equation:

$$-16v_1 + 8v_2 = 0$$

$$8v_2 = 16v_1$$

$$v_2 = 2v_1$$

Therefore, we can choose:

$$v = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

But PCA requires unit eigenvectors, so we normalize it.

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How to Compute Principal Components

Therefore, we can choose:

The length of the vector is:

$$\begin{aligned}\|v\| &= \sqrt{1^2 + 2^2} \\ &= \sqrt{5}\end{aligned}$$

Therefore:

$$v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Numerically:

$$v_1 = \begin{bmatrix} 0.447 \\ 0.894 \end{bmatrix}$$

This is the direction of **PC1**.

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How to Compute Principal Components

Part D: Calculate the Eigenvector for $\lambda_2 = 0$

Again:

$$(\Sigma - \lambda I)v = 0$$

Set:

$$\lambda = 0$$

Then:

$$\begin{bmatrix} 4 & 8 \\ 8 & 16 \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Using the first equation:

$$4v_1 + 8v_2 = 0$$

$$v_1 = -2v_2$$

Choose:

$$v = \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

Normalize:

$$\|v\| = \sqrt{(-2)^2 + 1^2} = \sqrt{5}$$

Therefore:

$$v_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

or approximately:

$$v_2 = \begin{bmatrix} -0.894 \\ 0.447 \end{bmatrix}$$

This is the direction of **PC2**.

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How to Compute Principal Components

Final PCA Directions

We have therefore obtained:

$$PC_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

$$PC_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

and:

$$\lambda_1 = 20, \quad \lambda_2 = 0$$

So:

$$PC_1 \rightarrow 100\% \text{ variance}$$

$$PC_2 \rightarrow 0\% \text{ variance}$$

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How to Compute Principal Components

What Have We Actually Found?

The eigenvector gives the direction of the new axis.

The eigenvalue tells us how much variation lies along that axis.

Therefore:

- Eigenvector = direction
- Eigenvalue = importance / variance

Important: The sign of an eigenvector is arbitrary

The vectors below represent the same principal direction. Reversing the sign only reverses the direction of the axis; it does not change the PCA solution.

$$\begin{bmatrix} 0.447 \\ 0.894 \end{bmatrix} \quad \begin{bmatrix} -0.447 \\ -0.894 \end{bmatrix}$$

Now that we have the principal directions v_1, v_2 , the next step is to project every observation onto those directions: $Z = XcV$

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How to Compute Principal Components

Step 6: Project the Data onto the Principal Components

We have now found the principal directions using eigenvectors.

The next question is *where does each observation lie along these new directions?*

Projection converts every original observation into its coordinates in the new principal-component coordinate system. If the centered/standardized data matrix is X_c , and the selected eigenvectors are placed into a matrix V , then: $Z = X_c V$

- X_c = centered data matrix
- V = matrix of eigenvectors
- Z = data represented in principal-component coordinates

The columns of Z are the principal-component scores.

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How to Compute Principal Components

Step 6: Worked Example

Continuing the previous example,

Our centered data was:

$$X_c = \begin{bmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{bmatrix}$$

The two normalized eigenvectors were:

$$v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

and

$$v_2 = \frac{1}{\sqrt{5}} \begin{bmatrix} -2 \\ 1 \end{bmatrix}$$

Therefore, the eigenvector matrix is:

$$V = \begin{bmatrix} | & | \\ v_1 & v_2 \\ | & | \end{bmatrix}$$

$$V = \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

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How to Compute Principal Components

1. Calculate the PCA Scores

We calculate:

$$Z = X_c V$$

Therefore:

$$Z = \begin{bmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} \frac{1}{\sqrt{5}} & -\frac{2}{\sqrt{5}} \\ \frac{2}{\sqrt{5}} & \frac{1}{\sqrt{5}} \end{bmatrix}$$

We calculate each element separately.

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Observation A

Original centered observation:

$$x_A = \begin{bmatrix} -2 & -4 \end{bmatrix}$$

PC1 score

$$\begin{aligned} PC1_A &= (-2) \left(\frac{1}{\sqrt{5}} \right) + (-4) \left(\frac{2}{\sqrt{5}} \right) \\ &= \frac{-2 - 8}{\sqrt{5}} \\ &= \frac{-10}{\sqrt{5}} \end{aligned}$$

$$PC1_A = -2\sqrt{5} \approx -4.472$$

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How to Compute Principal Components

PC2 score

$$PC2_A = (-2) \left(\frac{-2}{\sqrt{5}} \right) + (-4) \left(\frac{1}{\sqrt{5}} \right)$$
$$= \frac{4 - 4}{\sqrt{5}}$$

$$PC2_A = 0$$

Therefore:

$$A = (-4.472, 0)$$

in principal-component space.

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How to Compute Principal Components

Observation B

$$x_B = \begin{bmatrix} 0 & 0 \end{bmatrix}$$

Therefore:

$$PC1_B = 0$$

$$PC2_B = 0$$

So:

$$B = (0, 0)$$

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Observation C

$$x_C = \begin{bmatrix} 2 & 4 \end{bmatrix}$$

PC1 score

$$\begin{aligned} PC1_C &= 2 \left(\frac{1}{\sqrt{5}} \right) + 4 \left(\frac{2}{\sqrt{5}} \right) \\ &= \frac{2 + 8}{\sqrt{5}} \\ &= \frac{10}{\sqrt{5}} \end{aligned}$$

$$PC1_C = 2\sqrt{5} \approx 4.472$$

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How to Compute Principal Components

Observation C

PC2 score

$$PC2_C = 2 \left(\frac{-2}{\sqrt{5}} \right) + 4 \left(\frac{1}{\sqrt{5}} \right)$$
$$= \frac{-4 + 4}{\sqrt{5}}$$

$$PC2_C = 0$$

Therefore:

$$C = (4.472, 0)$$

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How to Compute Principal Components

Final Transformed Dataset

Our original data:

$$X_c = \begin{bmatrix} -2 & -4 \\ 0 & 0 \\ 2 & 4 \end{bmatrix}$$

has now become:

$$Z = \begin{bmatrix} -4.472 & 0 \\ 0 & 0 \\ 4.472 & 0 \end{bmatrix}$$

The original variables X_1, X_2 have been replaced by:

$$PC1, PC2$$

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How to Compute Principal Components

What Does This Mean Visually?

Originally, the observations were described using: (X1, X2)

After PCA, they are described using: (PC1, PC2)

So we can plot the transformed observations:

Observation	PC1	PC2
A	-4.472	
B	0	
C	4.472	

The points now lie entirely along the PC1 axis.

That is exactly what we would expect because: $\lambda_1 = 20$, $\lambda_2 = 0$

so PC1 contains 100% of the variance.

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Projection of a Single Observation

It is also useful to understand the calculation without the entire matrix.

For any centered observation:

$$x = \begin{bmatrix} x_1 & x_2 \end{bmatrix}$$

its score on PC1 is:

$$PC1 = xv_1$$

and its score on PC2 is:

$$PC2 = xv_2$$

So projection is essentially a **dot product** between the observation and each principal direction.

For observation A :

$$\begin{aligned} PC1_A &= [-2, -4] \cdot \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \\ &= -4.472 \end{aligned}$$

This number tells us **where observation A lies along the PC1 axis**.

Visualization of Data, Understanding and Interpretation

How to Compute Principal Components

Why Is Projection Important?

Eigenvectors tell us *which directions should we use?*

Projection tells us *where does each observation fall along those directions?*

This distinction is fundamental:

- Eigenvectors → new axes
- Projection → new coordinates

Once every observation has PC scores, we can finally create the visualization: PC1 on x-axis, PC2 on y-axis

Dimensionality Reduction Happens Here

Suppose the original dataset has $p = 100$ variables, but we retain only the first two eigenvectors:

$$V_2 = \begin{bmatrix} | & | \\ v_1 & v_2 \\ | & | \end{bmatrix}$$

$$\text{Then: } X_{n \times 100} V_{100 \times 2} = Z_{n \times 2}$$

Therefore: 100D → 2D

Visualization of Data, Understanding and Interpretation

Explaining Variance & Choosing k

How Many Principal Components Should We Keep?

Each principal component has an eigenvalue λ_i representing the amount of variance it captures.

The explained variance ratio (EVR) is:

$$EVR_i = \frac{\lambda_i}{\sum_{j=1}^p \lambda_j}$$

The cumulative explained variance after retaining k components is:

$$CEV_k = \frac{\sum_{i=1}^k \lambda_i}{\sum_{j=1}^p \lambda_j}$$

What is k ?

k = number of principal components retained

Visualization of Data, Understanding and Interpretation

Explaining Variance & Choosing k

Since dimensionality reduction requires:

$$k \ll p$$

we choose only the first k components that retain a sufficiently large proportion of the original variation.

Example:

$$\lambda = [6, 2, 1, 1]$$

Total variance:

$$6 + 2 + 1 + 1 = 10$$

Visualization of Data, Understanding and Interpretation

Explaining Variance & Choosing k

Components
PC1
PC1 – PC2
PC1 – PC3
PC1 – PC4

Thus, if the target is $\geq 90\%$ variance, choose: **$k = 3$**

Visualization of Data, Understanding and Interpretation

PCA Numericals

Extra Questions:

- 1) A dataset contains the following three observations for a single variable: $X=\{2,4,6\}$
 - Calculate the mean of the variable.
 - Subtract the mean from each observation to obtain the mean-centered values.
 - Verify that the mean of the centered data is zero.
- 2) A dataset contains three principal components with eigenvalues: $\lambda_1 = 8, \lambda_2 = 2, \lambda_3 = 1$
 - Calculate the percentage of variance explained by PC1.
 - Calculate the cumulative percentage of variance explained by PC1 and PC2.
 - If we want to retain at least 90% of the total variance, how many principal components should be retained?
- 3) A centered observation is represented by: $x = [2, 4]$.
 - Using the projection formula: $PC1=xv_1$, calculate the PC1 score of the observation.
 - The first principal component has the unit eigenvector:

$$v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Visualization of Data, Understanding and Interpretation

PCA Numericals

Answers:

1)

Given the dataset:

$$X = \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

Step 1: Calculate the mean

$$\bar{x} = \frac{2 + 4 + 6}{3} = 4$$

Step 2: Center each observation

$$x_c = x - \bar{x}$$

$$2 - 4 = -2$$

$$4 - 4 = 0$$

$$6 - 4 = 2$$

Therefore:

$$X_c = \begin{bmatrix} -2 \\ 0 \\ 2 \end{bmatrix}$$

Answer

$$X_c = [-2, 0, 2]$$

Visualization of Data, Understanding and Interpretation

PCA Numericals

Answers:

2)

Suppose PCA produces the following eigenvalues:

$$\lambda_1 = 8, \quad \lambda_2 = 2, \quad \lambda_3 = 1$$

Step 1: Calculate total variance

$$\lambda_{\text{total}} = 8 + 2 + 1 = 11$$

Step 2: Variance explained by PC1

$$EV R_1 = \frac{8}{11}$$

$$EV R_1 \approx 72.73\%$$

Step 3: Variance explained by PC1 + PC2

$$EV R_{1,2} = \frac{8 + 2}{11} = \frac{10}{11}$$

$$EV R_{1,2} \approx 90.91\%$$

Answer

If we want to retain at least 90% of the variance:

$$k = 2$$

Visualization of Data, Understanding and Interpretation

PCA Numericals

Answers:

3)

Suppose a centered observation is:

$$x = \begin{bmatrix} 2 & 4 \end{bmatrix}$$

and the first principal direction is:

$$v_1 = \frac{1}{\sqrt{5}} \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

The PC1 score is obtained by projection:

$$PC1 = xv_1$$

Substitute:

$$\begin{aligned} PC1 &= [2, 4] \begin{bmatrix} 1/\sqrt{5} \\ 2/\sqrt{5} \end{bmatrix} \\ &= \frac{2}{\sqrt{5}} + \frac{8}{\sqrt{5}} \\ &= \frac{10}{\sqrt{5}} \end{aligned}$$

$$PC1 = 2\sqrt{5} \approx 4.472$$

Answer

The observation's coordinate along the first principal component is:

$$PC1 \approx 4.472$$

PCA Looks for Linear Structure

PCA finds new axes that capture the maximum variance in the data.

This works well when important structure is approximately linear, but some datasets contain non-linear relationships that cannot be represented effectively by a straight-line projection.

Example: A Curved / Spring-Like Dataset

Imagine two classes arranged along a curved structure in 3D.

High-dimensional structure → 2D visualization

- PCA: may mix the classes together.
- t-SNE: can preserve the local neighborhoods and reveal the two groups.

PCA asks: “Which directions contain the most variance?”

t-SNE asks: “Which points are close to each other?”

Visualization of Data, Understanding and Interpretation

What Is t-SNE?



t-SNE stands for t-Distributed Stochastic Neighbor Embedding

It was developed by Laurens van der Maaten and Geoffrey Hinton as an improved version of SNE.

- The central idea is simple: points that are close together in high-dimensional space should remain close together in the low-dimensional representation.
- Instead of directly trying to preserve every numerical distance, t-SNE converts distances into probabilities of being neighbors.
- High-dimensional distances → Neighbor probabilities → 2D embedding

PCA	t-SNE
Finds directions of maximum variance	Finds similar neighbors
Linear transformation	Non-linear embedding
Global variance is central	Local relationships are central

Methodological Overview:

Step 1: Convert Distances into Probabilities

Suppose we choose one point x_i

We calculate its distance from every other point: $d(x_i, x_j)$

Nearby points should have a high probability of being neighbors. Distant points should have a low probability.

t-SNE/SNE uses a Gaussian distribution to convert these distances into probabilities:

$$p_{j|i} = \frac{\exp\left(-\frac{\|x_i - x_j\|^2}{2\sigma_i^2}\right)}{\sum_{k \neq i} \exp\left(-\frac{\|x_i - x_k\|^2}{2\sigma_i^2}\right)}$$

- Small distance $\rightarrow$ large similarity probability
- Large distance $\rightarrow$ small similarity probability
- The parameter σ_i controls the width of the Gaussian neighborhood.

Step 2: Build the Same Probabilities in 2D

Initially, the points are placed randomly in the lower-dimensional space.

For example: $x_i \in \mathbb{R}^{100}$ becomes $y_i \in \mathbb{R}^2$

We then calculate similarities between the 2D points using a corresponding probability distribution:

$$q_{ij} = \frac{(1 + \|y_i - y_j\|^2)^{-1}}{\sum_{k \neq l} (1 + \|y_k - y_l\|^2)^{-1}}$$

We now have two probability distributions:

- P = similarities in high-dimensional space
- Q = similarities in low-dimensional space
- Goal: Make $P \approx Q$

Step 3: Compare the Two Distributions

t-SNE uses Kullback–Leibler divergence (KL divergence) to measure how different the two probability distributions are. You can think of KL divergence as a measure of mismatch between probability distributions.

$$KL(P \parallel Q) = \sum_{i,j} p_{ij} \log \frac{p_{ij}}{q_{ij}}$$

Interpretation:

- $KL(P \parallel Q)$ large $\rightarrow$ high-dimensional and low-dimensional relationships are very different.
- $KL(P \parallel Q)$ small $\rightarrow$ the two representations are more similar.
- The ideal situation is: $KL(P \parallel Q) = 0$, when the distributions match.
- Main objective: minimize $KL(P \parallel Q)$

Visualization of Data, Understanding and Interpretation

What Is t-SNE?

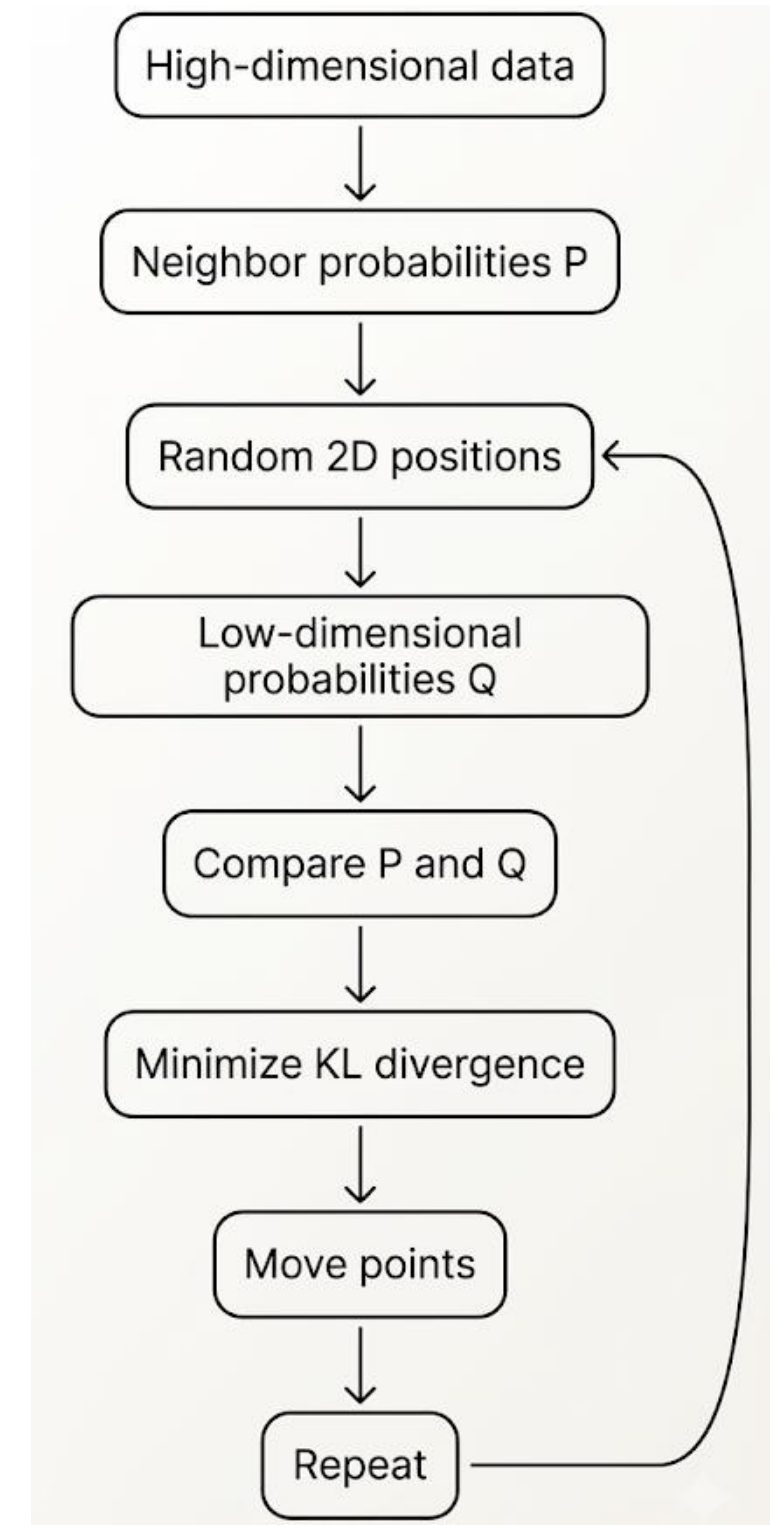
Step 4: Move the Points Until the Distributions Match

The points initially have random 2D positions. t-SNE repeatedly:

1. Calculates the low-dimensional similarities.
2. Compares them with the high-dimensional similarities.
3. Calculates the KL divergence.
4. Adjusts the 2D positions.
5. Repeats.

Eventually, points that were strong neighbors in the original space tend to become close together in the 2D representation.

t-SNE does not calculate the final 2D coordinates directly. It iteratively searches for a 2D arrangement that preserves neighborhood relationships.



Perplexity: Controlling the Neighborhood

The width of the Gaussian distribution is controlled by σ_i

But instead of directly choosing a different σ_i for every point, t-SNE uses a parameter called perplexity.

Conceptually:

- Perplexity $\approx$ effective neighborhood size
- Low perplexity focuses more strongly on very local relationships.
- Low perplexity $\rightarrow$ smaller neighborhoods
- High perplexity considers a broader neighborhood.
- High perplexity $\rightarrow$ larger neighborhoods
- The algorithm adjusts σ_i so that the resulting probability distribution has approximately the chosen perplexity.

t-SNE working: https://youtu.be/o_cAOa5fMhE?si=P-shHNEsnGu8zkML

Visualization of Data, Understanding and Interpretation

t-SNE vs. PCA



Strengths

- Handles non-linear structure
- Excellent at revealing local clusters
- Can separate groups that PCA may mix together
- Useful for exploring complex latent spaces

Limitations

- Computationally more expensive than PCA
- Results can be sensitive to perplexity
- Different parameter choices can produce different visualizations
- Cluster distances and sizes should not automatically be interpreted as meaningful global distances
- It is primarily a visual exploration technique, rather than a general-purpose dimensionality-reduction transformation

	PCA	t-SNE
Structure	Linear	Non-linear
Main focus	Variance	Local neighborhoods
Transformation	Explicit linear projection	Iteratively optimized embedding
Clusters	May overlap	Often separates local groups well
Speed	Fast	Slower

From Local Neighbors to a Global Map

t-SNE demonstrated that nonlinear dimensionality reduction can reveal local clusters that PCA may miss. However, t-SNE has important limitations:

- It can be computationally expensive.
- Results can be sensitive to parameter choices.
- It primarily emphasizes local neighborhood structure.
- The global arrangement of clusters may be difficult to interpret.

UMAP

Uniform Manifold Approximation and Projection (UMAP) is another nonlinear dimensionality-reduction method designed to preserve the structure of high-dimensional data while producing a low-dimensional representation.

- UMAP builds a representation of how observations are connected locally, then finds a low-dimensional layout that preserves those relationships.
- High-dimensional data → Neighborhood structure → Low-dimensional embedding

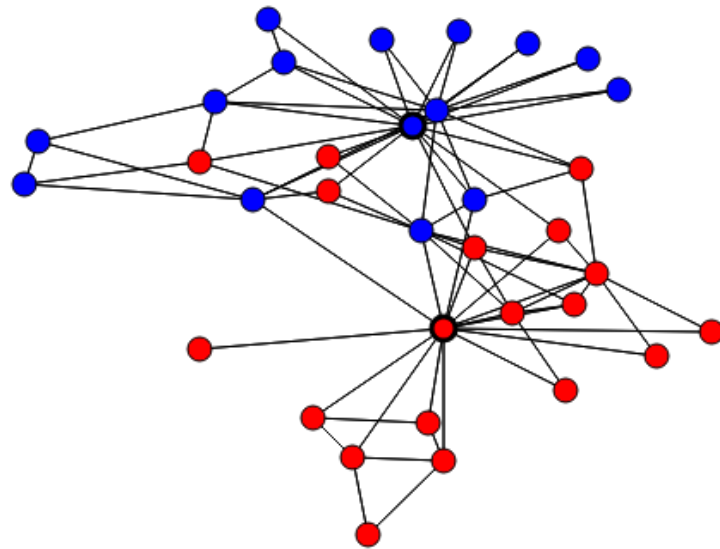
Visualization of Data, Understanding and Interpretation

What is UMAP?

Suppose we have points in a high-dimensional space.

For each point, UMAP asks *which other observations are its nearest neighbors?*

This produces a nearest-neighbor graph:



Each point becomes a node, and nearby observations are connected. UMAP then attempts to create a 2D representation where these important connections are retained.

Find neighbors → Build graph → Optimize 2D layout

Key difference from PCA

- PCA: Finds directions of maximum variance.
- UMAP: Finds relationships between neighboring observations.

Visualization of Data, Understanding and Interpretation

What is UMAP?

How UMAP Builds the Embedding

From High-Dimensional Neighborhoods to a 2D Map

1. Find nearest neighbors

For every observation, identify nearby observations.

$$x_i \rightarrow \{x_{j_1}, x_{j_2}, \dots, x_{j_k}\}$$

where k is controlled by:

`n_neighbors`

2. Construct a neighborhood graph

Nearby observations receive stronger connections.

Strong connection $\rightarrow$ very similar neighbors

Weak connection $\rightarrow$ less similar neighbors

Visualization of Data, Understanding and Interpretation

What is UMAP?

3. Find a low-dimensional representation

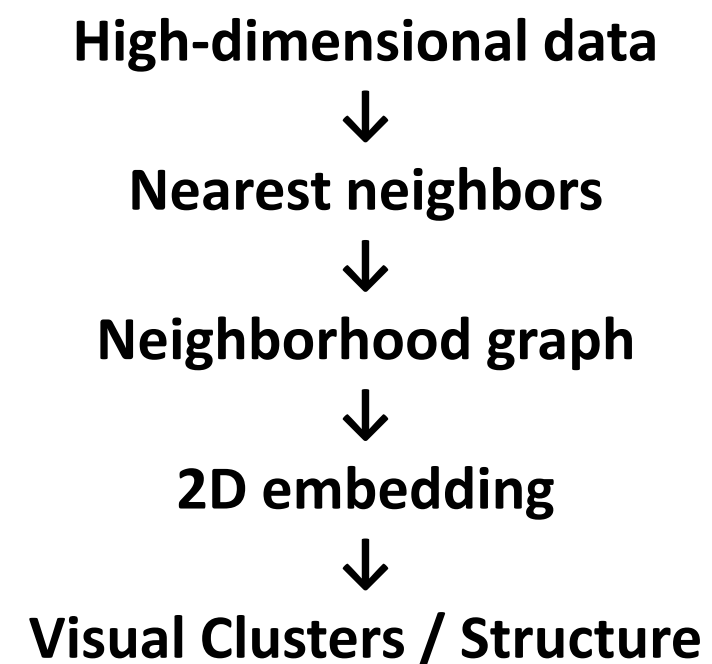
UMAP places the observations in 2D/3D:

$$\mathbb{R}^p \rightarrow \mathbb{R}^2$$

while attempting to preserve the important neighborhood relationships.

4. Optimize the layout

The points are repeatedly moved until the low-dimensional representation provides a good approximation of the high-dimensional neighborhood structure.



Visualization of Data, Understanding and Interpretation

What is UMAP?

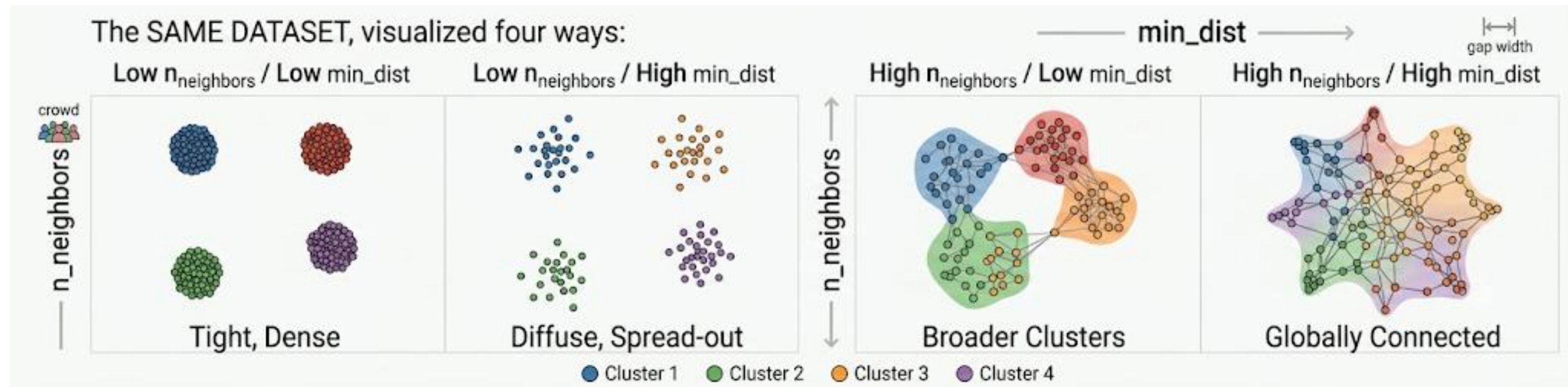
Parameters

nneighbours controls how much of the neighborhood UMAP considers.

- small $n \rightarrow$ more local structure
- large $n \rightarrow$ broader structure

min_dist controls how tightly points can be packed together in the low-dimensional representation.

- small $\text{min_dist} \rightarrow$ tighter clusters
- large $\text{min_dist} \rightarrow$ more spread-out points



Visualization of Data, Understanding and Interpretation

PCA vs. t-SNE vs. UMAP

	PCA	UMAP	t-SNE
Type	Linear	Nonlinear	Nonlinear
Main idea	Maximum variance	Neighborhood / manifold structure	Local similarity
New axes	Yes	No	No
Local structure	Moderate	Strong	Strong
Global structure	Generally preserved through linear projection	Often better represented	Often less reliable
Main parameters	Number of PCs	n_neighbours, min_dist	Perplexity, learning rate
Typical use	Compression + visualization	Exploring local + broader structure	Exploring local clusters

Visualization of Data, Understanding and Interpretation

Real-World Applications of Dimensionality Reduction

Where Do PCA, t-SNE and UMAP Actually Get Used?

Dimensionality reduction is particularly useful when each observation has many features, but we want to explore patterns, groups, similarities, or anomalies visually.

1. Image & Computer Vision

Images can be represented using hundreds or thousands of features.

Image features → PCA / t-SNE / UMAP → 2D embedding

Use:

- Visualize whether images form natural groups
- Identify similar images
- Detect unusual images
- Explore learned CNN/AI embeddings

Example: Plot thousands of image embeddings and color points by object class.

Visualization of Data, Understanding and Interpretation

Real-World Applications of Dimensionality Reduction

2. Text & Document Analysis

A document collection can be represented using a very high-dimensional vector space.

Use:

- Discover groups of similar documents
- Explore topics
- Identify unusual documents
- Visualize semantic relationships

Example: News articles → embedding → UMAP → clusters corresponding to different topics.

3. Genomics & Biological Data

A single biological sample can contain thousands of measured variables, such as gene-expression values. Gene expression → PCA → t-SNE / UMAP

Use:

- Identify groups of similar cells
- Explore cell populations
- Detect unusual observations
- Visualize biological states or transitions

Visualization of Data, Understanding and Interpretation

Real-World Applications of Dimensionality Reduction

4. Dimensionality Reduction in Machine Learning

Visualizing a Model's Latent Space:

Modern machine-learning models often transform raw inputs into high-dimensional embeddings.

For example: Image $\rightarrow$ Neural Network $\rightarrow$ $\underbrace{512D}_{\text{latent representation}}$

We cannot directly visualize the 512-dimensional representation. So $512D \rightarrow 2D$ using PCA, t-SNE, or UMAP. If similar examples appear close together: similar inputs $\rightarrow$ similar embeddings $\rightarrow$ nearby points

We can visually investigate:

- Class separation
- Clusters
- Outliers
- Misclassified examples
- Similarity between learned representations

Visualization of Data, Understanding and Interpretation

MCQ

Q1. Which statement best describes the primary objective of PCA?

- A. Preserve local neighborhood relationships
- B. Maximize variance along successive directions
- C. Construct a nearest-neighbor graph
- D. Minimize KL divergence

Visualization of Data, Understanding and Interpretation

MCQ

Q1. Which statement best describes the primary objective of PCA?

- A. Preserve local neighborhood relationships
- B. Maximize variance along successive directions
- C. Construct a nearest-neighbor graph
- D. Minimize KL divergence

Answer: B

Visualization of Data, Understanding and Interpretation

MCQ

Q2. A dataset has principal-component eigenvalues of 6,2,1, and 1. How many components are required to retain at least 80% of the total variance?

- A. 1
- B. 2
- C. 3
- D. 4

Visualization of Data, Understanding and Interpretation

MCQ

Q2. A dataset has principal-component eigenvalues of 6,2,1, and 1. How many components are required to retain at least 80% of the total variance?

- A. 1
- B. 2
- C. 3
- D. 4

Answer: B

Visualization of Data, Understanding and Interpretation

MCQ

Q3. In t-SNE, what does perplexity primarily control?

- A. Number of principal components
- B. Width of the visualization axes
- C. Effective neighborhood size
- D. Number of output dimensions

Visualization of Data, Understanding and Interpretation

MCQ

Q3. In t-SNE, what does perplexity primarily control?

- A. Number of principal components
- B. Width of the visualization axes
- C. Effective neighborhood size
- D. Number of output dimensions

Answer: C

Visualization of Data, Understanding and Interpretation

MCQ

Q4. Which operation converts a centered observation into its coordinate along a principal component?

- A. Covariance calculation
- B. Eigenvalue decomposition
- C. Matrix normalization
- D. Projection onto the eigenvector

Visualization of Data, Understanding and Interpretation

MCQ

Q4. Which operation converts a centered observation into its coordinate along a principal component?

- A. Covariance calculation
- B. Eigenvalue decomposition
- C. Matrix normalization
- D. Projection onto the eigenvector

Answer: D

Visualization of Data, Understanding and Interpretation

MCQ

Q5. Which characteristic most clearly distinguishes UMAP from PCA?

- A. UMAP uses nonlinear neighborhood-based structure
- B. UMAP requires exactly two variables
- C. UMAP maximizes total variance
- D. UMAP produces original feature values

Visualization of Data, Understanding and Interpretation

MCQ

Q5. Which characteristic most clearly distinguishes UMAP from PCA?

- A. UMAP uses nonlinear neighborhood-based structure
- B. UMAP requires exactly two variables
- C. UMAP maximizes total variance
- D. UMAP produces original feature values

Answer: A



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 2 : Advanced Visualization & Storytelling

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

By the end of this lecture, students should be able to:

- Explain how geospatial visualizations represent location, magnitude, and regional patterns using maps, symbol maps, choropleths, and cartograms.
- Interpret and communicate uncertainty using standard deviation, standard error, confidence intervals, confidence bands, and outcome-based visualizations.
- Design and evaluate dashboards for monitoring, balancing information density, cognitive load, and rapid detection of trends or exceptions.
- Understand coordinated multiple views and how brushing, linking, and shared interaction support exploratory analysis across different visual representations.

Visualization of Data, Understanding and Interpretation

Why Visualize Data Geographically?

(TB 1, Ch 15)

Geospatial visualization represents data according to its location in physical or geographic space.

A map becomes useful when where something occurs is an important part of the question.

It allows us to see:

- Spatial patterns: where values concentrate or spread
- Clusters and hotspots: regions with unusually high/low values
- Regional differences: how measurements vary across locations
- Spatial relationships: proximity, distance, boundaries, and connectivity
- Geographic context: relating data to roads, terrain, cities, borders, etc.



Visualization of Data, Understanding and Interpretation

From Location to Coordinates

To visualize geographic data, locations must first be represented numerically.

Geographic coordinates:

- Latitude:
 - Position north or south of the equator
 - Typically ranges from -90° to $+90^\circ$
- Longitude:
 - Position east or west of the prime meridian
 - Typically ranges from -180° to $+180^\circ$
- Altitude / elevation
 - Height relative to a reference surface

Together, coordinates allow geographic locations to be represented mathematically.

💡 Important Distinction!

A geographic coordinate system describes locations on the Earth's surface, while a visualization must eventually display those locations on a 2D screen or page. That creates the fundamental problem of map projection.

Visualization of Data, Understanding and Interpretation

The Map Projection Problem

The Earth is approximately spherical, but our displays are flat. Therefore, converting geographic coordinates into a map requires a map projection.

Every projection introduces some form of distortion.

A projection may distort:

- Shape
- Area
- Distance
- Direction

No single projection preserves all of these simultaneously.

Design principle: Choose the projection according to the purpose of the visualization.

For example:

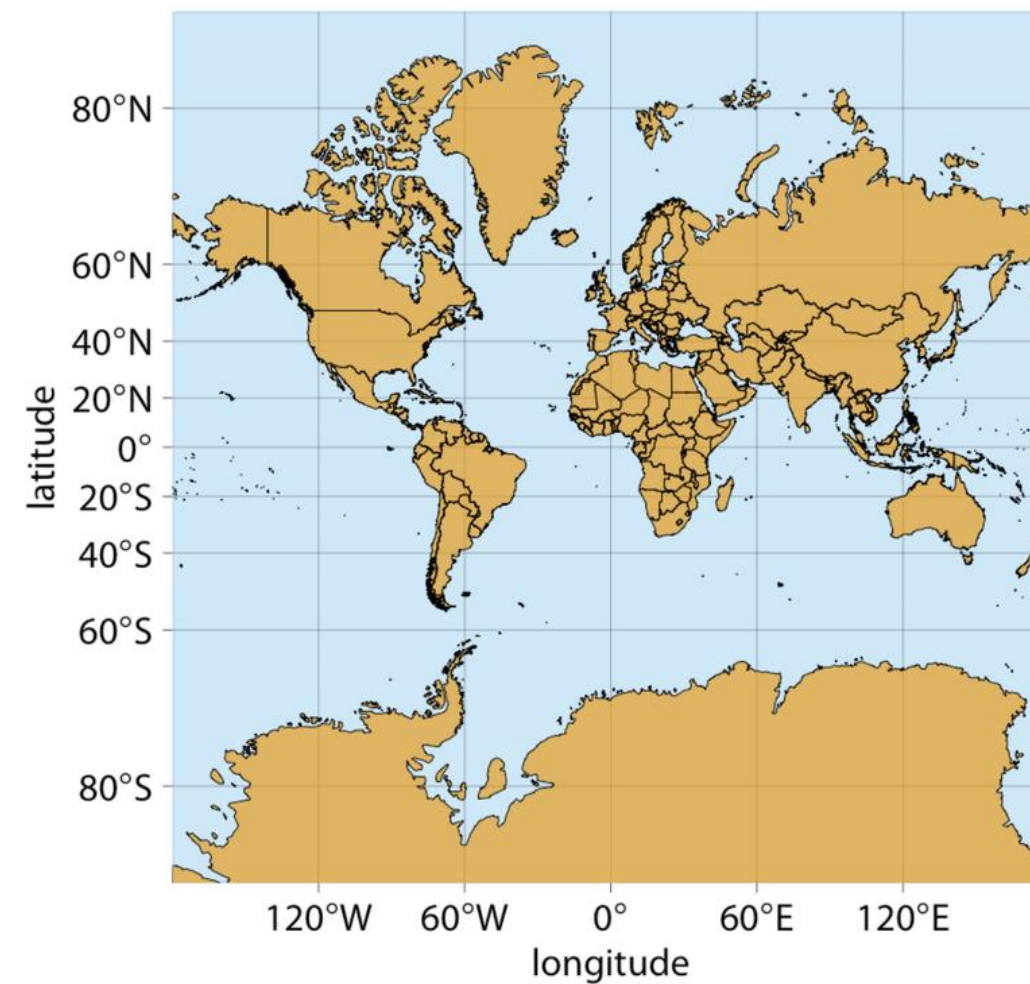
- Preserving area → useful when comparing regional quantities
- Preserving direction → useful for navigation
- Preserving shape → useful for certain geographic analyses

Visualization of Data, Understanding and Interpretation

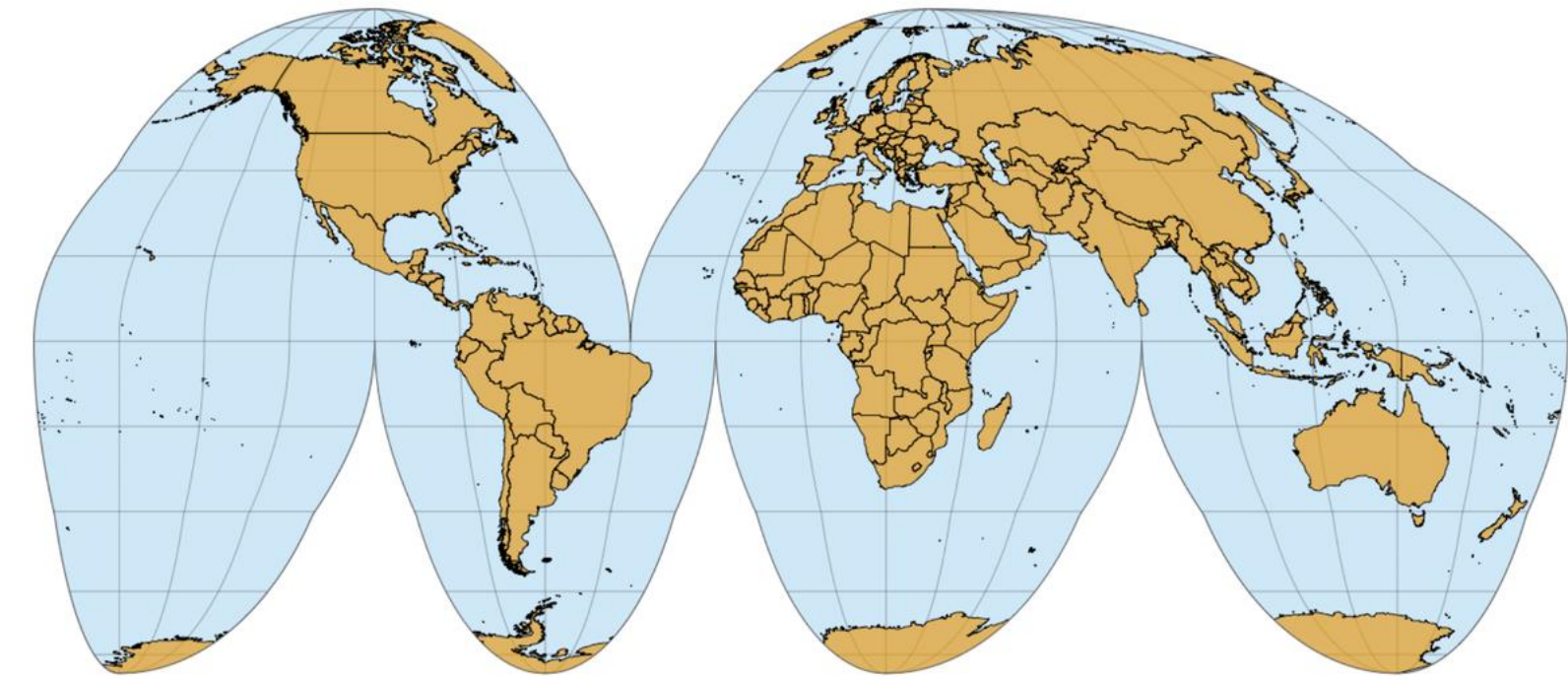
The Map Projection Problem



Orthographic Projection



Mercator Projection



Interrupted Goode Homolosine Projection

Visualization of Data, Understanding and Interpretation

Building a Map: Layers and Context

A useful map rarely consists of a single visual layer.

Multiple layers can provide geographic context:

- Base layer: terrain, satellite imagery, land/water
- Reference layers: roads, boundaries, cities, administrative regions
- Data layer: observations, measurements, events, highlighted regions

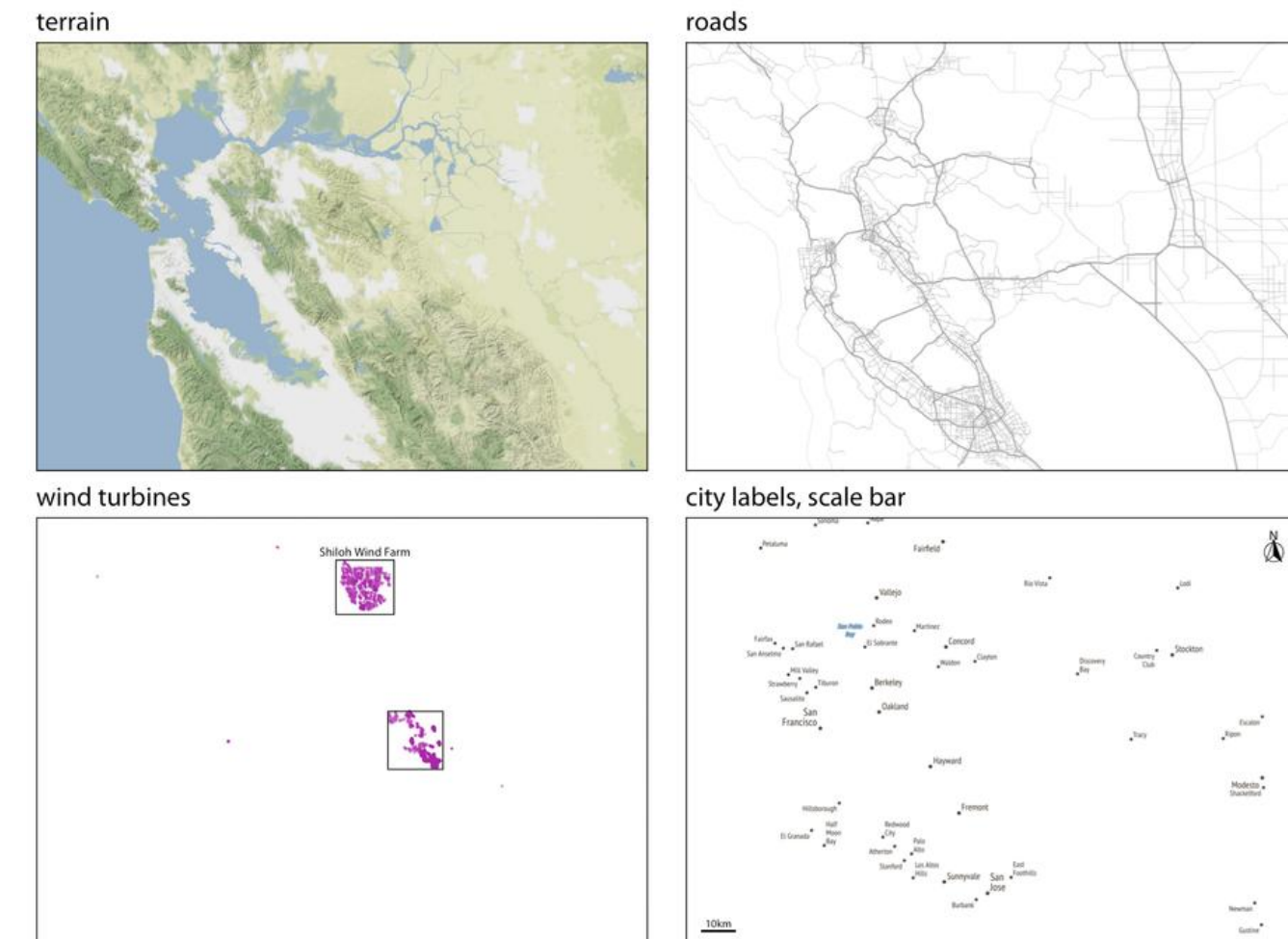
💡 *Key design principle*

- Every layer should serve the analytical task.
- Too many background layers can compete with the actual data.

A map can therefore fail through visual clutter, even when every individual layer is technically correct.

Example: for a traffic-accident map,

- Useful features: roads, accident locations, major intersections
- Potentially distracting features: unnecessary terrain, decorative satellite imagery, excessive labels



Visualization of Data, Understanding and Interpretation

Point Maps: Where Are the Observations?

A point map represents each geographic observation using its spatial position.

ENCODING: *position* → *geographic location*

Additional attributes can be encoded using:

- Color: category or quantitative value
- Size: magnitude
- Shape: category
- Transparency: density/overlap

Strengths

- Directly shows individual locations
- Easy to understand
- Preserves geographic relationships
- Useful for events and point observations

Limitations

- Overplotting when many observations overlap
- Difficult to compare exact quantities from size alone
- Too many visual attributes can create clutter

Example Applications:

- Earthquake locations
- Hospitals
- Crime incidents
- Sensor locations
- Wildlife observations

Visualization of Data, Understanding and Interpretation

Symbol Maps: Adding Another Dimension

A point can carry more information than simply its location.

A symbol map uses graphical properties of symbols to encode additional attributes.

Visual Property	Possible Encoding
Position	Location
Size	Magnitude
Colour	Category/Value
Shape	Type
Orientation	Direction

A map of cities could encode:

- Location → city position
- Circle size → population
- Color → air-quality category

This produces a multivariate spatial visualization.

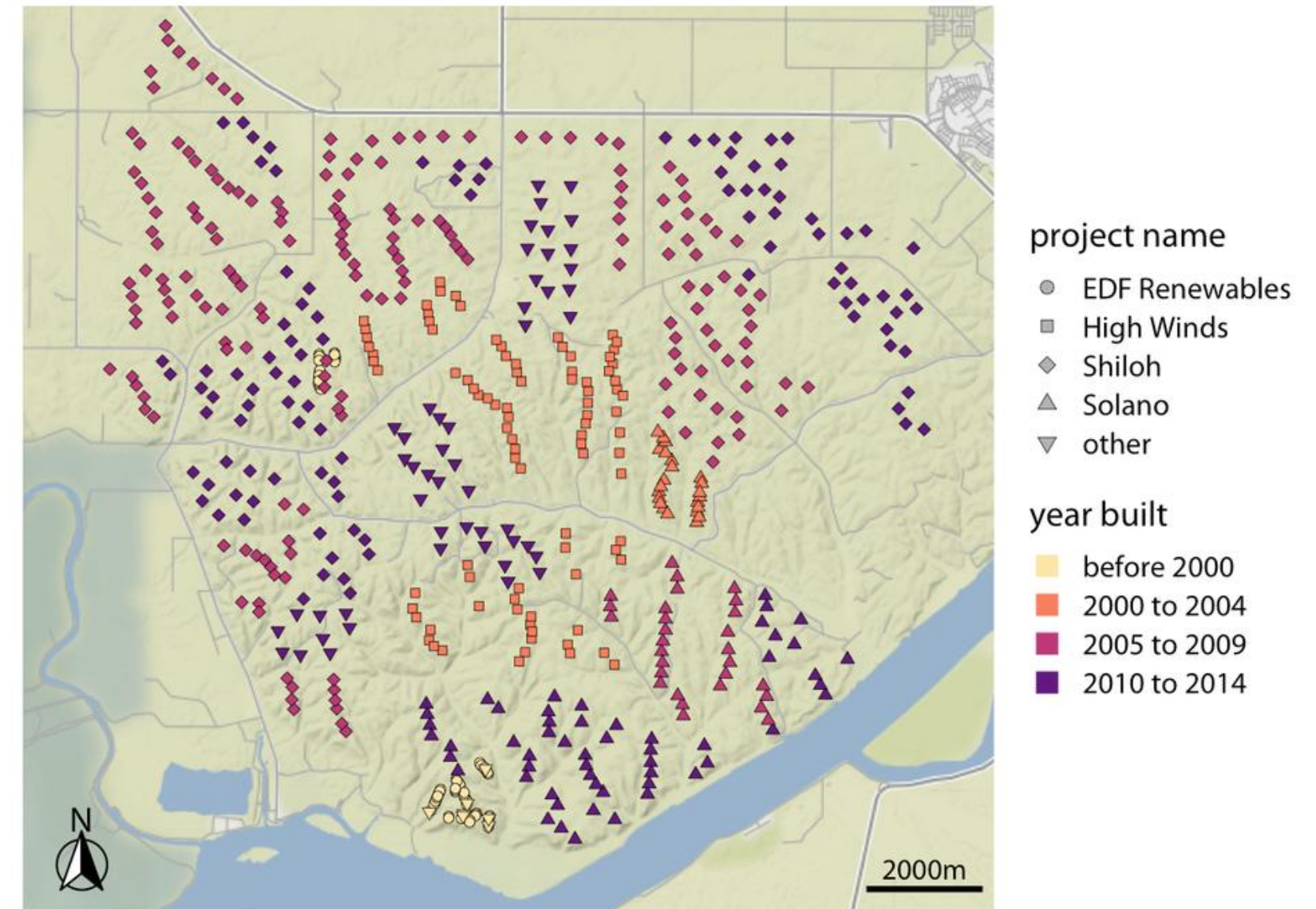
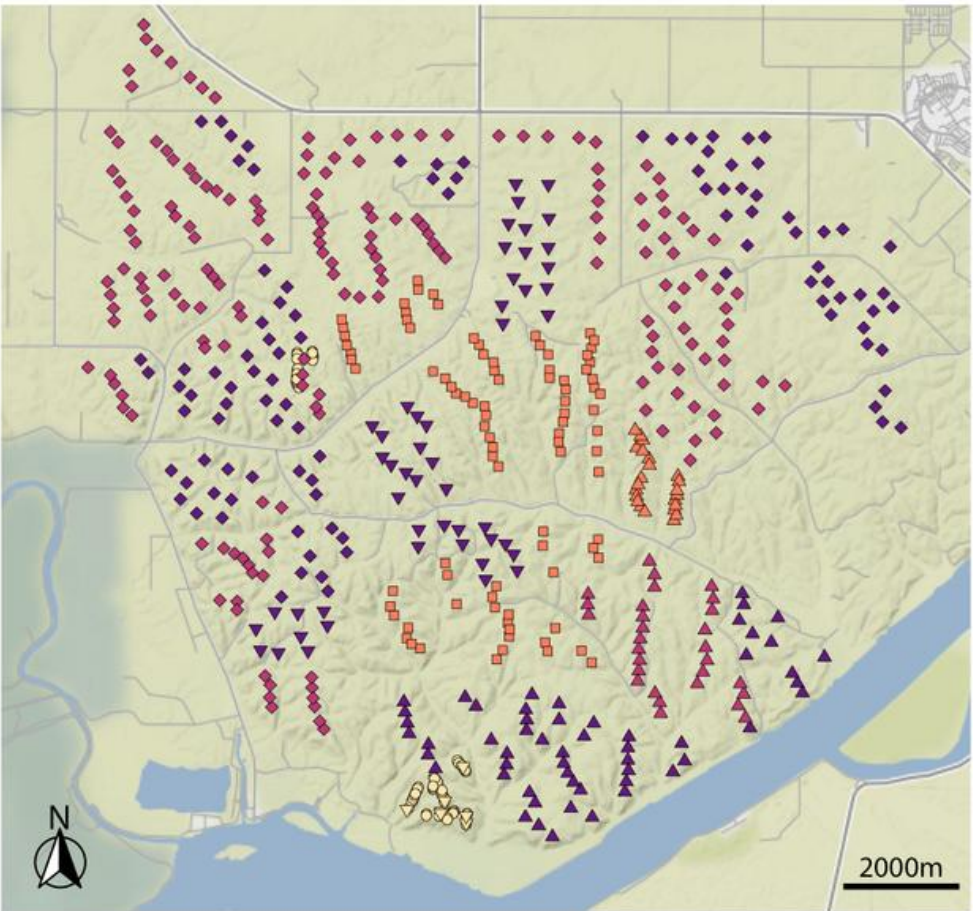
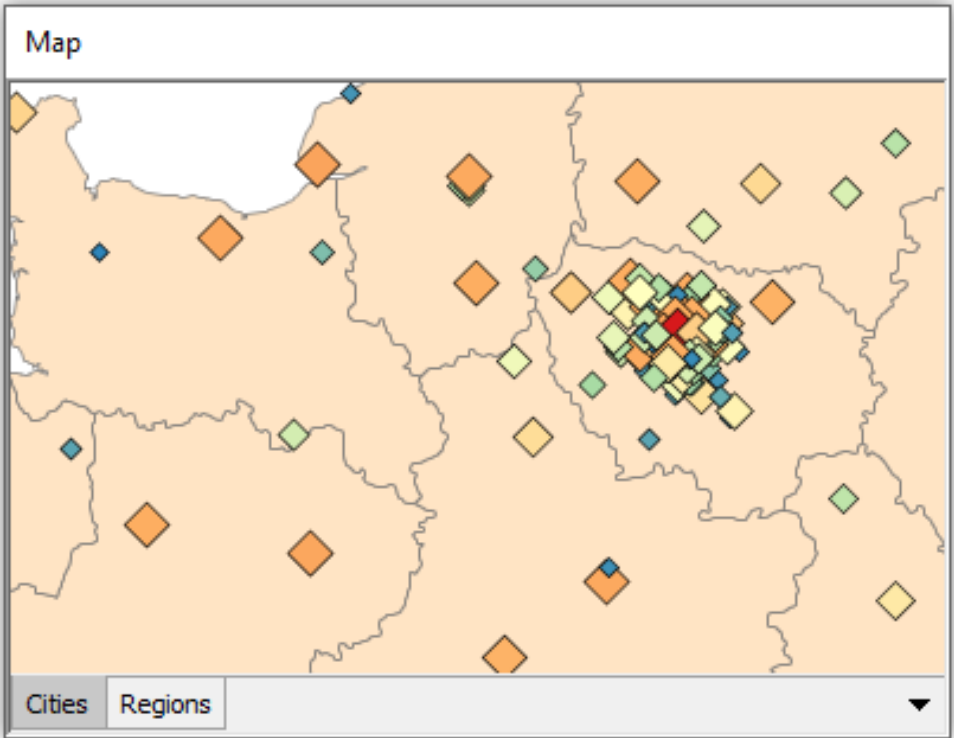
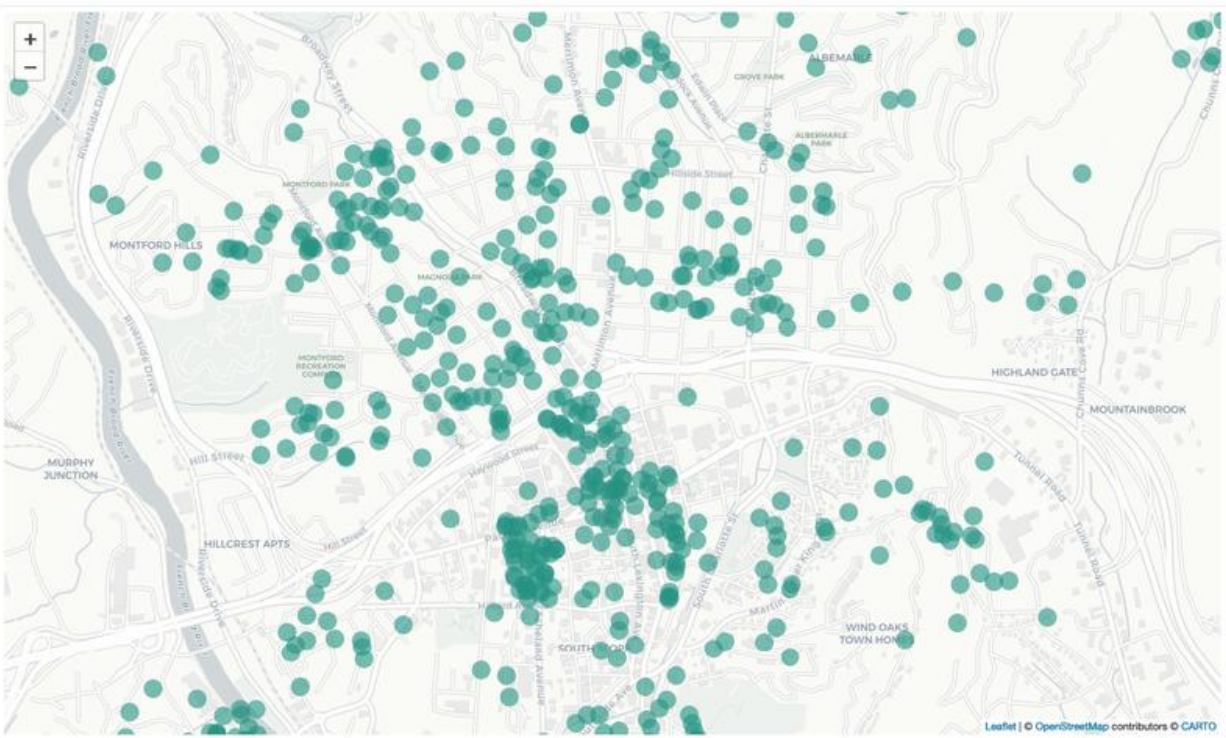
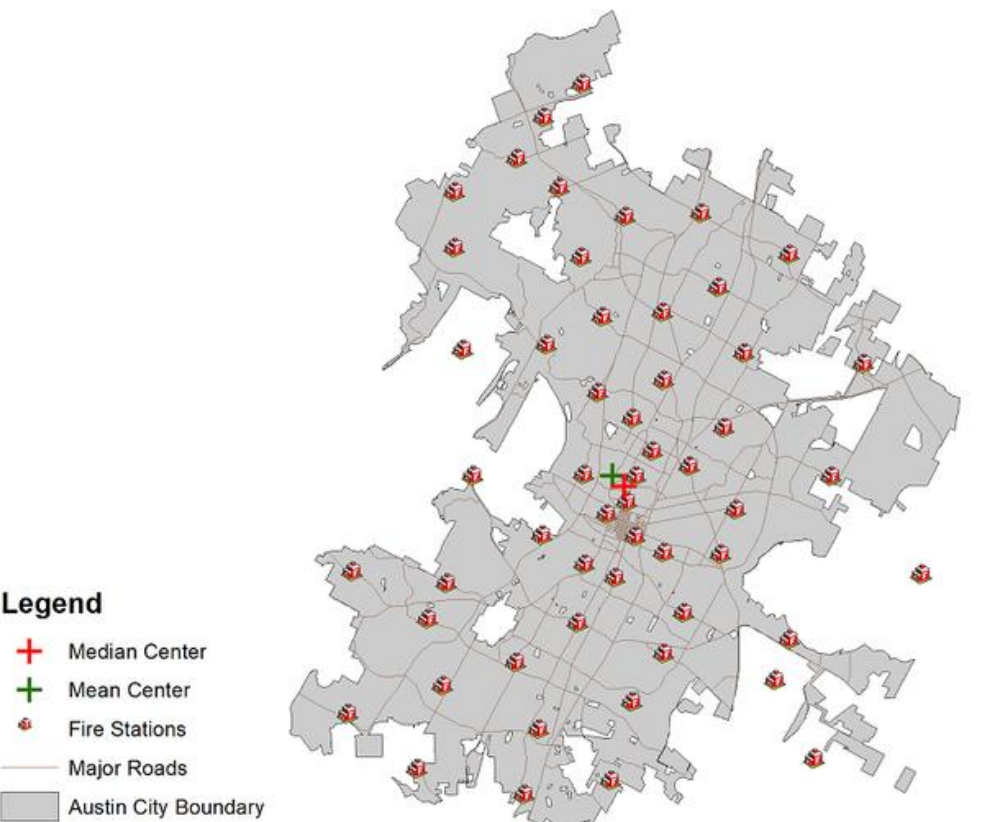
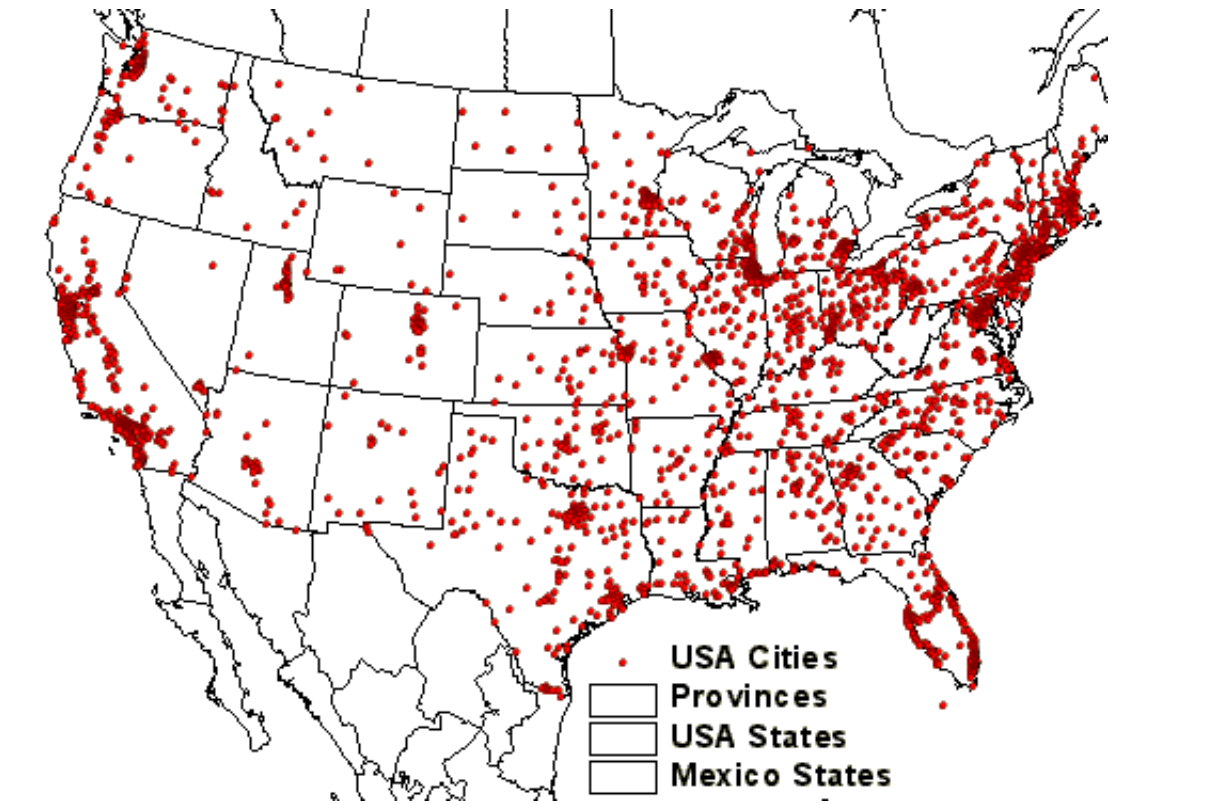


Figure: Location of individual wind turbines in the Shiloh Wind Farm. Each dot highlights the location of one wind turbine. Dots are colored by when the wind turbine was built, and the shape of the dots represents the various projects to which the individual wind turbines belong

Visualization of Data, Understanding and Interpretation

Examples



Visualization of Data, Understanding and Interpretation

Choropleth Maps

A choropleth map divides geographic space into predefined regions and uses color to represent a quantitative value.

Examples:

- population density by state
- unemployment rate by district
- rainfall by region
- disease incidence by country

ENCODING:

region → *geographic area*

colour → *quantitative value*

Why choropleths are powerful?

They allow us to see:

- regional patterns
- hotspots
- geographic clusters
- broad spatial differences

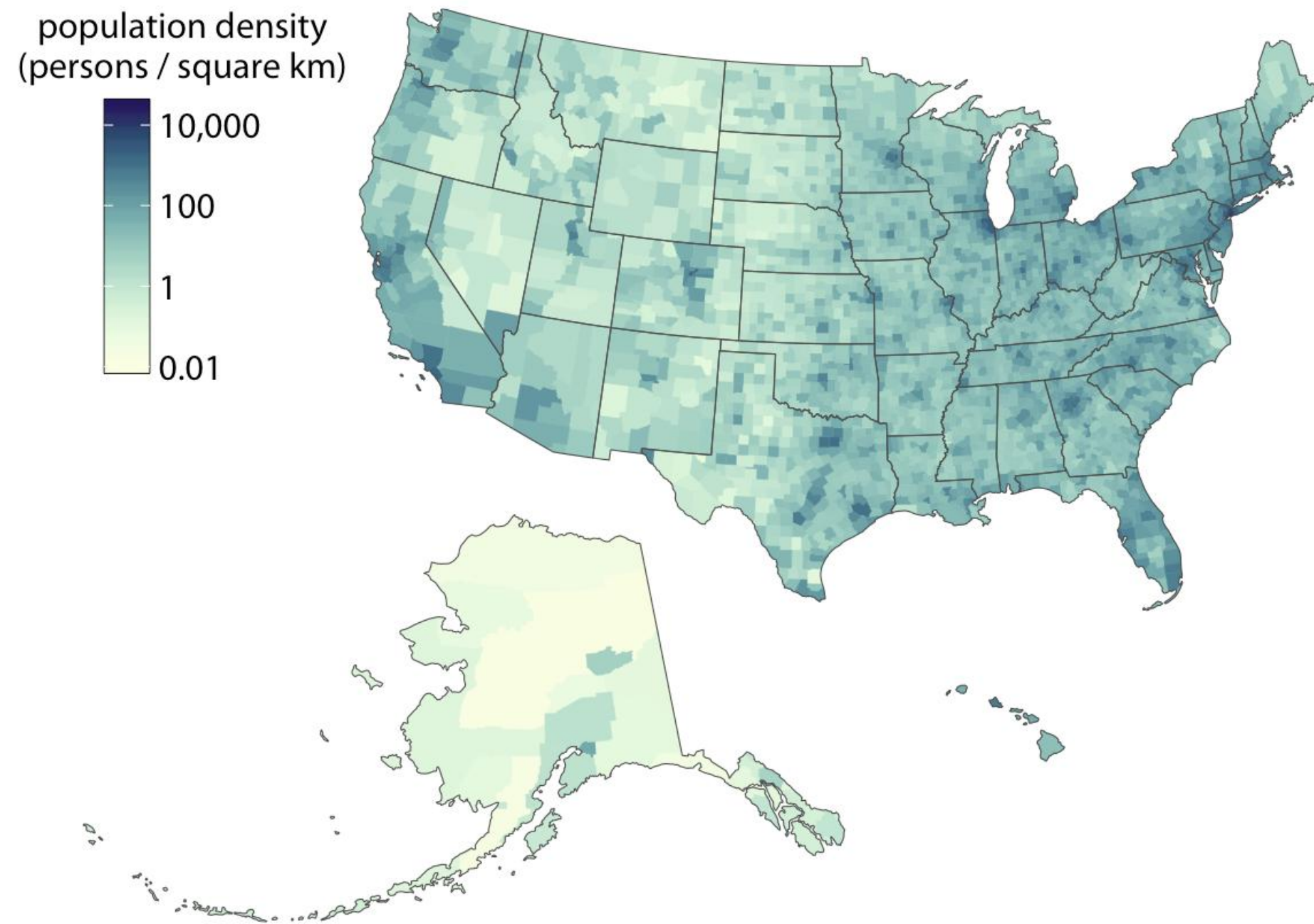


Figure: Population density in every U.S. county, shown as a choropleth map. Population density is reported as persons per square kilometer.

Visualization of Data, Understanding and Interpretation

The Choropleth Trap: Counts vs Rates

A major problem with choropleth maps is that raw counts can be misleading.

Suppose we map the number of hospitalizations by district. A district with more people will naturally tend to have more hospitalizations. But this does not necessarily mean that the district has a higher hospitalization rate. Instead, consider **normalization**.

For example:

Raw count: 5,000 hospitalizations

Rate: 5,000 hospitalizations / population

The second representation accounts for differences in population size.

💡 *General principle: when geographic regions differ substantially in population or area, consider whether the variable should be normalized*

Other problems:

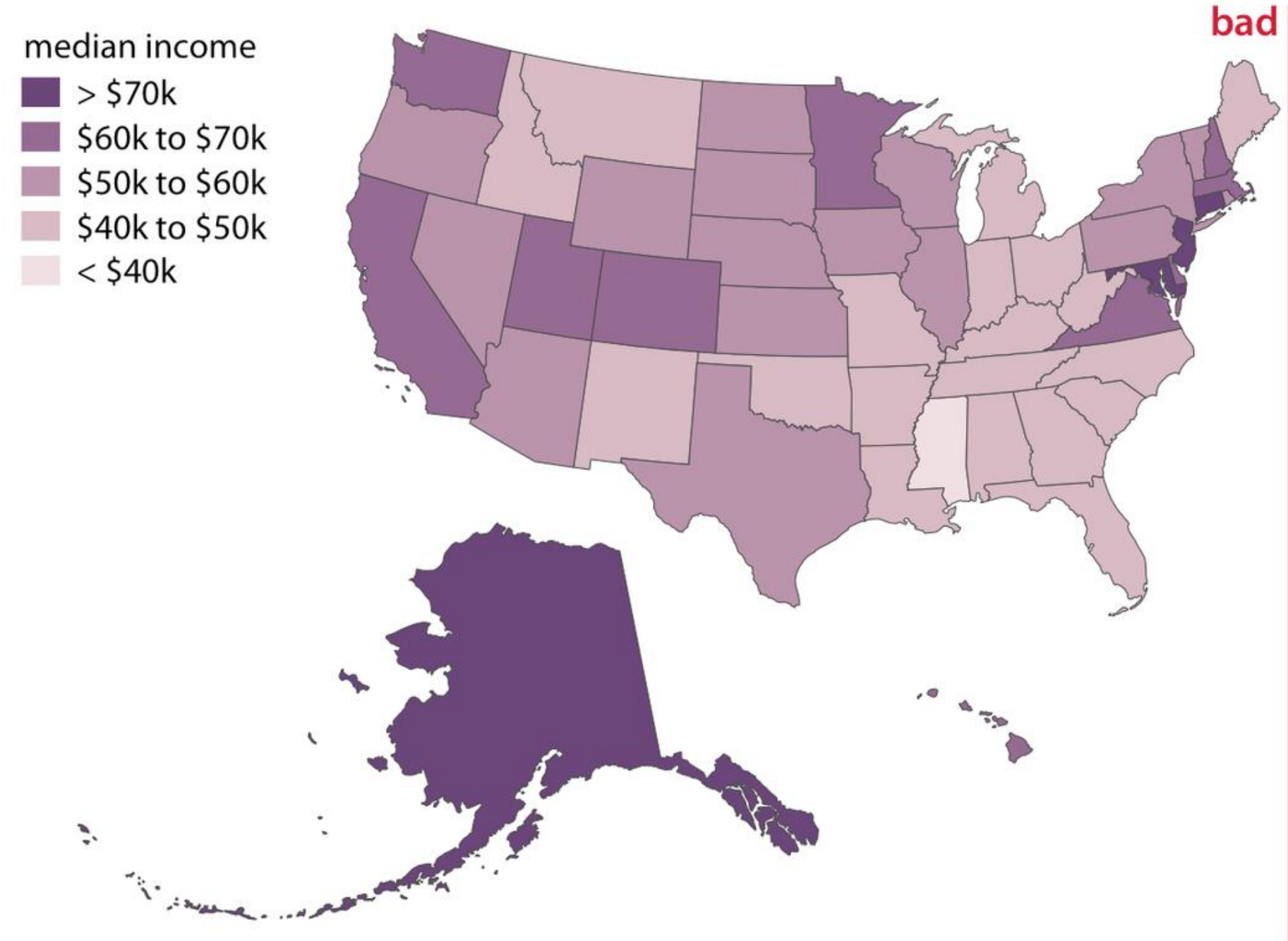
- Large regions can visually dominate
- Small regions may be difficult to see
- Color represents the value, not geographic area
- Regional boundaries can influence interpretation

Visualization of Data, Understanding and Interpretation

The Choropleth Trap: Counts vs Rates

Example:

- The adjacent figure represents the median income in every U.S. state, shown as a choropleth map.
- This map is visually dominated by the state of Alaska, which has a high median income but very low population density.
- At the same time, the densely populated high-income states on the East Coast do not appear very prominent on this map.
- In aggregate, this map provides a poor visualization of the income distribution in the U.S., and therefore labeled “bad.”



Visualization of Data, Understanding and Interpretation

Cartograms: Distorting Geography to Encode Data

A cartogram deliberately changes the geographic size or shape of regions so that another variable is represented through area.

For example: A country's displayed area can be made proportional to its population rather than its geographic area.

Why use a cartogram?

- A normal map emphasizes geographic space.
- A cartogram can emphasize data magnitude.
- This can make highly populated regions visually prominent even if their physical area is small.

Trade-off

Cartograms introduce a deliberate distortion:

- Gain: Important quantitative differences become visually prominent.
- Cost: Geographic shape and location become harder to recognize or compare.

💡 *Key question: do we want the viewer to compare geography or the encoded quantity?*

Visualization of Data, Understanding and Interpretation

Cartograms: Distorting Geography to Encode Data

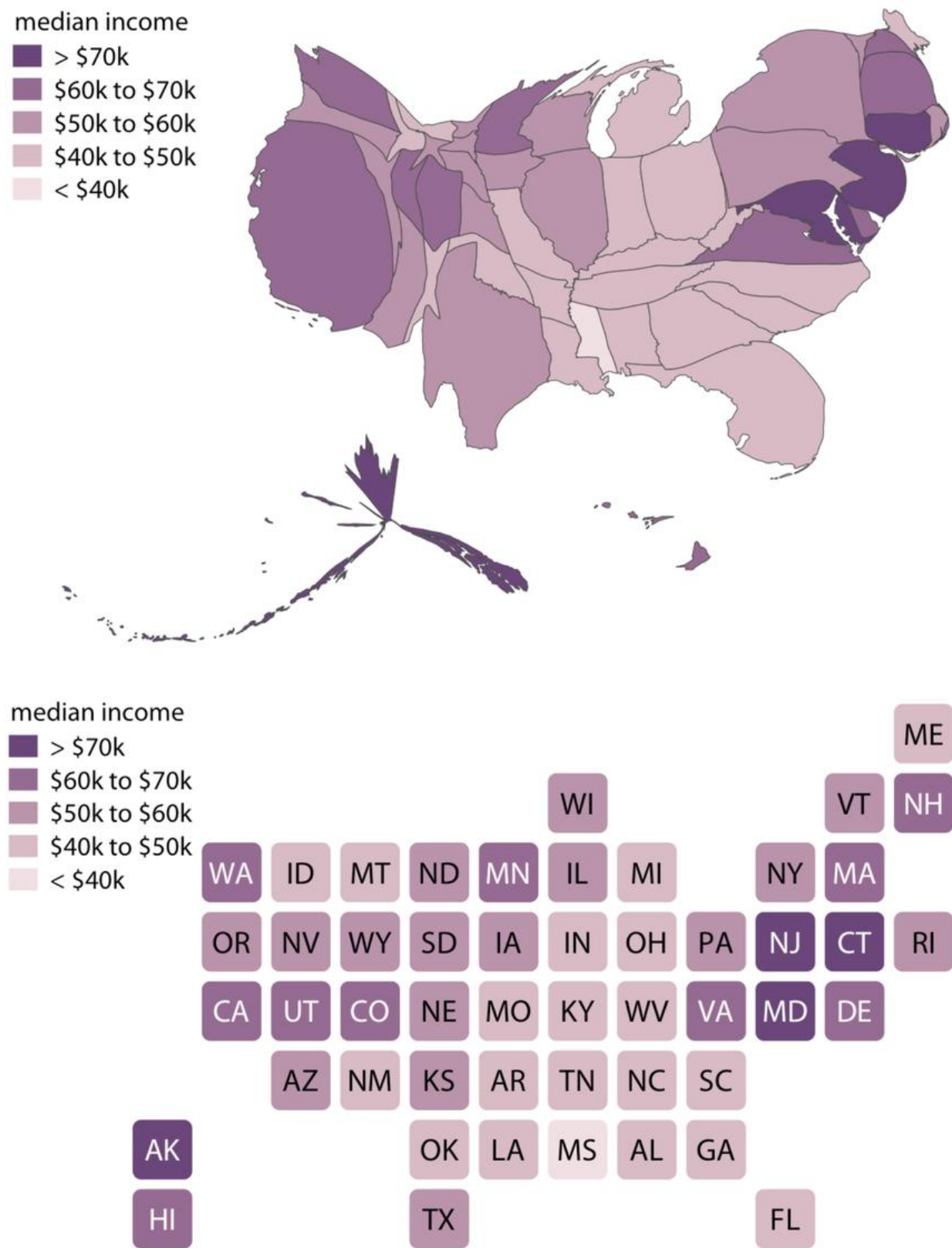


Figure 1: Cartogram
State areas are resized according to population, while color represents median income, emphasizing income differences across states.

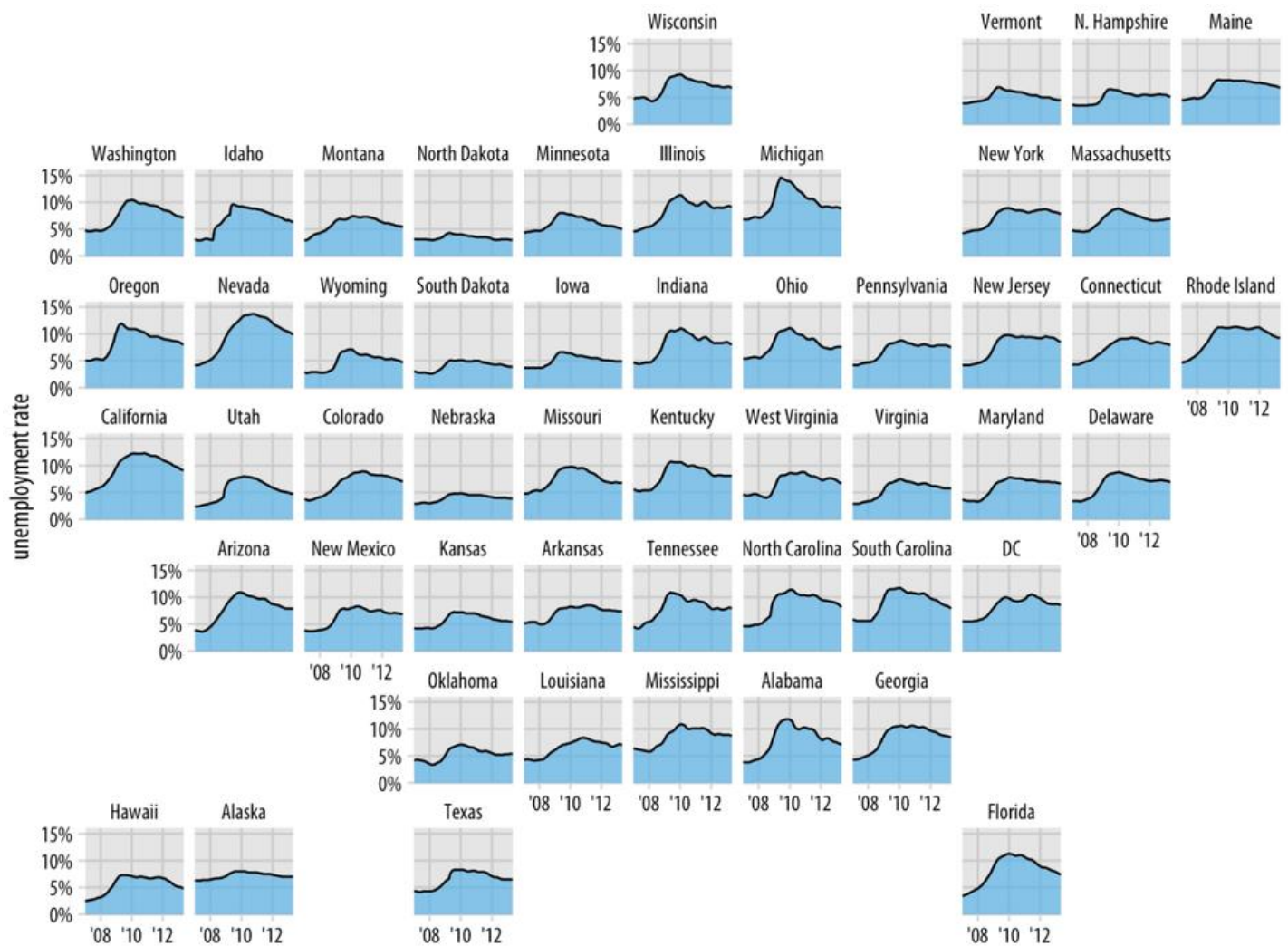


Figure 2: Cartogram Heatmap
Each state is given equal visual area and positioned approximately geographically, allowing median income to be compared without population-size distortion.

Figure 3: Geographically arranged small multiples
Each state displays its unemployment trend separately while preserving approximate geographic relationships for regional comparison.

Visualization of Data, Understanding and Interpretation

Choosing the Right Geospatial Visualization

1. What is the data?

- Individual locations → Point map
- Quantity by region → Choropleth
- Magnitude at locations → Symbol map
- Density of observations → Dot-density map
- Quantity should determine region size → Cartogram

2. What does the viewer need to do?

- Locate → preserve position
- Compare quantities → use effective quantitative channels
- Detect hotspots → emphasize spatial patterns
- Understand geographic context → preserve recognizable geography

3. What should not be sacrificed?

- Accurate interpretation
- Appropriate color scale
- Geographic context
- Readability
- Perceptual effectiveness

In a predictive model, different input features can contribute differently to the model's output.

Feature importance visualization communicates:

- Which features matter most
- Their relative importance
- How importance is distributed across features
- Which features deserve further investigation

Two levels of importance

- Global importance: how important is a feature across the model or dataset?
- Local importance: how important was a feature for one particular prediction?

💡 *Key idea: feature importance turns a high-dimensional model into a visual ranking of influential variables.*

Visualization of Data, Understanding and Interpretation

From Importance to Visual Encoding

Once we know which features are important, the next question is:
How should importance be visually encoded?

For quantitative importance, stronger visual channels include:

- Position on a common scale
- Length
- Size
- Color intensity as a supporting channel

Example

Instead of:

Feature A = dark purple

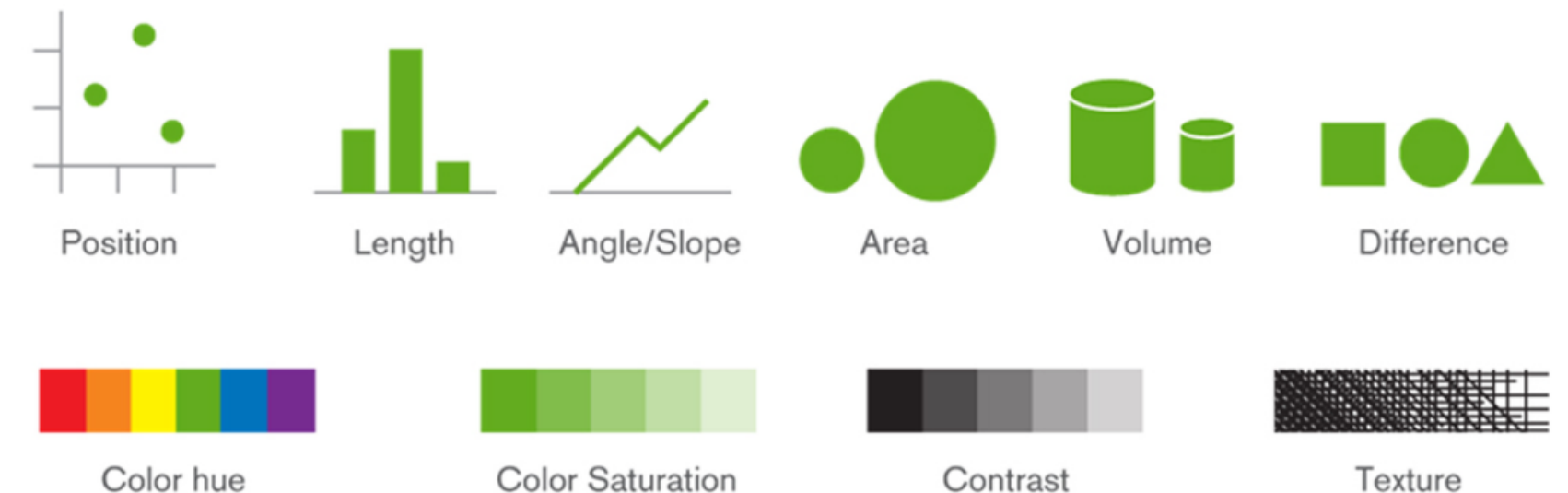
Feature B = light purple

Use:

Feature A = long bar

Feature B = short bar

because length/position provides a more direct quantitative comparison.



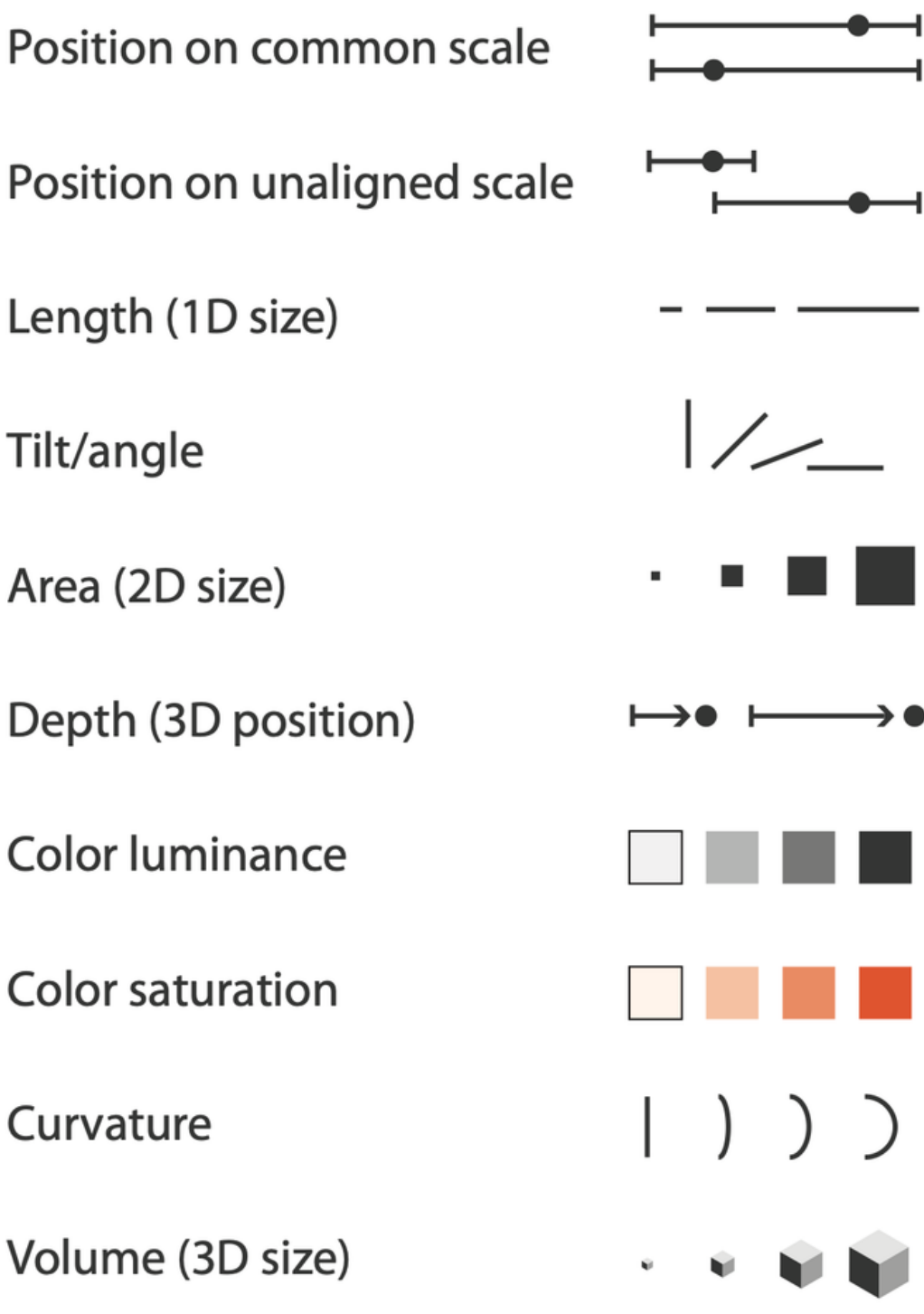
💡 *Principle: match the visual channel to the type of information being represented*

Visualization of Data, Understanding and Interpretation

From Importance to Visual Encoding

Channels: Expressiveness Types and Effectiveness Ranks

➔ **Magnitude** Channels: **Ordered** Attributes



➔ **Identity** Channels: **Categorical** Attributes



Visualization of Data, Understanding and Interpretation

From Importance to Visual Encoding



Figure 5.2. Marks are geometric primitives.

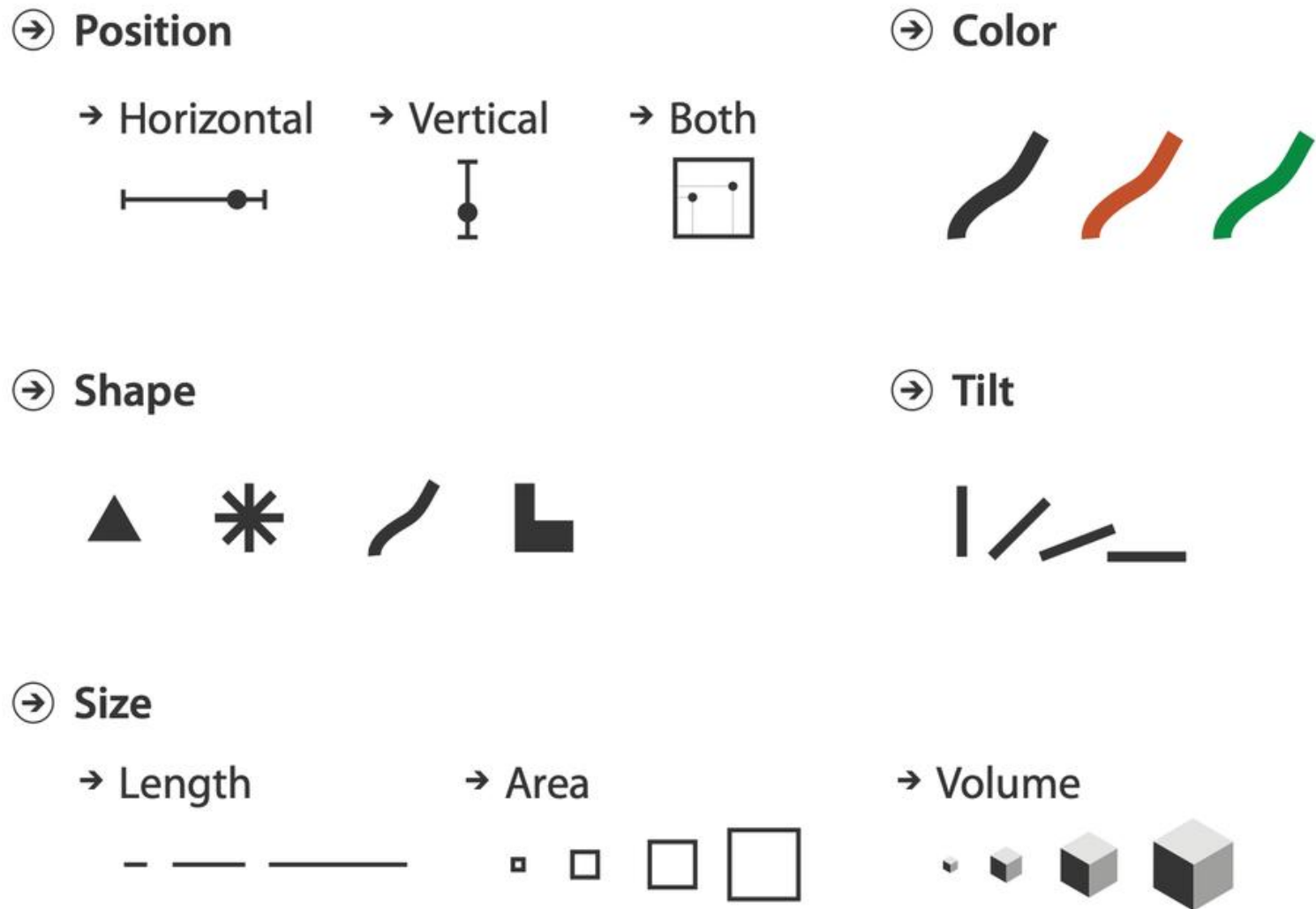


Figure 5.3. Visual channels control the appearance of marks.

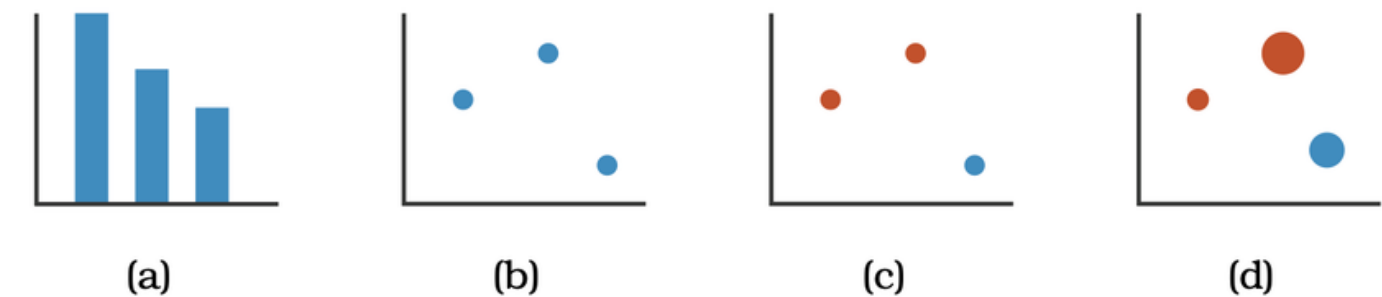
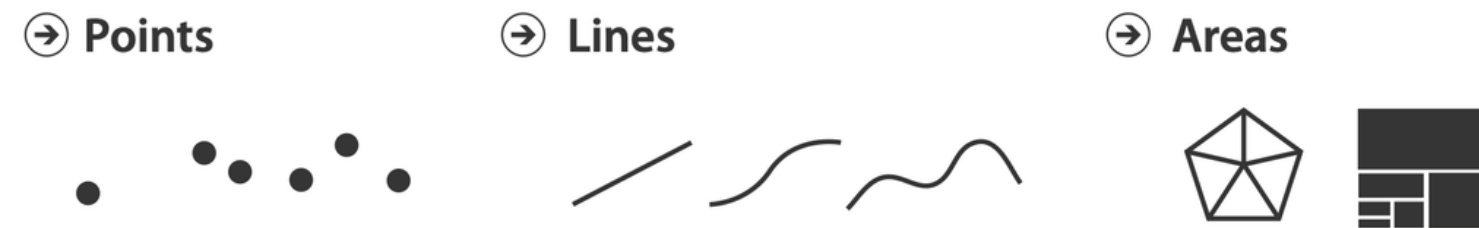


Figure 5.4. Using marks and channels. (a) Bar charts encode two attributes using a line mark with the vertical spatial position channel for the quantitative attribute, and the horizontal spatial position channel for the categorical attribute. (b) Scatterplots encode two quantitative attributes using point marks and both vertical and horizontal spatial position. (c) A third categorical attribute is encoded by adding color to the scatterplot. (d) Adding the visual channel of size encodes a fourth quantitative attribute as well.

Marks as Items/Nodes



Marks as Links

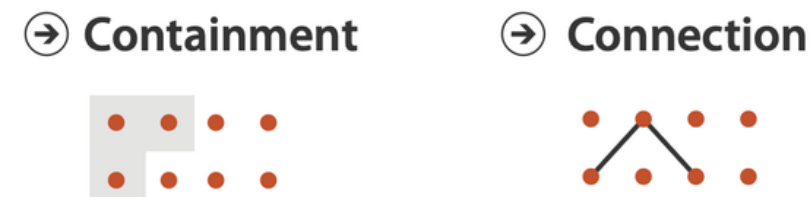


Figure 5.5. Marks can represent individual items, or links between them.

Visualization of Data, Understanding and Interpretation

Global vs Local Feature Importance

Feature importance can answer two different questions.

Global importance: which features generally influence the model?

Useful for:

- Understanding the model overall
- Feature selection
- Identifying dominant variables
- Comparing the model's reliance on different inputs

Local importance: which features influenced this particular prediction?

Useful for:

- Explaining individual decisions
- Investigating unusual predictions
- Understanding why two similar observations received different outputs

💡 *Global importance explains the model; local importance explains an individual decision*

Visualization of Data, Understanding and Interpretation

Positive and Negative Contributions

A feature can influence a prediction in different directions.

For example:

- Positive contribution → pushes prediction upward
- Negative contribution → pushes prediction downward
- Near-zero contribution → little influence

Visualization: use a diverging horizontal bar chart

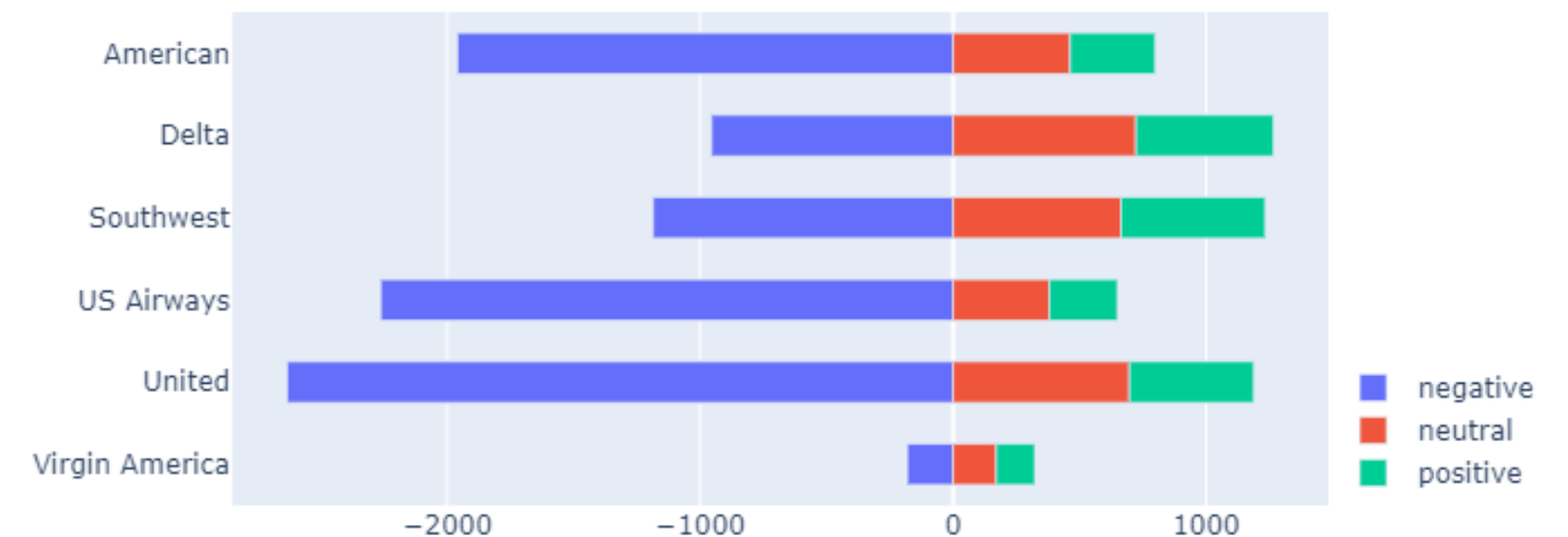
Why this matters?

A simple ranking such as:

Feature A = 0.8

Feature B = 0.7

doesn't necessarily communicate whether the features increase or decrease the prediction.



💡 *When direction matters, encode direction explicitly rather than showing magnitude alone*

Visualization of Data, Understanding and Interpretation

Feature Importance Is Not the Same as Feature Effect

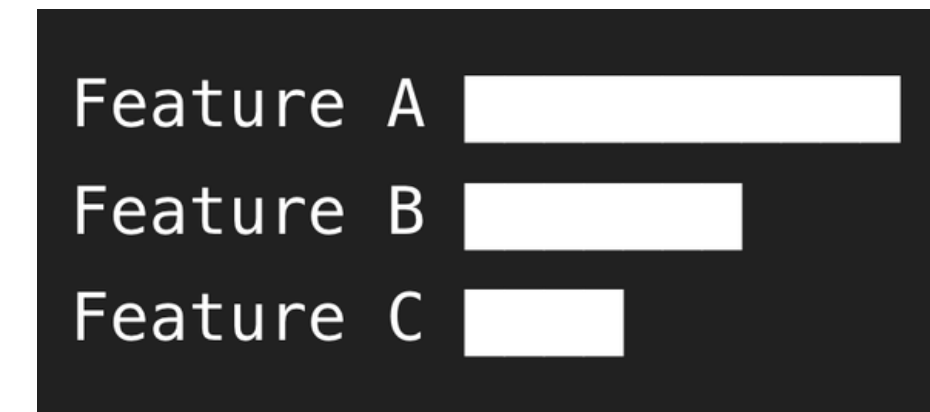
A feature can be important without having a simple relationship with the prediction.

For example:

High feature importance does not necessarily mean *higher feature value* → *higher prediction*

A model may use a feature:

- Nonlinearly
- Only within certain ranges
- In combination with other features
- Differently for different observations



Feature Importance

Therefore:

Importance asks: how much does the model rely on this feature?

Effect asks: how does changing this feature affect the output?

A ranking tells us what matters; it does not necessarily tell us how it behaves.



Feature Effect

Visualization of Data, Understanding and Interpretation

Uncertainty Visualization

(TB 1, Ch 16)

A plotted value is easily interpreted as precise, even when the underlying estimate is uncertain.

Uncertainty can arise from:

- Measurement limitations
- Sampling variability
- Limited data
- Model estimation
- Prediction
- Future outcomes

Why it matters

Ignoring uncertainty can make viewers:

- Overestimate precision
- Treat estimates as fixed values
- Make overly confident comparisons
- Misinterpret model outputs

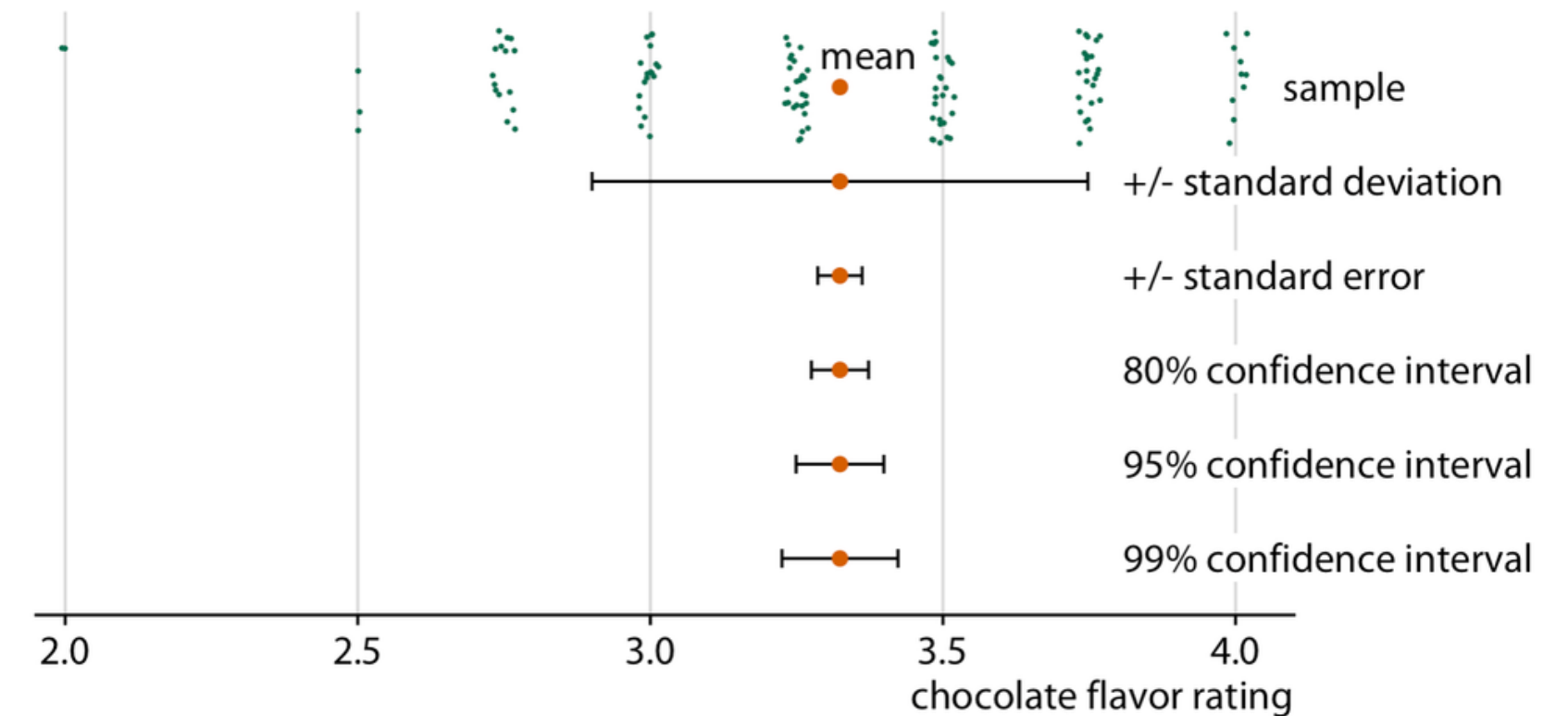


Figure 16.5: Relationship between sample, sample mean, standard deviation, standard error, and confidence intervals, in an example of chocolate bar ratings. The observations (shown as jittered green dots) that make up the sample represent expert ratings of 125 chocolate bars from manufacturers in Canada, rated on a scale from 1 (unpleasant) to 5 (elite). The large orange dot represents the mean of the ratings. Error bars indicate, from top to bottom, twice the standard deviation, twice the standard error (standard deviation of the mean), and 80%, 95%, and 99% confidence intervals of the mean. Data source: Brady Brelinski, Manhattan Chocolate Society



A visualization should communicate not only the estimated value, but also how uncertain that estimate is.

Visualization of Data, Understanding and Interpretation

Variation, Standard Error & Confidence Intervals

Not every interval around a value represents the same thing.

Standard deviation

Describes variation among individual observations.

How much do the observations differ from each other?

Standard error

Describes precision of an estimated parameter.

How precisely have we estimated the parameter?

Confidence interval

Represents uncertainty associated with an estimate under the frequentist framework.

Standard deviation $\neq$ standard error $\neq$ confidence interval

They may look visually similar, but they communicate different statistical concepts.

Visualization of Data, Understanding and Interpretation

Variation, Standard Error & Confidence Intervals

1. Standard Deviation — Variation in the data

Measures how much individual observations differ from one another.

Question: How spread out are the observations?

- Large SD → observations are widely dispersed
- Small SD → observations are clustered around the mean
- It describes variation, not how precisely we estimated the mean

2. Standard Error — Precision of the estimate

Measures how precisely a sample statistic—typically the sample mean—has been estimated.

Approximately:

$$SE = SD / \sqrt{n}$$

where n = sample size.

Therefore:

- Larger sample → smaller SE
- Smaller sample → larger SE
- Two datasets can have the same SD but different SE

Question: How precisely have we estimated the population mean?

Visualization of Data, Understanding and Interpretation

Variation, Standard Error & Confidence Intervals



3. Confidence Interval — Uncertainty in the estimate

A confidence interval gives a range associated with an estimate at a specified confidence level. For example, a 95% confidence interval is constructed so that, under repeated sampling, approximately 95% of such intervals would contain the true parameter. Question: How does the estimate behave under repeated sampling?

Measure	What it describes
Standard deviation	Variation among observations
Standard error	Precision of the estimated parameter
Confidence interval	Range associated with the estimate at a chosen confidence level

Visualization of Data, Understanding and Interpretation

Variation, Standard Error & Confidence Intervals

Calculation of SD, SE, CI:

Example: Suppose the test scores of 5 students are 8, 10, 12, 10, 10

Step 1: Calculate the mean

$$\text{Mean} = (8+10+12+10+10)/5 = 10$$

Step 2: Calculate the standard deviation

First calculate deviation from the mean

Score x	$x - \bar{x}$	$(x - \bar{x})^2$
8	-2	4
10	0	0
12	+2	4
10	0	0
10	0	0
Total		8

Visualization of Data, Understanding and Interpretation

Variation, Standard Error & Confidence Intervals

For a sample standard deviation:

$$s = \sqrt{\frac{\sum (x - \bar{x})^2}{n - 1}}$$

$$s = \sqrt{\frac{8}{4}}$$

$$s = 1.41$$

Individual scores typically vary by about **1.41 points** around the mean.

Visualization of Data, Understanding and Interpretation

Variation, Standard Error & Confidence Intervals

Step 3: Calculate Standard Error

Wilke gives the approximate relationship: $SE = \frac{SD}{\sqrt{n}}$

Therefore: $SE = \frac{1.41}{\sqrt{5}}$

$$SE = \frac{1.41}{2.236}$$

$$SE \approx 0.63$$

The estimated mean has a standard error of approximately **0.63 points**.

Notice: $SD = 1.41 > SE = 0.63$

That's expected because SD describes variation among individual observations, whereas SE describes uncertainty in the estimated mean.

Visualization of Data, Understanding and Interpretation

Variation, Standard Error & Confidence Intervals

Step 4: Approximate 95% Confidence Interval

For a simple introductory calculation, use Wilke's approximation:

$$\begin{aligned} 95\% CI &\approx \bar{x} \pm 2(SE) \\ &= 10 \pm 2(0.63) \\ &= 10 \pm 1.26 \end{aligned}$$

Therefore: $95\% CI \approx [8.74, 11.26]$

The estimated mean is: **10 ± 1.26** with an approximate 95% confidence interval of: **8.74 to 11.26**

Visualization of Data, Understanding and Interpretation

Variation, Standard Error & Confidence Intervals

Note: Sample size matters!

Imagine the same SD = 1.41, but the number of students are **100** instead of 5.

$$SE = \frac{1.41}{\sqrt{100}} = 0.141$$

Approximate 95% CI: $10 \pm 2(0.141)$

$[9.718, 10.282]$

So the SD can remain the same, while the SE and confidence interval become much narrower as sample size increases.

Visualization of Data, Understanding and Interpretation

Error Bars: The Basic Uncertainty Encoding

Error bars extend from an estimated value to indicate a range of uncertainty.

They can be used with:

- Bar charts
- Dot plots
- Scatter plots
- Point estimates
- Other 2D visualizations

Advantages

- Compact
- Show uncertainty for simultaneous estimates
- Easy to combine with existing charts

For example:

- ± 1 SD
- ± 1 SE
- 95% CI

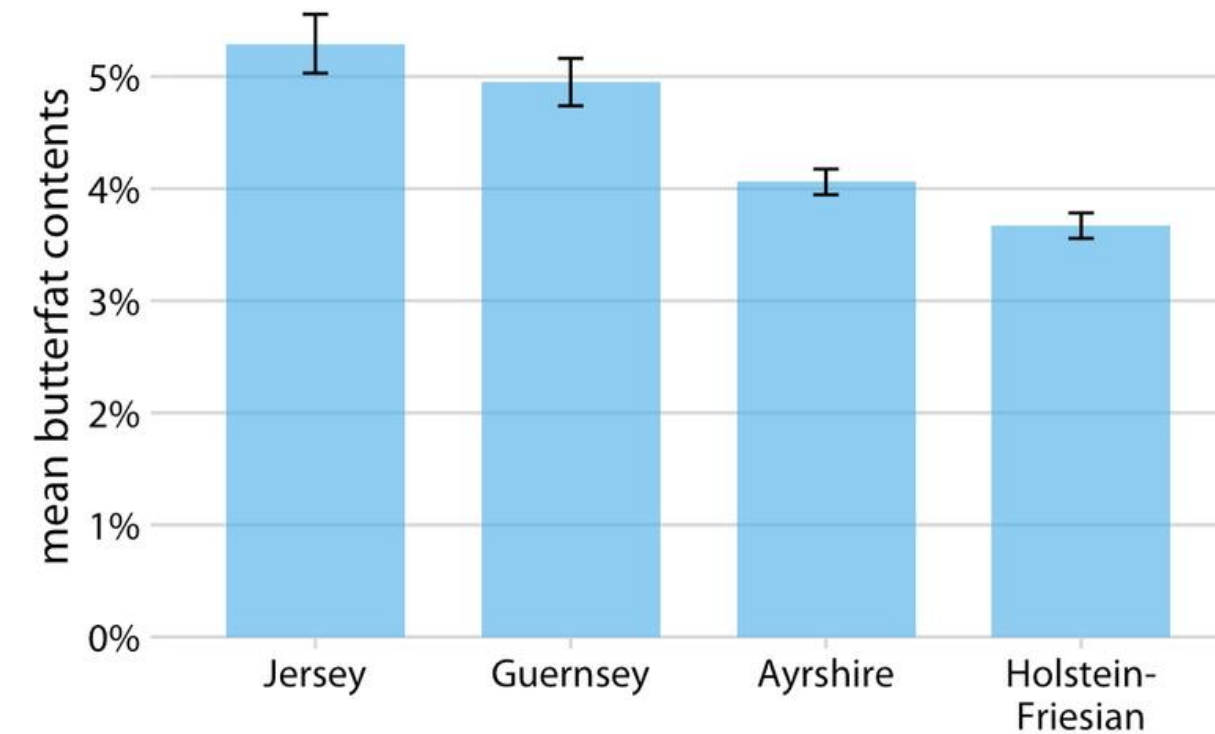


Figure 16.10: Mean butterfat contents in the milk of four cattle breeds. Error bars indicate $\pm$ one standard error of the mean. Visualizations of this type are frequently seen in the scientific literature. While they are technically correct, they represent neither the variation within each category nor the uncertainty of the sample means particularly well. See Figure 7.11 for the variation in butterfat contents within individual breeds. Data Source: Canadian Record of Performance for Purebred Dairy Cattle

Visualization of Data, Understanding and Interpretation

Confidence Intervals: Reading Uncertainty

A confidence interval gives a range associated with an estimate and a specified confidence level.
For example: 95% confidence interval means that under repeated sampling, intervals constructed using this procedure would contain the true parameter approximately 95% of the time.

💡 Important!

Do not interpret a particular 95% confidence interval as:
“There is a 95% probability that the true value is inside this particular interval.”

That is not the frequentist interpretation.

Visual interpretation

- Wider interval → greater uncertainty
- Narrower interval → greater precision

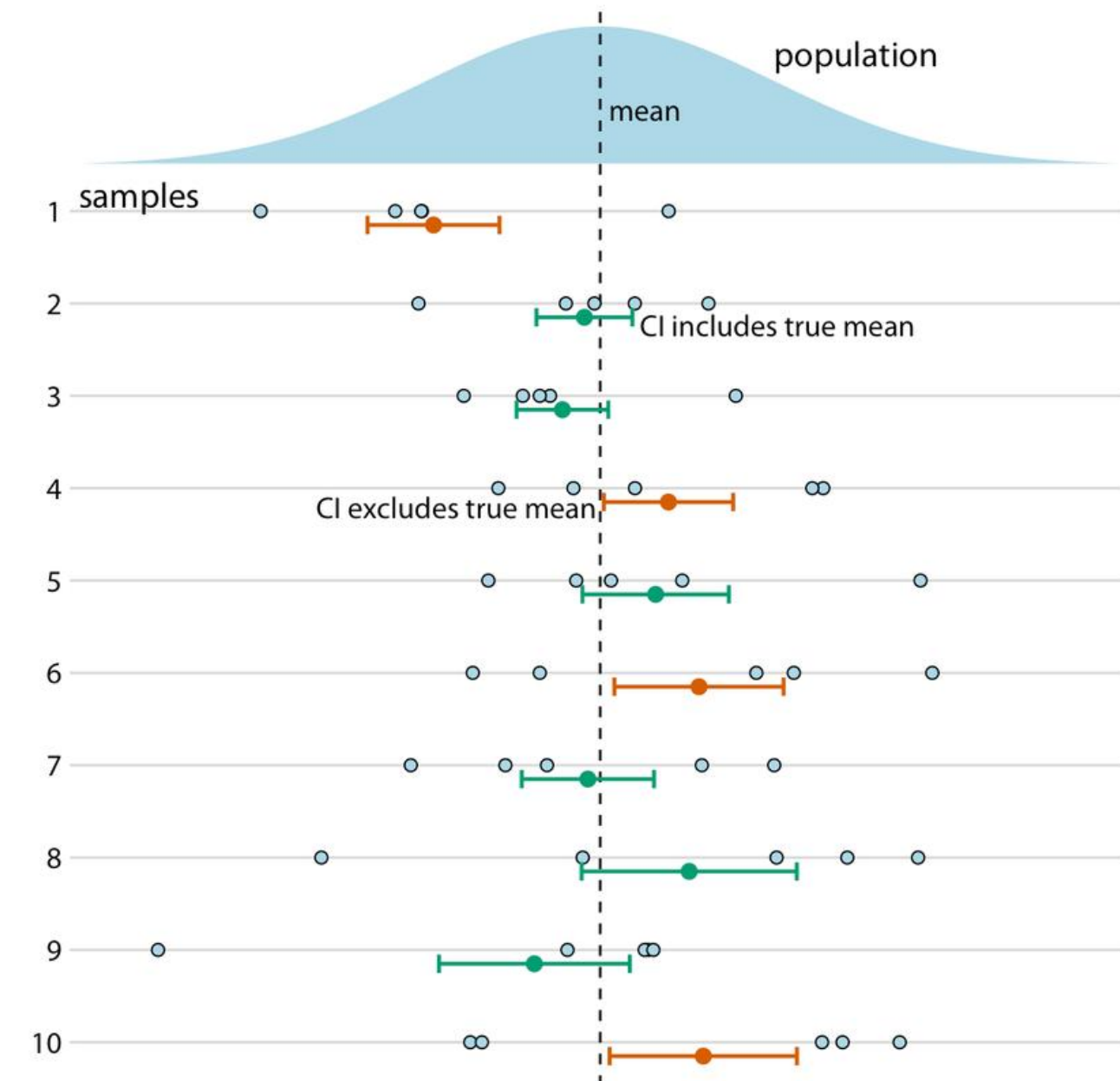


Figure 16.12: Frequency interpretation of a confidence interval. Confidence intervals (CIs) are best understood in the context of repeated sampling. For each sample, a specific confidence interval either includes or excludes the true parameter, here the mean. However, if we sample repeatedly, then the confidence intervals (shown here are 68% confidence intervals, corresponding to sample mean $\pm$ standard error) include the true mean approximately 68% of the time.

Visualization of Data, Understanding and Interpretation

Bayesian Uncertainty: Credible Intervals & Distributions

A Bayesian analysis represents uncertainty through a posterior distribution.
A credible interval describes a region of that posterior distribution.
For example: A 95% credible interval contains the central 95% of the posterior distribution.

Visualization

Rather than reducing the posterior to one interval, we can visualize the entire distribution.

Useful displays include:

- Density plots
- Violin plots
- Ridgeline plots
- Quantile dotplots

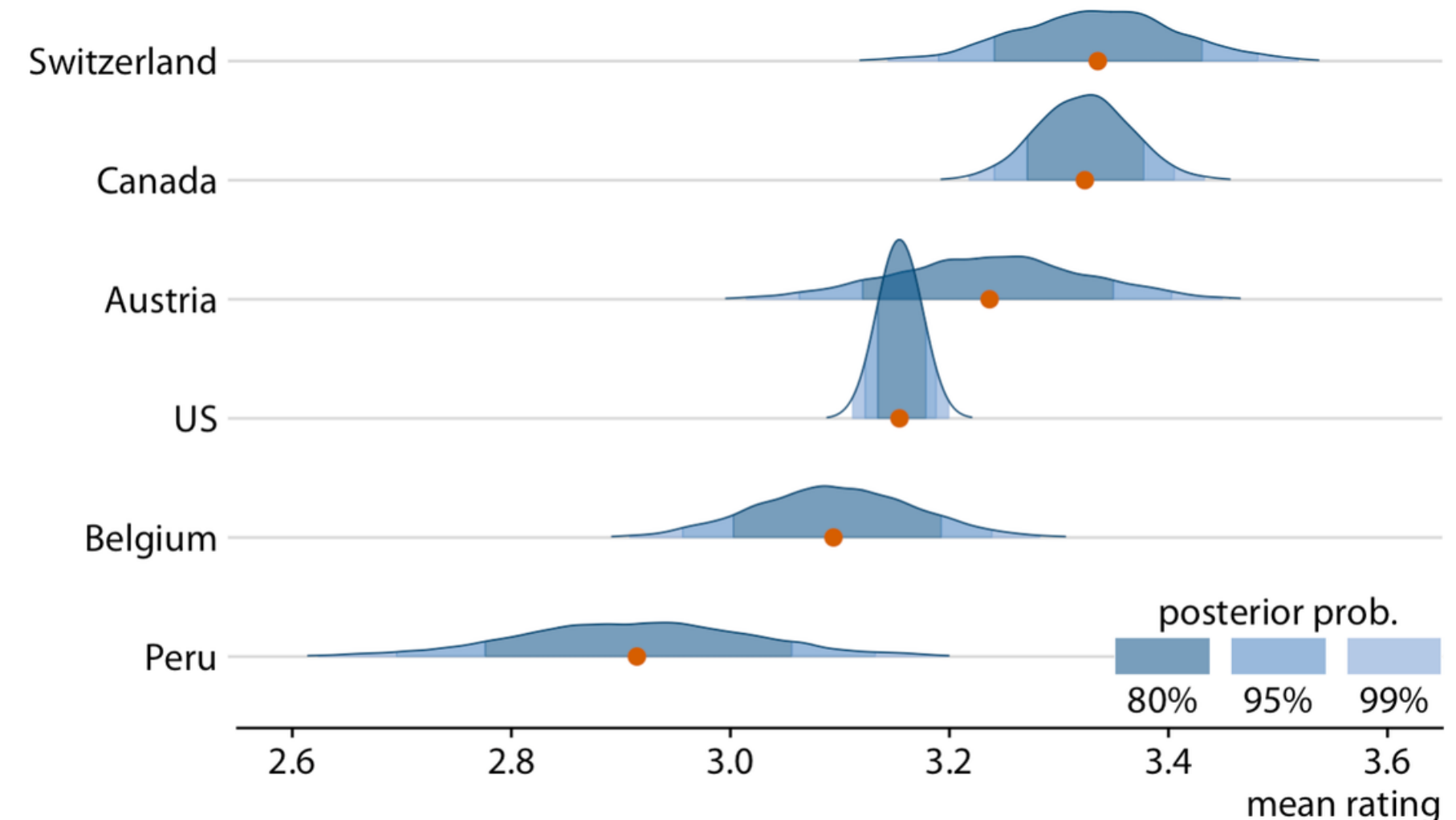


Figure 16.14: Bayesian posterior distributions of mean chocolate bar ratings, shown as a ridgeline plot. The red dots represent the medians of each posterior distribution. Because it is difficult to convert a continuous distribution into specific confidence regions by eye, I have added shading under each curve to indicate the center 80%, 95%, and 99% of each posterior distribution.

Visualization of Data, Understanding and Interpretation

Uncertainty in Trend Lines

A fitted trend is itself an estimate, so it has uncertainty.

A confidence band shows the range of plausible fitted values around a trend line.

Line → *estimated trend*

Band → *uncertainty around the estimated trend*

Why the band can curve

Even if the fitted model is a straight line, uncertainty exists in:

- Its intercept
- Its slope

Different plausible lines therefore produce an overall curved region.

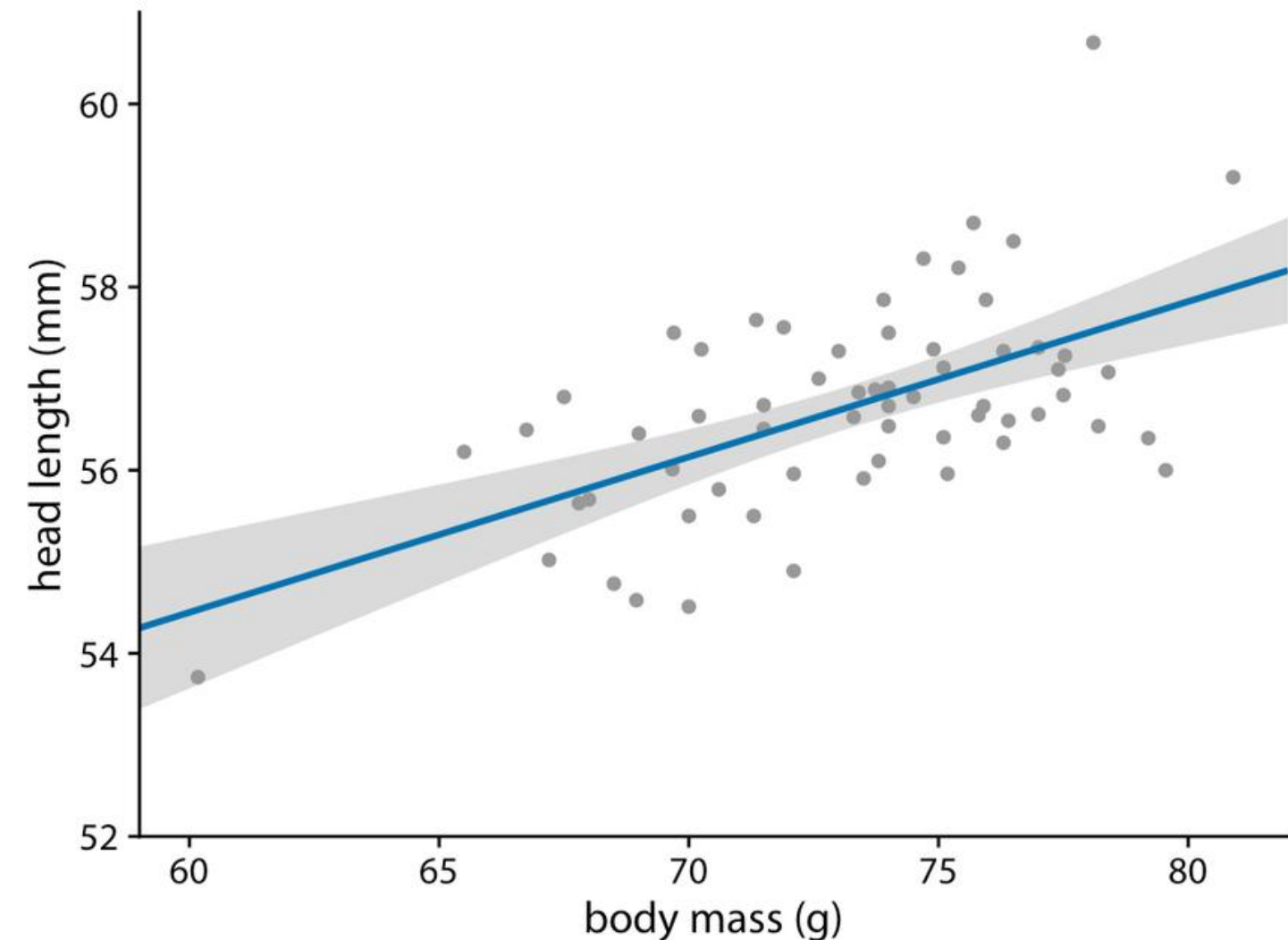


Figure 16.15: Head length versus body mass for male blue jays, as in Figure 14.7. The straight blue line represents the best linear fit to the data, and the gray band around the line shows the uncertainty in the linear fit. The gray band represents a 95% confidence level. Data source: Keith Tarvin, Oberlin College

Visualization of Data, Understanding and Interpretation

Communicating Uncertainty Through Possible Outcomes

Static uncertainty displays can sometimes be interpreted as deterministic.

For example, viewers may assume:

“The value cannot go outside this error bar.”

This is a deterministic construal error.

Alternative: **Hypothetical Outcome Plots (HOPs)**

A HOP repeatedly shows different plausible outcomes sampled from the uncertainty distribution.

Instead of saying:

“Here is the range of uncertainty.”

the visualization effectively shows:

“Here are many possible outcomes.”

Why useful?

- Makes probability more intuitive
- Emphasizes variation between plausible outcomes
- Useful in interactive/animated environments
- Can be effective for presentations and online visualizations

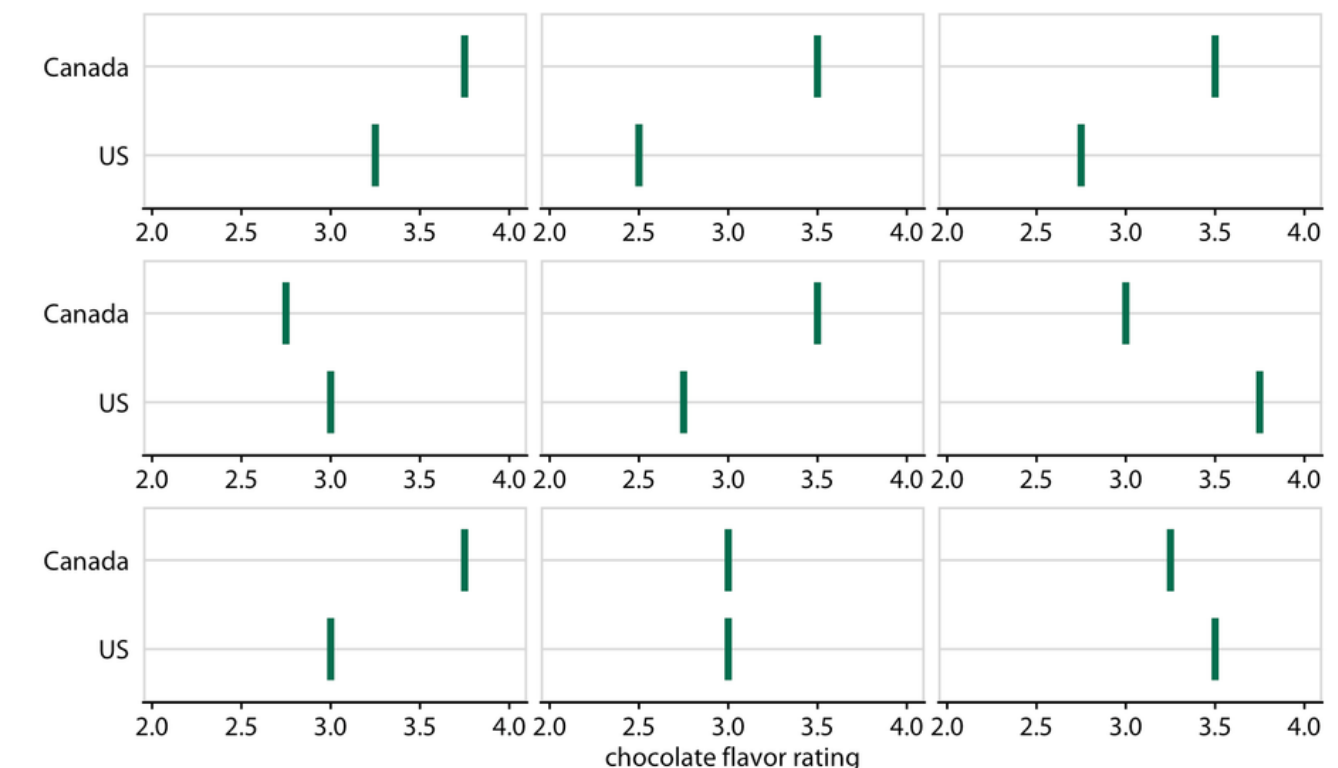


Figure 16.19: (for print edition) Schematic of a hypothetical outcome plot for chocolate bar ratings of Canadian and U.S. manufactured bars. Each vertical green bar represents the rating for one bar, and each panel shows a comparison of two randomly chosen bars, one each from a Canadian manufacturer and a U.S. manufacturer. In an actual hypothetical outcome plot, the display would cycle between the distinct plot panels instead of showing them side-by-side.

Visualization of Data, Understanding and Interpretation

Interactive Dashboards: What Is a Dashboard? (TB 3, Ch 12)

A dashboard is a visual display designed primarily for monitoring a system or situation. It brings multiple pieces of information together so that users can maintain awareness and respond to important changes.

Typical dashboard information

- Current status
- Key metrics
- Trends
- Exceptions
- Threshold violations
- Emerging patterns

Examples

- Business sales dashboard
- Factory monitoring
- Aircraft cockpit
- Healthcare monitoring
- Social-media analytics

💡 *A dashboard is not simply a collection of charts—it is designed around a monitoring task.*

Visualization of Data, Understanding and Interpretation

Interactive Dashboards: What Is a Dashboard?

Examples:



Visualization of Data, Understanding and Interpretation

Designing for Monitoring: What Should the User Notice?

Monitoring differs from ordinary exploratory analysis. The user is often:

- Performing another task simultaneously
- Checking the display intermittently
- Looking for exceptions
- Watching for trends
- Responding to changes

Therefore, a monitoring display should support rapid detection of:

1. Exceptions: Something has crossed a threshold.
2. Trends: Something is changing over time.
3. Emerging patterns: Something unusual is developing.
4. Actions: The information should support an appropriate response.

Cognitive challenge

Monitoring can involve:

- Task switching
- Vigilance
- Attention management
- Missed weak signals

Visualization of Data, Understanding and Interpretation

Dashboard Information Density

A dashboard must balance information availability with cognitive load.

Too little information

- Important context is missing
- Users must navigate elsewhere
- Monitoring becomes inefficient

Too much information

- Visual clutter
- Competing signals
- Slower search
- Important events become harder to detect



Provide enough information to support the task without overwhelming attention.

- Group related information
- Establish clear hierarchy
- Use consistent layouts
- Keep frequently compared information close together
- Avoid decorative elements that compete with data

Visualization of Data, Understanding and Interpretation

Dashboards as Composite Visualizations

A dashboard can combine different visualization types because different views answer different questions.

For example:

Map → Where?

Line chart → How is it changing?

Bar chart → Which category is largest?

Table → What is the exact value?

Example analytical dashboard:

Map + Scatterplot + Time Series

The views can represent different aspects of the same underlying dataset.

A composite visualization is useful when multiple views collectively answer the analytical question better than one view alone.

Visualization of Data, Understanding and Interpretation

Coordinated Multiple Views

A single visualization often cannot represent all relevant dimensions effectively.

Multiple views allow us to:

- Separate different analytical questions
- Use the best visual representation for each data aspect
- Reduce clutter
- Compare different perspectives
- Preserve different types of information simultaneously

Example

A dataset about city performance might use:

Map → geographic distribution

Scatterplot → relationship between two variables

Time series → temporal change

💡 *Multiple views distribute the cognitive and visual workload across several coordinated displays.*

Visualization of Data, Understanding and Interpretation

Coordinated Multiple Views

Coordinated Multiple Views (CMV) connect multiple visualizations so that interaction in one view affects the others.

Coordination can occur through:

- Shared data: All views represent the same underlying dataset.
- Shared encoding: The same variable is represented consistently across views.
- Shared navigation: Views can be explored together.
- Linked selection: Selecting an item in one view identifies the corresponding item elsewhere.

Example:

Select a state on the map → corresponding state is highlighted in the time series and scatterplot.

Visualization of Data, Understanding and Interpretation

Brushing & Linking

Brushing allows the user to select a subset of data in one view.

Linking propagates that selection to other views.

Example:

A user selects several points in a scatterplot

Selected observations → highlighted on the map → corresponding observations highlighted in the time series.

Why is this useful?

Without linking, the user must mentally match:

“Which points here correspond to those points there?”

With linking, the system performs the correspondence visually. This reduces the need to hold information in working memory.

💡 *Interaction turns separate visualizations into one coordinated analytical system*

Visualization of Data, Understanding and Interpretation

MCQ

Q1. A researcher maps the number of disease cases across districts. District A has 10,000 cases and District B has 5,000 cases, but District A also has four times the population of District B. Which visualization choice would most directly address the potential problem?

- A. Use a different categorical color for every district
- B. Normalize cases by population before using a choropleth
- C. Increase the geographic size of District B
- D. Replace the choropleth with a 3D map

Visualization of Data, Understanding and Interpretation

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Answer: B

Q2. Two studies estimate the same population mean and have approximately the same standard deviation. Study A has $n=25$, while Study B has $n=100$. Which statement is most consistent with the interpretation of standard error?

- A. Study A must have the smaller standard error because it has fewer observations
- B. Both studies must have identical standard errors because their SDs are similar
- C. Study B should have the smaller standard error because the estimate is based on more observations
- D. Study B should have the larger standard error because larger samples introduce more variability

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Answer: C

Visualization of Data, Understanding and Interpretation

MCQ

Q3. A visualization shows a fitted trend line surrounded by a 95% confidence band. Which interpretation is most appropriate?

- A. The band represents the range in which 95% of individual observations must fall
- B. The band represents uncertainty associated with the estimated trend
- C. The band guarantees that future observations will fall within its boundaries
- D. The width of the band directly represents the standard deviation of the observed data

Visualization of Data, Understanding and Interpretation

MCQ



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- B. The band represents uncertainty associated with the estimated trend
- C. The band guarantees that future observations will fall within its boundaries
- D. The width of the band directly represents the standard deviation of the observed data

Answer: B

Visualization of Data, Understanding and Interpretation

MCQ

Q4. A cartogram enlarges regions according to population while substantially distorting their geographic shapes. What is the primary trade-off introduced by this visualization?

- A. It removes the need for a color scale
- B. It makes geographic location completely irrelevant
- C. It improves population comparison at the cost of geographic fidelity
- D. It guarantees more accurate distance measurements between regions

Visualization of Data, Understanding and Interpretation

MCQ

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Answer: C

Visualization of Data, Understanding and Interpretation

MCQ

Q5. A sample contains 25 observations with a standard deviation of 10. Using the approximations provided by Wilke for SE and CI, if the sample mean is 50, what is the approximate 95% confidence interval?

- A. 45 to 55
- B. 48 to 52
- C. 40 to 60
- D. 46 to 54

Visualization of Data, Understanding and Interpretation

MCQ

Q5. A sample contains 25 observations with a standard deviation of 10. Using the approximations provided by Wilke for SE and CI, if the sample mean is 50, what is the approximate 95% confidence interval?

- A. 45 to 55
- B. 48 to 52
- C. 40 to 60
- D. 46 to 54

Answer: D



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Thank You



Visualization of Data, Understanding and Interpretation

Unit 2 : Advanced Visualization & Storytelling

(UE24CS342AA9)

Visualization of Data, Understanding and Interpretation

Dr.Prajwala TR (Course Instructor)

Dept. of CSE, PES University, Bangalore

Visualization of Data, Understanding and Interpretation

Acknowledgement

The slides are prepared from various resources from the Universities from abroad and India. Also, some material is taken from reliable resources from internet throughout this course.

Visualization of Data, Understanding and Interpretation

Lecture Overview

By the end of this lecture, students should be able to:

- Explain how storytelling transforms visualizations from collections of data into coherent analytical narratives.
- Construct a data story using audience, Big Idea, narrative structure, story arc, and logical sequencing.
- Apply horizontal logic, vertical logic, and reverse storyboarding to evaluate the coherence of a visualization-based presentation.
- Select and structure visual evidence to support a narrative while managing complexity, emphasis, memorability, and visual consistency.
- Critically evaluate AI-generated data narratives, recognizing the need for contextual validation, uncertainty assessment, and human judgment.

Visualization of Data, Understanding and Interpretation

From Showing Data to Telling a Story (TB 2, Ch 7-8)

A visualization can successfully show data without successfully communicating an insight. Effective communication requires deciding what matters, what to emphasize, and what order the audience should encounter the evidence.

A data story connects: *Evidence → Insight → Meaning → Action*

The objective is not to display everything available, but to make the intended interpretation clear and defensible.

Visualization Is Not Interpretation-Neutral

A visualization does more than represent numbers:

DATA → VISUAL ENCODING → PERCEPTION → INTERPRETATION → DECISION

- Different visual choices can make different aspects of the same dataset salient.
- Ordering, aggregation, scale, color and annotation influence what viewers notice first.
- Therefore, visualization is part of the interpretive process, not merely a final presentation layer.

💡 *The way data is shown influences what people think the data means*

Visualization of Data, Understanding and Interpretation

The Four Layers of a Data Story

1. DATA — What happened?

Raw observations and measurements.

2. EVIDENCE — What matters?

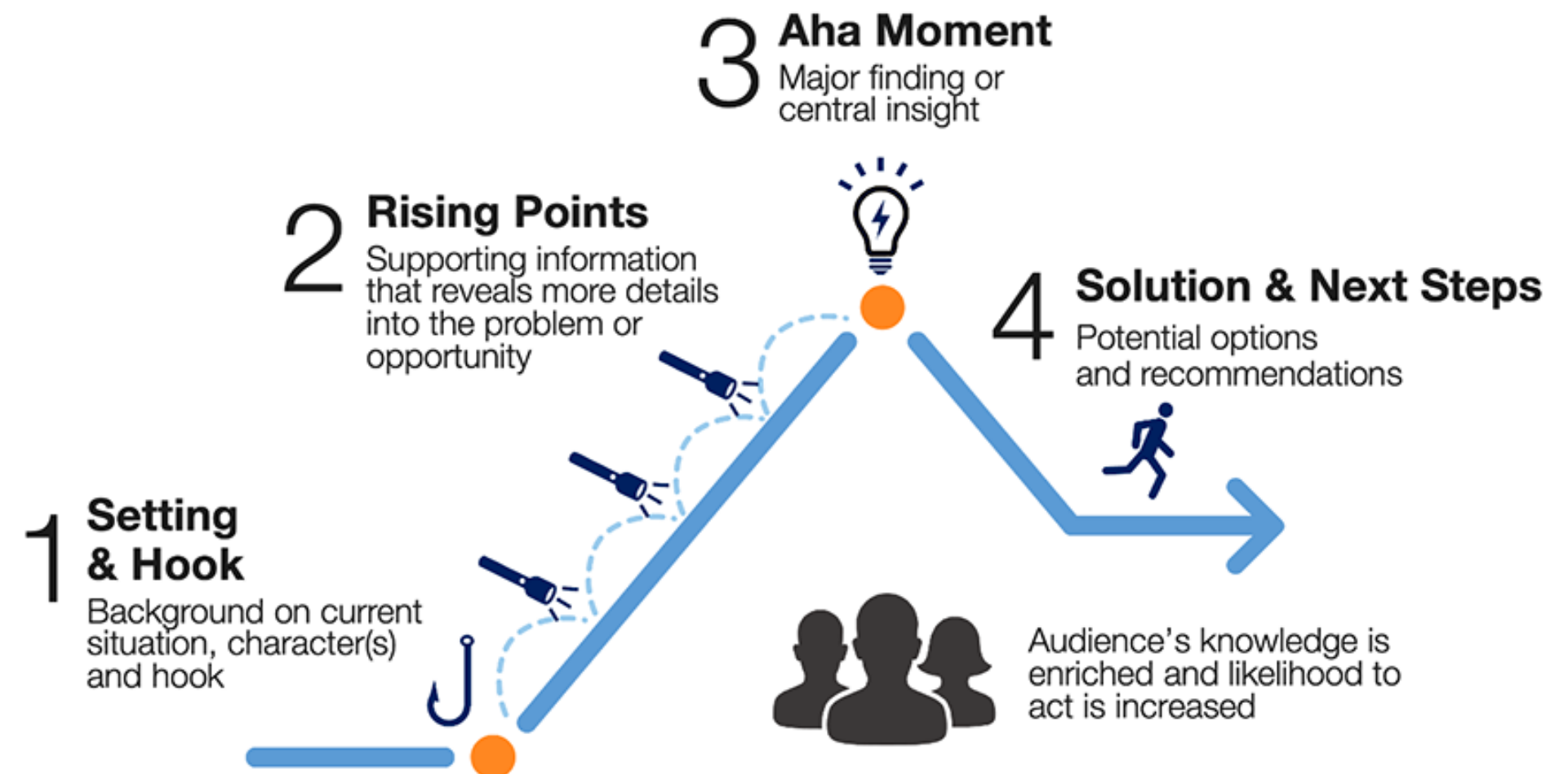
Patterns, comparisons, trends and anomalies.

3. NARRATIVE — What does it mean?

A structured explanation connecting the evidence.

4. ACTION — What should happen next?

A decision, recommendation or implication.



Visualization of Data, Understanding and Interpretation

Why Storytelling Matters in Data Visualization

- Data often arrives as a collection of disconnected facts rather than a clear message.
- Stories provide structure by connecting observations into a meaningful sequence.
- Narrative helps audiences understand what happened, why it matters, and what deserves attention.
- A well-structured story makes complex information easier to follow, remember, and communicate.
- It gives individual visualizations a clear purpose within a larger argument, rather than treating each chart as an isolated result.
- Storytelling can also help audiences move from understanding an insight to deciding what action should follow.
- However, storytelling should organize and clarify the evidence—not distort, exaggerate, or manufacture it.

Visualization of Data, Understanding and Interpretation

The Magic of Story

A list of facts:

Revenue ↑ 12%

Costs ↑ 8%

Churn ↑ 3%

Acquisition ↓ 5%

A story:

Growth is continuing — but rising acquisition costs and increasing churn are beginning to threaten the next phase of expansion.

Then:

- The facts have not changed.
- The relationship between the facts has changed.
- Narrative gives the audience a reason to connect individual observations.
- The story creates a path from observation → interpretation.

BEFORE: SHOWING DATA

Error Rate by Month & Site									
Month	Site A	Site B	Site C	Site D	Site E	Site F	Site G	Site H	Site I
Jan-20	0.062%	0.039%	0.755%	0.079%	0.000%	0.000%	0.069%	0.257%	0.133%
Feb-20	0.075%	0.260%	0.458%	0.039%	0.075%	0.073%	0.090%	0.324%	0.000%
Mar-20	0.041%	0.080%	0.335%	0.133%	0.000%	0.000%	0.054%	0.512%	0.000%
Apr-20	0.054%	0.058%	0.165%	0.052%	0.155%	0.024%	0.077%	0.197%	0.000%
May-20	0.009%	0.088%	0.234%	0.082%	0.060%	0.000%	0.084%	0.092%	0.000%
Jun-20	0.114%	0.043%	0.229%	0.161%	0.014%	0.000%	0.080%	0.043%	0.133%
Total	0.058%	0.094%	0.364%	0.090%	0.050%	0.015%	0.075%	0.239%	0.045%

AFTER: STORYTELLING WITH DATA

2020 monthly error rate by site

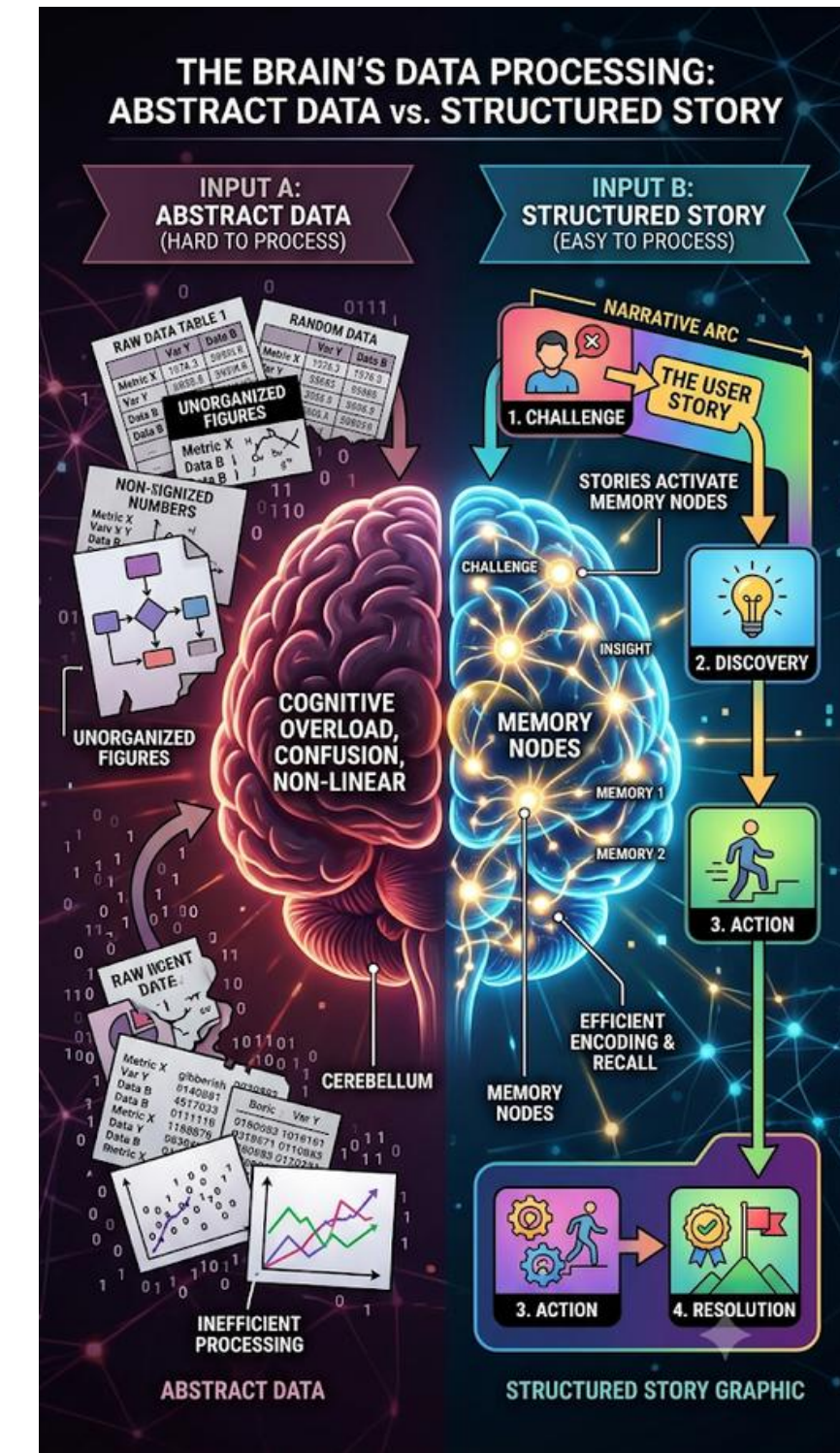


Visualization of Data, Understanding and Interpretation

Structuring a Coherent Data Story (TB 1, Ch 29)

- Most data visualization is performed for the primary purpose of communicating specific analytical insights to a target audience.
- To successfully convey an insight, we must present the audience with a clear, audience-relevant narrative rather than a random collection of charts.
- The need for a story does not mean making things up or putting a biased spin on technical engineering results.
- Stories play a fundamental role in human reasoning and memory by helping audiences retain information over much longer periods.

A coherent data story gives the audience a deliberate path from *observation* → *insight* → *implication*



Visualization of Data, Understanding and Interpretation

Storytelling Creates Ethical Responsibility

Every story involves choices:

What to show

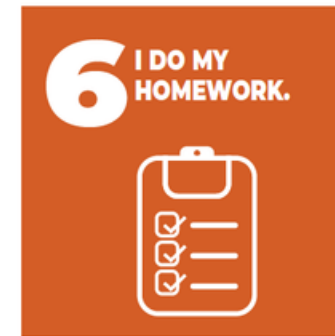
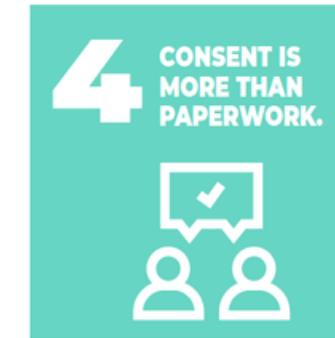
What to omit

What to emphasize

What order to use

What conclusion to foreground

These choices can influence interpretation.



Ethical storytelling means:

- Preserve the meaning of the underlying data.
- Do not hide relevant evidence merely because it weakens the story.
- Distinguish observation from interpretation.
- Distinguish correlation from causation.
- Represent uncertainty when it affects the conclusion.

A technically correct chart can still communicate a misleading story.



Visualization of Data, Understanding and Interpretation

What Makes a Visualization Misleading?

Selective framing: Choosing a time period or population that favors one conclusion.

Selective evidence: Showing only categories or observations supporting the desired claim.

Selective emphasis: Using color, size or annotation to make one interpretation dominate.

Missing context: Removing the baseline, comparison group or relevant reference point.

Overstated interpretation: Presenting correlation as causation or uncertainty as certainty.

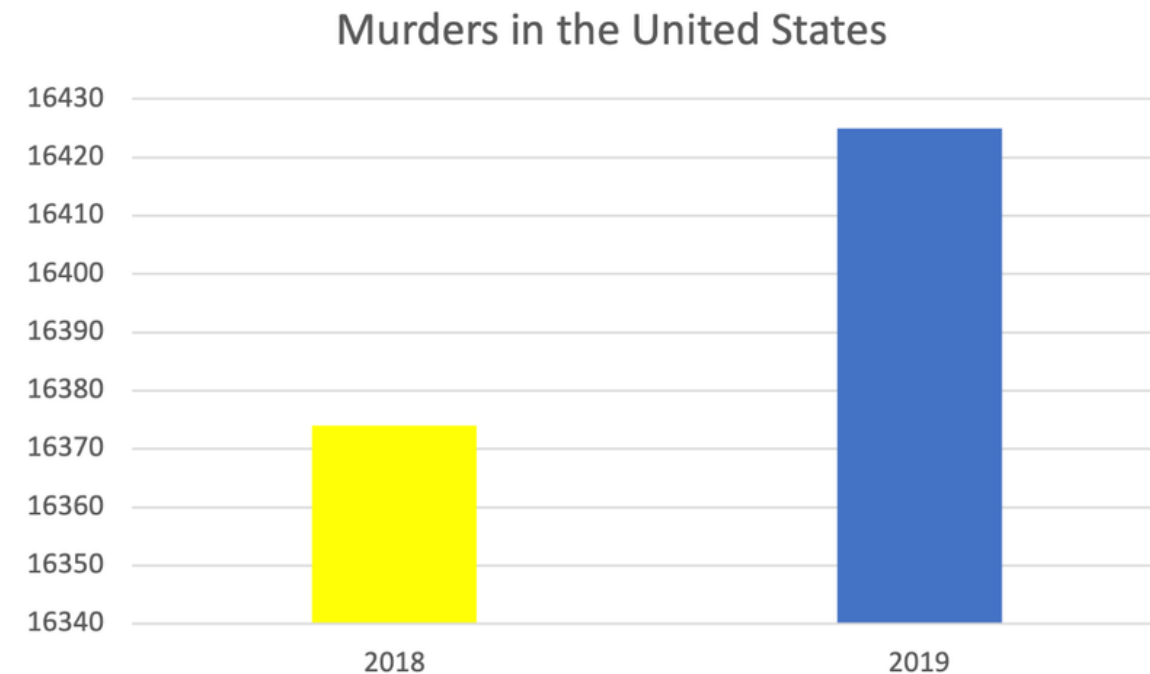


Fig 1: The y-axis does not start at 0.
Take a look at Fig 2 with the same data.

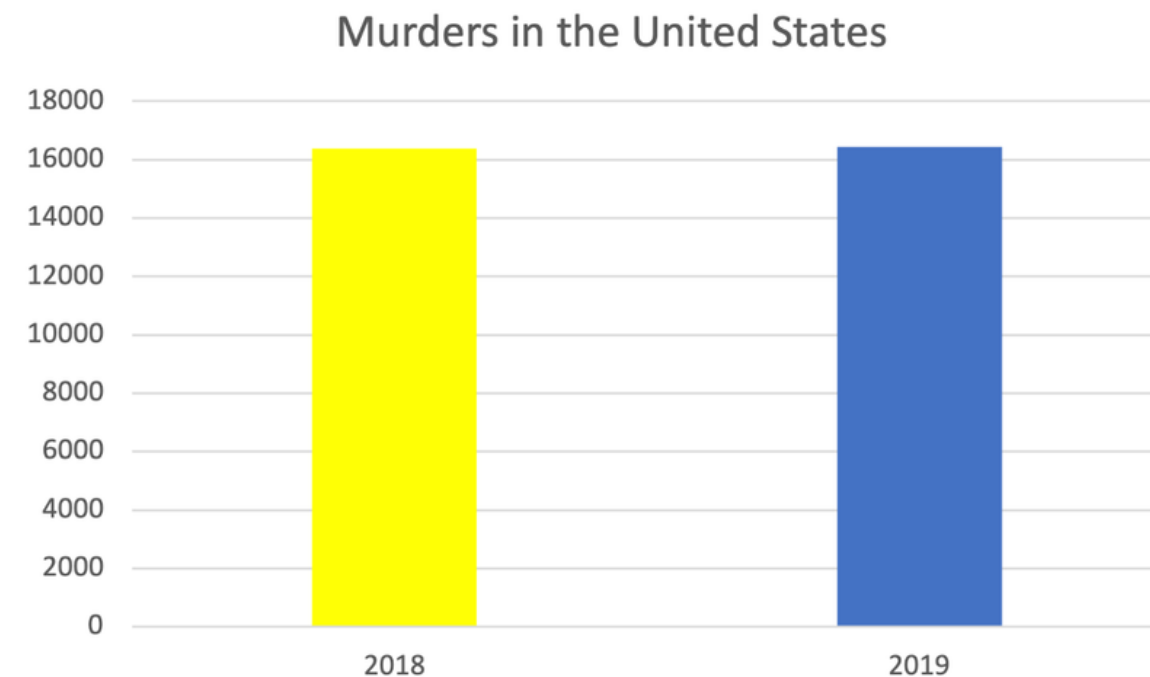


Fig 2: The y-axis starts at 0.
On this graph, its hard to tell which year had more murders unless you zoom in.

Visualization of Data, Understanding and Interpretation

The Audience Is the Protagonist



When constructing a data story, ask:

Who is the audience?

What do they already know?

What do they need to understand?

What decision must they make?

What action should follow?

The presenter is not the protagonist. The analysis exists to help the audience solve their problem, answer their question, or make their decision.

Example:

“Here are the results of our model.”

“Here is what the model tells us about which customers are most at risk, and what we should do next.”

Visualization of Data, Understanding and Interpretation

Constructing the Story: Start With the Big Idea

Before designing the first chart, write the **Big Idea**.

A useful big idea answers:

What should the audience know or believe after seeing this presentation?

Example:

Weak: “Quarterly sales analysis”

Strong: “Sales are growing, but almost all growth is concentrated in one region, creating a diversification risk.”

The Big Idea becomes the filter for everything that follows:

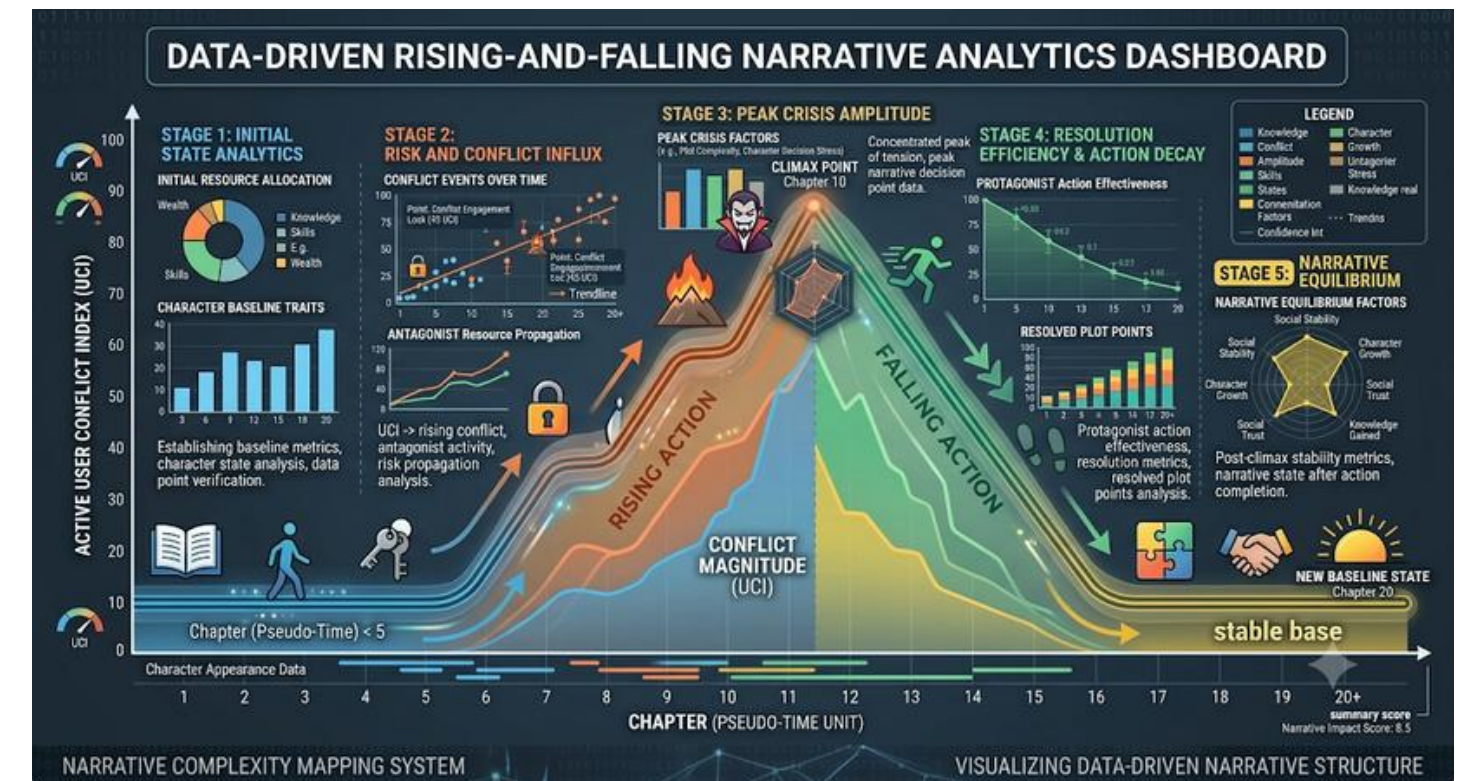
Big Idea → Evidence → Visuals → Narrative → Action

Visualization of Data, Understanding and Interpretation

The Anatomy of a Story Arc

- A story represents a set of observations and facts presented in a specific order to create a cognitive reaction in the audience.
- The cognitive reaction is built through a structured arc that establishes tension at the beginning and offers resolution at the end.
- We can structure modern visual presentations around the standard Opening-Challenge-Action-Resolution format.
- The opening introduces the current operational baseline, while the challenge presents the technical bottleneck or competitive threat.
- The action describes the model-driven data analysis, and the resolution provides the final clear recommendation to stakeholders.

A data story creates a journey from a situation or question to an evidence-based resolution.



Visualization of Data, Understanding and Interpretation

Narrative Structure: Give the Evidence an Order

A visualization may contain all the necessary evidence and still fail if the audience encounters it in the wrong order.

A narrative should establish:

1. Context

What is the situation?

2. Tension

What changed, went wrong, or needs attention?

3. Evidence

What does the data reveal?

4. Resolution

What does this mean?

5. Action

What should happen next?



The narrative is not decoration around the visualization; it is the structure that tells the audience how the evidence fits together.

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Beginning, Middle, End: Creating Tension and Resolution

A strong data story moves the audience from “*What is happening?*” to “*Why does it matter?*” to “*What should we do?*”

BEGINNING — Establish the Situation

Set the stage so everyone understands the current state.

- Setting: Where/when does the situation occur?
- Audience: Who is affected or responsible?
- Imbalance: What has changed or gone wrong?
- Desired state: What should ideally happen?
- Establish why the audience should care.

MIDDLE — Develop the Tension

Use evidence to explain the problem and build the case for action.

- Provide relevant background and comparisons.
- Show the data that demonstrates the problem.
- Explore what happens if no action is taken.
- Evaluate possible approaches or solutions.
- Use evidence to convince the audience that change is necessary.

END — Resolve and Act

Bring the evidence together into a clear conclusion.

- Answer the question raised at the beginning.
- Present the recommended solution or action.
- Explain the expected benefit or outcome.
- Make clear what the audience should do next.

Visualization of Data, Understanding and Interpretation

Narrative Flow: Which Order Should the Audience Experience?

Two common approaches:

A. Chronological

Past → Present → Future

Useful when:

- Sequence matters
- Events caused later developments
- The audience needs to understand how the situation evolved

Example:

Product launched → adoption increased → competition entered → growth slowed → recommendation

B. Lead With the Ending

Conclusion → Evidence → Supporting detail

Useful when:

- Audience time is limited
- Decision-makers need the answer quickly
- The recommendation is the most important information

Example:

Recommendation: reposition the product.

Why? → Growth has slowed.

Why? → Competitive pressure increased.

Visualization of Data, Understanding and Interpretation

Spoken vs. Written Data Stories

Live Presentation

The presenter supplies part of the narrative.

- Visuals can be progressively revealed.
- Spoken explanation provides context.
- Animation can guide the audience through iterations.
- Slides can remain relatively visual.



Written Report

The visualization must carry more of the narrative.

- Titles need to be informative.
- Labels and annotations matter more.
- The reader controls the pace.
- Each figure should be interpretable without relying entirely on spoken explanation.

Same data $\neq$ same communication design

Design the visualization for the way the audience will experience it.

Visualization of Data, Understanding and Interpretation

The Power of Repetition: Bing, Bang, Bongo

Repetition helps important information move from short-term awareness into longer-term memory.

- A central message is more likely to be remembered when the audience encounters it multiple times in a clear and consistent form.
- Use short, repeatable phrases to express the main takeaway throughout the presentation.

Knaflic presents the **Bing–Bang–Bongo** approach as a simple way to structure this repetition:

Bing — Introduce the Message

Tell the audience what you are going to tell them.

→ Executive summary / overview

Bang — Develop the Message

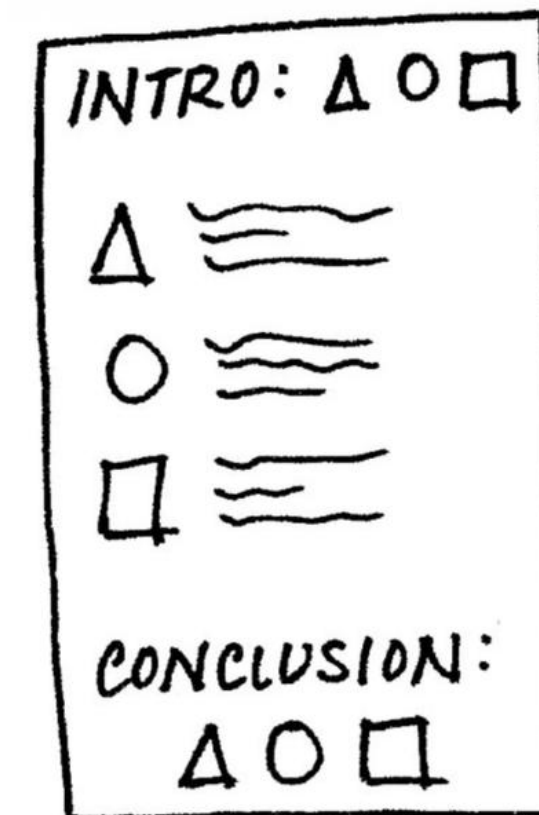
Show the evidence, analysis and supporting details.

→ Main body of the presentation

Bongo — Reinforce the Message

Summarize what the audience should remember and do.

→ Conclusion / recommendation



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Horizontal Logic

Horizontal logic refers to the continuity of the story as the audience progresses from one slide to the next.

- Each slide should contribute a distinct step in the overall argument.
- The sequence should follow a clear progression rather than presenting unrelated observations.
- Slide titles should communicate the main conclusion or message of each slide.
- Reading the titles alone should allow the audience to reconstruct the overall narrative.
- Each slide should naturally establish the context or question addressed by the following slide.

Descriptive sequence: *Sales Analysis → Regional Data → Customer Data → Pricing → Forecast → Conclusion*

Narrative sequence:

- Sales growth has slowed despite increased acquisition
- The slowdown is concentrated in the North region
- New customers account for most of the decline
- Higher acquisition costs are amplifying the problem
- The current trend is likely to continue without intervention
- Acquisition and onboarding should be redesigned

Visualization of Data, Understanding and Interpretation

Vertical Logic

Vertical logic concerns the relationship between the elements within an individual slide.

- The title, visualizations, annotations, and supporting text should communicate a consistent message.
- The slide title should clearly state the primary conclusion or intended takeaway.
- The visual evidence should provide direct support for that conclusion.
- Annotations and explanatory text should guide interpretation without introducing unrelated information.
- Every element should contribute to the central analytical message of the slide.
- Removing an element that does not support the message should not weaken the core argument.

A useful test: Can the audience understand the intended message by considering the slide title and visual evidence together?

💡 *Vertical logic ensures that all elements within a slide work together to communicate a single, coherent message*

Visualization of Data, Understanding and Interpretation

Horizontal & Vertical Logic

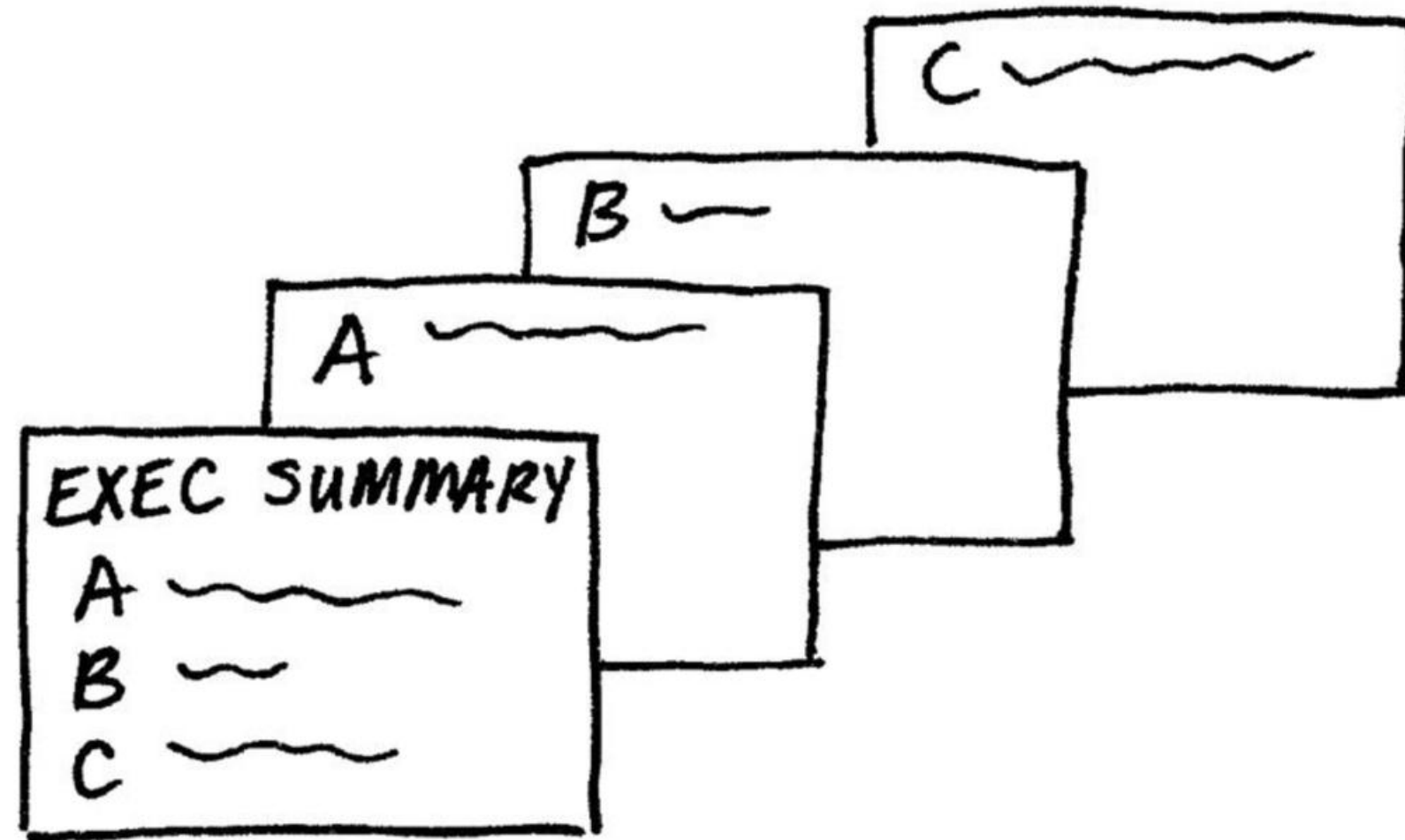


Fig 1: Horizontal Logic

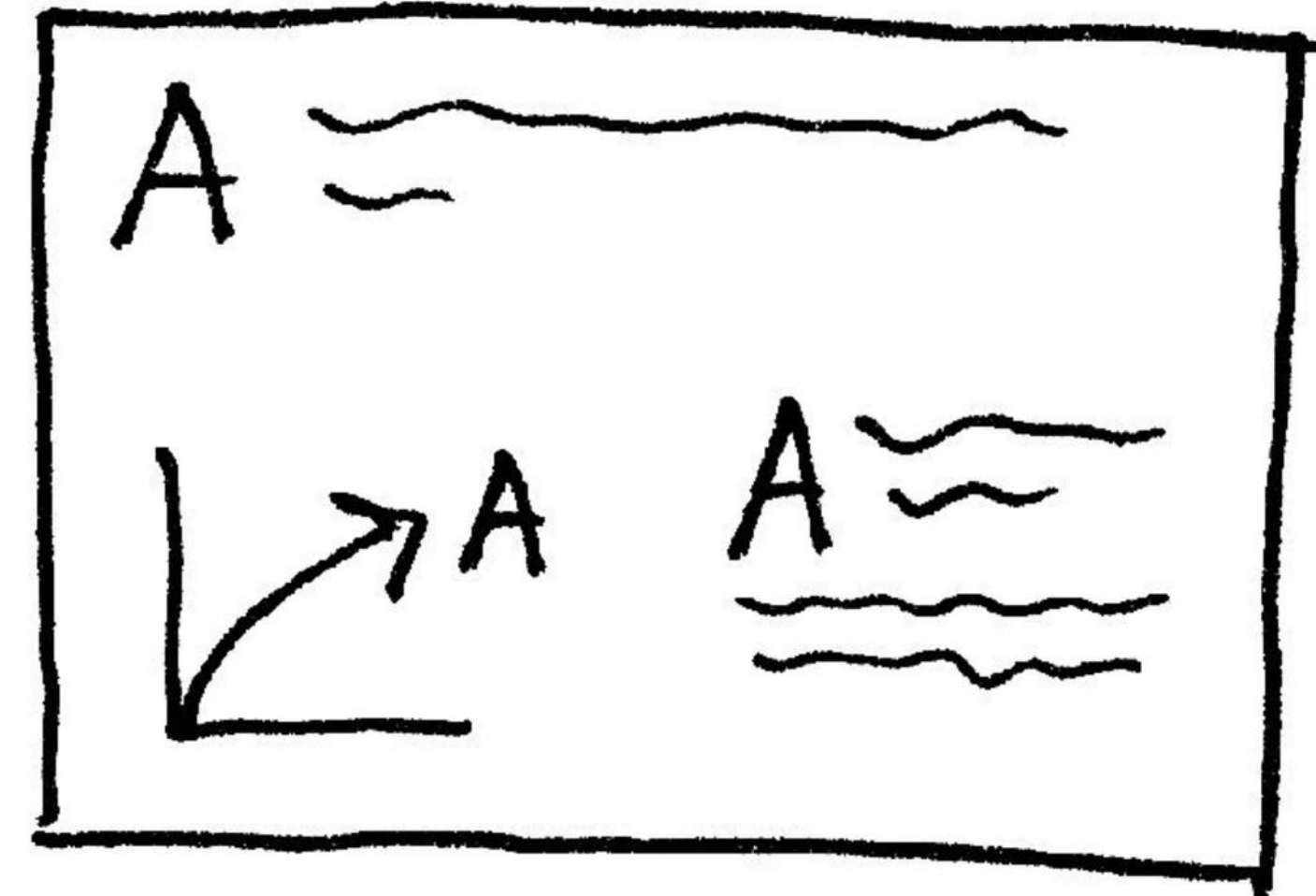


Fig 2: Vertical Logic

Visualization of Data, Understanding and Interpretation

Reverse Storyboarding + Fresh Perspective

Reverse Storyboarding

Instead of planning: *Story* → *Slides*

Work backwards: *Finished deck* → *Slide headlines* → *Overall story*

- For every slide, write down its main point.
- Then read those points in sequence.
- If they do not form a coherent narrative, something needs to be added, removed, or reordered.

Knaflic specifically describes reverse storyboarding as taking the final communication, writing down the main point of each page, and checking whether the resulting sequence resembles the intended storyboard.

Fresh Perspective

Finally, ask someone unfamiliar with the analysis to review it.

Ask them:

- What did you notice first?
- What do you think the main message is?
- What questions did you have?
- What would you remember tomorrow?

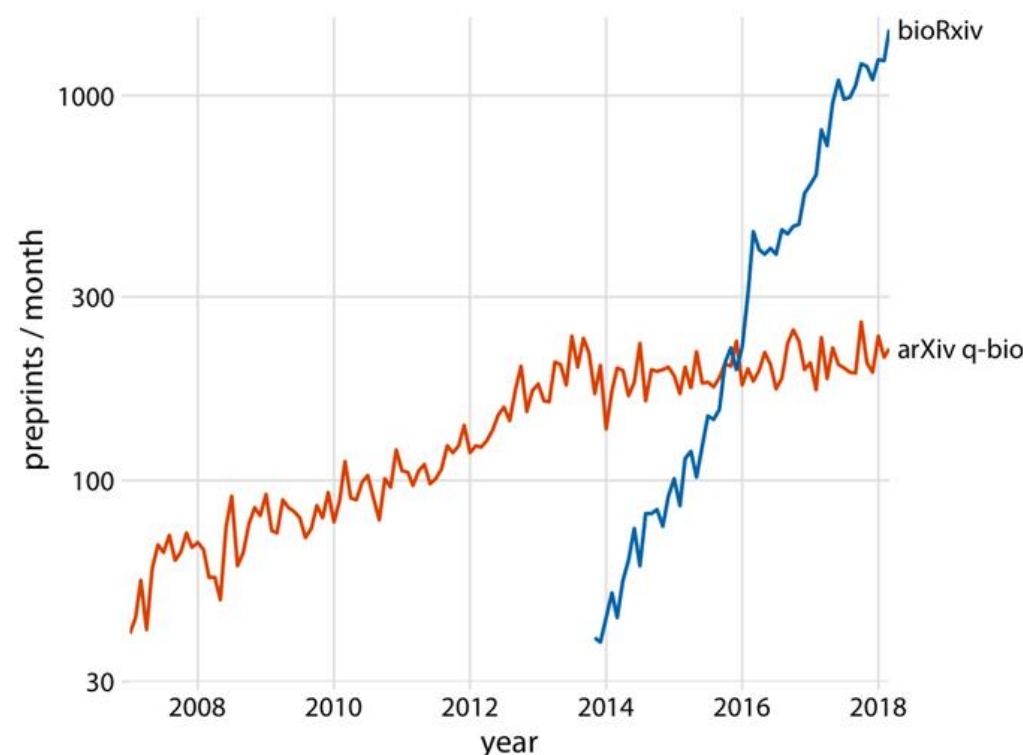
A fresh perspective helps reveal what the creator—who has become too familiar with the data—may no longer notice.

Now that we know how the story is structured, where do the visualizations themselves fit?

Visualization of Data, Understanding and Interpretation

Integrating Figures into the Narrative Structure

- A single static visualization can rarely convey an entire narrative arc on its own.
- We must design sequences of multiple figures where each chart represents a specific chapter in the overarching story.
- In a technical presentation, early slides establish the motivation and subsequent slides present the challenge and resolution.
- It is possible to condense an entire story arc into a single figure if it contains both a challenge and a resolution simultaneously.
- Condensing the narrative into one figure is highly appropriate for fast-paced environments like executive briefings or social media posts.

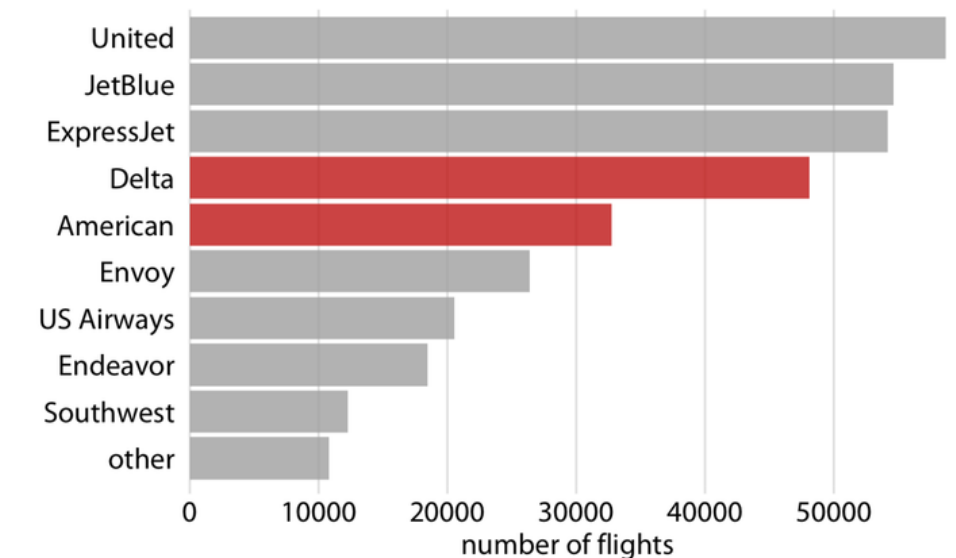
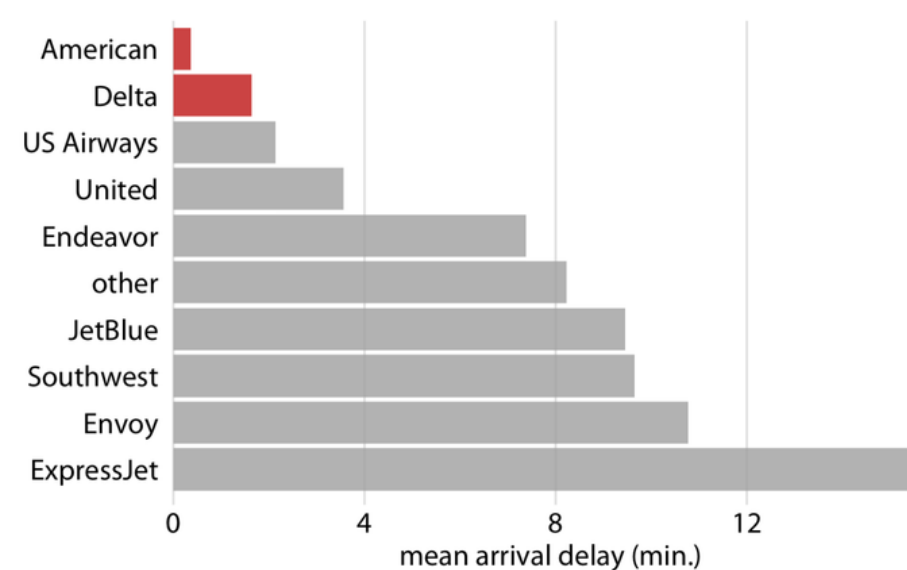
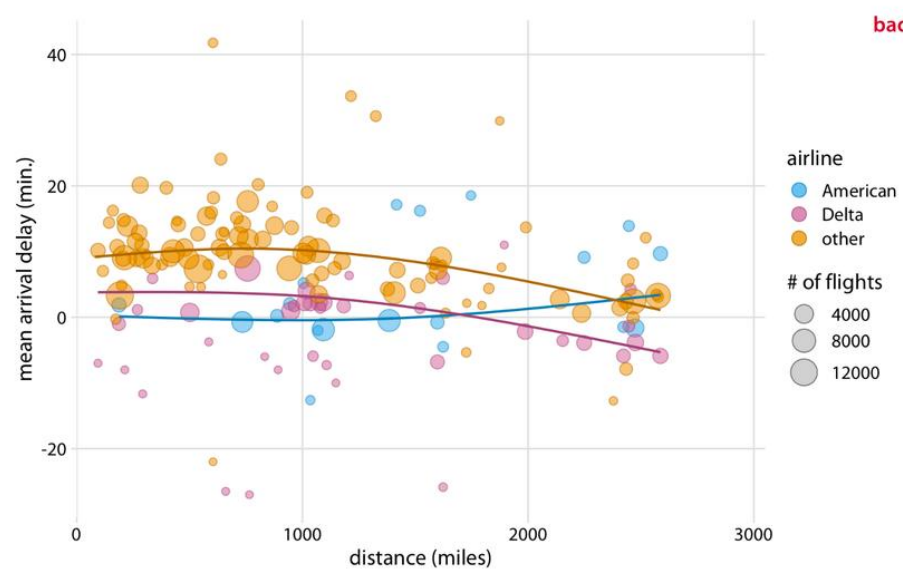


Biological preprint submission timeline of bioRxiv and arXiv q-bio to illustrate how a single multi-line graph can tell a complete story of exponential growth and leveling off.

Visualization of Data, Understanding and Interpretation

Choosing the Visual That Best Supports the Story

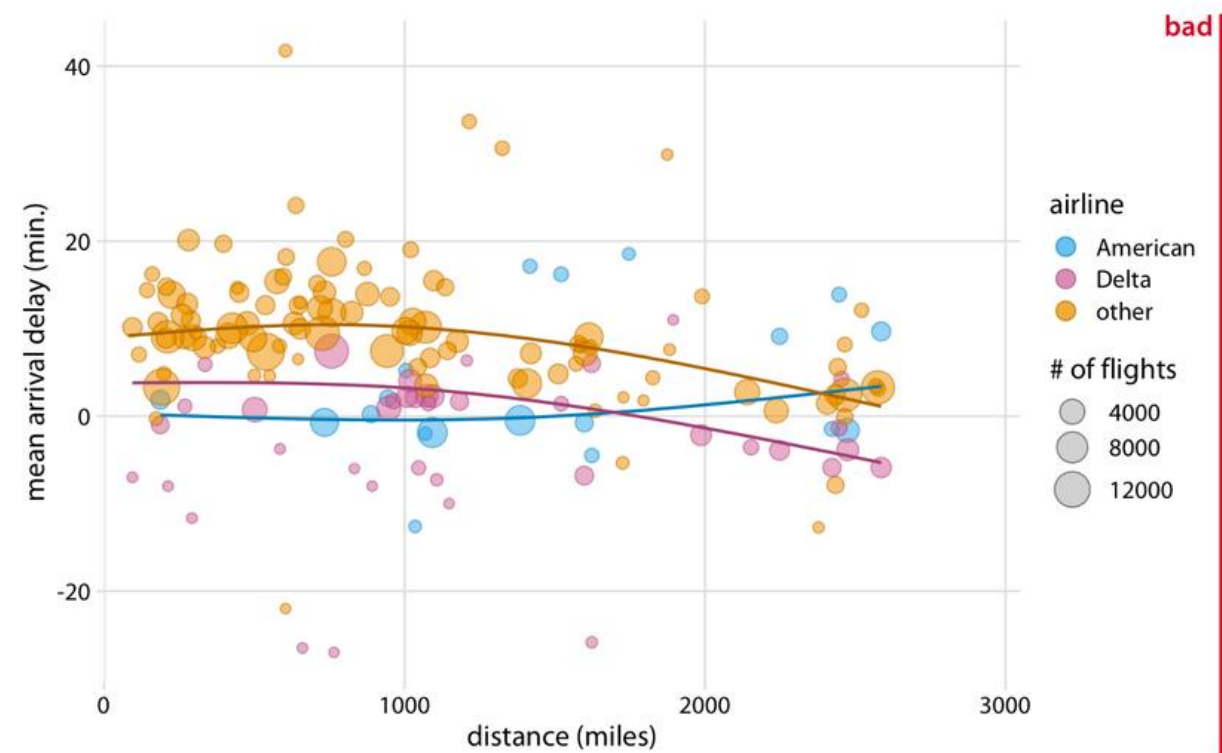
- A simple horizontal bar chart is often the absolute best way to communicate category-wise magnitude differences clearly.
- Shifting from a cluttered bubble plot to a sorted bar chart reduces cognitive processing time from minutes to seconds.
- If we are worried about hidden dimensions like flight volume, we can present a secondary bar chart directly alongside the primary figure.
- We must never let our desire to show off advanced graphing skills get in the way of delivering a clear, actionable point.
- Choosing the simpler visual format ensures that the operational message remains dominant over aesthetic styling.



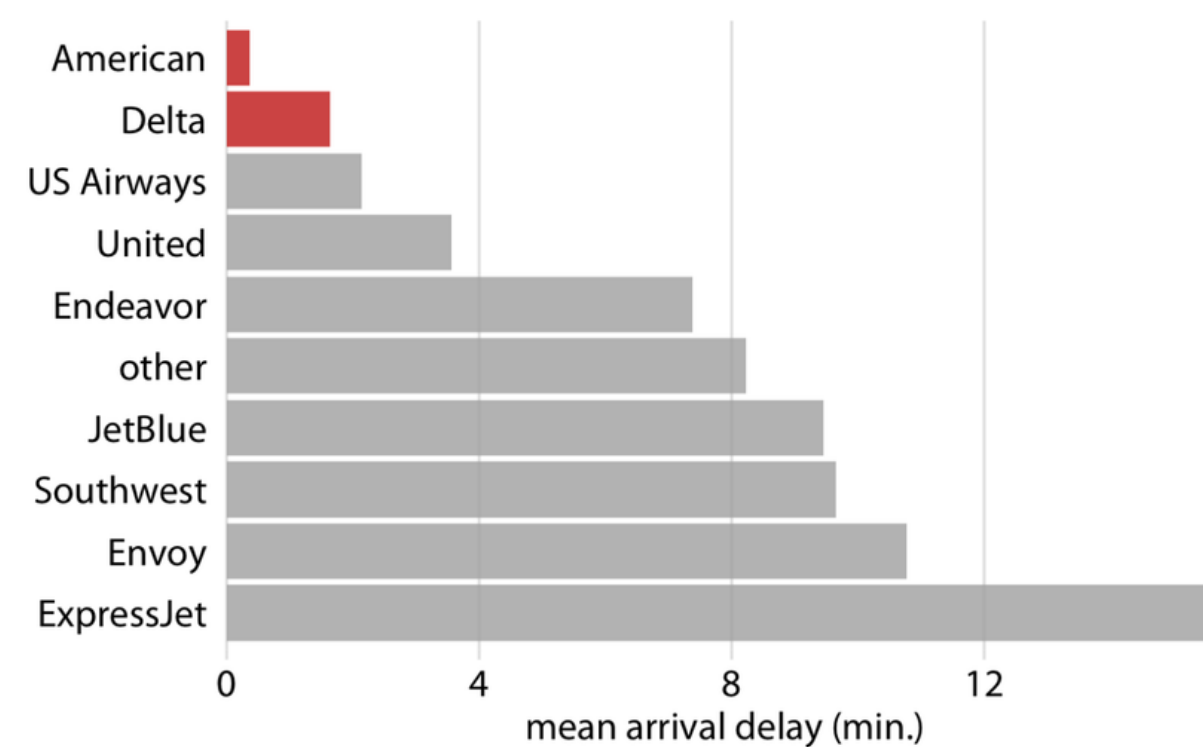
Visualization of Data, Understanding and Interpretation

Designing for "The Generals" - Simplifying Complex Displays

- Engineers must never assume that their target audience has the time or context to decode complex multi-faceted charts.
- High-resolution monitors make it easy to pile too many variables onto a single screen, resulting in severe visual noise.



The New York flights arrival delay bubble chart is incredibly impressive but takes several minutes to process.



Simplifying this figure into the direct airline bar chart communicates the core message in seconds.

Removing tangential variables and focusing on a singular message represents the primary design principle for busy stakeholders.

Visualization of Data, Understanding and Interpretation

The Fallacy of Visual Complexity in Model Interpretation

Modern data science libraries make it incredibly easy to construct highly complex, multi-dimensional visualizations that actually obscure key insights.

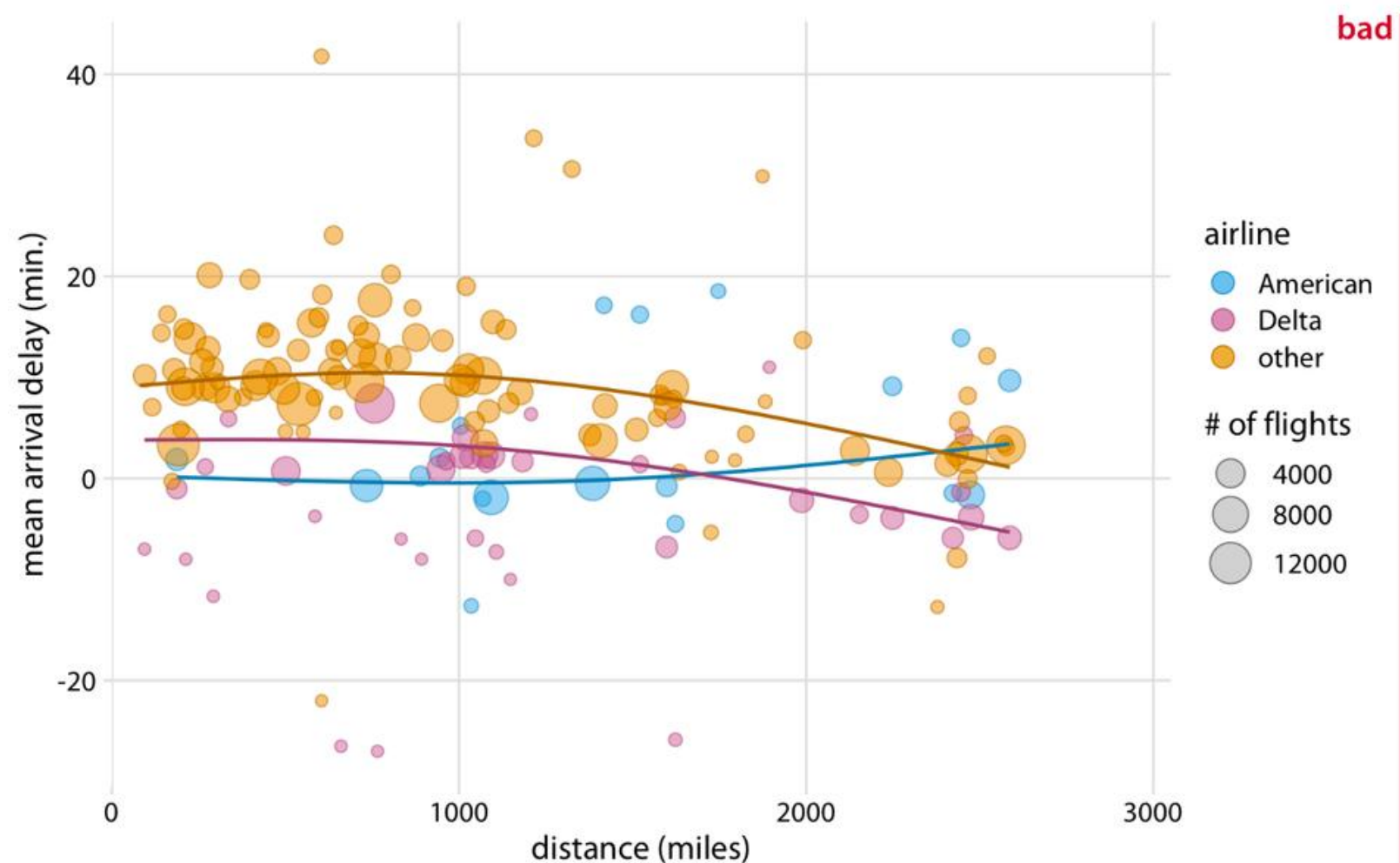


Figure demonstrates this common pitfall by plotting mean arrival delays against distance for thousands of individual flights.

This problematic figure attempts to encode airline carriers with color, flight volume with bubble sizes, and distance with spatial position simultaneously.

While this dense scatter plot looks visually impressive, it requires excessive cognitive effort to process and fails to communicate a direct message.

One must resist the temptation to pack every available database dimension into a single figure simply because their plotting libraries allow it.

Visualization of Data, Understanding and Interpretation

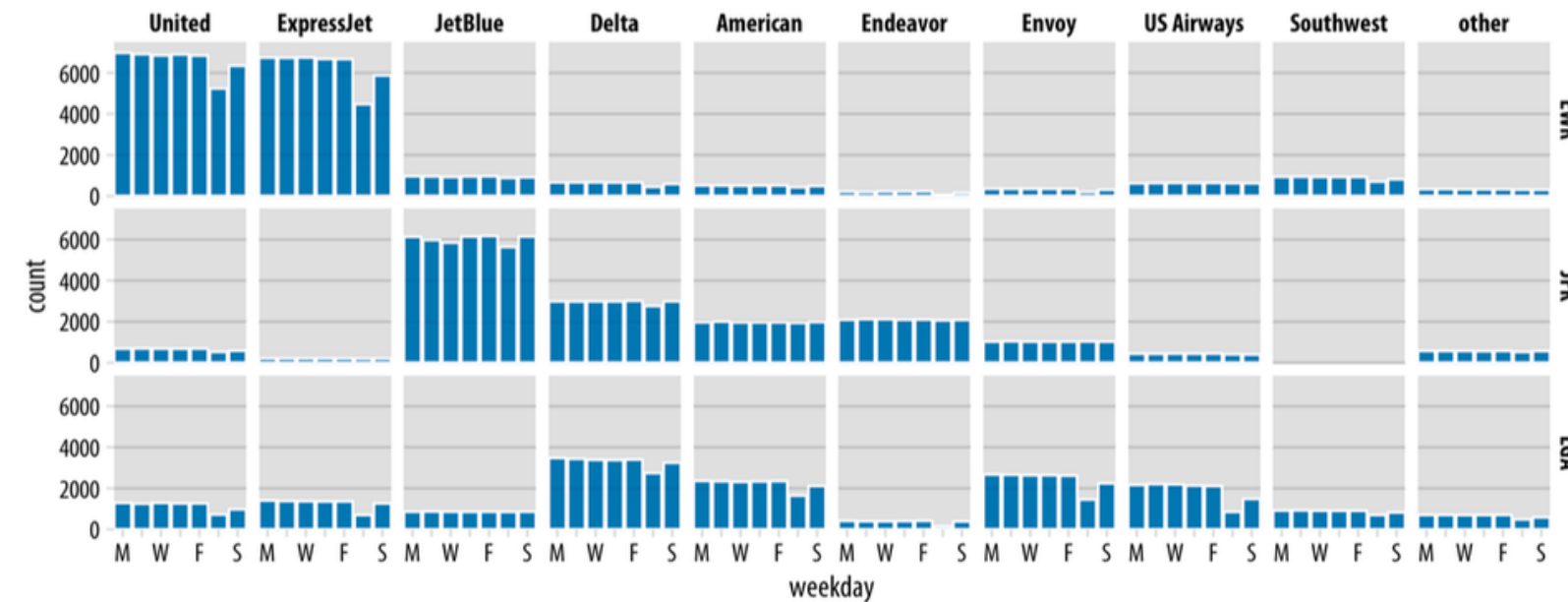
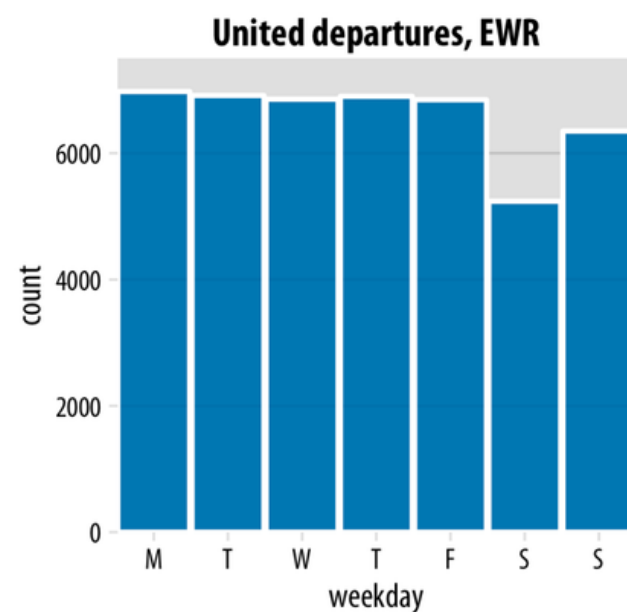
Simplifying Complex Displays Without Losing Information

- Modern visualization software makes it exceptionally easy to pile too many data dimensions onto a single coordinate grid.
- While highly complex, multi-faceted visualizations look impressive, they are highly unlikely to convey a clear story to decision-makers.
- Adding point size, point color, and regression lines simultaneously often creates visual noise that hides the main trend.
- If our primary goal is to compare two specific categories, we should not force the viewer to wade through hundreds of raw scatter points.
- Simplifying a chart does not mean lowering academic standards; it represents the ultimate refinement of visual communication.

Visualization of Data, Understanding and Interpretation

Progressive Complexity: Building Up to Small Multiples

- When we must present highly complex datasets, we should build up to the final visual layout progressively.
- Jumping straight to a highly complex grid of subplots will immediately overwhelm and disengage the audience.
- We can make complex visual grids highly digestible by first showing a single, easily understandable subplot by itself.
- Once the audience understands the axis mappings and scales of the simple chart, we can safely introduce the full multi-panel array.



- This step-by-step introduction ensures that the audience maintains complete comprehension and confidence throughout the presentation.

Visualization of Data, Understanding and Interpretation

Mapping a Derived Quantitative Path

A visualization may communicate not only individual values but also the relationship between successive quantities.

Example: Initial value → Change → Intermediate state → Final value

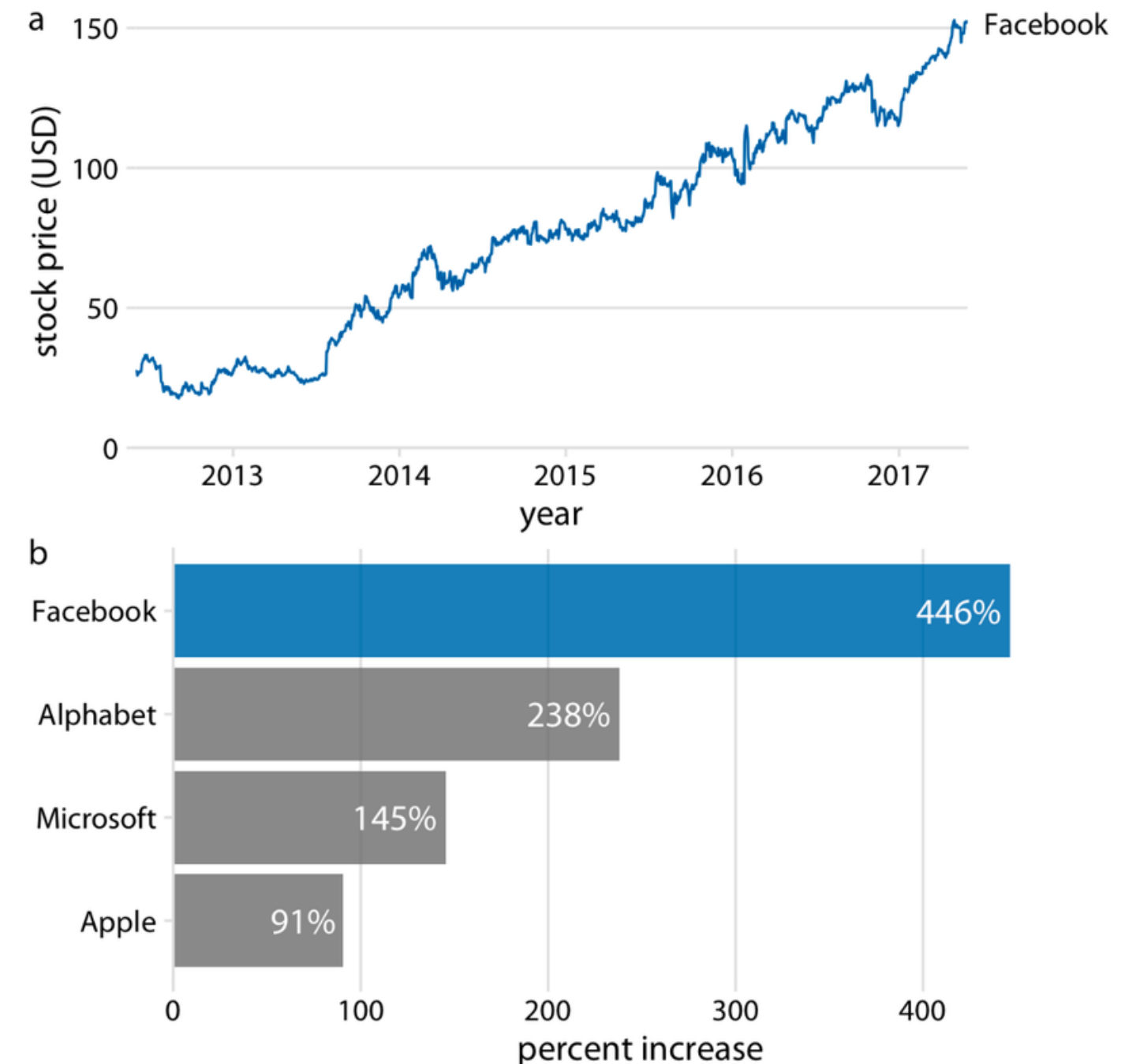
This can help the audience understand:

- How a final value was obtained
- Which factors contributed to the change
- Where gains or losses occurred
- How individual components combine into an outcome

Narrative Role

The visual path mirrors the reasoning path.

Instead of asking the audience to mentally reconstruct the calculation, the visualization can make the sequence explicit.

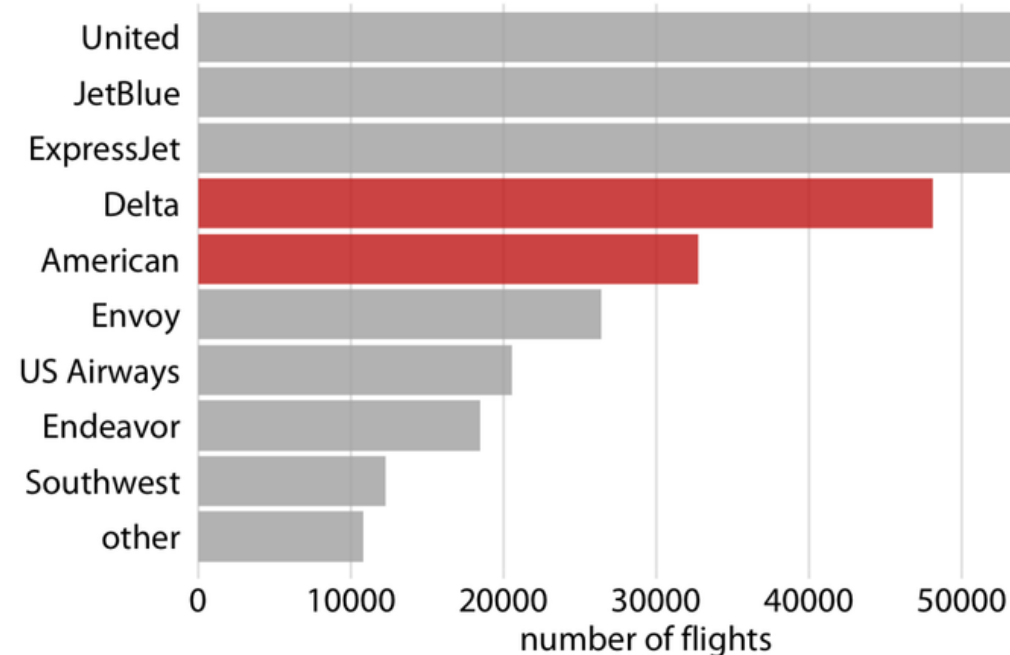
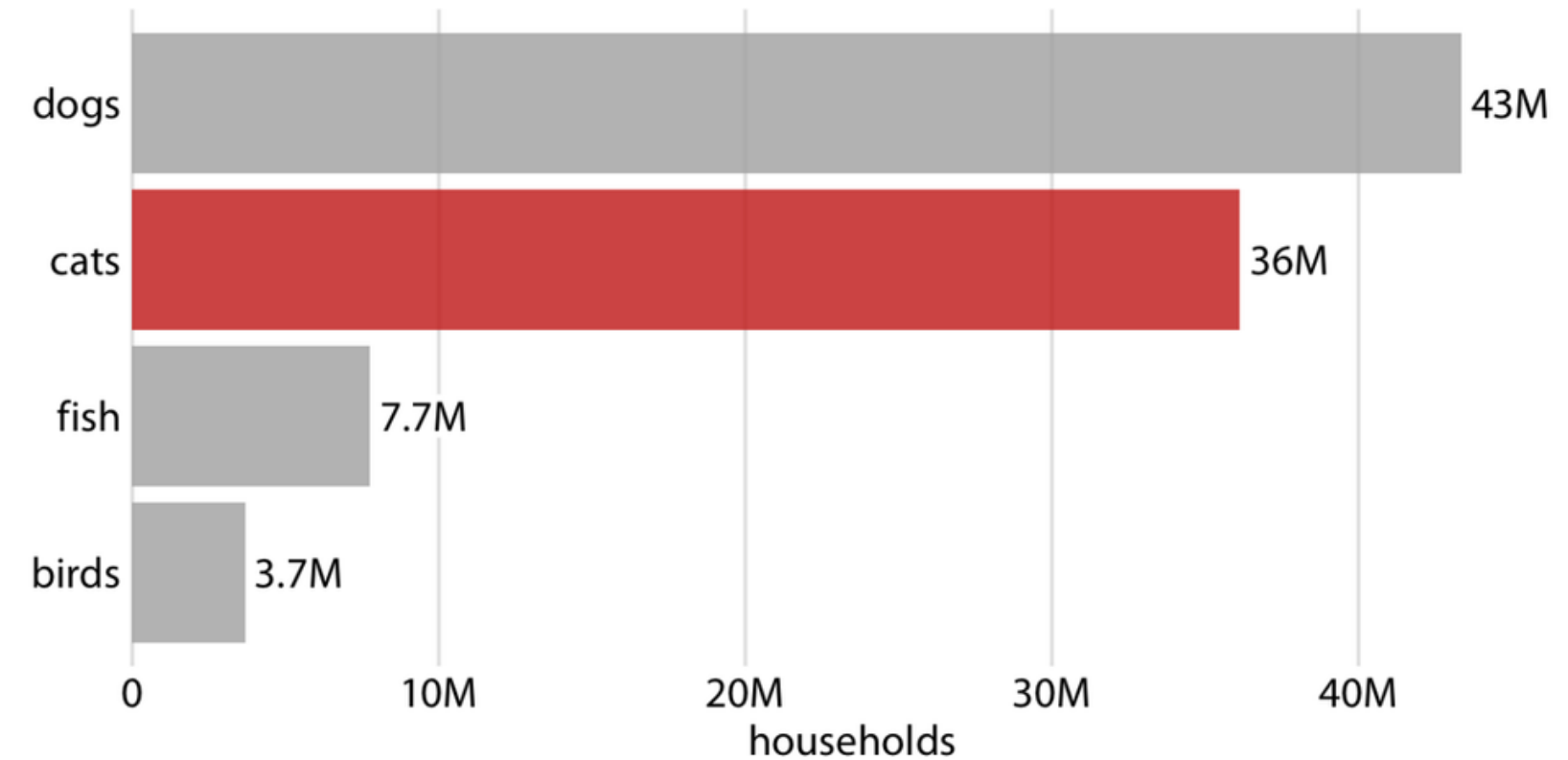


Part (a) displays raw stock prices over time, and Part (b) transitions to a derived bar plot showing total percent increase, guiding the audience along a logical derived path.

Visualization of Data, Understanding and Interpretation

Making Visualizations Memorable Without Sacrificing Clarity

- While simple and clean bar charts are easy to read, they can sometimes look highly generic and unmemorable.
- If we show an audience multiple similar-looking bar charts in quick succession, they will have a very hard time remembering individual points.
- We can make figures much more memorable by adding visual elements that directly reflect the core topic of the dataset.
- Incorporating distinct pictograms and isotype structures allows the audience to instantly recognize the subject matter of the graph.
- Memorable visualizations ensure that key statistical trends remain locked in the viewer's long-term memory after the presentation ends.

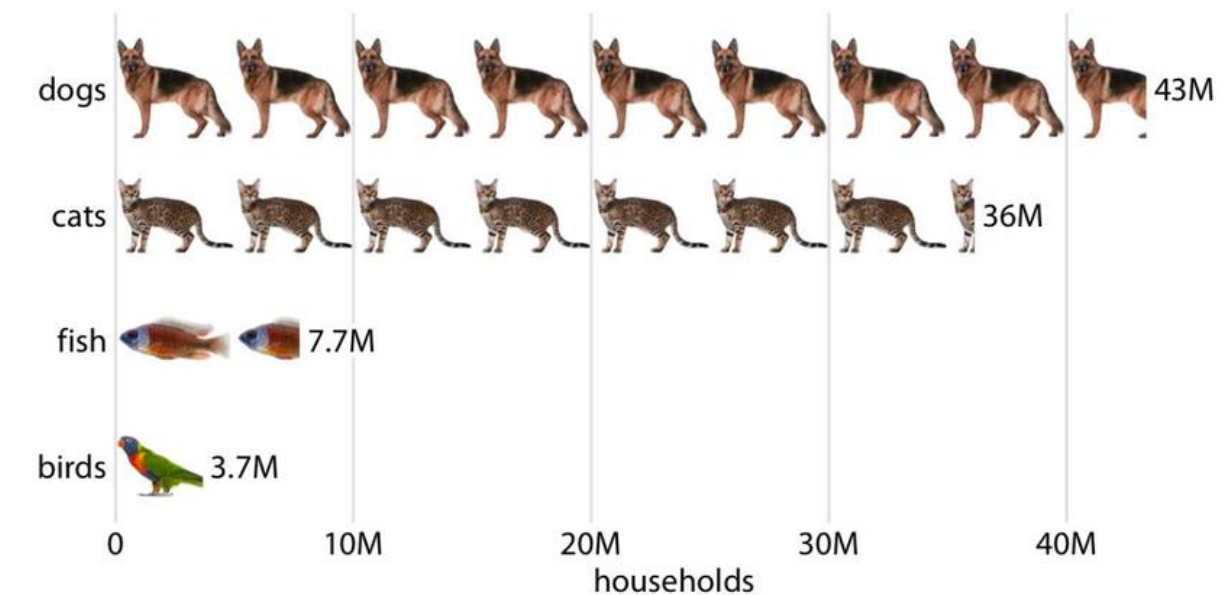
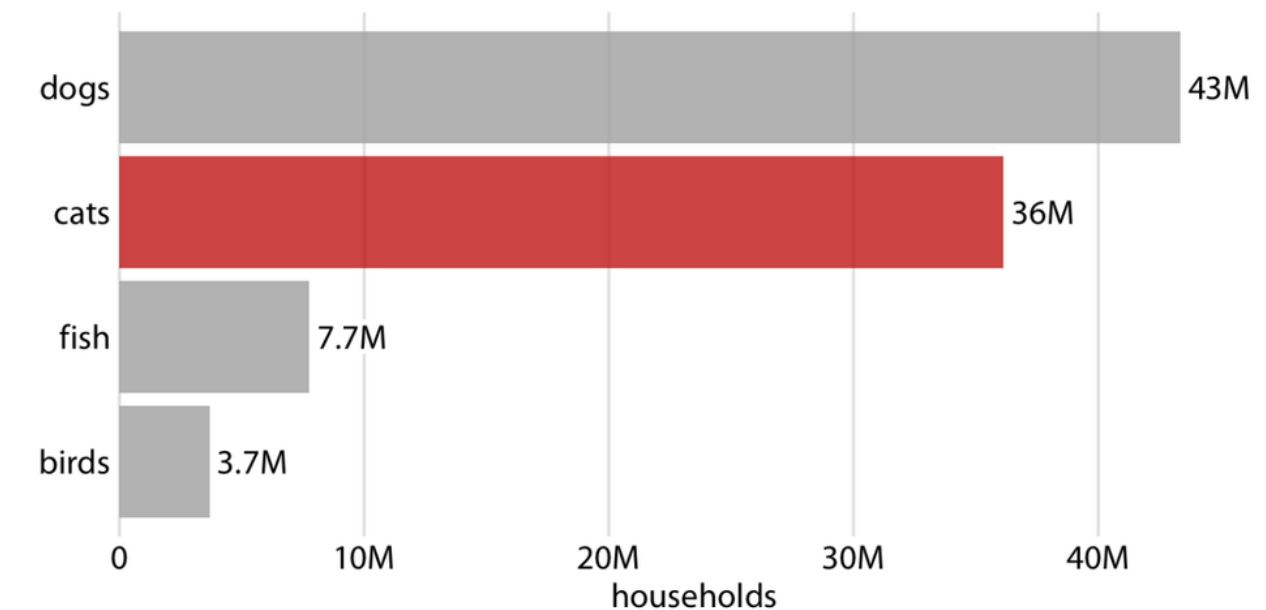


visual structure is virtually identical, making them difficult to distinguish.

Visualization of Data, Understanding and Interpretation

Making Visualizations Memorable Without Sacrificing Clarity

- An isotype graph replaces traditional solid bars with repeated copies of a highly recognizable topic icon.
- Each complete icon in an isotype graph represents a defined, scaled mathematical quantity within the dataset.
- Psychological research demonstrates that repeated icons inside the bars enhance visual engagement without creating cognitive distraction.
- We must avoid simply pasting decorative clip art around our axes, as this practice actively distracts from data-ink ratios.
- Isotype designs represent an elegant balance between technical, quantitative accuracy and creative, visual storytelling.

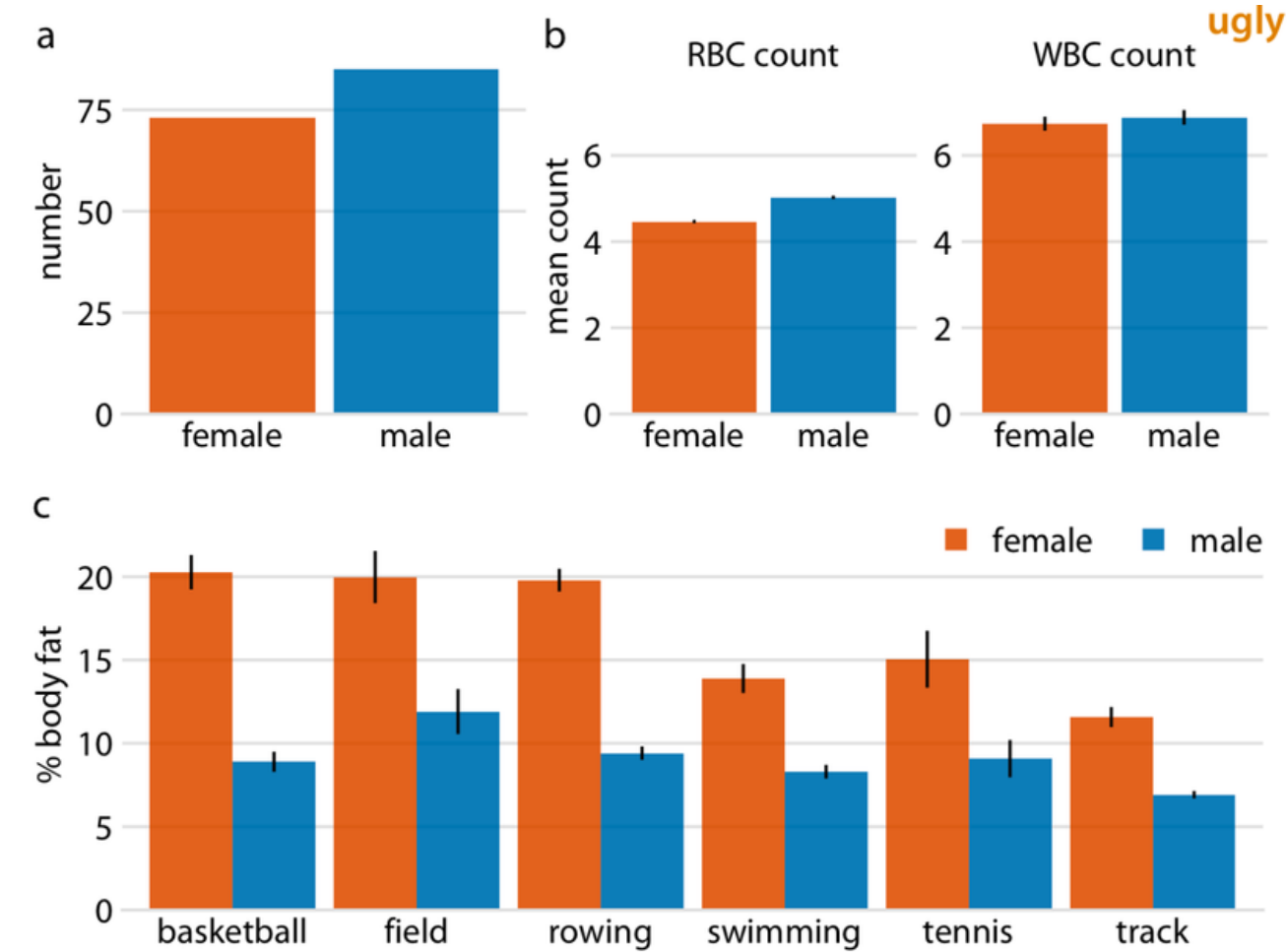


Replacing solid columns with scaled icons makes a chart instantly memorable.

Visualization of Data, Understanding and Interpretation

Visual Consistency and Preventing Repetitive Fatigue

- While consistency in color and typography is essential, we must avoid using the exact same chart type for every single analysis.
- If every slide features an identical bar chart, the different parts of our technical story will blur together in the viewer's mind.
- We should purposefully shift our visualization approach across slides by utilizing line graphs, scatter plots, and boxplots in sequence.
- This visual variation keeps the audience highly engaged and signals clearly where one analysis ends and a new one begins.
- Designing visually distinct slides prevents cognitive fatigue and helps the audience follow multi-part scientific arguments.



Demonstrating an overly repetitive and confusing compound layout where three sub-figures all utilize identical bar charts, making it difficult to process different results.

Visualization of Data, Understanding and Interpretation

Calibrating the Scale of Your Visual Story

- The total number of figures used in a presentation or report must be carefully calibrated to the specific communication venue.
- For brief communication channels like blog posts or social media, we should condense our story into a single, high-impact figure.
- For peer-reviewed scientific papers, we should limit our core presentation to between three and six key figures to avoid clutter.
- Any additional supporting figures should be moved to an appendix or supplementary materials section to protect the core narrative.
- Calibrating figure counts prevents wearing out our audience and ensures that critical insights receive maximum prominence.

FIGURE COUNT CALIBRATION: GUIDELINES FOR ACADEMIC COMMUNICATION	
COMMUNICATION TYPE	CALIBRATED FIGURE ALLOCATION & KEY OBJECTIVES
 Brief Communication (e.g., Blogs, Social Media)	 1 HIGH-IMPACT FIGURE (Maximize Clarity & Immediate Resonance) Condense the entire narrative into a single, compelling image. Maintain immediate reader attention and maximize message resonance with minimal effort.
 Scientific Papers & Detailed Reports (CORE PRESENTATION) (CORE PRESENTATION)	 3 to 6 KEY FIGURES (Avoid Clutter) A limited set of essential visualizations illustrating the main arguments. Must be carefully calibrated to the venue to avoid clutter and Protect the core narrative.
 SUPPLEMENTARY MATERIALS & APPENDIX (Protect Core Narrative)	 UNLIMITED SUPPORTING FIGURES (Provide Robustness) All additional data, detailed evidence, and robustness checks are moved here. Ensures the main text remains uncluttered and key insights prominent for the reader.
	WHY CALIBRATE? Prevents audience weariness and ensures that CRITICAL INSIGHTS receive MAXIMUM PROMINENCE . (Make “CRITICAL INSIGHTS” and MAXIMUM PROMINENCE” pop)

Source: Generalized Academic Communication Guidelines & Best Practices. Counts are indicative; specific context may vary.

Visualization of Data, Understanding and Interpretation

The Complete Visualization Process: From Context to Story

The Complete Storytelling with Data Process

1. *Understand the Context:* Who is the audience?, what do they need?, what decision or action matters?
2. *Choose an Appropriate Display:* Select the visual that best communicates the intended relationship.
3. *Eliminate Clutter:* Remove elements that do not contribute meaningful information.
4. *Direct Attention:* Use contrast, position, size and color strategically.
5. *Think Like a Designer:* Apply accessibility, alignment, hierarchy and aesthetics.
6. *Tell the Story:* Construct a clear narrative with Beginning → Middle → End and validate it through: Horizontal Logic → Vertical Logic → Reverse Storyboarding

Visualization of Data, Understanding and Interpretation

AI-Generated Data Reports and Summaries

Large language models can transform analytical outputs into natural-language summaries, reducing the effort required to communicate findings.

AI systems can assist with:

- Data summarization: converting large tables or analytical outputs into concise descriptions.
- Pattern description: identifying and describing apparent trends, differences, and relationships.
- Chart interpretation: generating textual explanations of visualizations.
- Report generation: combining multiple findings into structured reports.
- Executive summaries: converting detailed analyses into audience-oriented takeaways.
- Narrative drafting: organizing findings into a coherent sequence of observations and conclusions.

Example:

Analytical output: *Revenue increased 18% → Customer acquisition increased 25% → Churn increased 7% → Marketing expenditure increased 31%*

AI-generated summary: *Revenue growth was accompanied by substantially higher acquisition and marketing costs, while increasing churn may indicate emerging concerns about the sustainability of this growth.*

Visualization of Data, Understanding and Interpretation

What Does an AI-Generated Data Story Actually Do?

- AI systems can first identify patterns such as trends, differences, relationships, and anomalies in the data.
- They can then describe these patterns using concise natural-language statements.
- The system may also attempt to interpret what these patterns could mean in the given context.

A pattern can be observed without necessarily establishing its cause or significance.

Example:

Sales and marketing expenditure increased during the same period.

This does not necessarily mean- the increase in marketing expenditure caused sales to increase.

💡 *AI can efficiently describe patterns, but interpreting their meaning requires context and human judgment*

Visualization of Data, Understanding and Interpretation

Where AI-Generated Data Stories Can Go Wrong

Automation does not guarantee analytical validity. An AI-generated report may produce a fluent and convincing narrative while making errors at several levels.

1. Data-level errors

- Incorrect values or calculations
- Missing observations
- Misinterpreted variables
- Incorrect aggregation

2. Pattern-level errors

- Treating random variation as a meaningful trend
- Overlooking important subgroups
- Identifying patterns that are not statistically meaningful

3. Context-level errors

- Ignoring the domain context
- Using an inappropriate time period
- Missing relevant comparison groups
- Failing to account for external events

4. Narrative-level errors

- Overstating the importance of a finding
- Presenting correlation as causation
- Expressing uncertainty as certainty
- Producing a conclusion unsupported by the evidence

5. Communication-level errors

- Wrong emphasis for the intended audience
- Excessive detail or excessive simplification
- A persuasive narrative that hides important limitations

💡 *Fluent language is not evidence of analytical correctness*

Visualization of Data, Understanding and Interpretation

MCQ

Q1. A presentation contains accurate charts, but the slide titles are merely “Sales,” “Customers,” “Regions,” and “Forecast.” When the titles are read sequentially, no clear argument emerges. Which principle is primarily violated?

- A. Vertical logic
- B. Horizontal logic
- C. Progressive complexity
- D. Visual consistency

Visualization of Data, Understanding and Interpretation

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Answer: B

Visualization of Data, Understanding and Interpretation

MCQ

Q2. An AI system observes that two variables increased during the same period and generates the statement: “The increase in X caused the increase in Y.” What is the most important issue with this interpretation?

- A. The visualization may contain excessive graphical complexity.
- B. The system has converted an observed association into a causal claim.
- C. The narrative does not contain sufficient repetition.
- D. The selected variables should necessarily be represented using different charts.

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Visualization of Data, Understanding and Interpretation

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Q3. A designer creates a sequence of charts in which every chart uses identical colors, layouts, and visual forms, even though the charts address different analytical questions. Which principle provides the strongest reason to reconsider this approach?

- A. Consistency should never be used across multiple visualizations.
- B. Different analytical messages may require purposeful variation in visual form.
- C. Memorable visualizations should always use distinctive graphical elements.
- D. Complex analytical questions should always be represented using small multiples.

Visualization of Data, Understanding and Interpretation

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Answer: B

Visualization of Data, Understanding and Interpretation

MCQ

Q4. During reverse storyboarding, the presenter writes the main takeaway from every slide and reads these statements in sequence. The resulting sequence jumps repeatedly between unrelated findings. What does this most directly indicate?

- A. The individual visualizations contain insufficient detail.
- B. The presentation lacks sufficient visual consistency.
- C. The overall narrative structure requires revision.
- D. The presentation contains too many annotations.

Visualization of Data, Understanding and Interpretation

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Visualization of Data, Understanding and Interpretation

MCQ

Q5. A visualization initially presents twelve variables simultaneously. The analyst first shows the primary relationship, then introduces relevant comparisons, and finally provides the detailed multidimensional view. What storytelling principle does this approach primarily demonstrate?

- A. Horizontal logic
- B. Reverse storyboarding
- C. Progressive complexity
- D. Narrative repetition

Visualization of Data, Understanding and Interpretation

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Answer: C



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